Exam-day case list — viva guide by diagnosis

M.S. Ophthalmology practical examination · Tamil Nadu Dr M.G.R. Medical University · 16 October 2026

Every diagnosis on the department's colour-coded list, grouped as Glaucoma, Retina, Cornea and Other. Start with card E0, then read every card marked ★ — those patients have already arrived.

★ patient already arrived (new cards) or kept last year (glaucoma cards) · Viva card · Long case · Short case

Contents

No.	Card	Type	★ Note
E0	Tomorrow's case list sorted by diagnosis — and which card to read for each	Index	★ 19 already arrived
G1	Primary open-angle glaucoma, including the glaucomatous optic disc	Long + short case	★ Kept 6 times (one with pseudoexfoliation glaucoma)
G4	Primary angle-closure disease — PACS, PAC and PACG, including after laser iridotomy	Long case	★ Kept 4 times (one as the long case)
G2	POAG with trabeculectomy — bleb assessment and the failed bleb	Long case	—
G5	Neovascular glaucoma, including rubeosis iridis	Long case	★ Kept once
G7	CRVO with POAG, including hemi-CRVO with POAG	Short case	★ Kept twice (CRVO with POAG as a glaucoma case; hemi-CRVO with POAG as a fundus case)
ER1	Age-related macular degeneration, including disciform scar and drusen	Viva card	—
ER2	Retinitis pigmentosa and Stargardt disease	Viva card	★ Patient already arrived: S.No 9
ER3	Branch retinal vein occlusion and old CRVO, including the sclerosed venule	Viva card	★ Patient already arrived: S.No 46, 52
ER4	Diabetic retinopathy: severe NPDR with CSME, PDR, and the patient for PRP	Viva card	—
ER5	Uveal colobomas — retinochoroidal, iris and optic disc coloboma	Viva card	★ Patient already arrived: S.No 6
ER6	Full-thickness macular hole	Viva card	★ Patient already arrived: S.No 38
ER7	Pathological myopia and myopic maculopathy	Viva card	—
EC1	Keratoplasty cases: failed or opaque graft, after penetrating or therapeutic keratoplasty, and DSEK	Viva card	★ Patient already arrived: S.No 17, 32
EC2	Pseudophakic bullous keratopathy	Viva card	★ Patient already arrived: S.No 13, 16
EC3	Corneal opacity: nebula, macula, leucoma and adherent leucoma, including a leucomatous cornea with band keratopathy	Viva card	—
EC4	Bacterial corneal ulcer with hypopyon	Viva card	—
EC5	Corneal stromal dystrophies — granular dystrophy types 1 and 2 (Avellino), with lattice, macular and Fuchs as contrasts	Viva card	—
EC6	Corneal degenerations — climatic droplet keratopathy and band-shaped keratopathy, with arcus, Vogt girdle, Salzmann, Terrien and pellucid	Viva card	—
EC7	Keratoconus, including the listed "Lisch nodules" (neurofibromatosis type 1) and acute hydrops	Viva card	—
EO1	Optic atrophy, including post-neuritic optic atrophy	Viva card	★ Patient already arrived: S.No 2
EO2	Congenital anterior segment anomalies — microcornea, corectopia, and limbal dermoid and lipodermoid	Viva card	★ Patient already arrived: S.No 6

No.	Card	Type	★ Note
EO3	Aphakia and zonular (lamellar) cataract	Viva card	★ Patient already arrived: S.No 29
EO4	Ectopia lentis — subluxated and dislocated lens	Viva card	★ Patient already arrived: S.No 39
EO5	Ciliary staphyloma	Viva card	—

E0 Tomorrow's case list sorted by diagnosis — and which card to read for each

Index · 53 patients on the colour-coded list · ★ 19 already arrived · read the ★ cards first

How to use this file tonight

1. **First, every ★ card** — these patients have already arrived:
   - G1 (five POAG patients);
   - G4, G2, G7;
   - EC1, EC2;
   - ER2, ER3, ER6;
   - EO1, EO2, EO3, EO4.
2. **Then the remaining cards in each section.**
3. For each card read three parts:
   - "What to look for and present";
   - the say-it box;
   - the viva questions.

The traps and recall boxes are for the last revision in the morning.

1. **Glaucoma** uses the department's finished glaucoma cards (G1, G2, G4, G5, G7), already fact-checked. **Retina, cornea and other** use the new viva cards (ER, EC, EO).

Glaucoma — 16 patients

S.No	Diagnosis as listed	Card	Arrived
15	Both eyes POAG	G1	★
20	Both eyes POAG	G1	★
21	Both eyes POAG	G1	★
22	Both eyes POAG	G1	★
47	Both eyes POAG	G1	★
52	RE POAG (CD 0.9); LE old CRVO	G7, ER3	★
14	RE bleb; post-trabeculectomy	G2	★
23	RE PACG	G4	★
26	Both eyes POAG; after trabeculectomy	G1, G2	—
34	RE flat bleb after trabeculectomy; both eyes POAG; both eyes PDR; LE old BRVO	G2, G1, ER4, ER3	—
48	RE bleb + PCIOL, surgical PI; LE aphakia; both eyes 0.9 cupping	G2, EO3	—
25	Both eyes PACG	G4	—
31	Both eyes PACG	G4	—
4	LE NVG	G5	—
5	LE NVG	G5	—
24	LE NVG	G5	—

Retina — 17 patients

S.No	Diagnosis as listed	Card	Arrived
9	Both eyes RP + Stargardt disease	ER2	★
46	LE superotemporal BRVO with old sclerosed venule	ER3	★
38	LE full-thickness macular hole	ER6	★
1	Both eyes age-related macular degeneration	ER1	—
8	LE disciform scar	ER1	—
36	Both eyes disciform scar with scattered drusen	ER1	—
3	Both eyes retinitis pigmentosa	ER2	—
27	Both eyes retinitis pigmentosa	ER2	—

S.No	Diagnosis as listed	Card	Arrived
37	Both eyes optic atrophy, retinitis pigmentosa, Stargardt dystrophy	ER2, EO1	—
7	RE inferotemporal BRVO	ER3	—
43	RE superotemporal BRVO with sclerosed venule	ER3	—
11	Both eyes severe NPDR with CSME	ER4	—
42	Both eyes PDR with CSME	ER4	—
51	Both eyes PDR (for PRP)	ER4	—
18	Both eyes choroidal coloboma	ER5	—
45	Both eyes retinochoroidal coloboma	ER5	—
41	Both eyes myopic fundus with myopic maculopathy	ER7	—

Cornea — 14 patients

S.No	Diagnosis as listed	Card	Arrived
13	Pseudophakic bullous keratopathy	EC2	★
16	LE pseudophakic bullous keratopathy	EC2	★
17	RE PCIOL; LE DSEK; LE PKP	EC1	★
32	LE failed keratoplasty; opaque graft	EC1	★
10	LE failed keratoplasty	EC1	—
12	After therapeutic keratoplasty; failed corneal graft	EC1	—
33	RE opaque corneal graft	EC1	—
50	RE after PKP; opaque graft	EC1	—
19	RE leucomatous corneal opacity; LE adherent leucoma	EC3	—
53	LE bacterial corneal ulcer with hypopyon	EC4	—
28	Both eyes granular corneal dystrophy	EC5	—
35	Both eyes climatic droplet keratopathy; both eyes PCIOL	EC6	—
40	RE band-shaped keratopathy with leucomatous cornea	EC6, EC3	—
44	Both eyes keratoconus “with Lisch nodules” (verify)	EC7	—

Other — 6 patients

S.No	Diagnosis as listed	Card	Arrived
2	Both eyes post-neuritic optic atrophy	EO1	★
6	RE microcornea, corectopia; LE iris coloboma	EO2, ER5	★
29	RE aphakia; LE zonular cataract	EO3	★
39	LE dislocated lens	EO4	★
49	Lipodermoid + disc coloboma	EO2, ER5	—
30	Both eyes ciliary staphyloma	EO5	—

Not matched

Two patients were added by hand at the end of the name list (S.No 54 and 55). Their diagnoses are not on the colour-coded list — check on the day and use the card for that diagnosis.

Abbreviations used on the list

- Glaucoma and lens: POAG primary open-angle glaucoma; PACG primary angle-closure glaucoma; NVG neovascular glaucoma; PI peripheral iridectomy; PCIOL posterior chamber intraocular lens.
- Retina: CRVO and BRVO central and branch retinal vein occlusion; RP retinitis pigmentosa; NPDR and PDR non-proliferative and proliferative diabetic retinopathy; CSME clinically significant macular oedema; FTMH full-thickness macular hole.
- Cornea: KP keratoplasty; PKP penetrating keratoplasty; TKP therapeutic keratoplasty; DSEK Descemet’s stripping endothelial keratoplasty; CDK climatic droplet keratopathy.

- Eyes: RE, LE and BE — right eye, left eye, both eyes.

Read more: Source: the department's colour-coded patient list (glaucoma orange, cornea blue, retina green, other uncoloured) and the ticked arrival list. Patients appear by serial number only. Diagnoses are written as printed; verify unclear abbreviations on the day.

G1 Primary open-angle glaucoma, including the glaucomatous optic disc

★ Kept 6 times at JEH last year (one with pseudoexfoliation glaucoma) · Long case and short case

At a glance

- **Primary open-angle glaucoma (POAG)**: a chronic, progressive optic neuropathy with an **open, normal-looking angle** and reduced outflow; usually asymptomatic, found on routine examination or follow-up.
- Commonly kept: stable signs (cupped disc, measurable pressure, open angle). The **glaucomatous disc** is also a short case.
- Examiners test the **disc description in order**, intraocular pressure (IOP) with method, time and central corneal thickness (CCT), gonioscopy, target pressure and drugs.

Case format — history

Chief complaint: "Mr X, a 70-year-old man, has come for regular follow-up for glaucoma" — or "...came with complaints of defective vision in both eyes for 1 year".

House wording: "Patient was apparently normal ___ years ago, then had defective vision in both eyes, gradual in onset, progressive in nature, not associated with pain or headache. He was found to have increased pressure elsewhere and started on eye drops (1 drop at bedtime)."

What to ask	Why it matters
History of present illness: onset, progression; blurring at one time of day	Gradual, often silent until late; blurring matches the IOP peak
Frequent change of near glasses	Pressure on zonules and ciliary body impairs accommodation; field loss mistaken for poor vision
Night blindness	Defective dark adaptation; worse on miotics
Old records: baseline IOP, fields, optical coherence tomography (OCT)	Target pressure; progression
Negative history: no headache with vomiting, coloured haloes, redness, watering or pain	Rules out **angle closure** (haloes = corneal oedema in a pressure rise)
No trauma; no steroids (drops, nasal spray, tablets)	Rules out **angle-recession** and **steroid-induced glaucoma**
No sudden loss of vision	Rules out **central retinal vein occlusion**, an acute IOP rise, snuff-out after trabeculectomy
Past, drug and surgical: diabetes, hypertension, thyroid disease, migraine, Raynaud's, peripheral vascular disease, stroke, head injury, blood loss	Risk factors; vasospasm matters most in normal-tension glaucoma (NTG); shock or head injury can mimic it
Asthma, chronic obstructive pulmonary disease (COPD), heart failure, heart block; sulpha or other drug allergy, kidney disease	No beta-blocker; no carbonic anhydrase inhibitor (CAI)
Eye surgery or laser: cataract, trabeculectomy, refractive, peripheral iridotomy (PI)	Status (pseudophakia, bleb); refractive surgery thins the cornea: IOP reads low
Treatment history: "E/D (eye drops) timolol 0.5% BD (twice daily)"	Drug, strength, frequency; drops for **> 3 years** mean mitomycin C at trabeculectomy
Personal and family: glaucoma in relatives	Risk **4-fold in siblings, 2-fold in offspring**

Case format — examination

General and systemic (this case only)

- **Pulse, blood pressure, chest**: bradycardia, heart block or wheeze bars beta-blockers; low diastolic perfusion pressure is a risk factor.
- **Face and skin**: haemangioma (Sturge–Weber syndrome), neurofibromatosis, carotid–cavernous fistula: raised episcleral venous pressure.

Ocular examination — RE | LE template

Right eye (RE) and left eye (LE) columns.

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Central vision is lost last	6/18, pinhole 6/12
Extraocular movements (EOM)	Seniors' addition	Full

Structure	Look for in this case	How to record (example)
Intraocular pressure	Goldmann applanation tonometer (GAT) first; asymmetry **≥ 4 mm Hg** is significant	Tn (tension) by GAT 18 mm Hg at 10 am
Lids	Prostaglandin (PG) analogue: pigmentation, long lashes	Normal
Conjunctiva	Drug congestion, worse inferiorly; bleb	Clear
Cornea	Krukenberg spindle; pseudoexfoliation (PXF)	Clear
Anterior chamber	Van Herick (VH) grade; cells, flare	VH grade IV (ND, normal depth)
Iris	PXF; transillumination defects; PI	CPN (colour, pattern normal)
Pupil	Relative afferent pupillary defect (RAPD) = advanced	RTL 3 mm (round, reacting to light)
Lens	PXF on the capsule; glaucomflecken	PCIOL (posterior chamber intraocular lens)
Gonioscopy	Open angle; no peripheral anterior synechiae (PAS), recession or new vessels	Scleral spur (SS) seen in all quadrants
Fundus	Disc in slide order; other causes of field loss (pallor, tilted disc, drusen, pit)	Disc vertically oval, cup–disc ratio (CDR) 0.8, neuroretinal rim (NRR) thinning + (superior, inferior), nasalisation +, bayoneting +, peripapillary atrophy (PPA) +
Fields	Confrontation (seniors' addition)	Full in all quadrants

Special tests

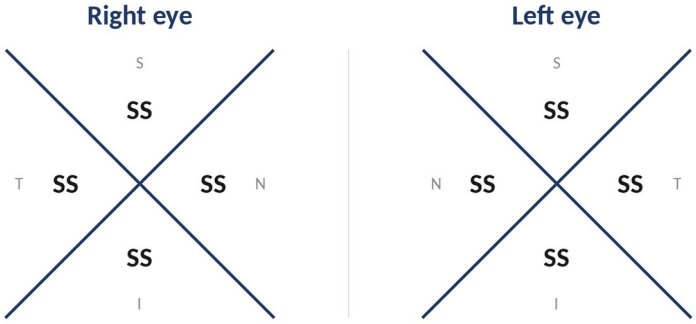
Applanation tonometry (card G8)

1. Do it first: gonioscopy lowers the reading; dilatation raises it by **4–5 mm Hg**.
2. GAT in each eye; note the time and the CCT (the GAT is accurate at **520 µm**).

Record: "Tn by GAT: RE 18, LE 16 mm Hg at 10 am."

Gonioscopy (card G9)

1. After tonometry, before dilatation.
2. Each quadrant: deepest structure seen and Shaffer grade (**4** = 35–40°, **3** = 25–35°, **2** = 20°, **1** = 10°, **slit** < 10°, **0** = closed).



Deepest structure seen in each quadrant (SS = scleral spur) — open angles in all quadrants

Disc examination

1. Dilate after tonometry and gonioscopy; direct ophthalmoscope first.
2. Slit lamp, **+90 dioptre (D)** lens: cup, rim against the **ISNT rule** (thickest Inferiorly, then Superiorly, Nasally, Temporally), notch, haemorrhage, laminar dots; red-free light for retinal nerve fibre layer (RNFL) defects.

Record: in slide order (Short-case version).

Confrontation fields (card G10)

1. Face to face, 60 cm apart; his right eye fixes your left, other eyes closed.
2. Bring a finger in from the periphery, midway, in each quadrant; compare with your own field.

Record: "Fields (confrontation test): full in all quadrants."

Summary and complete diagnosis

Mr ___, a ___-year-old man on ___ drops for glaucoma for ___ years, presented for follow-up with gradual, painless, progressive defective vision, without haloes, trauma or steroids. On examination: visual acuity ___ / ___; tension by GAT ___ / ___ mm Hg at ___; VH grade ___; RAPD ___; lens ___; open angles, deepest structure ___; vertical CDR ___ / ___ with ___; confrontation fields ___.

eye → disease → type / stage / grade → aetiology → complications or status → other eye

- **"Both eyes primary open-angle glaucoma with pseudophakia, with intraocular pressure under control on medical management."**
- "Right eye pseudoexfoliation glaucoma, severe stage, intraocular pressure not at target on maximal medical therapy; left eye pseudoexfoliation syndrome."

Differential diagnosis

Condition	For	Against
Ocular hypertension (OHT)	IOP > 21 mm Hg twice; open angle	Normal disc, rim and field
NTG	Glaucomatous disc and field	IOP always < 22 mm Hg on diurnal curve; more haemorrhages, notching, PPA
PXF glaucoma	Open angle; high IOP	Flakes on pupil margin and lens; bigger 24-hour swings; often one eye
Pigmentary glaucoma	Wide-open angle	Young myopic men; Krukenberg spindle; mid-peripheral transillumination defects
Steroid-induced glaucoma	Open angle; raised IOP	Steroid use; posterior subcapsular cataract
Angle-recession glaucoma	Open angle; raised IOP	Trauma; recession on gonioscopy; one eye

What you must know

Definition

Glaucoma: a chronic, progressive optic neuropathy with irreversible nerve damage and matching field defects, with or without raised IOP. **POAG**: open, normal-looking angle, reduced outflow, no other eye disease; onset after 40 (juvenile 3–40 years). Normal IOP **10–21 mm Hg**.

Risk factors: raised IOP; age (prevalence doubles each decade over 40); Black race; first-degree relatives; myopia; thin CCT; diastolic perfusion pressure < 55 mm Hg; diabetes, hypertension, thyroid disease; vasospasm.

Classification

Grading of glaucomatous optic nerve head damage, with the Z value for target pressure:

Stage	Disc	Field (mean deviation, MD, decibels, dB)	Z
Suspect	—	—	0
Mild (early)	Minimal cupping	Nasal step or paracentral scotoma; MD better than −6 dB	1
Moderate	NRR thinning	Arcuate scotoma; MD −6 to −12 dB	3
Severe	Marked cupping	Within the central 5°; MD worse than −12 dB	5
End-stage	Gross cupping	Small residual field	7

Pathogenesis

- **Outflow resistance**: greatest at the juxtacanalicular trabecular meshwork.
- **Mechanical**: raised IOP bows the lamina cribrosa back → axoplasmic transport blocked → ganglion cell apoptosis.
- **Vascular** (key in NTG): vasospasm, nocturnal hypotension, blood loss.
- **Poles first** (larger laminar pores): cupping often starts inferotemporally → early **upper arcuate scotoma**.
- **Genes**: myocilin (chromosome 1), optineurin (chromosome 10).

Investigations

Test	Why	Expected finding
Diurnal phasing: IOP 3-hourly for 24 hours	Baseline, peak, drop timing; NTG versus POAG	Swing **≥ 8 mm Hg** is significant (normal 3–6)
Pachymetry (CCT)	Correct the IOP reading	Thin **< 500 µm** reads falsely low; thick **> 570 µm** high
Humphrey 24-2 or 30-2	Stage and follow (card G10)	Paracentral, arcuate, nasal step; respects the horizontal meridian
OCT of RNFL; macular ganglion cell complex (GCC)	Pre-perimetric diagnosis; progression	Superior and inferior thinning; double hump lost; eyes differ by **> 9 µm**; GCC thinning

Management

Target pressure: the IOP at or below which further damage is unlikely.

- **TP = IP (1 − IP/100) − Z ± 2** (TP target pressure; IP initial pressure; Z from the staging table).
- Example: IP 30 mm Hg, moderate (Z 3): 30 × 0.7 − 3 = 18, so **16–20 mm Hg**.

- Or lower IOP by **20%** (OHT), **30%** (mild damage, NTG), **40%** (advanced).

Medical: start one drug, a **PG analogue**; review at 2 months; if not at target, switch before adding.

Class (IOP fall)	Drug and strength	Frequency	Main side effect / contraindication
PG analogue (25–32%)	**Latanoprost 0.005%** Travoprost 0.004% Bimatoprost 0.01%, 0.03%	Once, at bedtime	Hyperaemia, long lashes, iris pigmentation; avoid in uveitis
Beta-blocker (20–30%)	**Timolol 0.25%, 0.5%** Betaxolol 0.5% (cardioselective)	Twice daily	Bronchospasm, bradycardia; not in asthma, COPD, heart failure, heart block
Alpha-2 agonist (15–20%)	**Brimonidine 0.1%, 0.15%, 0.2%**	Three times, sometimes twice	Allergy, dry mouth, somnolence; not under 2 years
CAI (15–20%)	**Dorzolamide 2%** Acetazolamide 250 mg tablet	Three times 2–4 times	Stinging; tablets: paraesthesia, renal stones; not in sulpha allergy
Miotic (15–20%)	**Pilocarpine 0.5–4%**	2–4 times	Browache, induced myopia; not in uveitic or neovascular glaucoma

Laser trabeculoplasty, selective (SLT) or argon (ALT): when drops cannot be used reliably (cost, memory, instillation, intolerance). Suits POAG, pigmentary, exfoliation, steroid-induced glaucoma; not uveitic, neovascular or angle-recession glaucoma.

Trabeculectomy indications (card G2): not at target on **3 drugs**, or fields progressing on maximal therapy; side effects or poor compliance; baseline IOP **> 40 mm Hg**; one eye lost; difficult follow-up.

Cataract with glaucoma: expect a miotic pupil, synechiae, weak zonules (PXF), bleeding, a postoperative IOP rise.

- **Cataract alone**: IOP controlled on 1–2 drugs, no significant damage; risks an early IOP spike; later filtering surgery fails more.
- **Combined with filtering surgery**: many drugs, poor compliance, significant field loss, PXF, one-eyed. More complications, slower recovery.

Follow-up: IOP and compliance 3-monthly (mild–moderate) or monthly (advanced); fields 6-monthly or 3-monthly; gonioscopy yearly (a thickening lens can add angle closure).

Complications

- **Disease**: progressive field loss; blindness; sudden loss (acute IOP rise, vein occlusion, snuff-out).
- **Treatment**: surface toxicity, drug side effects; long-term drops lower surgical success.

Prognosis and counselling

- Damage is **irreversible**; treatment stops further loss. Central vision goes last.
- Chronic: drops and reviews for life; check compliance every visit.
- **Drop technique**: PG at bedtime, timolol in the morning; after each drop, lacrimal occlusion (press the inner corner) with eyes closed for **3 minutes**.
- Examine first-degree relatives (high risk).

Recent advances

- **SLT**: frequency-doubled neodymium-doped yttrium aluminium garnet (Nd:YAG) 532 nm laser targeting pigmented trabecular cells.
- **Microinvasive glaucoma surgery (MIGS)**: Schlemm's canal stents, goniotomy, canal dilatation, suprachoroidal stents.

Viva questions

Why did you ask / examine...?

Q Why did you ask about night blindness?
Dark adaptation is defective in open-angle glaucoma; miotics worsen it.

Q Why did you ask about frequent change of glasses?
Pressure on the zonules and ciliary body impairs accommodation; field loss is mistaken for poor vision.

Q Why did you ask about haloes and headache with vomiting?
To rule out angle closure; haloes mean corneal oedema.

Q Why did you ask about sudden loss of vision?
Acute IOP rise, central retinal vein occlusion, or snuff-out after trabeculectomy.

Q Why did you ask about asthma, heart disease and sulpha allergy?
No beta-blocker in asthma, COPD, heart failure or heart block; no CAI in sulpha allergy.

Q Why did you measure tension before gonioscopy and dilatation?
Gonioscopy lowers it; dilatation raises it by 4–5 mm Hg.

Diagnosis

Q What makes you suspect glaucoma?
Raised IOP; vertical CDR ≥ 0.7 or asymmetry > 0.2; rim breaking ISNT; RNFL defect; field defect.

Q Why is the vertical cup–disc ratio more important?
Early rim loss is at the poles; a normal cup is horizontally oval.

Q What is the significance of a disc haemorrhage?
Active disease: flame-shaped, inferotemporal, precedes field loss by months; commoner in NTG.

Q What are bayoneting, nasalisation and baring of circumlinear vessels?
Double angulation of a vessel; vessels pushed nasally; a vessel left bare as the rim recedes.

Q What are the laminar dot sign and beta-zone atrophy?
Laminar pores seen in an advanced cup; atrophy next to the disc with sclera visible.

Q How does NTG differ from POAG?
A diagnosis of exclusion: IOP below 22 mm Hg on diurnal testing; more haemorrhages, notching and PPA.

Q When will you suspect a lesion other than glaucoma?
Pallor more than cupping, dyschromatopsia, a field respecting the vertical meridian: order neuroimaging.

Investigations

Q What is the role of CCT?
The GAT is accurate at 520 µm; thin corneas read falsely low, thick ones high.

Q Why do diurnal phasing?
Baseline and peak; progression despite good readings; drop timing; NTG versus POAG.

Q Can the field be normal with a glaucomatous disc?
Yes: 25–30% of ganglion cells are lost before white-on-white perimetry shows it.

Q How does glaucoma show on OCT?
Superior and inferior RNFL thinning (7 o'clock right eye, 5 o'clock left first); flattened double hump.

Management and surgery

Q How do you calculate target pressure?
TP = IP (1 − IP/100) − Z ± 2; Z 0 suspect, 1 early, 3 moderate, 5 severe, 7 end-stage.

Q Why is a PG analogue given at bedtime?
Its effect peaks 10–14 hours later, with fewer vascular-dilatation symptoms.

Q Why is timolol best given in the morning?
It blunts the early-morning rise and avoids hypotension in sleep.

Q When will you advise trabeculectomy?
Not at target on 3 drugs or progressing; poor compliance or side effects; baseline > 40 mm Hg; one eye lost.

Complications

Q What are the systemic effects of timolol, and how do you reduce them?
Bradycardia, hypotension, bronchospasm, depression; lacrimal occlusion with eyes closed for 3 minutes.

Recent advances

Q What is SLT?
A 532 nm frequency-doubled Nd:YAG laser targeting melanin in trabecular cells.

Q What is MIGS?
Minimal-tissue pressure-lowering surgery for mild to moderate glaucoma: canal stents, goniotomy, suprachoroidal stents.

Examiner traps

- IOP without method, time and CCT, or after gonioscopy or dilatation.
- "Open angle" without gonioscopy; reversing Scheie and Shaffer (Scheie 4: only Schwalbe's line visible; Shaffer 4: wide open).
- "CDR 0.8" alone: give the vertical ratio, rim (ISNT), notch, haemorrhage, vessels, PPA, RNFL.
- Missing PXF at the pupil or lens: it changes diagnosis and cataract surgery.
- CCT in the clinical table: it is an investigation.
- Pallor more than cupping called glaucoma: needs neuroimaging.

Short-case version

Spot diagnosis: "Glaucomatous optic disc: vertically enlarged cup with neuroretinal rim thinning, suggestive of glaucomatous optic neuropathy."

Focused examination:

1. Pupils for RAPD before dilating.
2. Direct ophthalmoscope at arm's length (red glow), then close: media, disc.
3. Slit lamp with +90 D: cup, rim (ISNT), notch, haemorrhage, laminar dots; red-free light for RNFL defects.
4. Describe in slide order: size, disc shape, cup shape, margins, colour, cup–disc ratio, rim, vessels, arteriovenous ratio, peripapillary changes, RNFL defects.

Model: "Disc normal in size, vertically oval; cup vertically oval; margins well defined; pale; cup–disc ratio 0.9; circumferential rim thinning; centrally arising, dichotomously branching vessels; arteriovenous ratio 2:3; peripapillary atrophy; superior and inferior wedge-shaped RNFL defects; suggestive of glaucomatous optic neuropathy."

Three signs to show:

1. Vertical cup–disc ratio **≥ 0.7**, or asymmetry **> 0.2**.
2. Rim thinning or a **notch** breaking the ISNT rule.
3. **Bayoneting** and **nasalisation**.

Say-it script

SAY IT

My patient is a 70-year-old man on regular follow-up, on eye drops for glaucoma in both eyes for 2 years. He has gradual, painless, progressive defective vision in both eyes, with no coloured haloes, headache with vomiting, redness, trauma or steroid use.

Best-corrected visual acuity is 6/18 in the right eye and 6/9 in the left. Intraocular pressure by Goldmann applanation tonometry is 18 and 16 millimetres of mercury at ten in the morning. The anterior chambers are Van Herick grade four, there is no relative afferent pupillary defect, and both eyes are pseudophakic. Gonioscopy shows open angles, with the scleral spur seen in all four quadrants. The right disc has a vertical cup–disc ratio of 0.8 with superior and inferior rim thinning, nasalisation, bayoneting and peripapillary atrophy; the left has 0.6 with an inferior notch. Confrontation fields are full.

My diagnosis is both eyes primary open-angle glaucoma with pseudophakia, with intraocular pressure under control on medical management.

I would do pachymetry, Humphrey fields, optical coherence tomography and diurnal phasing, and set a target pressure. If it is met, I continue the prostaglandin analogue; if not, I switch or add a drug, then consider laser trabeculoplasty or trabeculectomy. I will counsel him on lifelong treatment, compliance and drop technique.

Quick recall

- Normal IOP **10–21 mm Hg**; asymmetry **≥ 4 mm Hg** is significant.
- Vertical CDR **≥ 0.7** or asymmetry **> 0.2**: suspect glaucoma.
- **ISNT**: rim thickest Inferiorly, then Superiorly, Nasally, Temporally.
- **25–30%** of ganglion cells are lost before a field defect.
- **TP = IP (1 − IP/100) − Z ± 2**; Z = 0, 1, 3, 5, 7.
- PG analogue first, at bedtime; timolol in the morning.

Mnemonic: PART B before dilating: Pupillary reflex, Angle, Rubeosis, Tension, BCVA (best-corrected visual acuity).

Read more: Baidya p.152–167, 195–198 · Namrata p.171–178 · Aravind 4.2, 4.4, 4.8, 4.15, glaucoma model case sheet p.661–667 · Glaucoma case format p.1–2

G4 Primary angle-closure disease — PACS, PAC and PACG, including after laser iridotomy

★ Kept 4 times at JEH last year (one as the long case) · Long case

At a glance

- **Primary angle closure**: the peripheral iris closes the angle, usually by **pupillary block**; staged as primary angle-closure suspect (**PACS**), primary angle closure (**PAC**) or primary angle-closure glaucoma (**PACG**); often seen after laser peripheral iridotomy (LPI).
- Kept: bilateral, stable, full of signs — shallow anterior chamber (AC), iridotomies, glaukomflecken, iris atrophy. Gonioscopy was the listed task in all 4.
- Examiners test: Van Herick grade, gonioscopy **with indentation**, the stage, the mechanism, imaging, LPI, the acute attack.

Case format — history

Chief complaint: "Mrs X, a 58-year-old woman, has come for regular follow-up for glaucoma, after laser to both eyes 3 years ago."

House wording: "Patient was apparently normal ___ years ago, after which she had episodes of coloured haloes and brow ache in dim light; laser was done to both eyes and she was started on timolol 0.5% eye drops twice daily."

What to ask	Why it matters
History of present illness: haloes, brow ache, "smoke-filled room" blurring — in the dark, reading, after dilating drops, with stress	**Intermittent closure**: a mid-dilated pupil gives maximum pupillary block
Sudden painful loss of vision with redness, headache, vomiting — or gradual painless blurring	**Acute attack** (now or past) versus **chronic** closure, mostly without symptoms
Laser and surgery (which eye, when); drops, compliance; old intraocular pressure (IOP) and field records	Status after LPI; control; progression
Use of glasses	**Hypermetropia**: short eye, thick lens
Negative history: no coloured haloes	Rules out **intermittent closure**; haloes also occur in incipient cataract (**Fincham test**)
No painful red, watering eye with headache and vomiting	Rules out an **acute congestive attack** and its mimics: **glaucomatocyclitic crisis**, uveitic glaucoma, herpetic keratouveitis
No trauma, no recent eye surgery	Rules out **traumatic** angle closure and **malignant glaucoma**
No diabetic eye disease, vein occlusion or retinal laser	Rules out **neovascular glaucoma** and ciliary body engorgement
No topiramate or sulpha drugs	Rules out drug-induced **ciliary effusion**
Past, drug and surgical: diabetes, hypertension	Repeated dilatation can start an attack
Asthma, heart or kidney disease; drug allergy (sulpha)	Choice of drugs (asthma: no beta-blocker)
Anticholinergic or adrenergic drugs (antihistamines, antidepressants, phenylephrine)	Precipitate angle closure
Personal and family: age, sex; glaucoma in the family	Middle-aged and older **women**; **first-degree relatives** at high risk

Case format — examination

General and systemic (this case only)

- Before drugs: wheeze (no beta-blocker), blood sugar (glycerol), weight (doses per kilogram).

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Hypermetropic correction	6/12; +2.00 D sph (dioptre sphere): 6/9
Conjunctiva	Attack: **circumciliary congestion**; on drops: inferior congestion	Quiet
Cornea	Attack: **epithelial and stromal oedema**; chronic: **Krukenberg spindle**	Clear
Anterior chamber	**Shallow**; **Van Herick (VH) grade**; cells and flare = recent attack	Shallow, VH grade II
Iris	**Iridotomy** (slide order); **sectoral atrophy**, whorling; posterior synechiae; **iris bombé** in an attack	Patent LPI at 11 o'clock
Pupil	Attack or old attack: **mid-dilated, vertically oval, poorly reacting**	RTL 3 mm (round, reacting to light)

Structure	Look for in this case	How to record (example)
Lens	**Glaukomflecken** (grey-white anterior subcapsular flecks); thick or **intumescent** lens	Glaukomflecken +
Intraocular pressure	Goldmann applanation tonometer (GAT) **before gonioscopy or dilatation**, with time; attack **50–100 mm Hg**	Tn (tension) by GAT 16 mm Hg at 10 am
Gonioscopy	Static, then indentation: iridotrabecular contact (ITC), peripheral anterior synechiae (PAS)	Cross diagram (below)
Fundus	**No dilatation before iridotomy**: direct or +90 D; attack: hyperaemic, oedematous disc; chronic: cupping	Undilated, +90 D: CDR (cup–disc ratio) 0.7
Fields	Confrontation	Full in all quadrants

Special tests

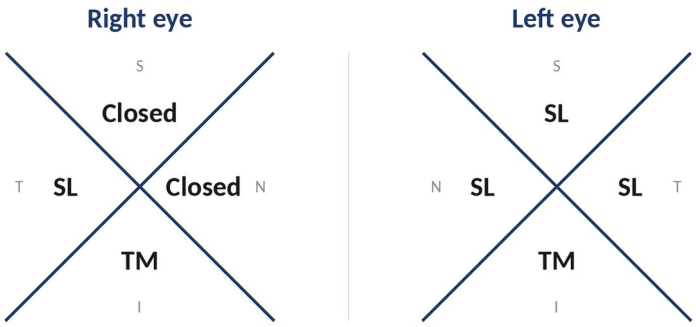
Anterior chamber depth (department script)

1. Pen-torch held **temporally**: whole iris lit = **grade 4**, normal depth; **2/3** lit = **grade 2**, shallow.
2. Van Herick: **slit-thin beam, 1 mm inside the temporal limbus, at 60°**; ratio of corneal thickness to the cornea–iris gap.
3. Grades (peripheral depth ÷ corneal thickness): **4** ≥ 1 (1:1, normal depth); **3** ¼ to ½; **2** ¼ (1:1/4); **1** < ¼; **0** iridocorneal contact. Grades 0–2: suspicious of angle closure. Depth **< ¼** corneal thickness: the angle may be occludable.

Record: "Slit-thin slit 1 mm inside the temporal limbus at 60°; ratio 1:1/4; Van Herick grade 2, suggestive of shallow anterior chamber with occludable angles."

Gonioscopy (technique: card G9)

1. Both eyes, before dilatation. Corneal oedema: wait, or clear it with **anhydrous glycerin**.
2. **Static**: Goldmann **two-mirror** lens without pressure (pressure distorts the angle). Posterior trabecular meshwork (TM) not seen = **ITC**; count the quadrants.
3. **Indentation**: Zeiss **four-mirror** lens, press: angle opens = **appositional**; stays shut = **synechial** (PAS; note extent). **Double hump** = plateau iris.



Deepest structure seen, before indentation — iridotrabecular contact in 3 quadrants in each eye (SL = Schwalbe's line)

Record: "Right eye (RE): ITC in 3 quadrants; indentation opens temporal to scleral spur (appositional); superior and nasal PAS. Left eye (LE): ITC in 3 quadrants, all opening on indentation."

Iridotomy (slide order)

1. **Margin**: ragged, irregular = laser (surgical iridectomy: sharp, well-defined).
2. **Site**: clock hour.
3. **Surrounding iris**: hypopigmented, with pigment dispersion.
4. **Retro-illumination test for patency**: light reflex seen = patent.

Record: "Iridotomy at 11 o'clock, ragged irregular edges, hypopigmented surrounding iris, light reflex on retro-illumination — suggestive of a patent laser peripheral iridotomy."

Summary and complete diagnosis

"Mrs ___, a ___-year-old hypermetropic woman, presented with ___. Visual acuity ___ / ___; AC shallow, VH grade ___ / ___; iridotomies ___; Tn by GAT ___ / ___ mm Hg at ___. Gonioscopy: ITC in ___ quadrants, PAS ___. CDR ___ / ___. Fields ___."

eye → stage (PACS / PAC / PACG) → mechanism → acute or chronic; LPI patent or not → IOP control and treatment → other eye

- **"Both eyes primary angle-closure glaucoma, status post laser peripheral iridotomy with patent iridotomies, with intraocular pressure under control on one medication (timolol 0.5% twice daily)."**
- "Both eyes primary angle-closure suspect: iridotrabecular contact in 3 quadrants, no peripheral anterior synechiae, normal intraocular pressure, discs and fields; not on treatment."

Differential diagnosis

Condition	For	Against
Primary open-angle glaucoma	Raised IOP, cupping	**Open angle**; normal AC depth
Plateau iris syndrome	Narrow angle, attacks	**Deep central AC**; flat iris; **double hump**; still occludable after a patent LPI; younger
Lens-induced closure	Shallow AC, high IOP	Intumescent or subluxated lens
Malignant glaucoma	Shallow AC, high IOP	After surgery; **diffuse** shallowing; ultrasound biomicroscopy (UBM): ciliary processes rotated forward
Neovascular glaucoma	Closed angle, high IOP	**New vessels** on iris or angle

Primary angle closure tends to be bilateral: a **wide-open angle in the fellow eye** points to another diagnosis.

What you must know

Definition

Primary angle closure: **appositional or synechial closure** of the angle caused by **pupillary block**, with or without raised IOP or optic neuropathy, acute or chronic; secondary closure is excluded.

Classification

The ISGEO classification, based on IOP, gonioscopy, disc and fields:

Stage	Gonioscopy	IOP and PAS	Disc and field
PACS	**ITC in ≥ 3 quadrants** (almost 270°)	Normal IOP, **no PAS**	Normal
PAC	ITC in ≥ 3 quadrants	**Raised IOP and/or PAS**	Normal
PACG	ITC in ≥ 3 quadrants	Raised IOP and/or PAS	**Glaucomatous optic neuropathy**: cupping, field loss

Mechanism — four sites: iris (**pupillary block**, most common) · ciliary body (**plateau iris**) · lens (thick, intumescent or subluxated) · behind the iris ("pushing": **malignant glaucoma**, cysts). Secondary "pulling": neovascular glaucoma, iridocorneal endothelial syndrome.

Gonioscopic grading — Shaffer:

Grade	Width	Deepest structure seen	Closure
4 wide open	35–40°	Ciliary body	Impossible
3 open	25–35°	Ciliary body	Impossible
2 moderately narrow	20°	Pigmented TM	Possible but unlikely
1 extremely narrow	10°	Only Schwalbe's line; maybe anterior TM	Possible
Slit	< 10°	None; no obvious iridocorneal contact	Portions appear closed
0 closed	—	Iridocorneal contact	Complete

Occludable angle: pigmented TM not visible (Shaffer 1 or 0) without indentation in at least **three quadrants**.

Spaeth — four parameters: iris-root insertion **A–E** · angle width **10–40°** · peripheral iris contour **S, R, Q** · TM pigmentation **0–4**.

Pathogenesis

- **Relative pupillary block**: lens–iris touch traps aqueous in the posterior chamber, bowing the peripheral iris onto the TM; worst at **mid-dilatation** (dim light).
- **Anatomy**: short axial length, small cornea, thick lens and anteriorly placed iris–lens diaphragm; the lens thickens and moves forward with age.
- Contact is first **appositional**, then **synechial**; outflow falls and IOP rises. Above **40 mm Hg** the sphincter is ischaemic; infarcted lens fibres form glaukomflecken.

Investigations

Test	Why	Expected finding
Gonioscopy with indentation	**Gold standard**; stage; appositional versus synechial	ITC ≥ 3 quadrants; PAS
Anterior segment optical coherence tomography (AS-OCT)	Non-contact, quick, 360°; lens vault	Occludable angle; **cannot see behind the pigmented iris epithelium**

Test	Why	Expected finding
UBM (50–100 MHz)	Mechanism: ciliary body, sulcus, zonules; works through corneal oedema	Iris bowing (block); double hump on indentation (plateau iris); cysts
Perimetry; central corneal thickness (after an attack)	Stage (PAC versus PACG); applanation accuracy	Glaucomatous field loss

Management

By stage: PACS — **LPI**, or observation with serial gonioscopy; PAC and PACG — **LPI plus medical or surgical** IOP control.

LPI in PACS if: frequent dilatation (diabetes, hypertension, age-related macular degeneration), an attack in the other eye, hypermetropia, intermittent symptoms, predisposing drugs, unreliable follow-up.

Neodymium:yttrium-aluminium-garnet (Nd:YAG) iridotomy (photodisruption; all iris colours)

1. **Pilocarpine 1%**; brimonidine or apraclonidine half an hour before; **proparacaine 0.5%**.
2. **Abraham lens** (+55 D button): steadies the eye, magnifies, concentrates energy.
3. **Site**: peripheral third, in a **crypt**, **11–1 o'clock** under the upper lid.
4. **2–5 mJ, 1–3 pulses**; optimal size **150–200 µm**.
5. **End point**: sudden gush of aqueous or pigment; the chamber **deepens**.
6. **After**: IOP at **1 hour** (spike: rise **≥ 8 mm Hg** or IOP **> 30 mm Hg**); **prednisolone acetate 1% four times a day for 5–7 days**; **gonioscopy at 1 week**.

Argon (thermal): anticoagulated patients; pre-treatment of thick dark irides. No LPI with corneal oedema, a flat AC, a totally synechial angle, uveitis or neovascular glaucoma.

Acute attack — step by step

1. **Mannitol 20% intravenously, 1–2 g/kg over 20–30 minutes**: miotics fail while IOP is very high.
2. If no nausea or vomiting: **acetazolamide 250–500 mg orally stat, then 250 mg four times a day**, or **glycerol 50% orally, 1–1.5 g/kg** (not in diabetes); **apraclonidine 0.5%**.
3. IOP **below 40 mm Hg**: **pilocarpine 2%** four times in 30 minutes, then 6-hourly.
4. **Timolol 0.5% twice daily** with **brimonidine 0.2% three times daily**.
5. **Prednisolone acetate 1% four times a day**; **ibuprofen 500 mg twice daily**; antiemetic.
6. Review **hourly**; gonioscopy of both eyes once the cornea clears. Indenting the central cornea may open the angle.
7. Quiet eye, clear cornea: **Nd:YAG iridotomy in both eyes**.
8. Hazy cornea, closure **< 2/3**: **surgical iridectomy** (YAG in the fellow eye). Closure **> 2/3** or persistent inflammation: **primary trabeculectomy**.

Plateau iris syndrome: LPI first; if still occludable, **argon laser iridoplasty**.

Medical after LPI: as in primary open-angle glaucoma (card G1).

Trabeculectomy (5-fluorouracil or mitomycin C): IOP uncontrolled despite LPI and drugs; progression on maximal tolerated therapy; **synechial closure > 270°**. Non-penetrating surgery is relatively contraindicated.

Lens extraction

- **Phacoemulsification with intraocular lens implantation** corrects hypermetropia, deepens the anterior chamber and opens the angle: IOP control in almost all patients with normal preoperative IOP, and in up to **80%** with raised preoperative IOP. *(Kanski — not in your practical books)*
- **EAGLE study**: clear lens extraction is more effective and more cost-effective than LPI in PAC with IOP **> 29 mm Hg**, or in PACG. *(Kanski — not in your practical books)*
- **PACS — ZAP trial**: LPI has a modest prophylactic effect; offer it only to those at highest risk of PACG. If significant ITC persists after LPI and there is symptomatic cataract, lens extraction usually opens the angle. *(Kanski — not in your practical books)*

Fellow eye: prophylactic LPI early. **Follow-up**: serial gonioscopy; high IOP with an open angle after LPI = **combined mechanism**: treat the open-angle part.

Complications

- **Of an attack**: glaukomflecken, iris atrophy, PAS, posterior synechiae, endothelial loss, cupping; **absolute glaucoma**.
- **Of LPI**: IOP spike, uveitis, hyphaema, focal cataract, glare; rarely **malignant glaucoma** (treated by Nd:YAG **hyaloidotomy**).

Prognosis and counselling

- PACS → acute closure **6–10%**; PAC or PACG **17–35%**. Fellow eye: angle closure in about **50% within 5 years**.
- A patent LPI is not a cure (plateau iris, PAS, combined mechanism): lifelong review.
- Counsel: no dilating or anticholinergic drugs before iridotomy; report haloes or pain at once; examine relatives.

Recent advances

- AS-OCT may detect occludability better than gonioscopy, but gonioscopy remains the **gold standard**; imaging is complementary, and UBM is still needed behind the iris.

Viva questions

Why did you ask / examine…?

Q Why did you ask about coloured haloes?
Corneal epithelial oedema acts as a diffraction grating (blue-green centre, yellow-red outside). Also: incipient cataract, mucus, vitreous opacities, tilted intraocular lens.

Q Why did you ask about diabetes and hypertension?
Repeated dilatation can start an attack: an indication for LPI.

Q Why did you ask about drugs?
Anticholinergics and adrenergics precipitate angle closure; topiramate and sulpha drugs cause ciliary effusion.

Q Why did you not dilate the pupil?
Contraindicated until iridotomies are done; I used a +90 D lens.

Q Why did you look for glaukomflecken?
A sign of a previous attack. **Vogt's triad**: glaukomflecken, sectoral iris atrophy, Krukenberg spindle.

Diagnosis

Q Define PACS, PAC and PACG.
All: ITC in ≥ 3 quadrants. PACS: normal IOP, disc and fields, no PAS. PAC: raised IOP and/or PAS, normal disc. PACG: plus glaucomatous optic neuropathy.

Q Plateau iris configuration versus syndrome?
Configuration: anterior ciliary processes hold the peripheral iris forward, without closure. Syndrome: the angle stays occludable despite a patent LPI.

Q Why is the pupil mid-dilated and fixed in an attack?
Above 40 mm Hg the sphincter is ischaemic; pain adds sympathetic overactivity.

Q Why is pain worse than in open-angle glaucoma?
Pain follows the speed of the rise, not the level.

Q What is the Fincham test?
A stenopaeic slit moved across the pupil: the glaucoma halo stays intact; the cataract halo breaks into colours.

Investigations

Q AS-OCT or UBM?
AS-OCT: non-contact, quick, 360°; cannot see behind the pigmented iris epithelium. UBM: 50–100 MHz, axial resolution 25 µm, 4–5 mm deep; sees the ciliary body, even through corneal oedema.

Q How are UBM and AS-OCT findings described?
UBM: echogenic, hypoechoic, echolucent; radial scan for the angle, transverse for clock-hour extent. AS-OCT: hyperreflective, hyporeflective.

Q Name the provocative tests.
Mydriatic, dark room (90 minutes), prone (45–60 minutes), prone dark room, phenylephrine–pilocarpine, triple. Mydriatic or dark room positive: rise > 8 mm Hg or closure on gonioscopy.

Management and surgery

Q Why does iridotomy work?
Aqueous bypasses the pupil; the gradient goes, the iris flattens and the chamber deepens.

Q Why Nd:YAG rather than argon?
Photodisruption works on all iris colours, with fewer late closures; argon needs pigment.

Q Why place the iridotomy at 11–1 o'clock?
The upper lid covers it; the upper nasal iris avoids diplopia and macular burns.

Q Why does lens extraction help in angle closure?
The lens thickens and moves forward with age. Phacoemulsification with intraocular lens implantation corrects hypermetropia, deepens the anterior chamber and opens the angle; in the EAGLE study, clear lens extraction was more effective than LPI in PAC with IOP > 29 mm Hg, or PACG. *(Kanski — not in your practical books)*

Q When would you do trabeculectomy?
IOP uncontrolled despite LPI and drugs, progression, synechial closure > 270°, or an attack with closure > 2/3.

Q Why repeat gonioscopy after LPI?
To prove the block is relieved (lens capsule seen through the iridotomy), map PAS and unmask plateau iris.

Complications

Q Complications of LPI?
IOP spike, uveitis, hyphaema, focal cataract, posterior synechiae, glare; rarely malignant glaucoma.

Recent advances

Q Newer imaging for angle closure?
AS-OCT and UBM. AS-OCT may detect occludability better than gonioscopy, but gonioscopy remains the gold standard.

Examiner traps

- Dilating before gonioscopy and iridotomy.
- "Patent" on retro-illumination alone: the lens capsule must be seen through it.
- Pressing during static grading, or skipping indentation.
- Staging without disc and fields: PAC and PACG differ only by optic neuropathy.
- Pilocarpine first when IOP is above 40 mm Hg.
- Forgetting the fellow eye.

Say-it script

SAY IT

My patient, a fifty-eight-year-old hypermetropic woman, has come for regular follow-up. Three years ago she had evening haloes and brow ache with raised pressure; laser was done to both eyes, and she uses timolol twice daily. She has no history of a painful red eye with vomiting, trauma, eye surgery, sulpha drugs or asthma.

Best-corrected visual acuity is 6/9 in each eye. The anterior chambers are shallow, Van Herick grade two. Each eye has an iridotomy at eleven o'clock with ragged edges, hypopigmented surrounding iris and a clear light reflex on retro-illumination, suggestive of patent laser peripheral iridotomies. Intraocular pressure by Goldmann applanation is 16 and 18 millimetres of mercury at ten in the morning. Gonioscopy shows iridotrabecular contact in three quadrants in each eye; on indentation the angles open, except superiorly and nasally in the right eye, where there are peripheral anterior synechiae. The undilated discs show cup–disc ratios of 0.7 and 0.6 with inferior rim thinning. Confrontation fields are full.

My diagnosis is both eyes primary angle-closure glaucoma, status post laser peripheral iridotomy with patent iridotomies, with intraocular pressure under control on one medication.

I would confirm with automated perimetry, look for plateau iris with ultrasound biomicroscopy, continue timolol, repeat gonioscopy at each review, and consider trabeculectomy if the disease progresses.

Quick recall

- ISGEO: **ITC in ≥ 3 quadrants**; PAC adds raised IOP and/or PAS; PACG adds optic neuropathy.
- Attack IOP **50–100 mm Hg**; pilocarpine only below **40 mm Hg**.
- Mannitol 20% **1–2 g/kg**; acetazolamide **250 mg** four times a day; glycerol **1–1.5 g/kg**.
- Nd:YAG LPI: **2–5 mJ, 1–3 pulses**, **150–200 µm**; spike = rise **≥ 8** or IOP **> 30 mm Hg**.
- Fellow eye **50% in 5 years**; PACS → PAC or PACG **17–35%**.
- Vogt's triad: glaukomflecken, sectoral iris atrophy, Krukenberg spindle.

Mnemonic: I CAN SEE TILL SCHWALBE'S LINE — Iris root, Ciliary body, Scleral spur, Trabecular meshwork, Schwalbe's line: the angle structures from the iris root forwards.

Read more: Baidya p.148–152, 441, 715 · Namrata p.178–184 · Aravind 4.2, 4.5, 4.6, 4.7, 4.17 · Kanski p.377

G2 POAG with trabeculectomy — bleb assessment and the failed bleb

On the department checklist (not kept last year) · Long case

At a glance

- **Primary open-angle glaucoma (POAG)** after **trabeculectomy**; the bleb has **failed** and the intraocular pressure (IOP) is above target.
- Checklist case: lasting signs (bleb, surgical iridectomy, cupped disc), long surgical viva.
- Tested: **bleb in house order**, Seidel's test, iridectomy patency, **internal ostium**, antimetabolites, the shallow chamber.

Case format — history

Chief complaint: "Mr X, a 56-year-old man, came with complaints of defective vision in the right eye (RE) for the past 1 year and the left eye (LE) for the past 6 months; the right eye was operated for raised pressure 6 months ago."

What to ask	Why it matters
History of present illness: which eye, when	Early and late complications differ
Why operated	Indication: failed medical therapy, poor compliance, progression
Mitomycin C (MMC) or 5-fluorouracil (5-FU) used?	More hypotony, bleb leak, bleb infection; more with MMC
After surgery: eye pressed, laser, injections, revision	What was tried for a failing bleb
Present drops	Restarting drops: a step for a failing filter
Negative history: no redness, watering, pain	Rules out **bleb infection** (earliest sign: hyperaemia around the bleb) and endophthalmitis
No sudden loss of vision	Rules out **snuff-out**, central retinal vein occlusion, acute IOP rise
Past, drug and surgical: years on drops before surgery	Long-term drops: more fibroblasts, fewer goblet cells; over 3 years → antimetabolite
Conjunctival or cataract surgery, keratoplasty, scleral buckle, uveitis	Poor prognosis; antimetabolite
Fellow eye	Tenon's cyst there is a risk factor; failed 5-FU surgery there → MMC
Asthma, heart failure, bradycardia; sulpha allergy; aspirin	Restarted drops: no beta-blocker; no carbonic anhydrase inhibitor in sulpha allergy. Aspirin off 5 days before surgery

Case format — examination

General and systemic (this case only)

- **Chest and pulse**: wheeze or bradycardia → no beta-blocker when drops restart.

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity (pinhole)	Fall since surgery: cataract, snuff-out	6/60, pinhole 6/12
Lids	Lid retraction; punctal occlusion (5-FU)	Normal
Conjunctiva	**Bleb** in slide order (Special tests); hyperaemia around it	"Flat, opaque bleb, 11 to 2 o'clock superiorly above the limbus; dilated tortuous vessels; no microcysts; subconjunctival fibrosis; no sutures — suggestive of a failed bleb"
Cornea	Dellen; 5-FU or MMC toxicity	Clear
Anterior chamber (AC)	Central and peripheral depth; cells; hyphaema	Van Herick (VH) grade IV (ND, normal depth)
Iris	**Peripheral iridectomy (PI)**: margin, site, surrounding iris, patency	"Sharp margin; 2 mm, 12 o'clock; iris normal; retro-illumination positive — patent surgical iridectomy"
Pupil	Relative afferent pupillary defect (RAPD)	Grade 3 RAPD
Lens	Cataract	Nuclear sclerosis grade 2
IOP	Goldmann applanation tonometry (GAT) before gonioscopy; time; target	Tn (tension) by GAT 26 mm Hg at 10 am

Structure	Look for in this case	How to record (example)
Gonioscopy	**Internal ostium** patent or blocked; PI; peripheral anterior synechiae (PAS)	Scleral spur seen in all quadrants; patent ostium and PI superiorly
Fundus	Disc, retinal nerve fibre layer (RNFL); macula (hypotony maculopathy); choroidal detachment	Cup–disc ratio (CDR) 0.9, diffuse rim loss, laminar dot sign
Fields	Confrontation	"Fields (confrontation test): ___"

Special tests

Bleb — in slide order

1. Patient looks down: **height**, **clarity**, **horizontal extent** (clock hours), **location over the limbus**.
2. High magnification: **vascularity**, **microcysts**, **subconjunctival fibrosis**, **sutures**; then Seidel's test.

Record: in this order, ending "suggestive of a good functioning / failing / failed bleb".

Feature	Good functioning bleb	Failing or failed bleb
Height	Medium: low and diffuse, or circumscribed and elevated	**Flat** (reduced)
Clarity, wall	**Translucent**, thin	**Opaque**; thick wall
Vascularity	**Avascular**	**Highly vascular, corkscrew** vessels
Microcysts	**Numerous**	**Absent**
Subconjunctival fibrosis	None	**Present**
IOP	Controlled	**Raised**

Seidel's test (bleb leak)

1. Fluorescein over the bleb: **0.25%** shows many leaks; **2%** shows subtle ones.
2. Cobalt blue light: aqueous leaking through the dye = **positive**.

Record: "Seidel's test negative."

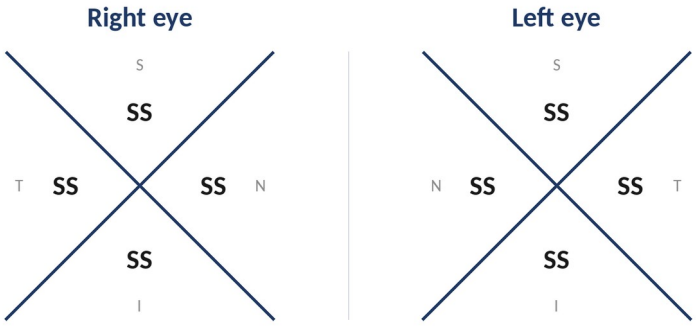
Retro-illumination (PI patency)

1. Light reflex clearly seen through the iridectomy = patent.
2. Margin **sharp, well defined = surgical**; **ragged, irregular**, with pigment dispersion = **laser**.

Record: "Retro-illumination positive: patent surgical PI at 12 o'clock."

Gonioscopy of the operated eye (technique: card G9)

1. After GAT, before dilatation: deepest structure seen in each quadrant.
2. Superiorly: internal ostium, PI, PAS.



Deepest structure seen in each quadrant — open angles in all quadrants (SS = scleral spur). RE: patent ostium and iridectomy superiorly

Summary and complete diagnosis

Mr ___, a ___-year-old man, had trabeculectomy (with ___) in the ___ eye ___ months ago; now on ___. Vision ___ / ___. Bleb ___ (slide order); Seidel's test ___. PI at ___ o'clock, patent. AC ___; lens ___. Tension by GAT ___ / ___ mm Hg at ___. Ostium ___. CDR ___ / ___. Fields ___.

eye → POAG → status post trabeculectomy (± antimetabolite) → bleb: functioning / failing / failed → IOP status → lens → other eye

- **"Right eye: primary open-angle glaucoma status post trabeculectomy with failed bleb and grade 2 nuclear sclerosis. Left eye: primary open-angle glaucoma and grade 2 nuclear sclerosis."**
- "Right eye primary open-angle glaucoma status post trabeculectomy with a good functioning bleb, intraocular pressure under control."

Differential diagnosis

Raised IOP after surgery:

Condition	For	Against
Failing bleb (salvageable)	Lower, thicker-walled, vascular; IOP rising	No microcysts, fibrosis obscuring details: no longer filters
Encapsulated bleb (type 4)	IOP raised; engorged vessels	Highly elevated dome; this bleb is flat
Blocked internal ostium	IOP raised; deep AC	Ostium patent on gonioscopy
Pupillary block, malignant glaucoma, delayed suprachoroidal haemorrhage	IOP raised	All flatten the AC; here AC normal, PI patent

What you must know

Definition

- **Trabeculectomy**: a **partial-thickness filtering operation**. Aqueous passes through a limbal fistula, around the edges of a scleral flap, into the subconjunctival space; it leaves by transconjunctival filtration or through lymphatics and vessels.
- **Failing bleb**: signs of failure appearing, IOP rising. **Failed bleb**: no longer filters (types 3 and 4), so IOP is elevated.
- For primary open-angle glaucoma itself, see card G1.

Classification

Filtering operations: full thickness, no flap (sclerectomy, trephination, Scheie's thermal sclerostomy, laser sclerostomy, iridencleisis, goniopuncture); partial thickness (**trabeculectomy**); non-penetrating (deep sclerectomy, viscocanalostomy, canaloplasty).

Bleb types (four)

Type	Appearance	Filters?
1	Thin, **polycystic**	Yes
2	**Flat, thin, diffuse**, relatively avascular; microcysts at high magnification	Yes
3	**Flat**, no microcysts, **engorged surface vessels**	No: IOP up
4	**Encapsulated (Tenon's cyst)**: localised, highly elevated dome of hypertrophied Tenon's capsule; engorged vessels	No: IOP up

Pathogenesis

- Fibroblasts proliferate → **subconjunctival fibrosis**, the **commonest cause of bleb failure**; blood under the flap adds scarring.
- Long-term drops: more conjunctival fibroblasts and inflammatory cells, fewer goblet cells.
- Aqueous damps fibroblasts; older eyes do better (atrophic Tenon's capsule).

Investigations

Test	Why	Expected finding
Seidel's test	Bleb leak	Negative
Ultrasound biomicroscopy	PI patency; function of the trabeculectomy	Patent PI and ostium

Management

Indications (POAG): failed maximal medical therapy with progressive damage; IOP above target on 3 drops; poor compliance or side effects; baseline IOP over 40 mm Hg; one eye lost to glaucoma; difficult follow-up.

Relative contraindications: likely to respond to less invasive treatment; **previously failed trabeculectomy**; scarred conjunctiva; active neovascularisation; uveitic glaucoma.

Key steps

1. Bridle suture; limbus- or fornix-based conjunctival flap.
2. Scleral flap based at the limbus, **3 mm radially × 4 mm circumferentially**, cut through 2/3 of scleral thickness.
3. MMC now, if used (table).
4. Dissect the flap into clear cornea; paracentesis.
5. Enter the AC behind the flap hinge; Kelly Descemet's membrane punch: **internal ostium 1.5 × 2 mm**.
6. **Peripheral iridectomy**, basal, wider than the ostium.
7. Flap: interrupted **10-0 nylon**; reform the AC with balanced salt solution (BSS).
8. Conjunctiva: **8-0 vicryl**, watertight.
9. **Atropine 1%**; subconjunctival betamethasone and gentamicin.

Antimetabolites go in the sub-Tenon's space: fibrosis is the commonest cause of failure.

Feature	Mitomycin C	5-fluorouracil
Action	Alkylating agent; all phases of the cell cycle; much more potent	Pyrimidine analogue; **S phase** only
Dose	**0.2–0.5 mg/ml for 1–5 min**	**50 mg/ml for 5 min**; subconjunctival **5 mg**
Method	Sponge **4.5 × 4.5 mm** under conjunctiva–Tenon's, flap edges pulled over it; rinse with BSS; tighter flap	30-gauge needle, **90°–180°** from the site; twice daily week 1, daily week 2: **21 injections (105 mg)**
Toxicity	More hypotony, bleb leak, bleb infection; **endothelial**; cystic overfiltering bleb (commonest)	**Epithelial**; punctal occlusion
Used in	Almost all; essential after failed trabeculectomy, conjunctival or cataract surgery, secondary glaucoma, the young or dark-skinned, drops over 3 years	MMC complications before; risk of hypotony maculopathy

Other modulation: steroids before and after surgery; non-steroidal anti-inflammatory drugs; collagen inhibitors (β-aminopropionitrile, penicillamine). Prevent toxicity: thorough irrigation, watertight closure.

Failing or failed filter

- **Blocked fistula** (early, deep AC): low-energy argon laser around the block; internal bleb revision; intracameral tissue plasminogen activator **6–12.5 µg** for fibrin or clot.
- **Failing bleb**: hourly steroids; **digital pressure** (IOP above **12 mm Hg** or target missed); argon laser suture lysis (**50 µm, 0.02–0.1 s, 250–1000 mW**) or releasable sutures.
- **Failed bleb** (late, fibrosis): restart drops; repeat filtering surgery with antimetabolite at a fresh site, or a drainage implant.
- **Encapsulated bleb**: settles in **2–4 months**; drops meanwhile.

This eye

1. Restart drops, **no beta-blocker** (asthma).
2. Above target: repeat trabeculectomy with **MMC** at a fresh superotemporal or superonasal site, or a **glaucoma drainage device** (card G3).
3. Cataract: temporal clear cornea spares the conjunctiva; combined surgery if IOP is uncontrolled.

Follow-up: bleb, leak, IOP, AC depth, disc and macula; fields as on card G1.

Complications

Early, besides the table: low IOP, formed AC → **hypotony maculopathy**; high IOP, deep AC → blocked fistula, failing bleb; also uveitis, hyphaema, dellen, loss of central vision.

Late: bleb infection and endophthalmitis, cataract, leaking bleb, Tenon's cyst, failure, malignant glaucoma, lid retraction, sympathetic ophthalmia. **Snuff-out**: sudden loss of vision after surgery in advanced glaucoma.

Shallow AC: commonest problem after surgery; causes PAS, endothelial damage, cataract.

Condition	IOP	Seidel	Fundus / other
Wound (conjunctival) leak	Low, soft eye	**Positive**	Poor bleb
Overfiltration	Very low	Negative	**Good bleb**; choroidal detachment possible
Serous choroidal detachment	Low	—	Drain if very deep
Pupillary block (incomplete PI)	High	—	PI not patent
Malignant (ciliary block) glaucoma	**High** (hard eye) or normal	Negative	**No bleb**; aqueous forced into the vitreous
Delayed suprachoroidal haemorrhage	High	—	Aphakic, highly myopic eyes at risk

— = not stated in the books.

- **Wound or bleb leak**: pressure dressing, Simmons shell, bandage contact lens, cyanoacrylate glue, aqueous suppressants; large defect → resuture or graft.
- **Overfiltration**: firm patching, shell, aqueous suppressants, atropine; then reform the AC.
- **Malignant glaucoma**: strong mydriatics → intravenous mannitol → neodymium:yttrium-aluminium-garnet (Nd:YAG) laser to the anterior hyaloid face through the PI → anterior vitrectomy via the pars plana.
- **Hypotony maculopathy**: prevent with a partial-thickness flap, tighter after MMC, and releasable sutures; 5-FU if at risk.
- **Bleb infection (blebitis)**: earliest sign **mild hyperaemia around the bleb**; delay → **bleb-related endophthalmitis**. Frequent topical antibiotics; steroids after **12–24 hours**.

Prognosis and counselling

- Worse: age under 40, previous failed filter, aphakia or pseudophakia, neovascular glaucoma, active uveitis, previous conjunctival surgery.

- Plain trabeculectomy lowers IOP by about **41%**.
- Counsel: surgery protects the remaining vision; lost vision does not return. Come at once if the eye turns red.

Recent advances

- **Photodynamic therapy**: blue light with a photosensitising agent kills fibroblasts.
- **Non-penetrating surgery**: AC not entered, so less overfiltration and hypotony.

Viva questions

Why did you ask / examine…?

Q Why did you ask when the operation was done?
Early complications (shallow chamber, blocked fistula, failing bleb) differ from late ones (leak, infection, Tenon's cyst, cataract).

Q Why did you ask whether an antimetabolite was used?
More hypotony, bleb leak and bleb infection, especially with mitomycin C.

Q Why did you ask about redness?
Mild hyperaemia around the bleb is the earliest sign of bleb infection; delay risks endophthalmitis.

Q Why did you ask about sudden loss of vision?
Acute IOP rise with corneal oedema, central retinal vein occlusion, or snuff-out in advanced glaucoma.

Q Why did you retro-illuminate the iridectomy?
To prove patency: an incomplete iridectomy causes pupillary block, with high IOP and a flat chamber.

Q Why did you do gonioscopy in an operated eye?
To see the internal ostium and iridectomy, and any block: iris, ciliary body, lens, vitreous or clot.

Diagnosis

Q What are the features of a good filtering bleb?
Avascular; numerous microcysts; low and diffuse, or circumscribed and elevated.

Q What are the signs of a failing bleb?
Reduced height, thicker wall, vascularisation, loss of microcysts, raised IOP.

Q What are the causes of failure of filtration?
Intraocular: blocked fistula, bleb failure, poor technique. Extraocular: subconjunctival fibrosis (commonest), racial and genetic factors.

Investigations

Q How do you detect a leaking bleb?
Fluorescein: 0.25% shows many leaks, 2% subtle ones (Seidel positive).

Management and surgery

Q Is "trabeculectomy" a misnomer?
Yes. The block need not contain trabecular meshwork; it is more a peripheral posterior keratectomy.

Q Why a peripheral iridectomy, and how big?
To stop the iris blocking the ostium and to prevent pupillary block; basal and wider than the ostium.

Q Why avoid 12 o'clock?
The anterior ciliary vessel enters the ciliary body 3–4 mm from the limbus and bleeds; scarring there also blocks a repeat operation.

Q Fornix-based or limbus-based conjunctival flap?
Equal success. Fornix: easier, better exposure, early limbal leak. Limbus: suture lysis, 5-FU, digital pressure safer early; lower late IOP.

Q When do you suture the scleral flap tighter?
Angle-closure glaucoma or past malignant glaucoma; aphakic or highly myopic eyes; nanophthalmos; raised episcleral venous pressure.

Q When do you advise digital pressure, and what are its risks?
When IOP rises above 12 mm Hg or misses target. Risks: subconjunctival haemorrhage, hyphaema, wound rupture.

Q Why use releasable sutures or laser suture lysis?
The flap can be closed tight against hypotony, then IOP titrated after surgery.

Complications

Q What is the commonest problem after trabeculectomy?
A shallow anterior chamber.

Q How do you tell wound leak, overfiltration and malignant glaucoma apart?
Leak: soft eye, poor bleb, Seidel positive. Overfiltration: good bleb, Seidel negative. Malignant glaucoma: no bleb, Seidel negative, IOP high (hard eye) or normal.

Q Which organisms cause bleb-related endophthalmitis?
Streptococcus species (most common), Haemophilus influenzae, Staphylococcus.

Q What is the snuff-out phenomenon?
Sudden loss of vision after trabeculectomy in advanced glaucoma.

Recent advances

Q What is new in wound modulation?
Photodynamic therapy: blue light with a photosensitising agent kills fibroblasts.

Examiner traps

- A highly elevated bleb called "good": an **encapsulated bleb** does not filter.
- "No leak" without **Seidel's test**: leak and overfiltration both give a low IOP.
- Not retro-illuminating the **iridectomy**: an incomplete one causes pupillary block.
- Gonioscopy without naming the **internal ostium**.
- A **beta-blocker** for this asthmatic.
- "Status post trabeculectomy" without **bleb status, IOP and lens**.

Say-it script

SAY IT

My patient, a fifty-six-year-old asthmatic man, has gradual defective vision in the right eye for one year and the left eye for six months. He was not compliant with drops, and the right eye was operated six months ago.

Vision is 6/60 in the right eye and 6/24 in the left, improving to 6/12 and 6/9 with pinhole. The right eye has a flat, opaque bleb from eleven to two o'clock superiorly, with dilated tortuous vessels, no microcysts and subconjunctival fibrosis. Seidel's test is negative. A surgical iridectomy at twelve o'clock is patent on retro-illumination. The anterior chambers are of normal depth. The right eye has a grade three relative afferent pupillary defect; both lenses have grade two nuclear sclerosis. Intraocular pressure by Goldmann applanation is twenty-six millimetres of mercury on the right and twenty-four on the left. Gonioscopy shows open angles, with a patent internal ostium on the right. The cup–disc ratio is 0.9 on the right and 0.75 on the left.

My diagnosis is right eye primary open-angle glaucoma status post trabeculectomy with failed bleb and grade two nuclear sclerosis; left eye primary open-angle glaucoma and grade two nuclear sclerosis.

I would document the fields, restart drops avoiding beta-blockers, and if the pressure stays above target, plan repeat trabeculectomy with mitomycin C at a fresh site or a glaucoma drainage device.

Quick recall

- **Good bleb**: avascular, numerous microcysts. **Failing**: lower, thicker-walled, vascular, microcysts lost.
- Commonest cause of failure: **subconjunctival fibrosis**; commonest problem after surgery: **shallow AC**.
- **Leak**: Seidel positive. **Overfiltration**: good bleb. **Malignant glaucoma**: no bleb; IOP high or normal.
- **MMC 0.2–0.5 mg/ml for 1–5 min**; **5-FU 50 mg/ml for 5 min**, subconjunctival **5 mg**.
- Flap **3 × 4 mm**; ostium **1.5 × 2 mm**.
- Digital pressure if IOP above **12 mm Hg**; Tenon's cyst settles in **2–4 months**.
- Earliest sign of bleb infection: **hyperaemia around the bleb**.

Mnemonic: none in your mnemonics file for this case.

Read more: Baidya p.151, 153–155, 164–166, 173, 177, 444 · Namrata p.172–173, 175–177 · Aravind 4.9, 4.18, 4.19, glaucoma model case sheet p.661–667

G5 Neovascular glaucoma, including rubeosis iridis

★ Kept once at JEH last year · Long case (rubeosis iridis is also a short case)

At a glance

- **Neovascular glaucoma (NVG)**: a severe secondary glaucoma. A fibrovascular membrane covers the iris and angle after long-standing retinal ischaemia — **proliferative diabetic retinopathy (PDR)**, **ischaemic central retinal vein occlusion (CRVO)**, **ocular ischaemic syndrome**.
- Kept for its signs: neovascularisation of the iris (NVI) and of the angle (NVA), ectropion uveae, very high intraocular pressure (IOP). Last year's case had applanation tonometry as its task.
- Examiners test: iris and angle **before dilatation**; the vessels in slide order; **stage**; **cause**; anti-vascular endothelial growth factor (anti-VEGF) → panretinal photocoagulation (PRP) → surgery; drugs to avoid.

Case format — history

Chief complaint: "Mr X, a 62-year-old man, came with complaints of pain, redness and defective vision in the right eye for the past 10 days, and sudden loss of vision in the same eye 3 months ago."

What to ask	Why it matters
History of present illness: pain, redness, haloes, headache, vomiting; bleeding into the eye	Acute angle-closure stage; **hyphaema** from fragile vessels
Earlier loss of vision — sudden or gradual?	Sudden: ischaemic CRVO (**"100-day glaucoma"**), central retinal artery occlusion (CRAO). Gradual: diabetic retinopathy, chronic uveitis, long-standing retinal detachment, radiation retinopathy
Retinal laser, intravitreal injections; known glaucoma, drops, old records	Prior treatment changes the plan; pre-existing primary open-angle glaucoma (POAG) raises IOP more
Glasses for hypermetropia; similar attacks	Primary angle-closure glaucoma (PACG); chronic angle closure can itself cause NVG (**important in India**)
Negative history: no trigger (dim light, reading, dilating drops, stress)	Against **primary acute angle closure**
No recurrent red eye treated with steroid drops	Rules out **uveitic glaucoma**
No injury	Rules out **old trauma** (angle recession)
Past, drug and surgical: diabetes	**Commonest cause** of NVI (a third of rubeotic eyes)
Hypertension, hypercoagulable states	Cause of the ischaemia
Carotid or heart disease	**Ocular ischaemic syndrome**; heart disease bars intravenous mannitol
Asthma; sulpha allergy	No beta-blocker; no carbonic anhydrase inhibitor (CAI)
Cataract surgery (capsule rupture, aphakia), vitrectomy	Lens, capsule and vitreous are **barriers** to angiogenic factors

Case format — examination

General and systemic (this case only)

- **Blood pressure, pulse, heart**: causes; rule out low blood pressure and heart disease before intravenous mannitol.
- **Diabetes, hypertension**: complications. **Chest**: asthma rules out beta-blockers.

Ocular examination — RE | LE template

Two columns: right eye (RE), left eye (LE). Pupil, iris, angle and tension **before any drop**.

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Poor, even no perception of light; corneal oedema adds blur	HM (hand movements)
Lids and conjunctiva	Lid oedema; **circumciliary congestion**	Circumciliary congestion +
Cornea	Epithelial and stromal **oedema**	Epithelial oedema +
Anterior chamber	Van Herick (VH) grade; **flare**, cells, **hyphaema**	VH grade ___; flare +
Iris	**NVI** in slide order; flat, glistening surface; peripheral anterior synechiae (PAS)	NVI 11 to 8 o'clock
Pupil	Mildly dilated, **oval, fixed**; **ectropion uveae**; relative afferent pupillary defect (RAPD)	Oval, fixed; ectropion uveae +; RAPD +
Lens	Cataract; pseudophakia with capsule dehiscence, or aphakia	PCIOL (posterior chamber intraocular lens)

Structure	Look for in this case	How to record (example)
Intraocular pressure	**Goldmann applanation tonometry (GAT)**; with corneal oedema, take IOP as in a higher range	Tn (tension) by GAT 48 mm Hg at 10 am
Gonioscopy	NVA crossing the scleral spur; PAS extent	Cross diagram
Fundus	Dilate with a cycloplegic (also treats); +90 dioptre (D) lens. Vitreous haemorrhage; disc hyperaemia, oedema (acute) or pallor, cupping (chronic); **the cause**: CRVO, PDR, CRAO	Disc pale, CDR (cup–disc ratio) 0.8; suggestive of old CRVO

Special tests

1. New vessels on the iris — before dilatation (department slide)

1. No drops; slit lamp at **high magnification**: pupillary margin first, then the iris surface and any iridotomy margin.
2. Describe: **appearance → character → course → arrangement → extent** (clock hours).

Record: "RE: **thin, friable, arborising, irregularly arranged** new vessels at the pupillary margin, extending over the iris from **11 o'clock to 8 o'clock** — suggestive of neovascularisation of the iris."

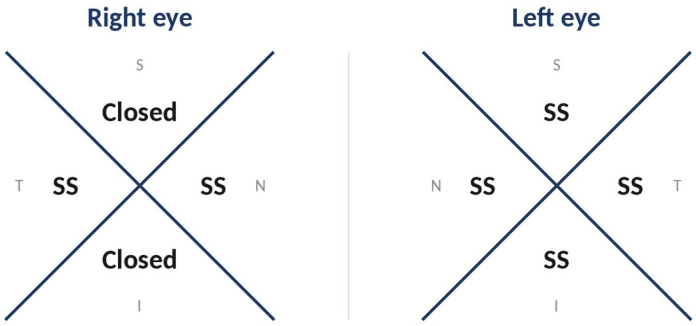
2. Applanation tonometry (card G8)

1. GAT in both eyes before dilatation; note the time.
2. Repeat once the corneal oedema clears.

Record: "Tn by GAT: RE 48, LE 14 mm Hg at 10 am."

3. Gonioscopy (card G9)

1. Goldmann **2-mirror** lens, both eyes, before dilatation; defer while the cornea is oedematous.
2. NVA: vessels run from the iris root **across the ciliary body band and scleral spur**, arborising over the trabecular meshwork. Present without NVI in about **12%**.
3. PAS extent in degrees (PRP can still reverse the IOP rise if synechial closure is **not more than 270°**); **4-mirror** indentation separates appositional from synechial closure.



Deepest structure seen in each quadrant (SS = scleral spur) — RE: closed superiorly and inferiorly by PAS; new vessels cross the scleral spur temporally and nasally. LE: open angles in all quadrants

Summary and complete diagnosis

Mr ___, a ___-year-old man with diabetes / hypertension, presented with pain, redness and defective vision in the ___ eye for ___, after sudden / gradual loss of vision ___ months ago. That eye: vision ___; congestion, corneal oedema, flare; new vessels from ___ to ___ o'clock; ectropion uveae; RAPD ___; tension by GAT ___ mm Hg at ___; NVA ___, PAS over ___°; fundus ___. Other eye: ___.

eye → disease → stage → aetiology → complications or status → other eye

- **"Right eye neovascular glaucoma, angle-closure stage, secondary to ischaemic central retinal vein occlusion, with uncontrolled intraocular pressure; left eye normal."**
- "Left eye neovascular glaucoma, open-angle stage, secondary to proliferative diabetic retinopathy, with pseudophakia; right eye proliferative diabetic retinopathy, post panretinal photocoagulation."

Differential diagnosis

Condition	For	Against
Primary acute angle closure	Painful red eye, haloes, high IOP	No NVI, NVA, fibrovascular membrane or ectropion uveae
Uveitic glaucoma	Flare, cells, raised IOP; dilated iris vessels	Keratic precipitates; dilated vessels are not rubeosis
Fuchs' heterochromic iridocyclitis	Fine angle vessels, spontaneous hyphaema	White, quiet eye; stellate keratic precipitates; heterochromia
Iridocorneal endothelial (ICE) syndrome	PAS, distorted iris, raised IOP	Corneal decompensation, corectopia, pseudopolycoria

Condition	For	Against
Old trauma (angle recession)	PAS, distorted iris, hyphaema	Injury; recessed angle; no NVI
Normal iris vessels (light or atrophic iris)	Visible vessels	Stromal, radial, uniform, unbranched; no leak

What you must know

Definition

NVG: severe secondary glaucoma; fibrovascular tissue proliferates over the angle and iris, giving very high IOP, after long-standing, severe retinal ischaemia. **Rubeosis iridis**: new vessels on the iris surface, whatever the angle or IOP. Also called haemorrhagic, thrombotic, congestive or **100th-day** glaucoma.

Causes: the three commonest are **diabetic retinopathy**, **ischaemic CRVO** and **ocular ischaemic syndrome**; in India also **chronic angle-closure glaucoma**. Others: RUBEOTIC (Quick recall); also CRAO, sickle cell, Coats', cataract surgery, radiation.

Classification

Clinicopathological stages:

Stage	NVI and gonioscopy	IOP	Other signs
Prerubeosis	No NVI or NVA; open angle	Normal	Iris angiography shows early leakage
Pre-glaucoma (rubeosis iridis)	NVI at the pupil margin; open angle, early NVA	Normal	Usually none
Open-angle glaucoma	Florid NVI; open angle, florid NVA under a fibrovascular membrane	Raised	Cells, flare, hyphaema
Angle-closure glaucoma	Vessels less apparent; PAS up to total closure	Persistently high; can reach **60 mm Hg**	Ectropion uveae, glistening iris, corneal oedema, disc damage

Some add a **burnt-out stage** (total synechial closure, few vessels).

Wand's classification: stage 1 vessels at the pupillary margin · 2 up to the collarette · 3 up to the angle · 4 crossing the scleral spur.

Pathogenesis

- **Retinal hypoxia** → angiogenic factors: **VEGF**, fibroblast growth factor, tumour necrosis factor-α, angiogenin.
- Lens, posterior capsule and vitreous are barriers (and supply vaso-inhibitory factors); capsule breach and vitrectomy remove them.
- Vessels bud from capillaries of the **minor arterial circle** at the pupillary margin; thin-walled and fenestrated, they leak and bleed.
- An invisible **fibrovascular membrane** covers the trabecular meshwork → **open-angle (pretrabecular) glaucoma**; its myofibroblasts contract → PAS, flat iris, **ectropion uveae** → **synechial angle closure**.

Investigations

Test	Why	Expected finding
Iris fluorescein angiography, fluorophotometry	Earliest sign, before vessels show	Leakage at the pupillary margin
Fundus fluorescein angiography (FFA)	Extent of ischaemia; plan PRP	Extensive capillary closure; leaking iris vessels
Fields; optical coherence tomography (OCT), nerve fibre layer	Known glaucoma	Progression
Systemic work-up	Cause; drug safety	Diabetes, hypertension, hypercoagulable state, heart or carotid disease

Management

Goals: control IOP, stop new vessels, treat the cause, relieve symptoms.

Step 1 — Medical. IOP below 40 mm Hg: topical aqueous suppressants. Above **40 mm Hg**: add intravenous mannitol.

Drug	Strength / dose	Frequency	Note
Timolol (beta-blocker)	0.5%	Twice daily	Not in asthma
Brimonidine (alpha-2 agonist)	0.2%	Three times	—
Dorzolamide (topical CAI)	2%	Three times	Not in sulpha allergy
Acetazolamide tablet	250 mg	Twice daily	Not in sulpha allergy
Mannitol 20% intravenous	1–2 g/kg over 20–30 min	Initial control	Not in heart disease, low blood pressure

Drug	Strength / dose	Frequency	Note
Prednisolone acetate	1%	6–8 times a day	Inflammation
Atropine (or homatropine 2%)	1%	Three times	Pain; cycloplegia is treatment

Avoid: miotics (more inflammation and pain), **prostaglandin analogues** and **adrenaline**.

Step 2 — Treat the cause once the cornea clears (gonioscopy and fundus are a must).

- **Bevacizumab 1.25 mg in 0.05 ml**, intravitreal through the pars plana (or intracameral, same dose). NVI regresses within **24–48 hours**, but the effect lasts only weeks, so it is an adjunct to PRP. Give it **2–3 days before** surgery: less bleeding, inflammation and failure.
- **PRP — the treatment of choice**: ablating ischaemic retina cuts VEGF; NVI regresses from 2 weeks, fully by **4–6 weeks**. It can still reverse the IOP rise while synechial closure is **not more than 270°**. Green 532 nm laser, **500 µm**, **0.1 s**; mild white burns one burn-width apart, out to the equator; **1200–1600 burns in 3–4 sittings**.
- **Hazy media**: trans-scleral panretinal cryotherapy, or bevacizumab alone.

Step 3 — Surgery when IOP stays high after retinal ablation (extensive PAS).

1. **Visual potential** (rare): after anti-VEGF, **trabeculectomy with mitomycin C (MMC) 0.2–0.4 mg/ml for 2 min** (or 5-fluorouracil 50 mg/ml for 5 min). Poor alone; with pre-operative bevacizumab and/or PRP, success may reach **95%**.
2. **Glaucoma drainage device** (**Ahmed valve**, Molteno, Baerveldt): often **primary** when filtering surgery is likely to fail.
3. **Poor visual potential with pain**: **cyclodestruction** — diode or neodymium:yttrium-aluminium-garnet (Nd:YAG) laser cyclophotocoagulation, cyclocryotherapy.
4. **Blind, painful eye**: **retrobulbar alcohol injection**, or **enucleation** after counselling.

Follow-up: every visit — undilated pupil margin and angle at high magnification, IOP, disc; both eyes.

Complications

Hyphaema, vitreous haemorrhage, optic atrophy. Late: **painful bullous keratopathy**, **complete synechial angle closure**, **intractable glaucoma**.

Prognosis and counselling

- Potentially blinding. Anti-VEGF with PRP controls IOP only in the open-angle stage; after extensive synechial closure, surgery is often needed and useful vision is rare.
- Control diabetes and hypertension.

Recent advances

- Intracameral bevacizumab alone can stabilise NVI and control IOP in the early open-angle stage.
- Newer: alpha-interferon, troxerutin, gene transfer of pigment epithelium-derived factor (PEDF).

Viva questions

Why did you ask / examine…?

Q Why did you ask about sudden loss of vision months earlier?
Ischaemic CRVO: NVG usually follows in 3–5 months (“100-day glaucoma”); range 2 months to 2 years.

Q Why did you ask about diabetes?
Diabetic retinopathy is the commonest cause of NVI: a third of rubeotic eyes have it.

Q Why did you ask about carotid and heart disease?
Ocular ischaemic syndrome is one of the three commonest causes; heart disease bars intravenous mannitol.

Q Why did you ask about cataract surgery or vitrectomy?
Lens, capsule and vitreous block angiogenic factors; removing them raises the risk.

Q Why did you look at the iris before dilating?
Early vessels are fine and at the pupillary margin; dilatation hides them.

Q Why gonioscopy when the iris is clean?
NVA can be present without NVI in about 12%.

Diagnosis

Q What is the first sign of NVG?
Clinically, vessel tufts at the pupillary margin; on fluorescein angiography, leakage there.

Q How do you tell new vessels from normal iris vessels?
New: superficial, irregular, varying size, branching, leaky, crossing the scleral spur. Normal: stromal, radial, uniform.

Q What is Wand’s classification?
Vessels at the pupillary margin (1), to the collarette (2), to the angle (3), crossing the scleral spur (4).

Q Why does ectropion uveae occur?
The contracting fibrovascular membrane pulls the posterior pigment layer round the pupil margin.

Q What closes the angle?
Myofibroblasts in the fibrovascular membrane contract: PAS and a flat iris.

Investigations

Q Why do new vessels leak fluorescein?
Gap junctions between endothelial cells and fenestrations in the basement membrane.

Q Where else does fluorescein leak from the pupillary margin?
Pseudoexfoliation (pupillary ruff defects) and Fuchs' heterochromic iridocyclitis.

Management and surgery

Q What is the treatment of choice, and how does it work?
PRP. Destroying retinal pigment epithelium and photoreceptors cuts oxygen demand, so fewer angiogenic factors are released.

Q What is the role of bevacizumab?
1.25 mg in 0.05 ml intravitreally; NVI regresses within 48 hours, so PRP and surgery bleed less. The effect lasts weeks.

Q Which drugs are contraindicated?
Miotics (more inflammation and pain), adrenaline, and prostaglandin analogues (more inflammation; the membrane blocks uveoscleral outflow).

Q Why atropine and a steroid?
They reduce inflammation and pain; cycloplegia is treatment even in angle closure.

Q Which operation would you choose?
Visual potential: trabeculectomy with MMC after anti-VEGF, or a drainage device. Poor potential: cyclophotocoagulation. Blind, painful: retrobulbar alcohol or enucleation.

Q When is PRP replaced by cryotherapy?
When hazy media prevent PRP: trans-scleral panretinal cryotherapy.

Q When will you do prophylactic PRP?
PDR; diabetic retinopathy with pupillary-margin leakage before lensectomy or vitrectomy; ischaemic CRVO with extensive ischaemia when follow-up is unreliable.

Complications

Q What are the late complications?
Painful bullous keratopathy, complete synechial angle closure, intractable glaucoma.

Recent advances

Q What newer treatments exist?
Anti-VEGF antibodies (bevacizumab), alpha-interferon, troxerutin, PEDF gene transfer.

Examiner traps

- Dilating, or any drop, before seeing the pupil margin and angle: early NVI is missed.
- No gonioscopy because the iris is clean: NVA occurs alone in 12%.
- Treating it as primary angle closure: miotics, prostaglandins and adrenaline are contraindicated.
- Lowering pressure without naming the cause: PRP is the treatment of choice; anti-VEGF lasts weeks.
- No stage or PAS extent: they decide whether PRP can still reverse the IOP rise (closure not more than 270°).
- Calling radial stromal vessels in a light iris "rubeosis".

Short-case version

Spot diagnosis: rubeosis iridis — new vessels at the pupillary margin and on the iris surface.

Focused examination:

1. Before any drop — **PART B**: Pupillary reflex, Angle, Rubeosis, Tension, Best-corrected visual acuity.
2. Slit lamp, **high magnification**, undilated pupil: pupillary margin, iris surface; describe in slide order.
3. **Gonioscopy**: NVA crossing the scleral spur; PAS extent.
4. **IOP** by GAT: normal IOP means the pre-glaucoma stage.
5. **Fundus** after dilatation for the cause: CRVO, PDR, CRAO.

Three signs to show:

1. New vessel **tufts at the pupillary margin**.
2. **Irregular, branching, non-radial** surface vessels.
3. **NVA crossing the scleral spur** (ectropion uveae when advanced).

"Right eye rubeosis iridis, pre-glaucoma stage, secondary to proliferative diabetic retinopathy."

Say-it script

SAY IT

My patient, a sixty-two-year-old man with diabetes and hypertension, came with pain, redness and defective vision in the right eye for ten days, with haloes and vomiting, three months after sudden loss of vision in that eye. There is no history of injury, eye surgery or steroid drops.

In the right eye, vision is hand movements, with circumciliary congestion, mild corneal oedema and flare. Before dilatation, I saw thin, friable, arborising, irregularly arranged new vessels at the pupillary margin from eleven to eight o'clock, with ectropion uveae and a relative afferent pupillary defect. Intraocular pressure by Goldmann applanation is forty-eight millimetres of mercury in the right eye and fourteen in the left at ten in the morning. Gonioscopy shows peripheral anterior synechiae superiorly and inferiorly, and new vessels crossing the scleral spur elsewhere. The fundus shows an old central retinal vein occlusion.

My diagnosis is right eye neovascular glaucoma, angle-closure stage, secondary to ischaemic central retinal vein occlusion, with uncontrolled intraocular pressure; left eye normal.

I would lower the pressure with timolol, brimonidine, dorzolamide, oral acetazolamide and intravenous mannitol, and give prednisolone acetate and atropine, avoiding miotics and prostaglandin analogues. Once the cornea clears, I would do fundus fluorescein angiography, give intravitreal bevacizumab, then panretinal photocoagulation. If the pressure stays high: a glaucoma drainage device or trabeculectomy with mitomycin C, or cyclophotocoagulation if the visual potential is poor.

Quick recall

- Commonest causes: **diabetic retinopathy**, **ischaemic CRVO**, **ocular ischaemic syndrome**.
- After CRVO: **"100-day glaucoma"** (usually 3–5 months).
- Vessels start at the **pupillary margin**; NVA without NVI in **12%**.
- **PRP**: treatment of choice; reverses the IOP rise only while synechial closure is **not more than 270°**; **1200–1600 burns**.
- **Bevacizumab 1.25 mg/0.05 ml**: NVI regresses in **24–48 hours**; **2–3 days** before surgery.
- IOP above **40 mm Hg**: mannitol 20%, 1–2 g/kg intravenously.
- **Avoid** miotics, prostaglandin analogues, adrenaline.

Mnemonic: RUBEOTIC (causes) — **R**etinopathy and **R**etinal vein occlusion (PDR, CRVO), **R**etinal detachment · **U**veitis · **B**ranch retinal vein occlusion · **E**ales' disease · **O**cular ischaemic syndrome · **T**rauma · **I**ntraocular tumours (choroidal melanoma) · **C**arotid–cavernous fistula.

Mnemonic: ABCD (medical treatment) — **A**tropine · **B**eta-blockers · **C**idamex (oral acetazolamide) · **D**examethasone (topical steroid).

Read more: Baidya p.176–180, 437 · Namrata p.192–197 · Aravind 4.9, 4.15

G7 CRVO with POAG, including hemi-CRVO with POAG

★ Kept twice at JEH last year (CRVO with POAG as a glaucoma case; hemi-CRVO with POAG as a fundus case) · Short case

The instruction you may get

- "Examine the fundus of this glaucoma patient and draw it."

Spot

- **Flame-shaped and dot-blot haemorrhages** radiating from the disc, with **dilated, tortuous veins**: four quadrants (**"blood and thunder"**) in central retinal vein occlusion (CRVO).
- **Hemiretinal vein occlusion** *(Kanski — not in your practical books)*: regarded by some as a CRVO variant, ischaemic or non-ischaemic, occluding the **superior or inferior branch** of the central retinal vein: **hemispheric** (a major branch at or near the disc) or **hemicentral** (one trunk of a dual-trunked central retinal vein). Branch-occlusion picture in that **hemisphere**. Neovascularisation of the iris: commoner than in branch, rarer than in central occlusion. Extensive ischaemia → manage as **ischaemic CRVO**.
- **Glaucomatous cupping**, same or fellow eye.
- Old eye: **optociliary shunts**, **sheathed veins**, **laser scars**, **epiretinal membrane**.

Focused examination

1. **Undilated, both eyes: relative afferent pupillary defect (RAPD)**: unreliable with **glaucomatous discs** or miotic pupils.
2. **Pupillary margin: neovascularisation of the iris (NVI)**, in clock hours.
3. **Intraocular pressure (IOP)**: over **22 mm Hg** needs a cause: open-angle glaucoma, neovascular glaucoma (NVG), angle closure.
4. **Undilated gonioscopy: neovascularisation of the angle (NVA)**, without NVI in **12%**; occludable angle.
5. **Lens**: nuclear sclerosis grade.
6. **Dilate. Distant direct ophthalmoscopy**: red glow; **media**.
7. **Disc**: size, shape, colour, margins, **cup–disc ratio**, rim; **glaucomatous cupping**; swelling or pallor; shunts or **new vessels on the disc (NVD)**.
8. **Vessels**: dilated, tortuous veins, all quadrants or one hemisphere; sheathing.
9. **Background**: each lesion by colour, shape, size in disc diameters, margins, number, location → "suggestive of…": flame (superficial) and dot-blot (deep) haemorrhages, cotton-wool spots.
10. **Macula**: foveal reflex, oedema, epiretinal membrane.
11. **Periphery, indirect ophthalmoscope**: **new vessels elsewhere (NVE)**.
12. **"Confirmed with a 90 D (dioptre) lens"**: tells shunts from NVD.
13. **Draw** both fundi: red haemorrhages, blue veins, yellow sclerosed vessels, black laser scars, green epiretinal membrane.

Say-it description

SAY IT

Mr X, a 65-year-old hypertensive man with primary open-angle glaucoma on drops, has sudden painless defective vision in the right eye for two months.

Vision: right eye 6/36, no better with pinhole; left eye 6/9. Non-contact tonometry: 18 and 16 millimetres of mercury.

Both eyes: conjunctiva, cornea, anterior chamber and iris normal, no new vessels; no relative afferent pupillary defect; grade 2 nuclear sclerosis; undilated gonioscopy: open angles, no new vessels.

Right fundus: good red glow, clear media. Disc vertically oval, pink, well-defined margins; cup–disc ratio 0.7, inferior rim thinning, nasalisation, suggestive of glaucomatous cupping. Superior veins dilated, tortuous. Multiple red flame-shaped lesions, a quarter to half a disc diameter, ill-defined margins, with red dots and blots, radiating from the disc into the superior hemisphere only, suggestive of superficial and deep retinal haemorrhages; no cotton-wool spots. Foveal reflex absent, retina thickened, suggestive of macular oedema. Indirect ophthalmoscopy: no peripheral new vessels. Confirmed with a 90 dioptre lens.

Left fundus: cup–disc ratio 0.8, inferior rim thinning, suggestive of glaucomatous cupping.

Right eye clinically **non-ischaemic hemi-central retinal vein occlusion** involving the superior hemisphere, with macular oedema, in a known case of primary open-angle glaucoma in both eyes, intraocular pressure under control on medical treatment; both eyes grade 2 nuclear sclerosis.

Differentials

Condition	Distinguishing point
Branch retinal vein occlusion	Collaterals cross the **horizontal raphe** (central: **at the disc**); more **retinal** than iris new vessels
Ocular ischaemic syndrome	Veins **not tortuous**; **mid-peripheral** haemorrhages; normal disc; carotid **stenosis**

Condition	Distinguishing point
Diabetic retinopathy	**Always bilateral**; **beaded** veins; **hard exudates**

Must know

- **Types (Central Vein Occlusion Study, CVOS)**: ischaemic; non-ischaemic, about **75%** (**34%** turn ischaemic by **3 years**); **indeterminate** (haemorrhage hides perfusion).

Feature	Non-ischaemic	Ischaemic
Vision	Moderate to severe loss	**Counting fingers or worse**
RAPD	Absent or mild	**> 0.7 log unit**
Fundus fluorescein angiography (FFA) non-perfusion	**< 10 disc areas**	**> 10 disc areas**
Electroretinogram (ERG)	Normal or mildly depressed	**b-wave < 60% of normal; b-to-a ratio < 1**

- **Grid laser**: helps branch occlusion, not central (diffuse leakage involving the fovea).
- **Glaucoma both ways**: open-angle glaucoma heads the ocular associations. CRVO raises IOP by NVG, **steroid-induced glaucoma** or angle closure (cause or effect).
- **"100-day glaucoma"**: NVI usually at **2–4 months**; ischaemic: **NVI 49%, NVG 29%** within **6 months**; **one-third** with NVI get NVG. Significant: **NVI over 2 clock hours**, or **any NVA**.
- **Investigations**: FFA once haemorrhages clear (about **3 months**); optical coherence tomography (OCT); blood pressure, fasting blood glucose, lipid profile; hypercoagulable screen if under 60.
- **Macular oedema**: intravitreal **anti-vascular endothelial growth factor (anti-VEGF)** first (**ranibizumab**, **aflibercept**, **bevacizumab**); steroids (**triamcinolone**, **dexamethasone implant**) cause cataract and **raise IOP**.
- **New vessels**: **prompt panretinal photocoagulation (PRP)**; **not prophylactic** unless ischaemic with follow-up impossible; anti-VEGF only temporises.
- **IOP and follow-up**: antiglaucoma drugs, then trabeculectomy or drainage valve. Review **monthly for 6 months**, then tapered (CVOS), with undilated gonioscopy each visit. See G1 and G5.

Viva questions

Q **Why does glaucoma predispose to CRVO?**
Retinal venous pressure must exceed IOP, so raised IOP causes stasis (Virchow's triad); the backward-bowed lamina cribrosa and nasalisation compress the vein.

Q **What will you look for at every follow-up visit?**
Vision, RAPD (conversion to ischaemic), IOP, undilated NVI and NVA, NVD, NVE, macular oedema (OCT), fellow eye.

Q **Why is RAPD unreliable here?**
Needs normal discs, pupils and fellow eye; ERG does not (b-wave: **80%** sensitivity and specificity).

Q **Why call it non-ischaemic?**
Vision better than counting fingers, no RAPD, no cotton-wool spots; FFA confirms (under 10 disc areas).

Q **Why not prophylactic PRP?**
CVOS: no prevention of NVI or NVA; later PRP regressed them better in untreated eyes (**56%** versus **22%**).

Q **Why is NVI commoner than NVD?**
Retina too ischaemic to form new vessels; anterior segment capillaries stay viable.

Q **Shunt or NVD?**
Shunts drain retinal blood to the choroid (good compensation, old occlusion); NVD are finer, forming a network.

Q **Ischaemic index?**
Non-perfused ÷ total retinal area; over 50% (about 10 disc areas): significant neovascular risk.

Q **Commonest cause of visual loss?**
Macular oedema.

Q **Commonest field defect?**
Central scotoma.

Q **Why gonioscopy in the fellow eye?**
An occludable angle, as the occlusion may follow angle closure.

Read more: Baidya p.264–273 · Namrata p.216–222 · Aravind 6.6, 4.8 · Kanski p.521

ER1 Age-related macular degeneration, including disciform scar and drusen

Retina · On the list: S.No 1, 8, 36 · ★ Patient already arrived: none

At a glance

- **Age-related macular degeneration (AMD)**: a progressive degeneration of the macula after 50 years. Early disease shows drusen and retinal pigment epithelium (RPE) changes; late disease is **geographic atrophy (GA, dry)** or **choroidal neovascularisation (CNV, wet)**.
- Dry (non-neovascular) AMD is 85–90% of cases, but wet (neovascular) AMD causes most of the severe visual loss.
- Kept because the signs are stable for years: drusen, GA and **disciform scars** do not change before the exam, and the patient has a real low-vision problem to discuss.
- **S.No 36**: "Both eyes late (end-stage) neovascular age-related macular degeneration with disciform scars and scattered soft drusen."
- **S.No 1**: "Both eyes age-related macular degeneration, non-neovascular, intermediate stage (large soft drusen with pigmentary change)" — stage each eye separately if one has GA or a scar.
- **S.No 8**: "Left eye disciform macular scar, probably end-stage neovascular AMD" — a one-eyed scar must be argued; fellow-eye drusen support AMD, a myopic or traumatic fundus does not.

What to look for and present

Look for	What to say or record
Age, onset and course	Over 50 years; gradual blur (dry) versus sudden distortion then loss (wet)
Distortion, central scotoma, difficulty reading and seeing faces	Metamorphopsia means fluid (active CNV); a dense fixed central scotoma means scar or GA
Previous injections, laser or photodynamic therapy (PDT)	Number, drug, date of the last one; a treated eye may end as a scar
Smoking, hypertension, family history, diet; aspirin or anticoagulants	Smoking doubles the risk (advise stopping); anticoagulants favour large haemorrhages
High myopia, eye injury, uveitis	For a one-eyed scar: myopic CNV, choroidal rupture, inflammatory CNV
Best-corrected visual acuity, distance and near; Amsler grid each eye	Scar eyes are usually 6/60 or less with eccentric fixation; record N notation for near
Lens, iris colour, intraocular pressure (IOP)	Cataract or pseudophakia; IOP before any intravitreal injection
Macula with +90 dioptre (D) lens: drusen	Size against the 125 µm vein at the disc margin; hard or soft, confluent; pigment clumps or loss
GA	Sharp-edged pale patch with large choroidal vessels visible; note foveal sparing
Signs of activity	Grey-green membrane, subretinal fluid, blood, hard exudate, pigment epithelial detachment (PED), RPE tear
Disciform scar	White-yellow raised fibrous mass with pigment at its edge; look at the margin for fresh blood or fluid
Disc, periphery, fellow eye	Myopic crescent, angioid streaks, punched-out spots, peripapillary atrophy; fellow-eye drusen
Tests to mention	Optical coherence tomography (OCT), fundus fluorescein angiography (FFA), indocyanine green angiography (ICGA), OCT angiography (OCTA), fundus autofluorescence (FAF)
Classification to quote	Beckman clinical classification; Age-Related Eye Disease Study (AREDS) categories 1–4; CNV by type, angiography and location

Diagnosis and summary to say

> **Say it**
>
> Mrs X, a 75-year-old former smoker with hypertension, has gradual painless loss of central vision in both eyes over 4 years, with distortion before the blur; she has had no injections. Best-corrected visual acuity is 3/60 and N36 in the right eye and 5/60 and N24 in the left, with a central scotoma on the Amsler grid in each eye; the anterior segments show early nuclear sclerosis and the intraocular pressures are normal. Each macula has a white-yellow, raised subfoveal fibrous scar about 2 disc diameters across with pigment at its edge, and scattered large soft drusen around it. There is no fresh blood, fluid or exudate at the margins, and the discs and periphery show no myopic, traumatic or inflammatory signs. My diagnosis is both eyes late (end-stage) neovascular age-related macular degeneration with disciform scars and scattered soft drusen. I would confirm that the scars are inactive on OCT, give AREDS2 supplements and smoking advice, prescribe low-vision aids, and reassure her that side vision will remain.
>
> S.No 1: Both eyes non-neovascular age-related macular degeneration, intermediate stage (large soft drusen with pigmentary change), AREDS category 3.
>
> S.No 8: Left eye disciform macular scar, probably end-stage neovascular AMD, with drusen in the right eye.

Viva questions and answers

Basics

Q Define AMD.
A degenerative disorder of the macula in people over 50 years, with drusen and RPE pigmentary change, progressing to GA or CNV, in the absence of another cause; visual acuity is not part of the definition.

Q What are the risk factors?
Age (strongest); smoking (doubles the risk, dose-related); family history and genes, especially complement factor H (CFH, chromosome 1q31) and the ARMS2–HTRA1 region (10q26); white race; hypertension and cardiovascular disease; obesity and high-fat diet; light iris; possibly sunlight and cataract surgery.

Q What are drusen and how are they sized?
Extracellular deposits between the RPE basement membrane and the inner collagenous layer of Bruch's membrane (lipid, complement, apolipoprotein E). Small (hard) ≤63 µm, medium 63–125 µm, large >125 µm, judged against a major vein at the disc margin (about 125 µm). Soft drusen are ill-defined and coalesce; reticular pseudodrusen lie above the RPE and carry a high risk.

Q Give the Beckman clinical classification.
Lesions within 2 disc diameters (DD) of the fovea: no ageing change (no drusen); normal ageing (only drupelets ≤63 µm); early AMD (medium drusen, no pigment change); intermediate AMD (large drusen or any AMD pigmentary change); late AMD (neovascular AMD or any GA).

Q Give the AREDS categories.
Category 1, none or few small drusen; 2, early (many small or few intermediate drusen, mild RPE change); 3, intermediate (extensive intermediate drusen, one or more large drusen, or GA sparing the foveal centre); 4, advanced (central GA or neovascular AMD in one eye).

Q Outline the pathogenesis.
Ageing RPE fails to clear outer-segment debris; lipofuscin builds up; basal deposits and drusen thicken Bruch's membrane and block diffusion; complement-driven inflammation and oxidative stress cause RPE and choriocapillaris loss (GA); hypoxic RPE releases vascular endothelial growth factor (VEGF), and new vessels breach Bruch's membrane (CNV).

Examination and signs

Q How do you examine the macula?
Distance and near acuity, Amsler grid each eye, then stereoscopic slit-lamp biomicroscopy with a +90 D or +78 D lens, red-free light for drusen, indirect ophthalmoscopy for the periphery, and always the fellow eye.

Q Describe the Amsler grid.
A 10 cm square of 5 mm boxes held at about 30 cm; each box subtends 1°, so it tests the central 10° from fixation. Wavy lines mean metamorphopsia and a missing area means a scotoma; it is given for weekly home checks of the better eye.

Q What is a disciform scar?
The end stage of neovascular AMD: a white-yellow, raised subretinal fibrovascular scar under the fovea with pigment clumps and surrounding atrophy. Vision is usually 6/60 or worse with a dense central scotoma, but the peripheral field is kept, so the patient never becomes totally blind.

Q What is the differential diagnosis of a disciform scar?
Myopic CNV (Fuchs spot, lacquer cracks, staphyloma, younger); polypoidal choroidal vasculopathy (PCV: Asians, orange nodules, peripapillary, massive bleeds); presumed ocular histoplasmosis-like picture (peripapillary atrophy, punched-out peripheral spots, no vitritis; in India usually multifocal choroiditis or punctate inner choroidopathy); traumatic choroidal rupture (crescent concentric to the disc); angioid streaks; old toxoplasma scar; end-stage Best disease.

Q What are the types of PED?
Drusenoid (confluent soft drusen, no CNV); serous (orange dome, sharp edge, transilluminates); fibrovascular (irregular, means occult type 1 CNV); haemorrhagic (dark dome from CNV or PCV).

Q What is an RPE tear?
A rip at the edge of a large vascularised PED, spontaneous or after anti-VEGF injection or PDT: a sharply bordered bare-choroid area beside a dark scrolled flap; FFA shows a bright bare area and blocked fluorescence from the flap.

Investigations

Q What does OCT show?
Dry: drusen as RPE bumps; GA as loss of RPE and outer retina with signal passing into the choroid (hypertransmission). Wet: intraretinal fluid, subretinal fluid, PED and subretinal hyperreflective material; type 1 CNV lies under the RPE, type 2 above it. A disciform scar is a dense hyperreflective subretinal mass with outer retinal loss and no fluid when inactive.

Q Describe FFA in classic and occult CNV.
Classic: well-defined lacy hyperfluorescence in early transit, leaking progressively so the borders blur late. Occult: fibrovascular PED with stippled hyperfluorescence at 1–2 minutes and late leakage, or late leakage of undetermined source. Drusen stain, GA gives a window defect, and a scar stains late.

Q When is ICGA needed?
The infrared, protein-bound dye sees through blood and pigment: polyps with a branching vascular network in PCV (gold standard), a hot spot and retinal–choroidal anastomosis in retinal angiomatous proliferation (RAP), occult CNV as a plaque or hot spot, and CNV hidden by haemorrhage.

Q What do OCTA and FAF add?
OCTA shows flow in the neovascular network (sea-fan or medusa shapes) without dye but cannot show leakage. FAF maps lipofuscin: GA is dark, and a bright rim around it predicts faster growth.

Q Classify CNV by type and location.
Type 1 sub-RPE (occult), type 2 subretinal (classic), type 3 intraretinal (RAP); PCV is an aneurysmal type 1 variant. Location: extrafoveal 200 µm or more from the foveal centre, juxtafoveal 1–199 µm, subfoveal.

Management

Q What are the AREDS and AREDS2 formulations?
AREDS: vitamin C 500 mg, vitamin E 400 international units (IU), beta-carotene 15 mg, zinc oxide 80 mg, cupric oxide 2 mg daily; cut progression to advanced AMD by about 25% and moderate visual loss by 19% over 5 years. AREDS2 replaced beta-carotene (lung cancer in smokers) with lutein 10 mg and zeaxanthin 2 mg; omega-3 added nothing and zinc 25 mg worked as well.

Q Who should take them?
Only AREDS category 3, or category 4 in one eye; so S.No 8 and 36 (late AMD in at least one eye) qualify. There is no benefit in early AMD or normal eyes.

Q Name the anti-VEGF drugs and doses.
Intravitreal, 3.5–4 mm behind the limbus: ranibizumab 0.5 mg/0.05 mL; bevacizumab 1.25 mg/0.05 mL (off-label, cheap); aflibercept 2 mg/0.05 mL (high-dose 8 mg); brolucizumab 6 mg/0.05 mL; faricimab 6 mg/0.05 mL (blocks VEGF-A and angiopoietin-2). Pegaptanib 0.3 mg is historical.

Q What regimens are used?
Three monthly loading injections, then either as needed (pro re nata, PRN, with monthly OCT) or treat-and-extend, lengthening the interval by 2 weeks while the macula stays dry, up to 12–16 weeks; aflibercept 2 mg is given every 8 weeks after loading.

Q Name the landmark wet AMD trials.
MARINA (occult or minimally classic, ranibizumab versus sham) and ANCHOR (predominantly classic, ranibizumab versus PDT): about 95% kept vision and a third gained 15 letters. CATT and IVAN: bevacizumab equal to ranibizumab. VIEW 1 and 2: aflibercept every 8 weeks equal to monthly ranibizumab. TENAYA and LUCERNE: faricimab up to every 16 weeks.

Q Describe PDT.
Verteporfin 6 mg/m² of body surface area infused over 10 minutes; 15 minutes from the start, a 689 nm diode laser at 600 mW/cm² for 83 seconds (50 J/cm²) over a spot 1,000 µm larger than the lesion; it closes new vessels by photothrombosis. Avoid sunlight for 48 hours; contraindicated in porphyria. Now used mainly with anti-VEGF for PCV.

Q What are the complications of intravitreal injection?
Endophthalmitis (about 0.02–0.05% per injection), IOP spike, lens touch and cataract, retinal tear or detachment, vitreous haemorrhage, RPE tear, subconjunctival haemorrhage; inflammation and occlusive vasculitis with brolucizumab; a theoretical stroke and heart-attack risk.

Q How do you manage a large submacular haemorrhage?
Within 1–2 weeks: pneumatic displacement with intravitreal expansile gas (sulphur hexafluoride or perfluoropropane), with or without tissue plasminogen activator, and face-down posture; or vitrectomy with subretinal tissue plasminogen activator and gas; then anti-VEGF for the underlying CNV.

Q How do you treat a disciform scar?
The scar itself cannot be reversed; treat an active edge (fluid or fresh blood on OCT) with anti-VEGF; otherwise low-vision rehabilitation, AREDS2 and Amsler monitoring for the fellow eye, smoking cessation and disability certification.

Q Which low-vision aids help?
Near: high-plus reading spectacles, hand-held, stand and illuminated magnifiers, closed-circuit television or electronic magnifiers, large print and good lighting. Distance: telescopes. Training in eccentric viewing to a preferred retinal locus. An implantable miniature telescope suits stable bilateral end-stage disease.

Q Is there treatment for GA?
Nothing restores vision. Intravitreal complement inhibitors, pegcetacoplan (C3) and avacincaptad pegol (C5), approved in the United States in 2023, slow GA growth by about 15–20% with monthly or alternate-month injections; risks are conversion to wet AMD and, with pegcetacoplan, occlusive retinal vasculitis. Lampalizumab failed.

Complications and prognosis

Q What predicts progression to late AMD?
Large drusen, pigment change, drusenoid PED, reticular pseudodrusen, late AMD in the fellow eye and smoking. On the AREDS simple scale (1 point per eye each for large drusen and pigment change), 5-year risk is 0.5%, 3%, 12%, 25% and 50% for 0 to 4 points.

Q What is the prognosis of wet AMD?
Untreated, most eyes reach 6/60 or worse within 2 years and end as a disciform scar. With prompt anti-VEGF, about 90–95% keep their vision; results depend on starting acuity, lesion size, fibrosis and adherence. When one eye already has late AMD, the fellow eye has roughly a 40% chance of late AMD over 5 years.

Q What is Charles Bonnet syndrome?
Vivid, formed visual hallucinations in a mentally normal person with severe visual loss, common with bilateral late AMD; the patient knows they are not real, and explanation and reassurance are the treatment.

Recent advances and favourite examiner questions

Q What is PCV and why does it matter in India?
Aneurysmal polyps at the end of a branching vascular network in the inner choroid, common in Asians; orange subretinal nodules, serosanguineous PEDs and massive subretinal haemorrhage; ICGA is diagnostic; treat with anti-VEGF with or without PDT (EVEREST II).

Q What is RAP?
Type 3 neovascularisation starting in the deep retinal capillary plexus; elderly, often bilateral; small intraretinal haemorrhages, oedema and soft drusen; ICGA hot spot with a hairpin retinal–choroidal anastomosis; responds to anti-VEGF.

Q Why has OCT replaced FFA for follow-up?
It is quick, non-invasive and measures fluid at every visit to decide retreatment; FFA is kept for diagnosis, unexplained visual loss, and with ICGA for suspected PCV or RAP.

Q What is new in wet AMD?
Faricimab and aflibercept 8 mg for longer intervals, biosimilar ranibizumab and aflibercept, the port delivery system for ranibizumab, gene therapy for continuous anti-VEGF production, and home OCT or hyperacuity monitoring.

Examiner traps

- Calling every macular scar AMD: check age, refraction, disc, periphery and fellow-eye drusen first (S.No 8 is one-eyed).
- Describing drusen as “soft” or “hard” without size against the 125 µm vein.
- Missing fresh blood or fluid at the edge of a scar: that edge is active and still treatable.
- Giving AREDS to early AMD, or beta-carotene to a smoker.
- Saying anti-VEGF will improve vision in a fibrotic disciform scar.
- Quoting the verteporfin dose per kilogram: it is 6 mg per square metre of body surface area.
- Forgetting near vision, the Amsler grid in each eye, and low-vision counselling.

- Dry 85–90% of AMD; wet causes most of the blindness.
- Drusen cut-offs 63 and 125 µm; a vein at the disc margin is about 125 µm.
- Beckman: early = medium drusen; intermediate = large drusen or pigment change; late = GA or CNV.
- AREDS2 daily: vitamin C 500 mg, vitamin E 400 IU, lutein 10 mg, zeaxanthin 2 mg, zinc 80 or 25 mg, copper 2 mg; categories 3 and 4 only.
- Doses: ranibizumab 0.5, bevacizumab 1.25, aflibercept 2, brolucizumab 6, faricimab 6 mg.
- A disciform scar is the end stage; treat only an active edge and rehabilitate.
- Scar mimics: “**M**any **P**atients **H**ave **T**he **A**MD **I**mitators” — myopic CNV, PCV, histoplasmosis-like, trauma, angioid streaks, inflammatory choroiditis.

Read more: Baidya p.301–317, 436, 707–708 · Namrata p.272–284 · Aravind 6.12 (p.414–421)

ER2 Retinitis pigmentosa and Stargardt disease

Retina · On the list: S.No 3, 9, 27, 37 · ★ Patient already arrived: S.No 9

At a glance

- **Retinitis pigmentosa (RP)**: an inherited, progressive rod-cone dystrophy (about 1 in 4,000): night blindness, then a shrinking field, with central vision lost late. Triad: **arteriolar attenuation, bone-spicule pigment, waxy disc pallor**.
- **Stargardt disease**: the commonest inherited macular dystrophy (autosomal recessive, ABCA4 gene): central loss in childhood or youth, a beaten-bronze macula, pisciform flecks and a dark choroid.
- Kept because the signs never change, both eyes match, and it tests pedigree drawing, systemic syndromes and electrophysiology.
- **S.No 3 and 27**: "Both eyes typical retinitis pigmentosa (rod-cone dystrophy) with arteriolar attenuation, mid-peripheral bone-spicule pigmentation and waxy disc pallor" — add macular oedema, cataract and the likely inheritance if present.
- **S.No 9 ★ and 37**: RP and Stargardt are separate diseases that rarely co-exist; a fundus showing both is usually **one** disorder, a **cone-rod dystrophy** or **advanced ABCA4 disease** (Stargardt spreading to the periphery). Write "Both eyes inherited retinal dystrophy with Stargardt-like maculopathy and retinitis pigmentosa-like peripheral pigmentation, probably ABCA4-related cone-rod dystrophy", adding "with consecutive optic atrophy" for S.No 37.

What to look for and present

Look for	What to say or record
First symptom and age at onset	Night blindness first means rod-cone (RP); central blur, glare and poor colour first mean Stargardt or cone-rod dystrophy
Bumping into objects, trouble driving at night	Field loss; ask about falls and work
Three-generation pedigree; consanguinity	Inheritance pattern; consanguinity favours recessive disease (RP or Stargardt)
Hearing, balance, extra digits, obesity, fainting, ataxia, numbness, fatty stools	Usher, Bardet–Biedl, Kearns–Sayre, Refsum, abetalipoproteinaemia, neuropathy-ataxia syndrome
Thioridazine or chloroquine; rubella, syphilis; injury, old detachment	Pseudo-RP (phenocopies)
Acuity distance and near, colour vision, refraction	Central acuity kept late in RP; myopia common; Stargardt acuity worse than the fundus suggests
Pupils, lids, eye movements	Relative afferent pupillary defect (RAPD) usually absent (symmetric disease); ptosis with ophthalmoplegia means Kearns–Sayre
Lens, cornea, intraocular pressure (IOP)	Posterior subcapsular cataract (PSC); keratoconus; open-angle glaucoma
Vitreous	Fine dust-like pigment cells; early posterior vitreous detachment
Disc	Waxy yellow pallor in RP; temporal pallor in cone-rod and late Stargardt; disc drusen
Vessels and periphery	Arteriolar attenuation; perivascular bone-spicules and retinal pigment epithelium (RPE) mottling in the mid-periphery; note sectoral or pericentral pattern
Macula	RP: cystoid macular oedema (CME), epiretinal membrane, atrophy. Stargardt: beaten-bronze or bull's-eye atrophy, yellow fish-tail flecks sparing the peripapillary retina
Tests to mention	Electroretinogram (ERG), electro-oculogram (EOG), Goldmann fields, optical coherence tomography (OCT), fundus autofluorescence (FAF), fundus fluorescein angiography (FFA), genetic testing
Classification to quote	RP: typical, atypical, syndromic; inheritance; early, mid and end stage. Stargardt: Fishman stages 1–4 and ERG groups 1–3

Diagnosis and summary to say

> **Say it**
>
> Mr X, a 28-year-old man born of a consanguineous marriage, has had difficulty seeing in dim light since childhood and has been bumping into objects for 5 years; his elder sister has the same problem, and he has no hearing loss, extra digits or neurological symptoms. Best-corrected visual acuity is 6/9 in each eye with mild myopia; the pupils react briskly with no RAPD, and both lenses show early posterior subcapsular opacity. Both fundi show waxy pallor of the disc, marked arteriolar attenuation and bone-spicule pigment clumps along the veins in all four mid-peripheral quadrants, with a dull foveal reflex. Fields are constricted to confrontation. My diagnosis is both eyes typical retinitis pigmentosa, probably autosomal recessive, with early posterior subcapsular cataract. I would confirm it with full-field ERG, Goldmann perimetry and OCT for macular oedema, and arrange genetic testing and counselling for the family.
>
> S.No 9: Both eyes inherited retinal dystrophy with Stargardt-like macular atrophy and flecks and retinitis pigmentosa-like peripheral pigmentation, probably ABCA4-related cone-rod dystrophy.
>
> S.No 37: Both eyes advanced inherited retinal dystrophy (retinitis pigmentosa-like periphery, Stargardt-like macular atrophy) with consecutive optic atrophy.

Viva questions and answers

Basics

Q Define RP.
A clinically and genetically varied group of inherited disorders in which rods degenerate first and cones and RPE later, causing night blindness, progressive field loss and finally central loss; it is a dystrophy, not an inflammation, despite the name.

Q What is the triad and which sign comes first?
Arteriolar attenuation (most consistent, 90–95%), mid-peripheral bone-spicule pigment, and waxy disc pallor (about half, late). The earliest visible change is fine dust-like pigment and RPE mottling in the mid-periphery.

Q Give the inheritance patterns and their prognosis.
Simplex (no affected relative) 40–50%, mostly recessive; autosomal dominant (AD) 15–25%; autosomal recessive (AR) about 20%; X-linked 5–15%; rarely digenic (PRPH2 with ROM1) or mitochondrial. Prognosis: AD best, then simplex, AR, and X-linked worst.

Q Name the commonest genes.
AD: RHO (rhodopsin). AR: USH2A; also ABCA4, PDE6B and RPE65. X-linked: RPGR, then RP2.

Q Why does pigment collect around the vessels?
As photoreceptors die the inner retina comes to lie on the RPE; contact with retinal vessels makes RPE cells migrate into the retina and cuff the vessels, giving the spicule pattern. It starts in the mid-periphery, where rods are densest. Arterioles narrow because the dying outer retina uses less oxygen, so inner retinal oxygen rises.

Q What are the atypical forms?
Sectoral (usually inferonasal, often AD, slow), pericentral or inverse, RP sine pigmento (an early stage), unilateral (exclude pseudo-RP), retinitis punctata albescens (white dots), and RP with Coats-like exudative vasculopathy.

Q What causes Stargardt disease?
Biallelic mutations in ABCA4 (chromosome 1p), a transporter in rod and cone outer-segment discs that clears used retinal. When it fails, bisretinoids such as A2E build up in the RPE as lipofuscin, the RPE dies and photoreceptors follow. A dominant look-alike is due to ELOVL4.

Examination and signs

Q Which syndromes with RP do you look for first?
Usher syndrome: RP with sensorineural deafness, the commonest syndromic form, AR; type 1 profound congenital deafness with absent balance, type 2 moderate deafness, type 3 progressive deafness. Bardet–Biedl syndrome: polydactyly, truncal obesity, hypogonadism, kidney disease, learning difficulty, often a bull's-eye macula.

Q Which forms are metabolic or mitochondrial?
Refsum disease (phytanic acid excess: neuropathy, ataxia, deafness, loss of smell, ichthyosis, heart disease); abetalipoproteinaemia or Bassen–Kornzweig syndrome (fat malabsorption, acanthocytes, ataxia, low vitamins A and E); Kearns–Sayre syndrome (mitochondrial deletion, onset before 20: ptosis, external ophthalmoplegia, heart block); neuropathy, ataxia and retinitis pigmentosa (NARP, mitochondrial ATP6 mutation).

Q Which of these can be treated?
Abetalipoproteinaemia: high-dose vitamins A and E with a low-fat diet halts the retinopathy. Refsum disease: a diet free of phytanic acid (no dairy or ruminant fat) and plasma exchange. Kearns–Sayre: an electrocardiogram and a pacemaker for heart block can save life.

Q Why is central vision lost in RP?
CME (reported in 10–70%), epiretinal membrane, macular atrophy, PSC cataract and, late, cone loss.

Q What other ocular associations occur?
PSC cataract (about 40–50%), myopia, keratoconus, open-angle glaucoma (about 3%), disc drusen, vitreous pigment cells, intermediate uveitis, and a golden tapetal reflex in female carriers of X-linked RP.

Q Why is the disc waxy pale in RP?
Consecutive optic atrophy: ganglion cells and axons are lost after the outer retina degenerates, and a glial membrane grows over the disc; with narrowed vessels this gives a yellow waxy look. RAPD is usually absent because both eyes are alike.

Q What is the differential diagnosis of pseudo-RP?
Congenital rubella and syphilis (salt-and-pepper), thioridazine and chloroquine toxicity, a settled exudative detachment, trauma, old panretinal laser, chronic uveitis, cancer-associated retinopathy, choroideremia, and gyrate atrophy (raised ornithine).

Q Describe the fundus in Stargardt disease.
At first normal or a dull foveal reflex while vision is already poor (often mislabelled as malingering); then an oval beaten-bronze atrophic macula, later bull's-eye or geographic atrophy; yellow-white pisciform (fish-tail) flecks at RPE level around the macula and out to the mid-periphery, sparing the peripapillary retina; temporal disc pallor late.

Q How does fundus flavimaculatus differ?
It is the same ABCA4 disease: flecks spread widely with the macula relatively spared, later onset and better vision. Stargardt disease is the macular-atrophy end of the spectrum.

Q Give the differential diagnosis of a bull's-eye macula.
Stargardt disease, cone and cone-rod dystrophy, chloroquine or hydroxychloroquine toxicity, Bardet–Biedl syndrome, neuronal ceroid lipofuscinosis, benign concentric annular dystrophy and central areolar choroidal dystrophy.

Investigations

Q What does the ERG show in RP?
The full-field ERG falls before the fundus changes: rod (scotopic) b-wave first, then 30 Hz cone flicker with a delayed implicit time; advanced disease is extinguished (below 10 µV). It also detects carriers. Other causes of an extinguished ERG: Leber congenital amaurosis, total detachment, ophthalmic artery occlusion, siderosis.

Q What does the EOG show?
It measures the standing potential of the RPE; the light-peak to dark-trough (Arden) ratio is normally about 2 and abnormal below 1.7. It is subnormal in RP but adds little to the ERG; a very low EOG with a normal ERG means Best disease.

Q Describe the fields in RP.
On Goldmann kinetic perimetry, scattered scotomas at 20–25° join into a ring; its outer edge spreads to the periphery and its inner edge closes on fixation, leaving tunnel vision. A field under 10° from fixation counts as blindness even with 6/6 acuity.

Q What do OCT and FAF show in RP?
OCT: loss of the ellipsoid zone and external limiting membrane outside a central island whose width follows the field; cystoid spaces and epiretinal membrane. FAF: a bright ring around the fovea marks the edge of working photoreceptors and shrinks with time; patchy dark areas in the periphery.

Q Which systemic tests would you order?
Guided by the history: audiometry, serum phytanic acid, lipid profile and blood film for acanthocytes, vitamins A and E, electrocardiogram, ornithine, syphilis serology, renal function, and a genetic panel.

Q What does FFA show in Stargardt disease?
A dark (silent) choroid in 60–80%: lipofuscin in the RPE masks choroidal fluorescence, so retinal vessels stand out on a black background. Fresh flecks block; old flecks and the macular atrophy give window defects.

Q What do FAF, OCT and ERG show in Stargardt disease?
FAF, the best test: bright flecks, dark atrophy, high background signal and peripapillary sparing. OCT: foveal outer retinal and ellipsoid loss, flecks as deposits at the RPE. ERG groups: 1, normal full-field with abnormal pattern ERG; 2, added cone loss; 3, added rod loss, with the worst prognosis.

Q Give the Fishman stages.
Stage 1: macular pigment change and flecks within 1 disc diameter of the fovea, ERG normal. Stage 2: flecks beyond the arcades, cone or rod responses may be low. Stage 3: flecks resorbed with macular choriocapillaris atrophy, ERG reduced. Stage 4: extensive RPE atrophy throughout the fundus, rod and cone ERG reduced, field restricted.

Management

Q How do you manage RP?
Explain the course; genetic counselling and examination of relatives; ultraviolet-blocking amber filters; low-vision aids (magnifiers, prisms or field expanders, bright torches) and mobility training; treat CME, cataract and glaucoma; review every 6–12 months with fields and OCT; advise on driving and work; disability certification.

Q What is the role of vitamin A?
Vitamin A palmitate 15,000 international units (IU) daily slightly slowed ERG decline in adults; avoid it in pregnancy, children, liver disease and ABCA4 disease, and check liver function and vitamin A levels yearly. High-dose vitamin E is harmful; docosahexaenoic acid and lutein may help a little.

Q How do you treat CME in RP?
Carbonic anhydrase inhibitors first: oral acetazolamide 250–500 mg daily or topical dorzolamide 2% three times daily; then intravitreal triamcinolone or dexamethasone implant. The cysts often leak little on FFA, and anti-vascular endothelial growth factor (anti-VEGF) drugs help less.

Q What precautions do you take in cataract surgery?
Operate for PSC, which usually helps; expect weak zonules (capsular tension ring), anterior capsule shrinkage (large capsulorhexis) and postoperative CME (topical non-steroidal drops); reduce microscope light exposure; warn of a guarded result.

Q How do you manage Stargardt disease?
No cure: avoid vitamin A supplements (including the high-dose vitamin formulas sold for macular degeneration), stop smoking, wear ultraviolet-blocking sunglasses and a hat, provide low-vision aids, screen siblings and counsel; treat rare choroidal neovascularisation with anti-VEGF.

Complications and prognosis

Q What is the prognosis in RP?
Slowly progressive; most become legally blind by field in middle age, while central acuity may last into the sixth decade in AD and simplex cases. X-linked males are worst, often blind by the third or fourth decade.

Q What is the prognosis in Stargardt disease?
Once acuity falls below 6/12 it drops quickly to about 6/60 and then stabilises; the peripheral field is kept, so total blindness does not occur. Early onset and rod involvement on ERG predict a worse course.

Recent advances and favourite examiner questions

Q What is voretigene neparvovec?
The first approved retinal gene therapy (United States Food and Drug Administration, 2017): an adeno-associated virus 2 vector carrying normal RPE65, injected under the retina after vitrectomy (1.5×10^{11} vector genomes in 0.3 mL per eye, eyes treated separately). It is for biallelic RPE65 dystrophy (Leber congenital amaurosis or RP) with enough surviving cells, and improves navigation in dim light.

Q What other new treatments are there?
Retinal prostheses (Argus II epiretinal implant, now discontinued), optogenetics, stem-cell and RPE transplants, antisense oligonucleotides, gene therapy trials for X-linked RPGR disease, and deuterated vitamin A for Stargardt disease.

Q The fundus shows both RP and Stargardt features. How do you present it?
Describe, do not label: macula (atrophy, flecks, beaten-bronze sheen), periphery (spicules), vessels, disc. Then offer one unifying diagnosis: cone-rod dystrophy or advanced ABCA4 disease (Fishman stage 4) rather than two diseases. If the department calls it "RP with Stargardt", accept the description, note the unusual pairing, and say ERG, FAF and genetic testing will settle it.

Q How does cone-rod dystrophy differ from RP?
Cones fail first: early central loss, photophobia, colour defect and day blindness, a bull's-eye macula and temporal disc pallor; night blindness and peripheral pigment come later. The ERG loses photopic responses more than scotopic, the reverse of RP. ABCA4 is the commonest gene for recessive cone-rod dystrophy.

Q Can ABCA4 cause an RP picture?
Yes. Severe ABCA4 genotypes give a pan-retinal rod-cone degeneration (recessive RP) with macular atrophy, so one gene explains "RP with Stargardt"; vitamin A is harmful in these patients.

Q What does optic atrophy add in S.No 37?
In RP it is waxy consecutive pallor from ganglion cell loss; in cone-rod dystrophy and late Stargardt it is temporal pallor from loss of the papillomacular bundle. Chalky pallor out of proportion to the retina suggests a separate optic neuropathy: check RAPD, colour vision and the visual evoked potential.

Examiner traps

- Calling every pigmented fundus RP: drugs, rubella, syphilis, old detachment and laser scars are mimics.
- Presenting "RP plus Stargardt" as two diseases without suggesting a unifying cone-rod or ABCA4 diagnosis.
- Not drawing a pedigree or not asking about hearing, balance, extra digits and heart block.
- Prescribing vitamin A to a Stargardt or ABCA4 patient.
- Missing CME or cataract as treatable causes of central loss.
- Saying RAPD is expected in RP, or calling waxy pallor "glaucomatous".
- Forgetting that a field under 10° is legal blindness even with 6/6 acuity.

- RP triad: attenuated arterioles, bone-spicules, waxy disc; earliest symptom night blindness.
- Inheritance prognosis: AD best, X-linked worst.
- ERG is diagnostic and falls before the fundus changes; extinguished below 10 μV.
- Stargardt: ABCA4, AR, beaten-bronze macula, pisciform flecks, dark choroid, peripapillary sparing.
- No vitamin A in Stargardt; vitamin A palmitate 15,000 IU only in adult typical RP.
- Voretigene neparvovec for biallelic RPE65 disease: subretinal, 1.5×10^{11} vector genomes per eye.
- Treatable syndromic RP: "**A**dd **R**etinol" — **A**betalipoproteinaemia (vitamins A and E) and **R**efsum (phytanic acid diet).

Read more: Baidya p.317–335, 423–430, 692 · Namrata p.244–251, 346–349 · Aravind 6.10 (p.398–406)

ER3 Branch retinal vein occlusion and old CRVO, including the sclerosed venule

Retina · On the list: S.No 7, 34, 43, 46, 52 · ★ Patient already arrived: S.No 46, 52

At a glance

- **Branch retinal vein occlusion (BRVO)**: thrombosis of a branch vein at an arteriovenous (AV) crossing, where a thickened artery compresses the vein inside their shared sheath; about three times commoner than **central retinal vein occlusion (CRVO)**. Hypertension is the main association.
- Kept because old occlusions are stable and full of signs (sectoral haemorrhage, collaterals, a **sclerosed white venule**, laser scars) and they test the Branch Vein Occlusion Study (BVOS), anti-vascular endothelial growth factor (anti-VEGF) treatment and the link with glaucoma.
- **S.No 7**: "Right eye inferotemporal branch retinal vein occlusion, non-ischaemic (or ischaemic), with (or without) macular oedema, no neovascularisation."
- **S.No 43 and 46** ★: "Right eye (S.No 43) or left eye (S.No 46) old superotemporal branch retinal vein occlusion with a sclerosed venule beyond the AV crossing and collaterals", adding macular oedema, new vessels or laser scars as found.
- **S.No 34**: "Left eye old branch retinal vein occlusion with both eyes proliferative diabetic retinopathy and glaucoma" — name the glaucoma (open-angle or neovascular) from gonioscopy; see also card ER4.
- **S.No 52** ★: "Left eye old central retinal vein occlusion with optociliary shunt vessels, sclerosed veins and macular pigmentary change; right eye primary open-angle glaucoma with a vertical cup–disc ratio of 0.9."

What to look for and present

Look for	What to say or record
Sudden painless blur, a field defect, distortion	Macular oedema or haemorrhage; nasal or peripheral BRVO is often silent
Floaters	Vitreous haemorrhage from new vessels in an old ischaemic occlusion
Pain, redness, haloes	Neovascular glaucoma (mainly after CRVO) or angle closure
Hypertension, diabetes, lipids, smoking, heart disease	The main systemic risks; most patients are over 60
Under 50, both eyes, or recurrent; clots, miscarriage, oral contraceptives, dehydration	Thrombophilia, vasculitis or hyperviscosity: full work-up
Glaucoma history and drops	Raised intraocular pressure (IOP) is the main ocular risk, especially for CRVO (S.No 52)
Previous laser or injections	Sectoral laser scars; number and date of injections
Acuity, relative afferent pupillary defect (RAPD), IOP in both eyes	RAPD is rare in BRVO; a marked RAPD after CRVO means ischaemia
Pupil margin and undilated gonioscopy	Neovascularisation of the iris (NVI) or angle (NVA), before dilating
Sector and the AV crossing	Wedge of flame haemorrhages and cotton-wool spots with its apex at the crossing; superotemporal commonest
Old BRVO signs	Sclerosed or sheathed venule, collaterals across the horizontal raphe, microaneurysms, telangiectasia, hard exudate
New vessels	Neovascularisation elsewhere (NVE) at the edge of non-perfused retina, or of the disc (NVD)
Old CRVO signs	Optociliary shunts on the disc, sheathed veins in all quadrants, macular pigment or oedema, disc pallor
Fellow eye	Hypertensive or diabetic retinopathy, glaucomatous cupping
Tests and classification	Optical coherence tomography (OCT), fundus fluorescein angiography (FFA) and OCT angiography (OCTA); ischaemic BRVO over 5 disc diameters of non-perfusion, ischaemic CRVO over 10 disc areas

Diagnosis and summary to say

Say it

Mrs X, a 64-year-old woman with hypertension, noticed sudden painless blurring of the lower half of her vision in the left eye 2 years ago and received three injections elsewhere; she now has mild distortion. Best-corrected visual acuity is 6/6 in the right eye and 6/18 in the left; the pupils react normally with no RAPD, the IOP is 16 mmHg in each eye, and undilated gonioscopy shows no new vessels. In the left eye the superotemporal vein is white and sheathed beyond an AV crossing one disc diameter from the disc, with collaterals crossing the horizontal raphe, dot haemorrhages, microaneurysms and hard exudates in that sector, and retinal thickening at the fovea; there are no new vessels or laser marks. The right fundus shows arteriolar narrowing and AV nicking. My diagnosis is left eye old superotemporal branch retinal vein occlusion with a sclerosed venule and collaterals, with macular oedema, and both eyes hypertensive retinopathy. I would do OCT and FFA for oedema and non-perfusion, treat centre-involving oedema with anti-VEGF injections, watch for new vessels, and control her blood pressure with her physician.

S.No 7: Right eye inferotemporal branch retinal vein occlusion, non-ischaemic, with macular oedema.

S.No 52: Left eye old central retinal vein occlusion with optociliary shunts; right eye primary open-angle glaucoma, cup–disc ratio 0.9.

Viva questions and answers

Basics

Q Define BRVO and give its types.
Occlusion of a branch of the central retinal vein, almost always at an AV crossing. Types: major (a first-order branch draining a quadrant), macular (a small macular branch) and peripheral; a hemiretinal occlusion at the disc drains half the retina and behaves like CRVO. By perfusion: non-ischaemic or ischaemic.

Q Describe the pathogenesis.
At most crossings the artery lies in front of the vein in a common adventitial sheath; a thick arteriosclerotic artery narrows the vein, causing turbulence, endothelial damage and thrombosis (Virchow's triad). The raised venous pressure causes haemorrhage, oedema and ischaemia, and ischaemic retina releases VEGF.

Q Why is the superotemporal quadrant affected most?
It has the most AV crossings, and temporal occlusions drain the macula and so cause symptoms; nasal occlusions are usually silent and missed.

Q What are the risk factors?
Age over 60, hypertension (strongest), diabetes, hyperlipidaemia, smoking, obesity, glaucoma and a short axial length; in the young, thrombophilia, hyperhomocysteinaemia, antiphospholipid syndrome, oral contraceptives, vasculitis (Behçet disease, sarcoidosis, tuberculosis) and dehydration.

Q How do you define ischaemic BRVO and CRVO?
On FFA, BRVO is ischaemic with more than 5 disc diameters of capillary non-perfusion (BVOS), and CRVO with more than 10 disc areas (Central Vein Occlusion Study, CVOS). Ischaemic CRVO also has acuity below 6/60, a dense RAPD, many cotton-wool spots and an electroretinogram (ERG) b:a ratio below 1.

Examination and signs

Q What are the signs of an old BRVO?
Haemorrhages fade over 6–12 months, leaving a sclerosed or sheathed venule, collaterals, microaneurysms, telangiectasia, hard exudate rings, macular pigment change or epiretinal membrane and a narrowed sheathed arteriole; ischaemic eyes may show NVE, vitreous haemorrhage or tractional detachment.

Q What is a sclerosed venule and what does it mean?
The occluded vein beyond the crossing becomes a white thread, or a vessel with white lines along both walls, because its wall thickens and its lumen narrows or closes after thrombosis. It marks an old occlusion (months to years), usually with a poorly perfused sector; FFA shows delayed or absent filling and late staining of the vein wall.

Q How do you tell it from vasculitis?
In BRVO the sheathing starts at an AV crossing and stays in one sector with collaterals and old haemorrhage. Periphlebitis (Eales disease, sarcoidosis, tuberculosis) affects younger patients with patchy sheathing in several quadrants, often peripheral, with vitreous cells and no relation to crossings.

Q How do you tell collaterals from NVE?
Collaterals are existing capillaries that enlarge to bypass the block: they lie within the retina, are smooth and of larger calibre, often cross the horizontal raphe, carry no fibrous tissue and do not leak on FFA, so they must never be lasered. NVE are fine lacy fronds on the retinal surface at the edge of non-perfused retina, may carry fibrous tissue and leak heavily.

Q Why does BRVO cause retinal new vessels and CRVO iris new vessels?
In BRVO a moderately ischaemic sector next to healthy retina grows NVE (about 40% of ischaemic eyes). In ischaemic CRVO the whole retina is too damaged to grow vessels, but VEGF reaches the front of the eye, so NVI and NVA dominate.

Q What are the signs of an old CRVO?
Optociliary shunt vessels on the disc (the footprint of a past CRVO), sclerosed sheathed veins in all quadrants, narrowed arterioles, a few residual haemorrhages, macular pigment mottling, chronic cystoid oedema or epiretinal membrane, disc pallor after ischaemia, and possibly NVD, NVE, NVI or laser scars.

Q Give the differential diagnosis of optociliary shunts.
Old CRVO, optic nerve sheath meningioma (with visual loss and optic atrophy, the classic triad), chronic papilloedema, glaucoma, optic nerve glioma, and congenital shunts.

Q Give the differential diagnosis of an acute BRVO.
Diabetic and hypertensive retinopathy, ocular ischaemic syndrome, radiation retinopathy, retinal arterial macroaneurysm and vasculitis; a sectoral wedge with its apex at a crossing decides.

Investigations

Q How do you work up an elderly patient?
Blood pressure, fasting and postprandial blood sugar, glycated haemoglobin (HbA1c), lipid profile, complete blood count, erythrocyte sedimentation rate, renal function and electrocardiogram; IOP and gonioscopy in both eyes.

Q What extra tests for a young or bilateral case?
Thrombophilia screen (prothrombin and partial thromboplastin times, protein C, protein S, antithrombin III, factor V Leiden), homocysteine, antiphospholipid antibodies, antinuclear antibody, serum angiotensin-converting enzyme, chest X-ray, Mantoux test, syphilis and human immunodeficiency virus serology, and serum protein electrophoresis.

Q When and why do you do FFA?
After the haemorrhages clear (about 3 months): delayed venous filling, blocked fluorescence, the area of non-perfusion (ischaemic or not), leaking macula versus ischaemic macula (enlarged foveal avascular zone), collaterals (no leak) versus NVE (leak), and late vein-wall staining. Wide-field FFA maps peripheral non-perfusion.

Q What does OCT show?
Acute: cystoid spaces, diffuse thickening, subretinal fluid and shadowing from haemorrhage. Old: epiretinal membrane, foveal thinning and inner retinal thinning in the ischaemic sector. It is the main tool for following oedema; OCTA shows capillary dropout without dye.

Management

Q What did the BVOS show?
Grid argon laser for perfused macular oedema of at least 3 months with acuity 6/12 or worse: about two-thirds gained 2 lines versus about one-third of controls at 3 years. Sector scatter laser once new vessels appeared roughly halved vitreous haemorrhage (about 60% to 30%), but laser before new vessels appear is not advised.

Q Why not give prophylactic sector laser?
Only about 40% of ischaemic BRVOs ever grow new vessels, and treating them when they appear works as well, so prophylaxis over-treats the majority. Exception: when follow-up cannot be assured.

Q Describe grid laser for BRVO oedema.
After 3 months and an FFA showing leakage with a perfused macula: 532 nm green, 50–100 µm spots, 0.05–0.1 s, light-grey burns one burn-width apart over the leaking area of the affected sector, not inside the edge of the foveal avascular zone, not beyond the arcade, and not on haemorrhage or collaterals.

Q Describe sector laser for new vessels.
Scatter burns over the whole non-perfused sector seen on FFA: 500 µm, 0.1 s, moderate grey-white, one burn-width apart, from 2 disc diameters outside the fovea to the periphery, sparing collaterals and the macula.

Q What is the first-line treatment of macular oedema now?
Intravitreal anti-VEGF for centre-involving oedema with acuity 6/12 or worse: ranibizumab 0.5 mg, aflibercept 2 mg or bevacizumab 1.25 mg, monthly until stable (usually 3–6 doses), then as needed or treat-and-extend. Second line: dexamethasone implant 0.7 mg, or grid laser for persistent oedema.

Q Give the results of BRAVO, CRUISE and SCORE.
Monthly ranibizumab for 6 months gained about 18 letters versus 7 with sham in BRVO (BRAVO) and about 15 versus 1 in CRVO (CRUISE). SCORE: intravitreal triamcinolone 1 mg was no better than grid laser in BRVO but beat observation in CRVO (27% versus 7% gained 15 letters); 4 mg caused more cataract and raised IOP. Aflibercept gave similar gains (VIBRANT, GALILEO, COPERNICUS).

Q When do you prefer the dexamethasone implant?
Pseudophakic eyes, patients who cannot attend monthly, recent stroke or heart attack, pregnancy, or poor anti-VEGF response. The effect peaks at about 2 months and lasts 4–6 months (GENEVA); about a quarter need drops for raised IOP, and phakic eyes develop cataract.

Q How do you treat neovascularisation after CRVO?
Prompt full panretinal photocoagulation (PRP) once NVI reaches 2 clock hours or any NVA, NVD or NVE appears, with anti-VEGF for quick regression and IOP control; review monthly for the first 6 months. CVOS found prophylactic PRP did not reliably prevent NVI; it is given only when follow-up is impossible.

Q When is vitrectomy needed?
Non-clearing vitreous haemorrhage, tractional detachment, and epiretinal membrane or vitreomacular traction with persistent oedema.

Complications and prognosis

Q What are the complications of BRVO?
Macular oedema (the commonest cause of poor vision), macular ischaemia, NVE with vitreous haemorrhage, tractional or rhegmatogenous detachment and epiretinal membrane; NVI is rare (about 1%).

Q What is the prognosis?
BRVO does well: over half reach 6/12 or better, many as collaterals form; macular ischaemia or foveal haemorrhage limit recovery. In CRVO presenting acuity predicts outcome; about a third of non-ischaemic eyes become ischaemic within 3 years, and ischaemic eyes carry a high risk of neovascular glaucoma (the "90-day glaucoma"). The fellow eye risk is about 1% a year.

Q What is the link with primary open-angle glaucoma (POAG)?
Raised IOP and posterior bowing of the lamina cribrosa compress the central retinal vein, so glaucoma or ocular hypertension is the commonest ocular risk factor for CRVO and hemiretinal occlusion. CRVO in turn causes neovascular glaucoma, and steroid treatment can raise IOP.

Q How does POAG change the plan in S.No 52?
Measure IOP and examine the disc and angle in both eyes; gonioscopy for NVA in the CRVO eye; treat both eyes to a target pressure, since controlling IOP may lower the risk of a second occlusion; avoid steroids in a steroid responder; follow fields and disc.

Recent advances and favourite examiner questions

Q Why does grid laser help BRVO but not CRVO?
In BRVO the oedema is sectoral, with a healthy perifoveal circulation and collaterals across the raphe; in CRVO leakage is diffuse in all four quadrants around the fovea, so CVOS grid laser dried the angiogram without improving vision.

Q What do OCTA and wide-field imaging add?
OCTA shows non-perfusion and collaterals without dye; ultra-wide-field FFA maps peripheral non-perfusion, which predicts oedema recurrence and guides targeted laser.

Q What treatments have failed in retinal vein occlusion?
Systemic anticoagulants and haemodilution (no proven benefit), tissue plasminogen activator, laser chorioretinal venous anastomosis, radial optic neurotomy and AV sheathotomy (cutting the shared sheath at the crossing).

Q Why is S.No 34 a trap?
It combines an old BRVO with proliferative diabetic retinopathy (PDR) and glaucoma: tell sectoral collaterals from diabetic NVE, look for NVA on gonioscopy, and plan PRP for the PDR while sparing collaterals.

Examiner traps

- Lasering collaterals as if they were new vessels.
- Calling a sheathed venule "vasculitis" without relating it to an AV crossing.
- Dilating before checking RAPD, the pupil margin and gonioscopy for NVI and NVA.
- Advising prophylactic sector laser in ischaemic BRVO (BVOS says treat only when new vessels appear).
- Quoting 10 disc areas for ischaemic BRVO: it is over 5 for BRVO and over 10 for CRVO.
- Forgetting IOP and glaucoma in the fellow eye of a CRVO (S.No 52), and systemic work-up in the young.
- Doing FFA through dense fresh haemorrhage instead of waiting about 3 months.

- BRVO occurs at AV crossings; superotemporal commonest; hypertension the main risk.
- Ischaemic: BRVO over 5 disc diameters, CRVO over 10 disc areas of non-perfusion.
- BVOS: grid laser for perfused oedema after 3 months with acuity 6/12 or worse; sector laser only after new vessels appear.
- Anti-VEGF first line for oedema; dexamethasone implant 0.7 mg second line.
- Sclerosed venule = old occlusion; optociliary shunts = old CRVO.
- Collaterals do not leak; new vessels do.
- Old vein occlusion signs: "**S**heaths, **C**ollaterals, **S**hunts, **S**cars" — sclerosed vessels, collaterals, optociliary shunts, laser scars.

Read more: Baidya p.258–273 · Namrata p.216–229 · Aravind 6.6 (p.355–362)

ER4 Diabetic retinopathy: severe NPDR with CSME, PDR, and the patient for PRP

Retina · On the list: S.No 11, 34, 42, 51 · ★ Patient already arrived: none

At a glance

- **Diabetic retinopathy (DR)**: a retinal microangiopathy with leakage (oedema, exudate) and closure (ischaemia, new vessels), graded as non-proliferative (NPDR) or proliferative (PDR), with diabetic macular oedema (DME) possible at any stage; a leading cause of blindness in working-age adults.
- Kept because it is common, both eyes are full of signs, and it tests grading (the 4-2-1 rule, high-risk characteristics, clinically significant macular oedema or CSME), laser settings, anti-vascular endothelial growth factor (anti-VEGF) trials and systemic control.
- **S.No 11**: "Both eyes severe non-proliferative diabetic retinopathy with clinically significant macular oedema" (state centre-involving or not on optical coherence tomography, OCT).
- **S.No 42**: "Both eyes proliferative diabetic retinopathy (with or without high-risk characteristics) with clinically significant macular oedema."
- **S.No 51**: "Both eyes proliferative diabetic retinopathy with high-risk characteristics, for panretinal photocoagulation (PRP)"; add "partial PRP done" if scars are present.
- **S.No 34**: "Both eyes proliferative diabetic retinopathy with glaucoma (open-angle or neovascular)" plus a left eye old branch retinal vein occlusion (see card ER3).

What to look for and present

Look for	What to say or record
Type and duration of diabetes, treatment, glycated haemoglobin (HbA1c)	Duration is the strongest predictor; type 1 tends to PDR, type 2 to macular oedema
Hypertension, kidney disease, lipids, anaemia, pregnancy, smoking	All worsen retinopathy; kidney disease travels with PDR
Gradual blur or distortion	Macular oedema or ischaemia
Floaters or sudden loss; painful red eye	Vitreous haemorrhage; neovascular glaucoma
Previous laser, injections, vitrectomy, cataract surgery	Number of PRP sittings, last injection, operated eye
Acuity distance and near; relative afferent pupillary defect (RAPD)	Baseline for treatment; RAPD suggests ischaemia or detachment
Pupil margin, undilated gonioscopy, intraocular pressure (IOP)	Neovascularisation of the iris (NVI) or angle (NVA) means PRP now; glaucoma in S.No 34
Lens	Cataract limits the view and the laser
Haemorrhages, microaneurysms, cotton-wool spots, hard exudate, quadrant by quadrant	Count quadrants for the 4-2-1 rule
Venous beading and loops, intraretinal microvascular abnormalities (IRMA)	Markers of severe ischaemia
New vessels at the disc (NVD) or elsewhere (NVE): size in disc areas, fibrous tissue, preretinal or vitreous haemorrhage	Decide high-risk or not
Macula with +90 dioptre (D) lens	Thickening or exudate within 500 µm of the centre; circinate rings
Fibrous bands, tractional detachment, laser scars	Advanced disease; extent of earlier PRP
Tests to mention	OCT, fundus fluorescein angiography (FFA), wide-field imaging, B-scan ultrasound if hazy; HbA1c, renal function, lipids
Classification to quote	Early Treatment Diabetic Retinopathy Study (ETDRS) and international scale; Diabetic Retinopathy Study (DRS) high-risk criteria; CSME; OCT patterns of DME

Diagnosis and summary to say

> **Say it**
>
> Mr X, a 52-year-old man with type 2 diabetes for 14 years on insulin, with poor control and hypertension, has had gradual blurring in both eyes for 8 months and floaters in the right eye for 2 weeks; he has had no laser or injections. Best-corrected visual acuity is 6/24 in the right eye and 6/18 in the left; there is no RAPD, no new vessels on the irides, and the IOP is 14 mmHg in each eye. Both fundi show dot-blot haemorrhages, microaneurysms and cotton-wool spots in all four quadrants with venous beading; the right eye has new vessels on the disc of about one-third disc area with a small preretinal haemorrhage, and the left has a half-disc-area frond of new vessels on the superotemporal arcade without haemorrhage. Both maculae show central thickening with a circinate ring of hard exudate within 500 µm of the fovea. My diagnosis is both eyes proliferative diabetic retinopathy with clinically significant macular oedema, with high-risk characteristics in the right eye. I would do OCT and FFA, give intravitreal anti-VEGF for the centre-involving oedema, start PRP in the right eye first, and work with his physician on sugar, blood pressure and lipids.
>
> S.No 11: Both eyes severe non-proliferative diabetic retinopathy with clinically significant, centre-involving macular oedema.
>
> S.No 51: Both eyes proliferative diabetic retinopathy with high-risk characteristics, for panretinal photocoagulation.

Viva questions and answers

Basics

Q Give the ETDRS-based international classification.
No apparent retinopathy; mild NPDR (microaneurysms only); moderate NPDR (more than microaneurysms, less than severe); severe NPDR (4-2-1 rule, no new vessels); PDR (NVD or NVE, preretinal or vitreous haemorrhage). DME is graded separately as absent, non-centre-involving or centre-involving.

Q State the 4-2-1 rule.
Severe NPDR if any one of: more than 20 intraretinal haemorrhages in each of 4 quadrants; definite venous beading in 2 or more quadrants; prominent IRMA in 1 or more quadrants. Two or more features make it very severe. The 1-year risk of PDR is about 50% for severe and 75% for very severe NPDR.

Q Define CSME.
ETDRS, any one of: retinal thickening at or within 500 µm of the centre of the macula; hard exudate at or within 500 µm of the centre with adjacent thickening; thickening of 1 disc area or more, any part within 1 disc diameter of the centre. It is a biomicroscopic definition, independent of vision.

Q What are the DRS high-risk characteristics?
NVD of one-quarter to one-third disc area or more (standard photograph 10A), with or without haemorrhage; any NVD with vitreous or preretinal haemorrhage; NVE of half a disc area or more with vitreous or preretinal haemorrhage. Untreated, about 26% of such eyes fall below 5/200 in 2 years; PRP cuts this by more than half.

Q Outline the pathogenesis.
Hyperglycaemia acts through the polyol (aldose reductase), advanced glycation end-product, protein kinase C and oxidative pathways; pericyte loss, basement membrane thickening and endothelial damage cause microaneurysms and leakage, while capillary closure causes ischaemia that drives VEGF release, IRMA and new vessels.

Q How do you tell IRMA from NVE?
IRMA are shunt or remodelled vessels inside the retina at the edge of non-perfusion; they do not cross major vessels and leak little on FFA. NVE lie on the retinal surface, may cross vessels, form fronds and leak heavily.

Q Why is NVD more dangerous than NVE?
There is no internal limiting membrane over the disc and the vitreous is attached there, so disc vessels bleed and contract readily; NVD also means widespread ischaemia.

Examination and signs

Q How do you tell a microaneurysm from a dot haemorrhage?
Both are small red dots; on FFA a patent microaneurysm is a bright dot that may leak, while a haemorrhage blocks fluorescence.

Q Compare hard exudates and cotton-wool spots.
Hard exudates are yellow, waxy, sharp-edged lipoprotein in the outer plexiform layer, often in circinate rings around leaking microaneurysms (a sign of leakage). Cotton-wool spots are fluffy white swellings of the nerve fibre layer from blocked axoplasmic flow after arteriolar closure (a sign of ischaemia).

Q What is advanced diabetic eye disease?
Vitreous or subhyaloid haemorrhage, fibrovascular proliferation, tractional detachment (concave, immobile), combined tractional-rhegmatogenous detachment, and NVI with neovascular glaucoma.

Q What glaucoma may S.No 34 have?
Neovascular glaucoma (NVI and NVA, open angle at first, then synechial closure), coexisting primary open-angle glaucoma (commoner in diabetics), steroid-induced glaucoma after injections or implants, or ghost-cell glaucoma after vitreous haemorrhage; gonioscopy decides.

Q Which ocular conditions protect against DR?
High myopia, extensive chorioretinal scarring, retinitis pigmentosa and optic atrophy reduce retinal oxygen demand; one-sided carotid stenosis does the opposite and worsens retinopathy on that side.

Investigations

Q How does OCT classify DME?
Patterns: diffuse spongy thickening, cystoid oedema, serous neurosensory detachment, and vitreomacular traction or a taut posterior hyaloid. It measures central subfield thickness (normal about 250–300 µm, machine-specific) and shows prognostic markers: disorganisation of the retinal inner layers (DRIL), ellipsoid zone loss and hyperreflective foci.

Q What does FFA add?
Focal leaking microaneurysms versus diffuse capillary leakage, macular ischaemia (enlarged irregular foveal avascular zone, which predicts poor results), capillary non-perfusion and leaking new vessels; wide-field FFA finds peripheral disease. OCT angiography (OCTA) shows non-perfusion and new vessels without dye.

Q When is B-scan ultrasound needed?
When vitreous haemorrhage hides the fundus: to look for tractional or rhegmatogenous detachment and the state of the posterior vitreous, which decides the timing of vitrectomy.

Management

Q What are the indications for PRP?
High-risk PDR (treat now) and any NVI or NVA; also non-high-risk PDR or very severe NPDR when follow-up is unreliable, in type 2 diabetes, before cataract surgery, in pregnancy, or when the fellow eye did badly.

Q Describe your PRP technique.
Consent, dilate, topical anaesthesia; a wide-field contact lens that magnifies the spot about 2 times (so set about 250 µm to get 500 µm on the retina); 532 nm green laser; 500 µm spot, 0.1 s, power raised from about 150–200 mW to a moderate grey-white burn, one burn-width apart; 1,200–1,600 burns in 3–4 sittings, inferior retina first in case haemorrhage later hides it.

Q Where do the burns go?
From 500 µm outside the nasal disc margin and 2 disc diameters (3,000 µm) from the foveal centre, outside the temporal arcades, out to the equator and beyond; avoid the macula, papillomacular bundle, major vessels, haemorrhages, fibrovascular tissue and the long ciliary nerves at 3 and 9 o'clock.

Q Why split PRP into sittings?
Treating everything at once risks severe pain, worse macular oedema, choroidal effusion, exudative detachment, and angle closure from forward rotation of the ciliary body.

Q How do you judge the response?
Review about 3 months after completion: regressing vessels thin, whiten and turn fibrotic, and leakage falls. Persistent or new vessels need fill-in laser between and in front of old scars, anti-VEGF, or vitrectomy.

Q How do you treat centre-involving DME?
Anti-VEGF first line when vision is reduced: ranibizumab 0.5 mg, aflibercept 2 mg or bevacizumab 1.25 mg, monthly for 4–6 doses then as needed or treat-and-extend. Switch to a dexamethasone implant 0.7 mg (or fluocinolone 0.19 mg) for poor response, in pseudophakes, or when monthly visits are impossible. With acuity of 6/7.5 or better, close observation is acceptable (Diabetic Retinopathy Clinical Research Network, DRCR, Protocol V).

Q Is macular laser still used?
Yes, for non-centre-involving CSME with focal leaking microaneurysms 500–3,000 µm from the centre: 50–100 µm, 0.05–0.1 s, light burns that whiten the microaneurysm; a modified grid for diffuse leakage; or deferred, added to anti-VEGF (Protocol I). Not for an ischaemic macula.

Q What did Protocols S and T show?
Protocol S: ranibizumab 0.5 mg was non-inferior to PRP for vision in PDR at 2 years, with less field loss, less DME and fewer vitrectomies, but only with strict follow-up (CLARITY: aflibercept beat PRP at 1 year). Protocol T: in DME the three drugs were similar when vision was 6/12 or better; with 6/15 or worse, aflibercept did best at 1 year.

Q PDR with CSME: what comes first?
The macula, or both together: anti-VEGF for the oedema, then PRP soon after (within 1–2 weeks in high-risk eyes), because PRP alone worsens oedema; anti-VEGF also speeds regression of the new vessels.

Q What are the indications for vitrectomy?
Non-clearing vitreous haemorrhage (1–3 months; sooner in type 1 diabetes, bilateral haemorrhage, an only eye, or detachment on B-scan); tractional detachment involving or threatening the macula; combined tractional-rhegmatogenous detachment; dense premacular haemorrhage; active proliferation despite full PRP; DME with traction; haemorrhage with NVI. Give anti-VEGF 3–7 days before to reduce bleeding.

Q What do the landmark glycaemic control trials teach?
Diabetes Control and Complications Trial (DCCT, type 1): intensive control cut the onset of retinopathy by 76% and progression by 54%, with lasting benefit (metabolic memory) despite possible early worsening. United Kingdom Prospective Diabetes Study (UKPDS, type 2): intensive glucose control cut microvascular end points by 25%; tight blood pressure control cut retinopathy progression by about a third. ACCORD and FIELD: fenofibrate slowed progression and reduced laser need.

Q What are the screening and follow-up intervals?
Type 1: 5 years after diagnosis (not usually before puberty), then yearly. Type 2: at diagnosis, then yearly. Pregnancy: before conception or in the first trimester, then each trimester. Follow-up: no DR yearly; mild NPDR 6–12 months; moderate 4–6 months; severe 2–3 months; non-high-risk PDR about 2 months or treat.

Complications and prognosis

Q What are the complications of PRP?
Early: pain, blur, worse macular oedema, choroidal effusion, exudative detachment, angle closure, vitreous haemorrhage. Errors: foveal burn, Bruch's membrane rupture with choroidal neovascularisation, corneal, iris or lens burns. Late: constricted field (may bar driving), night blindness, poorer contrast and colour, loss of accommodation and a dilated pupil, contraction of fibrovascular tissue with tractional detachment, and scar expansion.

Q How do you plan cataract surgery in a diabetic?
Control sugar and blood pressure; treat DME first with anti-VEGF or steroid, or inject at the end of surgery; complete PRP for PDR before surgery (or soon after if the view is poor), since surgery accelerates retinopathy and rubeosis; large capsulorhexis and a 6 mm acrylic intraocular lens (IOL), avoiding silicone (in case silicone oil is needed) and multifocal IOLs with maculopathy; topical non-steroidal drops for 4–6 weeks; close review.

Q What is the prognosis?
Timely PRP halves severe visual loss and anti-VEGF improves vision in most DME eyes. The outlook is poor with macular ischaemia, a foveal exudate plaque, long-standing cystoid oedema, tractional detachment, neovascular glaucoma or poor systemic control; PDR also marks a high risk of kidney and cardiovascular disease.

Recent advances and favourite examiner questions

Q Why is anti-VEGF alone for PDR risky in India?
Vessels regress only while injections continue; patients who miss visits can return with tractional detachment or neovascular glaucoma, so PRP stays the durable standard when follow-up is uncertain.

Q What is the crunch phenomenon?
Rapid contraction of fibrovascular membranes with worsening tractional detachment after anti-VEGF (or heavy PRP) in eyes with extensive proliferation; avoid it by operating within a few days of the injection.

Q What did ETDRS conclude about early PRP and aspirin?
Early scatter laser is not advised for mild or moderate NPDR but may be considered for severe NPDR or early PDR, especially in type 2 diabetes; aspirin neither helps nor worsens retinopathy or haemorrhage. Focal laser halved moderate visual loss in CSME (about 24% to 12% at 3 years).

Q What is new?
Faricimab (VEGF-A and angiopoietin-2) and aflibercept 8 mg for longer intervals; biosimilars; the fluocinolone implant lasting up to 3 years; pattern scan laser (PASCAL) with 10–30 ms burns for faster, less painful PRP; subthreshold micropulse laser for DME; and anti-VEGF in moderate to severe NPDR (Protocol W cut progression but gave no visual gain).

Examiner traps

- Calling PDR "high-risk" without sizing new vessels in disc areas or noting haemorrhage.
- Using CSME and "centre-involving DME" interchangeably: CSME is clinical (ETDRS); centre involvement is on OCT.
- Starting PRP in an eye with centre-involving oedema without treating the macula.
- Dilating before checking the pupil margin and doing gonioscopy for NVI and NVA.
- Confusing IRMA with NVE, or old laser scars with chorioretinitis (laser scars are regular and spare the macula and disc).
- Giving PRP settings without spot size, duration, spacing, number of burns and sittings.
- Forgetting HbA1c, blood pressure, kidneys and lipids.

- 4-2-1: haemorrhages in 4 quadrants, beading in 2, IRMA in 1 = severe NPDR.
- CSME: thickening or exudate with thickening within 500 µm of the centre, or 1 disc area of thickening within 1 disc diameter.
- High-risk PDR: NVD of a quarter to a third disc area; any NVD with haemorrhage; NVE of half a disc area with haemorrhage.
- PRP (house): 500 µm, 0.1 s, one burn-width apart, 1,200–1,600 burns in 3–4 sittings.
- Anti-VEGF first for centre-involving DME; Protocol S: ranibizumab non-inferior to PRP if follow-up is assured.
- DCCT 76% and 54%; UKPDS 25% for glucose, about a third for blood pressure.
- High-risk PDR factors, "**N**ew **D**isc **S**ize **B**leed": new vessels, at the disc, large size, bleeding; three of four means treat now.

Read more: Baidya p.231–252, 436–438 · Namrata p.229–243, 308–313 · Aravind 6.3–6.4 (p.330–348), case sheet p.649–650

ER5 Uveal colobomas — retinochoroidal, iris and optic disc coloboma

Retina · On the list: S.No 6, 18, 45, 49 · ★ Patient already arrived: S.No 6

At a glance

- **What it is:** a congenital gap in the uvea and the tissues it induces, from failure of the embryonic (optic) fissure to close in the 5th–7th week. *Typical* = inferonasal, along the fissure line; *atypical* = anywhere else.
- **Spectrum on tomorrow's list:** iris (keyhole pupil), retinochoroidal (choroidal) and optic disc coloboma — often together, often with microphthalmos and microcornea.
- **Why examiners keep it:** embryology, Ida Mann classification, the peculiar retinal detachment (RD), the excavated-disc differential and the syndromes.
- **S.No 18 and S.No 45:** "Both eyes typical retinochoroidal coloboma, Ida Mann type __ (disc involved or spared, macula involved or spared), retina attached; with or without microphthalmos and nystagmus."
- **S.No 6 (patient already here):** "Left eye typical (inferonasal) iris coloboma with keyhole pupil; right eye microcornea with corectopia — bilateral congenital anomaly in the coloboma spectrum; fundus and disc coloboma and RD to be excluded in both eyes."
- **S.No 49:** "(Side) eye epibulbar lipodermoid (dermolipoma) with optic disc coloboma — Goldenhar syndrome (oculo-auriculo-vertebral spectrum) to be excluded."
- **Rule:** every iris coloboma earns a dilated fundus examination of both eyes; every coloboma earns a look at the ears, heart, face and family.

What to look for and present

Look for	What to say or record
Complaint and age	Child or young adult with poor vision since early childhood, squint, nystagmus, glare or an "odd pupil". A recent painless drop in vision means RD until proved otherwise.
Family and birth history	Bilateral cases are often autosomal dominant — examine parents and siblings; consanguinity; maternal infection (a toxoplasma scar mimics a macular coloboma).
Systemic history and general examination	Heart disease, choanal atresia, deafness, genital and ear anomalies (CHARGE); preauricular tags, hemifacial microsomia, spine (Goldenhar); cleft lip or palate and polydactyly (Patau); anal atresia (cat-eye).
Globe and cornea	Microphthalmos (short axial length); microcornea (horizontal diameter under 10 mm in an adult); pear-shaped cornea over an iris coloboma. In S.No 6 measure both corneas.
Iris and pupil	Site (typical = inferonasal); complete (stroma and pigment layer absent) or incomplete (one layer, seen on transillumination); to the root (keyhole) or margin only (notch); corectopia; heterochromia.
Lens	Flattened or notched inferior equator from absent zonules ("lens coloboma"); cataract; subluxation.
Fundus — the coloboma	Inferonasal, oval or shield-shaped; glistening white bare sclera; sharp, often pigmented edge; depressed floor; retinal vessels dipping over the rim; choroidal vessels in the floor; size in disc diameters (DD); upper limit relative to disc and macula.
Disc	Outside the coloboma and normal, outside but abnormal, or itself colobomatous. Enlarged white bowl decentred inferiorly with a thin inferior rim = disc coloboma; central glial tuft with radial vessels = morning glory.
Retinal detachment	Breaks at the margin and inside the gap; RD outside only, inside only, or both; loss of the excavated contour, grey mobile retina over the white floor; full periphery with indentation.
Vision, refraction, motility	Best-corrected visual acuity (BCVA); cycloplegic refraction (myopia, astigmatism, anisometropia); amblyopia; sensory strabismus; nystagmus.
Field	Superior absolute field defect matching the inferior coloboma; enlarged blind spot with a disc coloboma.
Tests to mention	B-scan ultrasonography; optical coherence tomography (OCT) of margin, macula and disc; axial length; ultrasound biomicroscopy (UBM) for a ciliary coloboma; echocardiography, hearing and renal assessment.
Classification you will be asked	Ida Mann types I–VII (relation to the disc); Gopal types I–VI of disc involvement; Gopal pattern of breaks and RD.

Diagnosis and summary to say

SAY IT

Mr X, 19, has had poor vision in both eyes since childhood, with no recent change. Both eyes are small with microcornea and a horizontal pendular nystagmus; best-corrected vision is 6/36 in the right eye and 6/24 in the left. Each fundus shows a large, sharply margined, glistening white defect below and nasal to the disc, with pigmented edges, bare sclera in a depressed floor and retinal vessels dipping over its rim. In the right eye the defect includes the disc and extends above it; in the left eye it reaches the upper border of the disc. Both maculae are spared, the retina is attached and no break is seen at either margin. The ears, heart and face are normal.

Diagnosis (S.No 18 and S.No 45): both eyes typical retinochoroidal coloboma — right eye Ida Mann type I, left eye type II — with microphthalmos, microcornea and nystagmus; maculae spared; retina attached.

S.No 6: left eye typical iris coloboma with keyhole pupil; right eye microcornea with corectopia; both fundi to be examined for coloboma.

S.No 49: right eye epibulbar lipodermoid with optic disc coloboma — Goldenhar syndrome to be excluded.

Viva questions and answers

Basics

Q What does "coloboma" mean, and which colobomas share one embryology?
Greek for "curtailed" — a congenital gap in an ocular tissue. Iris, ciliary body, zonular, retinochoroidal and disc colobomas all come from incomplete closure of the embryonic fissure; a lid coloboma is a separate defect of the lid folds.

Q Describe the embryology of the embryonic (optic) fissure.
In the 4th week the optic vesicle invaginates into a two-layered optic cup whose inferior wall is deficient — the embryonic fissure, through which the hyaloid artery enters. Closure starts about day 33 at the equator and runs forwards and backwards, complete by the 7th week (7–14 mm stage). Failure at any point leaves a coloboma there.

Q Why are typical colobomas inferonasal?
The fissure lies on the inferior aspect of the cup and stalk, slightly nasal to the vertical meridian; so typical iris, choroidal and disc colobomas lie inferonasally and the field defect is superior.

Q Why is bare sclera seen in a choroidal coloboma?
At the unclosed fissure the outer layer of the cup fails to form retinal pigment epithelium (RPE); RPE induces the choroid and inner sclera, so the choroid is absent and the sclera thin and ectatic. The retina is reduced to a thin glial "intercalary membrane" through which white sclera shows.

Q What are typical and atypical colobomas?
Typical — along the fissure line (inferonasal). Atypical — any other site; a macular "coloboma" is often the scar of intrauterine inflammation such as toxoplasmosis rather than a fissure defect.

Q Classify iris coloboma.
Site — typical (inferonasal) or atypical; depth — complete (stroma and pigment epithelium absent) or incomplete (one layer, seen on transillumination); extent — total (to the root, keyhole pupil) or partial (margin only, notch). A strand across the gap is a bridge coloboma.

Q How common is coloboma and how is it inherited?
About 0.5–2.2 per 10,000 births; bilateral in 60–70%. Most are sporadic; familial cases are usually autosomal dominant with variable penetrance. Genes include PAX6, PAX2, SOX2 and CHD7 (microphthalmia–anophthalmia–coloboma spectrum).

Examination and signs

Q Give the Ida Mann classification of retinochoroidal coloboma.
Type I — includes the disc and extends above it; II — reaches the upper border of the disc; III — below the disc, separated from it by normal retina; IV — disc coloboma alone; V — isolated gap along the fissure line with normal retina above and below; VI — pigmentary disturbance only; VII — extreme peripheral coloboma.

Q What is Gopal's classification of the disc in fundus coloboma?
I — disc outside the coloboma and normal; II — outside but abnormal; III — outside and itself colobomatous; IV — inside but normal; V — inside and colobomatous (commonest, about 44%); VI — disc not identifiable. Types I–III see better and are often myopic; IV–VI go with microphthalmos.

Q Describe an optic disc coloboma.
An enlarged disc holding a sharply defined, glistening white, bowl-shaped excavation decentred inferiorly; thin or absent inferior rim with the superior rim spared; normal vessels from the edges; no glial tuft. It may join a retinochoroidal coloboma and can cause serous macular detachment.

Q How do you tell disc coloboma from morning glory disc anomaly?
Coloboma — excavation within the disc, decentred inferiorly, no glial tuft, little pigment, normal vessels, often bilateral and familial, with iris and fundus colobomas. Morning glory — disc within a central funnel-shaped excavation, white glial tuft, pigmented annulus, many straight radial vessels, usually unilateral, linked to transsphenoidal basal encephalocele; serous RD in 26–38%.

Q How do optic disc pit and peripapillary staphyloma differ?
Optic pit — small round grey-white depression, usually temporal, in a normal-sized disc; about 1 in 11,000; serous maculopathy in 25–75%, in the 3rd–4th decade. Peripapillary staphyloma — deep cup-shaped excavation around a relatively normal disc, with no glial tuft or vascular anomaly.

Q What field defect does a typical coloboma produce?
An absolute superior defect matching the inferior gap, often unnoticed because it is congenital; a disc coloboma adds an enlarged blind spot. Central vision depends on sparing of the macula and papillomacular bundle.

Q What is a keyhole pupil and what mimics it?
A round pupil continuous with an inferonasal iris gap reaching the root; it still reacts through the remaining sphincter. Mimics: surgical iridectomy (usually superior, cut edges), iridodialysis (peripheral D-shaped gap after trauma), Axenfeld–Rieger anomaly and iridocorneal endothelial syndrome (corectopia, polycoria).

Q What is "lens coloboma"?
A misnomer: zonules are missing in that sector, so the lens equator flattens or notches for lack of tension. True loss of lens tissue is rare.

Q In S.No 6, how do you present the right microcornea and corectopia with the left iris coloboma?
As one bilateral developmental anomaly. Dilate both eyes for fundus and disc coloboma, measure corneas and axial lengths, check intraocular pressure and gonioscopy, and exclude Axenfeld–Rieger anomaly (posterior embryotoxon, iris strands) as the cause of the corectopia.

Investigations

Q Which investigations would you order?
B-scan (excavation, RD in hazy media); OCT of macula, disc and margin; axial length; UBM for ciliary coloboma and before lens surgery; wide-field photography; perimetry. Systemic — echocardiography, hearing test, renal ultrasound, magnetic resonance imaging (MRI) of the brain if neurological signs, gene panel.

Q What does OCT show at the margin of a coloboma?
Near the edge the retina splits: inner layers continue over the gap as the intercalary membrane, while outer layers turn back and fuse with the RPE at the margin — the "locus minoris resistentiae", where breaks form. OCT shows small breaks, schisis and shallow detachment of the membrane.

Q What systemic work-up does S.No 49 need?
Examine ears (tags, microtia), face (hemifacial microsomia), lids (upper lid coloboma), motility (Duane syndrome) and spine; hearing test, spine X-ray for hemivertebrae, echocardiography; MRI of the orbit if the lipodermoid extends deep.

Management

Q Outline the management of a retinochoroidal coloboma.
Cycloplegic refraction and glasses; amblyopia and squint treatment; low-vision aids; dilated review every 6–12 months with RD symptoms explained; prophylactic laser; cataract surgery when needed; protective polycarbonate glasses if one eye is poor; genetic counselling.

Q How is prophylactic laser done, and does it work?
Two to three rows of moderate burns (200–300 µm, 0.1–0.2 s, 532 nm green or diode; indirect delivery under anaesthesia in children) on normal retina just outside the margin, sparing disc and macula — temporally below the inferotemporal arcade. No burns inside the coloboma (no RPE). In one Indian series RD occurred in 2.9% of lasered versus 24.1% of untreated eyes.

Q Describe the patterns of RD in coloboma.
A break in the intercalary membrane alone lifts only the membrane; breaks in the membrane and at the margin give a clinical RD; a peripheral break alone gives an RD that stops at the coloboma edge. Gopal: type I — RD not entering the coloboma; type II — RD extending into it (A subclinical, B break inside, C inside and outside, D peripheral only, E none seen).

Q Why is vitrectomy preferred to scleral buckling for coloboma RD?
Breaks are posterior, at the edge or inside the coloboma and hard to see; cryotherapy cannot create adhesion without RPE and choroid; the sclera is thin. Pars plana vitrectomy (PPV) removes traction, finds breaks under magnification and allows endolaser along the whole margin to wall off the coloboma.

Q Which tamponade, and how do these eyes do?
Usually silicone oil (perfluoropropane, C3F8, gas is an alternative). Re-detachment after oil removal is common, so oil is often kept long term. A buckle alone suits only an RD from a peripheral break that does not enter the coloboma.

Q What are the problems in cataract surgery in these eyes?
Microcornea and microphthalmos (crowded chamber; scleral tunnel helps); zonular weakness in the coloboma sector (capsular tension ring or segment); unreliable biometry with an ectatic posterior wall. Glare from an iris coloboma: tinted or printed-iris contact lens, McCannel or Siepser pupilloplasty, or an iris-print intraocular lens (IOL) at surgery.

Complications and prognosis

Q List the complications of uveal coloboma.
RD (23–42% with retinochoroidal coloboma); early cataract and subluxation; amblyopia, anisometropia and sensory strabismus; nystagmus; choroidal neovascularisation (CNV) at the margin; serous macular detachment and macular hole with disc coloboma; microphthalmos with orbital cyst; secondary glaucoma.

Q Where do the breaks lie in coloboma RD?
Most within 2 DD of the margin — about 54% at the margin, 25% within the coloboma and 8% at the macula; the rest peripheral. Several breaks often coexist.

Q How is CNV at the coloboma margin treated?
It arises where Bruch's membrane ends at the margin. Give intravitreal anti-vascular endothelial growth factor (anti-VEGF) — ranibizumab 0.5 mg or bevacizumab 1.25 mg — monthly until inactive, then as needed.

Q What decides the prognosis?
Sparing of the macula and papillomacular bundle (Ida Mann III, V and VII and Gopal I–III do best), amblyopia control and early detection of RD. Surgery reattaches most eyes, but vision is limited by the anomaly itself.

Recent advances and favourite examiner questions

Q Expand CHARGE.
Coloboma, Heart defects, Atresia of the choanae, Retarded growth and development, Genital hypoplasia, Ear anomalies and deafness; most cases carry a CHD7 mutation.

Q Which syndromes are associated with uveal coloboma?
CHARGE; trisomy 13 (Patau — microphthalmos, cleft lip and palate, polydactyly) and trisomy 18; cat-eye syndrome; 13q deletion; Aicardi syndrome; Walker–Warburg; Meckel–Gruber; basal cell naevus syndrome; Goldenhar; papillorenal syndrome (PAX2).

Q What is cat-eye syndrome?
Partial tetrasomy of chromosome 22q11: a vertical inferior iris coloboma gives a cat-like pupil, with anal atresia, preauricular tags or pits, and heart and renal defects.

Q What is Goldenhar syndrome?
The oculo-auriculo-vertebral spectrum (first and second branchial arch defect): epibulbar dermoid or lipodermoid, upper lid coloboma, Duane syndrome, preauricular tags, microtia, hemifacial microsomia and hemivertebrae; disc or choroidal coloboma may coexist — hence S.No 49.

Q What else can look like a fundal coloboma?
Healed toxoplasma or toxocara scar; myopic chorioretinal atrophy and staphyloma; morning glory and peripapillary staphyloma; laser or cryotherapy scars; North Carolina macular dystrophy and torpedo maculopathy (macular lesions).

Examiner traps

- Calling a superior, straight-edged iris gap a coloboma — that is usually a surgical iridectomy; typical colobomas are inferonasal.
- Presenting an iris coloboma (S.No 6) without dilating both eyes; examiners expect a search for fundus and disc coloboma and RD.
- Saying the field defect is inferior — the coloboma is inferior, so the field defect is superior.
- Missing an RD because the floor already looks white — look for loss of the excavated contour, mobile grey retina and breaks at the margin.
- Offering cryotherapy and a buckle for a break inside the coloboma, or laser inside it — there is no RPE or choroid to form an adhesion.
- Confusing disc coloboma with morning glory (glial tuft, radial vessels, disc inside the excavation).
- Forgetting the ears, heart, face, spine and family, or treating a lid coloboma as a fissure defect.

- Typical coloboma = embryonic fissure fails to close (5th–7th week) → inferonasal gap → superior field defect.
- No RPE → no choroid → bare sclera under a thin intercalary membrane.
- Ida Mann: I above disc, II to upper disc border, III below disc, IV disc only, V isolated gap, VI pigment only, VII extreme periphery.
- RD in 23–42%; breaks at the margin (locus minoris resistentiae) or in the membrane; PPV, endolaser and silicone oil.
- Barrage laser outside the margin, sparing disc and macula, cuts the RD rate.
- Disc coloboma: excavation within the disc; morning glory: disc within the excavation.
- Mnemonic: **CHARGE** — Coloboma, Heart, Atresia choanae, Retardation, Genital, Ear.

Read more: Baidya p.338–342 (fundal coloboma), p.636–642 (disc anomalies), p.2–3 (Goldenhar, lipodermoid) · Namrata p.318–324 · Aravind 7.6 (p.474–477)

ER6 Full-thickness macular hole

Retina · On the list: S.No 38 · ★ Patient already arrived: S.No 38

At a glance

- **What it is:** a full-thickness macular hole (FTMH) — a defect at the foveola interrupting every neural retinal layer from the internal limiting membrane (ILM) to the retinal pigment epithelium (RPE).
- **Cause:** mostly idiopathic (primary), from vitreofoveal traction during perifoveal posterior vitreous detachment (PVD); secondary holes follow trauma, high myopia, cystoid macular oedema (CME) and other disease.
- **Who:** women over 60 (about 2:1), prevalence about 3 per 1,000; the fellow eye develops a hole in 10–15% within 5 years.
- **Why examiners keep it:** a slit-lamp sign case (Watzke–Allen), two staging systems (Gass and the International Vitreomacular Traction Study, IVTS), and a surgery that now closes about 90% of holes.
- **S.No 38 (patient already here):** "Left eye idiopathic full-thickness macular hole, Gass stage __, IVTS small, medium or large (minimum diameter __ µm) with or without vitreomacular traction (VMT); right eye normal or with vitreomacular adhesion; lens status __."
- **Before you present:** know the duration of symptoms, the hole size on optical coherence tomography (OCT), the vitreous status, the lens, the fellow eye and whether she can posture.

What to look for and present

Look for	What to say or record
Symptoms and duration	Central blur, metamorphopsia, a gap in the centre of letters, reading difficulty, often noticed on covering the good eye. Under 6 months predicts better closure and vision.
Age, sex, cause	Woman over 60 = idiopathic. Young patient: ask about blunt trauma, high myopia, eclipse or laser exposure, lightning. Also diabetic or vein-occlusion oedema, uveitis, previous retinal detachment surgery.
Fitness to posture	Neck or back disease, obesity, breathlessness; planned air travel or high-altitude trips; one-eyed patient.
Visual acuity	Usually 6/18–6/60 once full thickness; large, long-standing holes fall below 6/60. Vision worse than the hole explains — look for cataract, age-related macular degeneration or glaucoma.
Lens	Nuclear sclerosis matters: most phakic eyes develop cataract within 1–2 years of vitrectomy.
The hole on biomicroscopy	Round, punched-out red defect at the foveola, about one-third of a disc diameter in stage 3; grey cuff of subretinal fluid; yellow dots in the base; operculum on the hyaloid; Weiss ring if complete PVD; epiretinal membrane.
Watzke–Allen test	Thin vertical slit beam across the hole with a macular lens: patient sees a **break** in the line = positive (FTMH). Narrowing or bending only = pseudohole, lamellar hole or cyst.
Laser aiming beam test	A 50 µm aiming beam placed in the hole disappears; seen again on intact retina.
Amsler grid	Central distortion with lines pinched towards the centre, small central scotoma; used to watch the fellow eye.
Fellow eye	Vitreomacular adhesion or traction on OCT, an early (stage 1) hole, or complete PVD (low risk).
Periphery	Lattice or breaks to be treated before surgery; iatrogenic breaks are a known complication.
OCT	Confirms full thickness; minimum linear diameter (narrowest, parallel to RPE) and base diameter; VMT; epiretinal membrane; cystic edges; ellipsoid zone.
Classification	Gass 1A, 1B, 2, 3, 4; IVTS by size (≤250, 250–400, >400 µm), by VMT (with or without), by cause (primary or secondary).

Diagnosis and summary to say

SAY IT

Mrs X, 66, noticed distortion and then a central blur in her left eye over four months. Best-corrected vision is 6/6 in the right eye and 6/36 in the left, and the left lens shows early nuclear sclerosis. At the left macula there is a round, punched-out red defect about one-third of a disc diameter across, with a grey cuff of subretinal fluid and yellow dots in its base, and the Watzke–Allen test shows a broken beam. Optical coherence tomography shows a full-thickness defect with a minimum diameter of 450 micrometres, cystic edges and the posterior hyaloid still attached at the hole edge. The right macula is normal and both peripheries are healthy. She can lie face-down and has no travel plans.

Diagnosis (S.No 38): left eye idiopathic full-thickness macular hole, Gass stage 3, large with vitreomacular traction by the International Vitreomacular Traction Study classification, of about four months' duration; left eye early nuclear cataract; right eye normal.

Viva questions and answers

Basics

Q Define a full-thickness macular hole.
A foveal lesion with full-thickness interruption of all neural retinal layers from the ILM to the RPE (IVTS definition). Idiopathic holes are round and caused by vitreofoveal traction.

Q What are the risk factors?
Age over 60–65 and female sex (about 2:1) are the only proven systemic risks. Ocular: VMT in the eye, a hole in the other eye, high myopia, blunt trauma, eclipse exposure, CME.

Q Explain the pathogenesis.
Gass blamed tangential traction of the prefoveal cortex. OCT shows that holes begin during perifoveal PVD: the vitreous stays stuck to the foveola and pulls anteroposteriorly, splitting the inner fovea into a pseudocyst (1A), breaking the outer retina (1B), then opening the roof (2). Tangential pull then retracts the edges centrifugally (3). Little tissue is lost, which is why closure restores vision.

Q Give the Gass staging.
1A — impending hole, yellow spot (foveolar detachment); 1B — occult hole, yellow ring; 2 — small full-thickness hole under 400 µm, often eccentric or crescentic; 3 — full-thickness hole of 400 µm or more with a cuff of fluid and an operculum, vitreous not fully detached; 4 — any full-thickness hole with complete PVD (Weiss ring).

Q How does the IVTS classify the vitreomacular interface?
Vitreomacular adhesion (VMA) — perifoveal vitreous separation with the cortex still attached within a 3 mm radius of the fovea and no retinal change; VMT — the same with distortion of the foveal surface or retina, but no full-thickness defect. Both are graded focal (≤1500 µm) or broad (>1500 µm), isolated or with other macular disease.

Q How does the IVTS classify a full-thickness hole, and how does it match Gass?
By minimum width — small ≤250 µm, medium >250–400 µm, large >400 µm; by vitreous — with or without VMT; by cause — primary (from VMT) or secondary. Gass 1 = VMT; 2 = small or medium FTMH with VMT; 3 = medium or large FTMH with VMT; 4 = FTMH without VMT.

Q What is an operculum?
A disc of condensed vitreous cortex and glial tissue on the detached hyaloid over the hole; about 40% contain photoreceptor elements. Most are pseudo-opercula — the hole forms mainly by retraction, not tissue loss.

Q What is a "stage 0" hole?
Vitreomacular adhesion in the fellow eye of a patient with a hole; it carries a higher risk of a hole forming than in an eye with complete PVD.

Examination and signs

Q Describe the Watzke–Allen test.
With a fundus contact lens or a 90 dioptre lens, a thin vertical then horizontal slit beam is placed across the centre of the lesion. A patient with an FTMH sees the line broken (gap); with a pseudohole, lamellar hole or cyst the line is intact but thinned or bent.

Q What is the laser aiming beam test?
A 50 µm helium–neon aiming beam is placed in the hole: the patient cannot see it, but sees it again on surrounding retina. It is thought to be more sensitive and specific than Watzke–Allen.

Q Give the differential diagnosis.
Lamellar macular hole, macular pseudohole with epiretinal membrane, CME with a large central cyst, VMT with foveal pseudocyst, solar retinopathy, central serous chorioretinopathy, foveal drusen, choroidal neovascularisation (CNV), macular atrophy and haemorrhage.

Q How do lamellar hole and pseudohole differ from FTMH on OCT?
Lamellar — irregular inner foveal defect with intraretinal splitting between outer plexiform and outer nuclear layers, photoreceptors intact, often yellowish epiretinal proliferation. Pseudohole — epiretinal membrane with a central opening, steep heaped edges and near-normal foveal thickness, no tissue loss. Both lack a cuff of fluid, and Watzke–Allen is negative.

Q What does the Amsler grid show?
Central distortion with lines bowed towards the centre and sometimes a small central scotoma; it documents early holes and monitors the fellow eye.

Q What is the typical vision by stage?
Stages 1–2 about 6/9–6/24; stages 3–4 about 6/30–6/120, large old holes being worst. Vision much worse than the hole explains should prompt a search for another cause.

Investigations

Q What does OCT measure, and which indices predict outcome?
Minimum linear diameter (narrowest point, used for IVTS size), base diameter at the RPE, height, VMT, epiretinal membrane and ellipsoid zone. Macular hole index (height ÷ base) ≥0.5, hole form factor (sum of arm lengths ÷ base) >0.9, and tractional hole index (height ÷ minimum diameter) >1.4 predict closure and better vision; a minimum diameter under about 310 µm does well.

Q What do fluorescein angiography and autofluorescence show?
Angiography: an early, well-defined window defect that does not enlarge and does not leak — this separates it from CME and CNV. Autofluorescence: a bright foveolar spot because the macular pigment that normally masks the RPE is displaced.

Q Which other tests may help?
Microperimetry (dense central scotoma, fixation stability), multifocal electroretinography (reduced central response) and B-scan when media are hazy to judge the vitreous.

Q How is closure described on postoperative OCT?
Type 1 — tissue bridges the hole (U-shaped best, then V); type 2 — edges flattened but bare RPE remains (W or "flat-open"), with poor vision and risk of reopening.

Management

Q How do you manage by stage?
Stage 1 (VMT) — observe with OCT and Amsler every 2–4 months; about half progress to a full-thickness hole, the rest abort when the vitreous releases. Stages 2–4 — vitrectomy. Selected small holes with VMT — pharmacological or pneumatic vitreolysis.

Q What is ocriplasmin?
Recombinant truncated human plasmin; a single intravitreal 125 µg injection cleaves the vitreoretinal interface. MIVI-TRUST trials: VMT release in about 26% versus 10% with placebo; FTMH closure in about 40% of holes ≤400 µm with VMT. Best in small holes, focal adhesion, no epiretinal membrane. Side effects: transient vision loss, dyschromatopsia, electroretinogram changes; now little used.

Q What is pneumatic vitreolysis?
An office intravitreal injection of 0.3 mL pure perfluoropropane (C3F8) gas to separate the vitreous from the fovea; it releases most VMT, but closes only a minority of full-thickness holes and carries a risk of retinal tears.

Q Describe the operation.
Three-port 23–25 gauge pars plana vitrectomy (PPV); induction of PVD (triamcinolone to see the hyaloid, active suction); ILM staining with brilliant blue G 0.025–0.05%; pinch-and-peel of ILM 1–2 disc diameters around the hole; indented peripheral search and laser to breaks; fluid–air exchange; non-expansile gas (sulphur hexafluoride, SF6 20%, lasts about 2 weeks; or C3F8 14%, about 6–8 weeks); face-down positioning.

Q Why peel the ILM?
It removes residual cortex and tangential traction, takes away the scaffold for epiretinal membrane, and stimulates glial repair of the hole. Closure rises to about 90–98% and late reopening falls.

Q How long must the patient lie face-down?
Traditionally 7–10 days; most surgeons now ask for 3–5 days, longer for holes over 400 µm. Holes under 400 µm often close with simply avoiding lying on the back. Patients who cannot posture may be given silicone oil, but closure and vision are better with gas.

Q What do you tell the patient about the gas?
No air travel, high-altitude trips or nitrous oxide anaesthesia until the bubble has gone — expansion raises the intraocular pressure and can occlude the central retinal artery. Vision is poor until the bubble falls below the visual axis.

Q How do you manage large or failed holes?
Inverted ILM flap — a hinged flap of ILM is left at the edge and folded over or into the hole as a scaffold (closure about 98% versus 88% with plain peel in large holes). Others: free autologous ILM plug, lens capsule or amniotic membrane plug, autologous retinal graft, platelet concentrate, longer-acting gas.

Q Would you combine cataract surgery?
In phakic patients over about 50, combined phacovitrectomy is reasonable because 80–98% develop nuclear cataract within a few years; aim slightly myopic, since gas causes a small myopic shift. Cataract surgery later can reopen a closed hole, especially if CME develops.

Complications and prognosis

Q List the complications of macular hole surgery.
Cataract (commonest); raised intraocular pressure from gas; peripheral breaks (3–17%) and retinal detachment (RD; 1–5%, usually inferior); visual field loss (10–20% in older series, from air infusion drying the nerve fibre layer); reopening (2–10%); dye toxicity (indocyanine green); endophthalmitis (rare).

Q What predicts the visual outcome?
Hole size (minimum diameter under 400 µm, and especially ≤250 µm, best), duration under 6 months, better preoperative vision, intact ellipsoid zone, favourable OCT indices, idiopathic rather than traumatic or myopic cause, and closure at the first operation.

Q What results do you quote?
Anatomical closure about 90–100% for recent idiopathic holes; vision improves in 80–90%, and about 65% reach 6/12 or better; up to 10% lose vision.

Q What is the risk to the fellow eye?
About 10–15% over 5 years; very low once that eye has a complete PVD, higher with VMA or VMT on OCT. Teach Amsler monitoring and repeat OCT.

Recent advances and favourite examiner questions

Q How is a traumatic macular hole different?
Young patient after blunt trauma (sudden vitreous traction and contusion); irregular, often larger, PVD often absent; commotio or choroidal rupture may coexist. Many close on their own within 3–6 months, so observe first, then operate (PPV, ILM peel or flap, gas).

Q What is special about a myopic macular hole?
It arises in a posterior staphyloma with foveoschisis; it may lead to macular hole RD, especially with axial length over 30 mm. The ILM is stiff and the staphyloma resists closure, so an inverted flap, longer tamponade or silicone oil, or a macular buckle is used.

Q What is the CLOSE surgical size grading?
A newer surgical extension of the IVTS sizes: large 400–550 µm, extra-large 550–800 µm, giant over 800 µm — peel or flap for large, inverted flap for extra-large, and other techniques (retinal graft, hydrodissection) for giant holes.

Q Which dye stains which tissue?
Brilliant blue G (0.025–0.05%) — ILM; trypan blue (0.15%) — epiretinal membrane; triamcinolone — vitreous; indocyanine green stains ILM well but is the most toxic to RPE and nerve fibres.

Examiner traps

- Calling every red foveal spot a hole — do the Watzke–Allen test and OCT; a pseudohole with epiretinal membrane is the commonest mimic.
- Mixing Gass stages with IVTS size: IVTS uses the minimum linear diameter at its narrowest point, not the base diameter.
- Defining stage 4 by size — stage 4 means complete PVD with a Weiss ring, whatever the size.
- Offering laser to the hole itself; laser is only for peripheral breaks.
- Forgetting posture, air travel, nitrous oxide and the one-eyed patient before advising gas.
- Missing the fellow eye and not warning a phakic patient that cataract will follow.
- Blaming the hole for vision that is worse than its size explains, without looking for cataract, macular degeneration or glaucoma.

- FTMH = ILM-to-RPE defect at the foveola; women over 60; anteroposterior traction during perifoveal PVD.
- Gass: 1A spot, 1B ring, 2 under 400 µm, 3 at least 400 µm, 4 with complete PVD.
- IVTS: small ≤250, medium 250–400, large >400 µm; with or without VMT; primary or secondary.
- Watzke–Allen positive = broken beam; aiming beam disappears in the hole.
- PPV, ILM peel and gas close about 90%; inverted flap for large and failed holes.
- Fellow eye 10–15% in 5 years; cataract follows vitrectomy in most phakic eyes.
- Mnemonic for Gass: **Spot, Ring, Small, Big, Bare** — the last stage bare of vitreous.

Read more: Baidya p.287–293 · Namrata p.252–260 · Aravind p.323 (angiography of macular holes)

ER7 Pathological myopia and myopic maculopathy

Retina · On the list: S.No 41 · ★ Patient already arrived: none

At a glance

- **High myopia:** a spherical equivalent of −6.00 dioptres (D) or more, or an axial length over 26 mm (normal adult 22.5–24.5 mm; each extra 1 mm adds about 3 D).
- **Pathological myopia:** excessive axial elongation with degenerative change in the posterior segment — myopic maculopathy of META-PM (Meta-Analysis for Pathologic Myopia) category 2 or more, a "plus" lesion, or a posterior staphyloma. It is progressive, often −6 to −25 D, with a strong genetic and Asian predilection.
- **Why examiners keep it:** a fundus full of named signs (tessellation, crescent, lacquer cracks, Foster–Fuchs spot, staphyloma), a modern classification, and questions that range into choroidal neovascularisation (CNV), retinal detachment (RD), glaucoma, refractive surgery and cataract surgery.
- **S.No 41:** "Both eyes pathological myopia (right eye −__ D, left eye −__ D; axial lengths __ and __ mm) with myopic maculopathy, META-PM category __ (+ lacquer cracks or Fuchs spot if present), posterior staphyloma, tilted discs with temporal crescents and peripapillary atrophy; no active CNV; retina attached."
- **Always state:** best-corrected acuity, the category in each eye, any plus lesion, activity of CNV, the periphery and the intraocular pressure (IOP).

What to look for and present

Look for	What to say or record
Myopia history	Age at first glasses, rate of progression, current power, contact lenses, refractive surgery, family history (genetic tendency).
Recent symptoms	Sudden central blur or distortion = CNV until proved otherwise; gradual distortion = traction maculopathy; flashes, floaters, curtain = RD; glare = cataract.
Syndromic clues	Tall with long limbs (Marfan), joint laxity and cleft palate (Stickler), hearing loss; prematurity.
Vision and refraction	Best-corrected visual acuity (BCVA) with the actual prescription, near vision, retinoscopy (against movement; dull, slow reflex in very high myopia), keratometry and axial length.
Anterior segment and IOP	Deep anterior chamber, large pupil; nuclear or posterior subcapsular cataract; pigment on endothelium (pigment dispersion); IOP and its measurement errors.
Disc	Large, tilted, oval disc; temporal crescent (myopic conus) with peripapillary atrophy; supertraction crescent on the nasal side; shallow cup that is hard to judge.
Posterior pole	Tessellated (tigroid) fundus; diffuse or patchy atrophy; lacquer cracks; Foster–Fuchs spot; haemorrhage or a grey CNV membrane; macular atrophy.
Staphyloma	Edge seen as a curved line where vessels bend; whole posterior pole bowed backwards; type by extent (Curtin).
Vitreous and periphery	Syneresis, posterior vitreous detachment (PVD) with Weiss ring; lattice, snail-track, white without pressure, atrophic holes, horseshoe tears — examine with indentation.
Special tests	Amsler grid (distortion, scotoma); optical coherence tomography (OCT) for foveoschisis, CNV and choroidal thickness; angiography when CNV is suspected.
Classification	META-PM categories 0–4 plus lacquer cracks, CNV and Fuchs spot; Curtin types of staphyloma; ATN grading (atrophy, traction, neovascularisation).

Diagnosis and summary to say

Say it

Mr X, 48, has worn glasses for myopia since the age of 8, with steady progression into his twenties and a slow fall in vision over the past few years, but no recent distortion or flashes. Best-corrected vision is 6/18 in the right eye with minus 14 dioptres and 6/24 in the left with minus 16 dioptres; axial lengths are 29.4 and 30.2 millimetres. The anterior chambers are deep and there is early nuclear sclerosis in both eyes, with normal pressures. Both discs are tilted with large temporal crescents and peripapillary atrophy. Both fundi are tessellated with a posterior staphyloma, yellow lacquer cracks and well-defined patches of chorioretinal atrophy at the posterior pole, and a pigmented Foster–Fuchs spot at the left fovea. There is no haemorrhage or fluid, the retina is attached, and the periphery shows lattice degeneration without holes.

Diagnosis (S.No 41): both eyes pathological myopia with myopic maculopathy — category 3 (patchy chorioretinal atrophy) of the international meta-analysis grading of pathologic myopia, with lacquer cracks, a left Foster–Fuchs spot (old scarred choroidal neovascular membrane), posterior staphyloma, tilted discs with temporal crescents, and lattice degeneration; retina attached; early nuclear cataract both eyes.

Viva questions and answers

Basics

Q Define myopia and grade it.
Parallel rays focus in front of the retina with accommodation at rest. Low up to −3 D, moderate −3 to −6 D, high −6 D or more (or axial length over 26 mm).

Q How does pathological myopia differ from simple (school) myopia?
Simple myopia starts in school years, stabilises by 18–20 years, rarely exceeds −6 D and is a variant of normal. Pathological myopia keeps progressing into adult life, reaches −6 to −25 D or more, and causes degenerative change of the sclera, choroid, Bruch's membrane and retina.

Q What is the pathogenesis of the fundus changes?
Excess growth is posterior: the sclera thins and stretches, dragging the choroid, Bruch's membrane and retinal pigment epithelium (RPE). The choroid thins (often under 100 µm under the fovea), the choriocapillaris is lost, Bruch's membrane cracks and the posterior wall bulges — giving atrophy, lacquer cracks, CNV and staphyloma.

Q Give the META-PM classification.
Category 0 — no myopic macular lesion; 1 — tessellated fundus only; 2 — diffuse chorioretinal atrophy; 3 — patchy chorioretinal atrophy; 4 — macular atrophy. "Plus" lesions: lacquer cracks, myopic CNV, Foster–Fuchs spot. Pathological myopia = category 2 or more, any plus lesion, or a posterior staphyloma.

Q Name the secondary causes and syndromes of high myopia.
Stickler (COL2A1), Marfan, Ehlers–Danlos, homocystinuria, Knobloch syndrome, congenital stationary night blindness, retinopathy of prematurity, microspherophakia (lenticular myopia), and buckling surgery or keratoconus (secondary).

Examination and signs

Q List the fundus signs of pathological myopia.
Disc — large, tilted, temporal crescent, peripapillary atrophy, supertraction crescent nasally; posterior pole — tessellation, diffuse and patchy atrophy, lacquer cracks, Foster–Fuchs spot, CNV, staphyloma, foveoschisis, macular hole; periphery — lattice, snail-track, white without pressure, atrophic holes; vitreous — syneresis and early PVD.

Q Why is the fundus tessellated?
Thinning and depigmentation of the RPE and choroid let the large orange choroidal vessels and the dark spaces between them show through, giving a tiger-stripe (tigroid) pattern.

Q What is a myopic crescent?
A white or pigmented crescent, usually temporal, where the choroid and RPE stop short of the disc margin as the globe stretches, exposing sclera (peripapillary atrophy). It is acquired, unlike a congenital inferior crescent, which stays unchanged. On the nasal side the retina may be pulled over the disc (supertraction crescent).

Q What are lacquer cracks?
Yellow-white, irregular, branching lines at the posterior pole — mechanical breaks in the Bruch's membrane–RPE–choriocapillaris complex. A new crack may bleed (a "simple" haemorrhage without CNV that clears). They predispose to patchy atrophy and CNV. On indocyanine green angiography (ICGA) they are late hypofluorescent lines; on fundus fluorescein angiography (FFA), window defects.

Q What is the Foster–Fuchs spot?
A dark, raised pigmented spot at the macula — RPE hyperplasia over a regressed myopic CNV; seen in about 5–10% of pathological myopes. Vision is poor because the scar is foveal.

Q Describe and classify posterior staphyloma.
An outpouching of the posterior wall whose curvature is sharper than the surrounding wall; its edge is seen where vessels bend. Curtin described 10 types: I — posterior pole (commonest), II — macular, III — peripapillary, IV — nasal, V — inferior; VI–X compound. Three-dimensional magnetic resonance imaging (MRI) groups them as wide macular, narrow macular, peripapillary, nasal and inferior.

Q What is myopic traction maculopathy?
Traction from vitreous, internal limiting membrane (ILM), vessels and a stiff staphyloma splits the macula: foveoschisis (inner or outer), foveal detachment, lamellar hole, full-thickness macular hole and macular hole RD. Seen in about 9–34% of highly myopic eyes with staphyloma; OCT is essential.

Q What is a dome-shaped macula?
An inward bulge of the macula within a staphyloma, from relatively thick sclera under the fovea; it can cause serous foveal detachment and CNV.

Investigations

Q Which investigations would you order?
Cycloplegic refraction in the young; keratometry; axial length by optical biometry; OCT with enhanced depth or swept source (schisis, CNV, choroidal thickness, dome-shaped macula); OCT angiography; FFA or ICGA when CNV is suspected; autofluorescence for atrophy; B-scan or MRI for staphyloma; visual fields; indented peripheral examination.

Q How do FFA and OCT show myopic CNV?
FFA: early well-defined hyperfluorescence with mild late leakage (a classic, type 2 membrane). OCT: a hyperreflective lesion above the RPE with little subretinal fluid; OCT angiography shows the vascular net without dye. A lacquer-crack bleed shows no leakage.

Management

Q Outline management of S.No 41.
Best spectacle or contact lens correction; Amsler self-monitoring with urgent review for new distortion; OCT every 6–12 months; treat CNV with anti-vascular endothelial growth factor (anti-VEGF) injections; laser to symptomatic tears; manage cataract and glaucoma; low-vision aids; genetic and lifestyle counselling for children.

Q How do you treat myopic CNV?
Intravitreal anti-VEGF — ranibizumab 0.5 mg, aflibercept 2 mg or bevacizumab 1.25 mg: one injection, then monthly review and re-injection only for activity (1 + as needed). Few injections are needed (median 2–4 in the first year). Photodynamic therapy with verteporfin is now second-line, and thermal laser is obsolete.

Q What did RADIANCE show?
Ranibizumab 0.5 mg, re-treated by vision stability or by disease activity, gained about 10.5 letters at 3 months versus about 2 with verteporfin photodynamic therapy, and about 14 letters by 12 months with a median of only 2–4 injections. MYRROR showed a similar gain (about 12 letters at 24 weeks) with aflibercept 2 mg.

Q How do you manage myopic traction maculopathy?
Observe stable schisis with good vision. Operate for falling vision, foveal detachment or a lamellar hole: pars plana vitrectomy (PPV) with fovea-sparing ILM peel (to avoid a postoperative hole) ± gas; a macular buckle for deep staphyloma; macular hole RD needs PPV, inverted ILM flap and gas or silicone oil.

Q What are the refractive options in high myopia?
Spectacles (slight undercorrection; minification and aberrations); contact lenses (larger image, wider field); laser-assisted in situ keratomileusis (LASIK) or small-incision lenticule extraction (SMILE) only within corneal-thickness limits (residual stromal bed at least 250–300 µm, usually up to about −8 to −10 D); phakic intraocular lens (IOL) for higher powers; refractive lens exchange (RLE) mainly in presbyopes.

Q What is the implantable collamer lens (ICL), and what are its problems?
A posterior chamber phakic IOL sitting in the sulcus, correcting up to about −18 D. Needs adequate anterior chamber depth and endothelial count; vault 250–750 µm. Problems: anterior subcapsular cataract (low vault), pupillary block or angle closure (high vault — older models need iridotomy; newer ones have a central port), endothelial loss, pigment dispersion.

Q Why is RLE risky in a young high myope?
It removes accommodation and raises the RD risk (about 2–8% over years), most in young men with long eyes and no PVD. Prefer a phakic IOL until presbyopia.

Q What are the problems in cataract surgery in high myopia?
Biometry — staphyloma makes A-scan unreliable, so use optical biometry and modern formulas (Barrett Universal II, Kane, Haigis, or SRK/T with Wang–Koch axial adjustment) to avoid a hyperopic surprise; low or minus IOL powers; very deep, fluctuating chamber with lens–iris diaphragm retropulsion; weak zonules; needle perforation risk with peribulbar block, so use topical or sub-Tenon; higher RD risk afterwards, especially with capsule rupture or laser capsulotomy.

Q How do you rehabilitate low vision from myopic maculopathy?
Refraction first; near work without glasses or with high adds; hand, stand and electronic magnifiers; telescopes for distance; good lighting, contrast and large print; eccentric viewing training; tints for glare; phone accessibility tools; disability certification and counselling.

Complications and prognosis

Q List the complications of pathological myopia.
Macular — CNV, atrophy, traction maculopathy, macular hole; RD; early nuclear and posterior subcapsular cataract; glaucoma; vitreous degeneration and haemorrhage; amblyopia (anisometropic); strabismus (myopic strabismus fixus).

Q What is the RD risk in myopes and why?
About 40% of RDs occur in myopes. Reasons: a stretched, thin retina; more lattice (in 8% of people but 40% of RDs) and atrophic holes; earlier PVD; macular holes; vitreous loss in cataract surgery; diffuse chorioretinal atrophy.

Q Why is glaucoma hard in high myopes?
Myopia raises the risk of primary open-angle glaucoma (POAG) and pigment dispersion, but tilted oval discs, shallow cups and peripapillary atrophy hide cupping; OCT normative databases exclude long eyes (false “red”); staphyloma and atrophy cause field defects; Schiötz underestimates IOP because of low scleral rigidity, and a cornea thinned by refractive surgery reads low on applanation.

Q What is the prognosis?
Slowly progressive; atrophy (category 3–4) and CNV are the main causes of permanent central loss. Anti-VEGF restores vision in CNV, but chorioretinal atrophy around the scar often erodes it after 5–10 years.

Recent advances and favourite examiner questions

Q What is the ATN classification?
A 2019 extension of META-PM: Atrophy A0–A4 (as categories 0–4); Traction T0–T5 (none, inner or outer schisis, both, foveal detachment, full-thickness hole, macular hole RD); Neovascular N0, N1 (lacquer cracks), N2a (active CNV), N2s (scar or Fuchs spot).

Q What is peripapillary intrachoroidal cavitation?
A yellow-orange crescent just below the disc in highly myopic eyes — a cavity within the choroid on OCT; it is associated with field defects that mimic glaucoma.

Q What is myopic strabismus fixus (heavy eye syndrome)?
Progressive esotropia and hypotropia in very long eyes: the globe herniates up and out between the superior and lateral recti, displacing the lateral rectus downwards. Treat by the Yokoyama procedure — joining the superior and lateral rectus bellies — with medial rectus recession if needed.

Q How can myopic progression be slowed in children?
Low-dose atropine (0.01–0.05% at night), at least 2 hours outdoors a day, less near work, myopia-control spectacle lenses, dual-focus contact lenses and orthokeratology.

Examiner traps

- Calling every tessellated fundus pathological — tessellation alone is category 1; pathological myopia needs category 2 or more, a plus lesion or staphyloma.
- Missing active CNV: a new grey lesion or haemorrhage with recent distortion needs OCT that day; a lacquer-crack bleed is the main mimic.
- Describing the temporal crescent as "glaucomatous" or missing the tilted disc when judging cupping.
- Forgetting to examine the periphery with indentation and to ask about flashes and floaters.
- Offering clear lens exchange to a young high myope without mentioning the RD risk.
- Forgetting staphyloma in biometry and a sharp-needle block in a long eye.
- Trusting a Schiötz reading in a high myope.

- High myopia: −6 D or more, or axial length over 26 mm; each 1 mm ≈ 3 D.
- META-PM: 0 none, 1 tessellated, 2 diffuse, 3 patchy, 4 macular atrophy; plus lesions: lacquer cracks, CNV, Fuchs spot.
- Foster–Fuchs spot = pigmented scar of old myopic CNV (5–10%).
- Myopic CNV: small, type 2, little fluid; anti-VEGF 1 + as needed (RADIANCE, MYRROR).
- 40% of RDs occur in myopes; examine with indentation.
- Biometry and blocks are the cataract traps; RLE carries RD risk.
- Mnemonic for the fundus: **"Tired Discs Lack Fresh Spots"** — Tessellation, Disc crescent and tilt, Lacquer cracks, Fuchs spot, Staphyloma.

Read more: Baidya p.206–211 · Namrata p.267 · Aravind p.217–218 (myopia and glaucoma), p.374 (retinal detachment in myopes), p.440 (myopic crescent)

EC1 Keratoplasty cases: failed or opaque graft, after penetrating or therapeutic keratoplasty, and DSEK

Cornea · On the list: S.No 10, 12, 17, 32, 33, 50 · ★ Patient already arrived: S.No 17, 32

At a glance

- **Keratoplasty** replaces diseased host cornea with donor cornea: full thickness (**penetrating keratoplasty, PK**; written PKP on the list) or partial thickness (**deep anterior lamellar keratoplasty, DALK**; **endothelial keratoplasty, EK**).
- On the list "KP" means keratoplasty and "TKP" therapeutic keratoplasty. On this card keratic precipitates are always written in full.
- **Rejection** is an immune attack on a graft that was clear and is often reversible; **failure** is permanent loss of clarity from any cause. A quiet opaque graft is a failed graft.
- Why kept: stable signs; one patient covers graft examination, rejection, failure, eye banking and regrafting.
- Diagnosis lines:
  - S.No 10, 32, 33, 50: "Left (or right) eye failed penetrating keratoplasty with an opaque, oedematous graft, sutures in situ or removed, vascularised in __ quadrants; probably late endothelial failure (or after rejection)."
  - S.No 12: "__ eye failed graft after therapeutic keratoplasty for infective keratitis; eye quiet, no active infection."
  - S.No 17: "Left eye after PK and Descemet's stripping endothelial keratoplasty (DSEK), probably DSEK for a failed PK; graft clarity grade __; right eye pseudophakia (posterior chamber intraocular lens, PCIOL)." Confirm the order of surgeries from the records.

What to look for and present

Look for	What to say or record
Original disease	Scar after keratitis or trauma, bullous keratopathy, keratoconus, dystrophy (can recur), herpes simplex (recurs; needs oral acyclovir)
Course after surgery	Never cleared = **primary failure**; cleared, then slow painless haze = **late endothelial failure**; cleared, then sudden red, painful eye = **rejection**
Number of grafts; drops	Each regraft does worse; rejection often follows stopping or tapering steroid
Lids, lashes, tears, sensation	Trichiasis, exposure and dry eye cause epithelial defects and graft infection
Graft size and centration	Measure with the slit beam; a graft over 8 mm or an eccentric graft lies near limbal vessels
Host–graft junction	Apposition, step or override, gape, leak (Seidel test), vessels or synechiae reaching it
Sutures	Interrupted (usually 16), continuous or combined; loose, broken, exposed knots, mucus, vessels along a suture, abscess
Graft clarity	Grade 4 (clear) to Grade 1 (opaque); thickness on slit section; Descemet's folds
Vascularisation	Superficial or deep, quadrants, reaching the graft; deep vessels in 2 or more quadrants = high risk
Rejection signs	Ciliary flush, epithelial rejection line, subepithelial infiltrates, **Khodadoust line** with oedema behind it, anterior chamber cells
Epithelium (fluorescein)	Persistent defect, dendrite (herpes), infiltrate (infection)
Endothelium; DSEK lenticule	Keratic precipitates, guttae; lenticule edge, attachment, interface haze
Anterior chamber, iris, lens	Depth, peripheral anterior synechiae (PAS) to the junction, iridectomy, aphakia or intraocular lens (IOL) type, haptic or vitreous touching the graft
Intraocular pressure (IOP)	Tono-Pen or digital; Goldmann is unreliable on an irregular graft; glaucoma is a major cause of failure
Visual potential	Projection of rays in four quadrants; B-scan ultrasonography if no fundal view
Fellow eye	Keratoconus, Fuchs' guttae or a dystrophy confirms the original disease

Diagnosis and summary to say

SAY IT

"Mrs X, a 45-year-old woman, had penetrating keratoplasty in the left eye 4 years ago for a scar after a corneal ulcer. She saw well for about 2 years and then noticed gradual, painless blurring over 6 months, without redness or photophobia. The left eye has a central 7.5 mm graft with all sutures removed, a well-apposed junction, diffuse stromal oedema with Descemet's folds, a grade 1 opaque graft and superficial vessels reaching the junction in two quadrants. There is no epithelial defect, rejection line or keratic precipitate; the anterior chamber is quiet and IOP is 18 mm Hg by Tono-Pen. Projection of rays is accurate in all quadrants and the right eye is normal."

Diagnosis: Left eye failed penetrating keratoplasty (late endothelial failure) with an opaque, oedematous graft and two-quadrant superficial vascularisation; good projection; high-risk candidate for regraft.

S.No 12: "__ eye failed graft after therapeutic keratoplasty, eye quiet, no recurrence of infection."

Viva questions and answers

Basics

Q What is keratoplasty and what are its types?
Replacement of diseased host cornea by donor cornea. PK replaces the full thickness; lamellar keratoplasty replaces only the diseased layer: anterior (superficial lamellar, and DALK down to Descemet's membrane) or posterior, that is EK (DSEK, Descemet's stripping automated endothelial keratoplasty (DSAEK), Descemet's membrane endothelial keratoplasty (DMEK)).

Q What are the indications for keratoplasty?
Optical: scars, bullous keratopathy, keratoconus, dystrophies, failed grafts. Tectonic: perforation, descemetocele, severe thinning. Therapeutic: infection not responding to maximal medical treatment. Cosmetic: a white scar in a blind eye (rare). In India the commonest indication is a scar after keratitis; in the West, endothelial disease.

Q Why does a corneal graft survive better than other allografts?
Immune privilege: no blood vessels or lymphatics, few antigen-presenting cells centrally, Fas ligand on the endothelium, and anterior chamber-associated immune deviation (ACAID).

Q How do DSEK, DSAEK and DMEK differ?
In all three the host Descemet's membrane and endothelium are stripped. DSEK transplants a thin layer of posterior stroma with Descemet's membrane and endothelium, cut by hand; DSAEK cuts it with a microkeratome; DMEK transplants Descemet's membrane and endothelium only (about 15 µm): best vision, no hyperopic shift, least rejection, but harder to unfold and detaches more often.

Q What is DALK and the big-bubble technique?
DALK replaces all the stroma and keeps the host endothelium, for keratoconus and stromal scars with a healthy endothelium. In Anwar's technique, after trephining to 75–80% depth, air is injected into the deep stroma to separate Descemet's membrane. No endothelial rejection; risks are Descemet's perforation (convert to PK), double anterior chamber and interface haze.

Q How do you choose the donor size?
Usual graft 7.5–8 mm, punched from the endothelial side 0.25–0.5 mm larger than the host bed (better seal, fewer PAS, more endothelium); same size in keratoconus (less myopia); 0.75–1.0 mm larger in therapeutic, aphakic and paediatric grafts.

Examination and signs

Q How do you grade graft clarity?
Grade 4: optically clear, fine iris detail seen; Grade 3: faint haze, iris detail visible; Grade 2: marked haze, iris detail only faintly seen; Grade 1: opaque, no view of anterior chamber structures.

Q What is graft rejection and when can you diagnose it?
A cell-mediated (type IV) reaction against donor antigens, with inflammation and oedema in a graft that had been clear for at least 2 weeks (1 week in a regraft). About 30% of PK eyes have at least one episode, mostly within the first year.

Q Describe the types of rejection and their signs.
Epithelial: a raised rejection line (Krachmer's line) moving in from the junction, staining with fluorescein, in a quiet eye. Subepithelial: small white nummular infiltrates in the donor only (Kaye's dots in Baidya; Krachmer spots in Kanski). Stromal: limbal flush, stromal haze and oedema. Endothelial: flush, keratic precipitates scattered or in a Khodadoust line, cells and graft oedema; the most serious type.

Q What is the Khodadoust line?
A line of keratic precipitates (lymphocytes) on the donor endothelium that starts at a vascularised edge, loose suture or synechia and moves centrally over days; the graft behind it is oedematous and the graft ahead clear. It marks endothelial rejection.

Q How do you tell rejection from failure and other causes of a hazy graft?
Rejection: graft clear first, then sudden symptoms, flush, keratic precipitates on the donor only; reverses with steroid. Late failure: gradual, painless, quiet. Herpes: keratic precipitates on host and donor, reduced sensation. Infection: infiltrate, hypopyon. Epithelial downgrowth: grey membrane, raised IOP.

Q What are interrupted, continuous and combined sutures?
All are 10-0 nylon, after four cardinal sutures at 12, 6, 3 and 9 o'clock. Interrupted (16): easier, allow selective removal; used in vascularised, inflamed, therapeutic and paediatric grafts. Continuous: faster, even tension, less inflammation, adjustable, but a break loosens the whole wound. Combined: both.

Q What do loose sutures signify and when are sutures removed?
A loose or broken suture attracts vessels, mucus and bacteria and can trigger rejection or an abscess: remove it at once under antibiotic and steroid cover. Tight interrupted sutures may be removed selectively from about 3 months to reduce astigmatism; all are usually out by 12–18 months (by about 6 months in children).

Investigations

Q Which investigations do you do in a failed or rejecting graft?
Pachymetry (rising thickness is the earliest sign of endothelial failure and tracks treatment); specular microscopy, or confocal microscopy if oedematous; anterior segment optical coherence tomography (AS-OCT) for lenticule attachment and angle; topography; B-scan; scraping if infection is suspected.

Q Which donors are not used?
Death of unknown cause; rabies, Creutzfeldt–Jakob disease and other slow-virus disease; human immunodeficiency virus (HIV), hepatitis B and C, syphilis; septicaemia; leukaemia, lymphoma, retinoblastoma and other intraocular tumours; donors under 1 year. Remove the eye within about 6 hours of death.

Q What storage media do you know?
Moist chamber at 4°C, up to 48 hours; McCarey–Kaufman (MK) medium (TC-199, 5% dextran, gentamicin) at 4°C, 2–4 days; Optisol GS at 4°C, up to 14 days; organ culture at 31–37°C, up to 4 weeks; glycerol or gamma-irradiated corneas for tectonic use. Pink-red MK medium is normal; yellow or turbid means contamination.

Q What endothelial count do you want in donor tissue?
At least 2,000 cells/mm² for optical PK and about 2,200–2,500 for EK (handling loses cells). A normal adult has 2,500–3,000; the cornea decompensates below about 400–500. DALK and tectonic tissue need no good endothelium.

Management

Q How do you treat endothelial rejection?
Admit; prednisolone acetate 1% hourly, dexamethasone ointment at night, a cycloplegic; subconjunctival dexamethasone; if severe, a single pulse of intravenous methylprednisolone 500 mg or oral prednisolone 1 mg/kg/day; lower IOP; taper over months. Treatment within 8 days of onset gave about 92% survival against 55% when delayed. Oedema beyond about 2 months means failure.

Q How do you treat epithelial and subepithelial rejection?
Prednisolone acetate 1% about 3-hourly, halved every 3 days. They usually settle but show the host is sensitised, so watch for endothelial rejection.

Q What are the risk factors for rejection?
Host: young age, deep vessels in 2 or more quadrants, regraft or previous rejection, herpes simplex, inflammation, atopy, glaucoma, PAS, surface disease. Technical: large or eccentric graft, therapeutic graft, loose or exposed sutures, suture removal, combined surgery. After surgery: stopping steroid early, poor compliance.

Q How do you protect a high-risk regraft?
Quieten the eye and control IOP first; ABO-matched donor; central graft under 8 mm with buried knots; topical steroid for 12–18 months, then low dose; oral cyclosporine (15 mg/kg/day for 2 days, then 7.5 mg/kg/day, adjusted to trough level) or mycophenolate mofetil 750 mg twice daily with blood monitoring; DALK or EK where the disease allows.

Q What are the causes of graft failure?
Primary failure, oedema from day one that never clears: poor donor endothelium, storage problems, rough handling, faulty trephination, early infection. Late failure: rejection (about 30%), slow endothelial loss, glaucoma, infection, persistent epithelial defect, recurrence of disease, trauma.

Q What causes glaucoma after keratoplasty and how do you treat it?
Angle collapse (tight sutures, aphakia), PAS, retained viscoelastic, pupillary block, steroid response, pre-existing glaucoma. Use beta-blockers and alpha-agonists; avoid topical carbonic anhydrase inhibitors in a borderline graft (they weaken the endothelial pump); prostaglandins with care. Surgery: trabeculectomy with mitomycin C, a tube (speeds endothelial loss) or cyclophotocoagulation.

Q How do you manage astigmatism and visual rehabilitation after PK?
Astigmatism averages 4–5 D. With sutures in: topography-guided removal of tight interrupted sutures in the steep meridian, or adjusting a continuous suture. Afterwards: spectacles; rigid gas-permeable (RGP) contact lens, best for irregular astigmatism; relaxing incisions; wedge resection; toric IOL; excimer laser. Treat amblyopia in children.

Q What do you do for a failed graft?
Treat the cause (IOP, inflammation, surface, herpes prophylaxis), then regraft when quiet: repeat PK if the stroma is scarred, or EK under the failed PK if only the endothelium has failed (as at S.No 17). After a therapeutic graft, wait until the eye has been quiet for some months. After two or three failures, consider a keratoprosthesis.

Complications and prognosis

Q List the complications of PK.
Early: wound leak, flat chamber, epithelial defect, raised IOP, primary failure, endophthalmitis (0.2–0.7%). Late: rejection, astigmatism, glaucoma, suture problems, keratitis, recurrence, wound dehiscence after trauma (the junction stays weak for life), cataract, late endothelial failure.

Q Which diseases recur in a graft?
Herpes simplex (oral acyclovir 400 mg twice daily for at least a year after grafting), fungus after a therapeutic graft (junction, anterior chamber or vitreous), and stromal dystrophies such as Reis–Bücklers, lattice, granular and macular.

Q What are the complications of DSEK and DMEK?
Graft detachment (commonest; rebubble with air), pupillary block from air (inferior iridectomy, partial air release), primary graft failure, endothelial rejection, interface haze, epithelial ingrowth, hyperopic shift of about 1 D after DSEK, glaucoma.

Q What is Brightbill's prognostic grading?
Grade I excellent: keratoconus, central avascular scars, lattice and granular dystrophy. II good: bullous keratopathy, Fuchs' and macular dystrophy, interstitial keratitis. III fair: vascularised cornea, active herpes or bacterial keratitis, failed grade II graft. IV guarded: active fungal keratitis, congenital glaucoma, paediatric graft, staphyloma, severe dry eye. V poor: severe chemical and thermal burns.

Q What is the prognosis of a regraft?
Each regraft does worse: the host is sensitised and the eye has more vessels and glaucoma. Failure from rejection rose from 8% after a first graft to 40% after two or more in one series.

Recent advances and favourite examiner questions

Q What did the Collaborative Corneal Transplantation Studies (CCTS) show?
In high-risk eyes (deep vessels in 2 or more quadrants or previous rejection), human leucocyte antigen (HLA) matching did not reduce rejection; ABO matching might; intensive steroid, compliance and close follow-up mattered most.

Q What are the rejection rates after the different keratoplasties?
DMEK 1–3%, DSAEK 8–14%, DALK 2–10% (no endothelial rejection), low-risk PK 5–17%, high-risk PK up to 68%.

Q What is a keratoprosthesis and when is it used?
An artificial cornea when any graft will fail: multiple failures, heavy vascularisation, severe limbal stem cell deficiency. Boston type I (polymethyl methacrylate optic in a donor corneal carrier) for a wet eye with a good blink; Boston type II or osteo-odonto-keratoprosthesis for dry, keratinised eyes (Stevens–Johnson syndrome, pemphigoid). Lifelong bandage lens, antibiotics and IOP checks; risks are glaucoma, retroprosthetic membrane, melt and endophthalmitis.

Q What are the principles of therapeutic keratoplasty?
Excise all infected tissue with a 1 mm clear margin; oversize the donor by 0.75–1 mm; interrupted sutures; peripheral iridectomy; anterior chamber wash; send the button for culture and histopathology; continue antimicrobials; no topical steroid for about 2 weeks after a fungal graft. The aim is a quiet globe; an optical regraft follows later.

Q What newer techniques are used for endothelial failure?
DMEK; Rho-kinase inhibitor drops (ripasudil 0.4%, netarsudil 0.02%); injected cultured endothelial cells; femtosecond laser-shaped wounds.

Examiner traps

- Calling every opaque graft "rejection": rejection needs a previously clear graft and inflammation; a quiet opaque graft is **failure**.
- Not asking how good the vision was after surgery and when it fell: this one question separates primary failure, late failure and rejection.
- Skipping IOP, or using Goldmann on an irregular graft; say "Tono-Pen" and that glaucoma is a leading cause of failure.
- Missing loose sutures and the vessels along them: both a sign and a cause of rejection.
- Using "KP" loosely: on the list it means keratoplasty; say keratic precipitates in full.
- For S.No 17, not explaining why one eye has both PK and DSEK, and not checking the order in the records.
- Offering a regraft without checking projection, B-scan and IOP.

- Rejection is immune and often reversible; failure is permanent; rejection is one cause of failure.
- Clear for 2 weeks first; Khodadoust line = endothelial rejection; hourly prednisolone acetate 1%, best within 8 days.
- Donor at least 2,000 cells/mm^2; MK medium 2–4 days, Optisol up to 14 days; oversize 0.25–0.5 mm, same size in keratoconus.
- Rejection: DMEK 1–3%, DSAEK 8–14%, high-risk PK up to 68%.
- After repeated failures: keratoprosthesis.
- High-risk graft, mnemonic **RIVAL**: Regraft, Inflammation (and herpes), Vascularisation, Age (young), Large or eccentric graft.

Read more: Baidya p.75–90, 109–110, 699, 728–729 · Namrata p.134–140 · Aravind 2.17 (p.125–133)

EC2 Pseudophakic bullous keratopathy

Cornea · On the list: S.No 13, 16 · ★ Patient already arrived: S.No 13, 16

At a glance

- **Pseudophakic bullous keratopathy (PBK)**: chronic corneal oedema with epithelial bullae after cataract surgery with an intraocular lens (IOL), because the endothelium can no longer pump the cornea dry. After aphakia it is called aphakic bullous keratopathy.
- A leading indication for keratoplasty; today treated mainly by **endothelial keratoplasty (EK)**.
- Why kept: stable signs; tests the endothelial pump, specular microscopy, the IOL, and the choice between Descemet's stripping endothelial keratoplasty (DSEK), Descemet's membrane endothelial keratoplasty (DMEK) and penetrating keratoplasty (PK).
- Always examine the other eye: guttae there mean **Fuchs' endothelial dystrophy** was the underlying cause.
- Diagnosis lines:
  - S.No 16: "Left eye pseudophakic bullous keratopathy with posterior chamber IOL in the bag (or anterior chamber IOL), diffuse stromal and epithelial oedema with bullae and Descemet's folds, no stromal scarring, intraocular pressure (IOP) normal, projection accurate; right eye __, no guttae."
  - S.No 13: the eye is not stated on the list; record which eye or both.

What to look for and present

Look for	What to say or record
Cataract surgery: when, technique, complications, IOL type	Complicated surgery, vitreous loss, long surgery and an anterior chamber IOL (ACIOL) cause PBK
Vision after surgery	Good, then failing = late decompensation; never good = surgical trauma, Descemet's membrane detachment (DMD) or toxic anterior segment syndrome (TASS)
Blur worst on waking	Overnight oedema (lids closed, no evaporation): early endothelial failure
Pain, watering, photophobia; later less pain	Ruptured bullae expose nerves; pain settles once subepithelial fibrosis forms
Glaucoma, uveitis, herpes, trauma	Each damages the endothelium further
Epithelium	Microcysts and bullae (best by retroillumination), ruptured bullae staining with fluorescein; bandage contact lens (BCL) in place
Stroma and Descemet's	Thickening, ground-glass haze, Descemet's folds; scarring, vascularisation and pannus in old cases
Endothelium	Guttae, keratic precipitates, a detached Descemet's membrane near the incision, vitreous or haptic touching the cornea
Pattern of oedema	Near the wound: DMD, wound burn, vitreous wick; peripheral with clear centre after intracapsular extraction: Brown–McLean syndrome
Anterior chamber and IOL	Depth, cells, vitreous; ACIOL (type, tilt, haptics in angle, patent iridectomy); posterior chamber IOL in bag or sulcus, decentred, chafing iris
Pupil and iris	Peaked pupil (vitreous wick), atrophy, peripheral anterior synechiae (PAS)
IOP	Tono-Pen; Goldmann reads falsely low on an oedematous cornea
Visual potential	Glow, projection of rays, B-scan ultrasonography; cystoid macular oedema (CMO) may coexist
Fellow eye	Guttae (Fuchs'), hammered-silver endothelium (iridocorneal endothelial syndrome), vesicles (posterior polymorphous dystrophy)
Tests to mention	Pachymetry, specular microscopy (fellow eye if this one is hazy), anterior segment optical coherence tomography (AS-OCT), B-scan

Diagnosis and summary to say

> **Say it**
>
> "Mr X, a 68-year-old man, had cataract surgery in the left eye 2 years ago. Vision was good for a year and then became blurred, worst on waking, with bouts of pain and watering for 6 months. The left eye has diffuse stromal oedema with epithelial bullae, one ruptured bulla staining with fluorescein, and Descemet's folds, without stromal scarring or vascularisation. A posterior chamber IOL lies in the bag, the anterior chamber is quiet without vitreous, and IOP is 16 mm Hg by Tono-Pen. Projection of rays is accurate and the B-scan is normal; the right eye is phakic with a few central guttae."
>
> **Diagnosis: Left eye pseudophakic bullous keratopathy with posterior chamber IOL in the bag, no stromal scarring, good visual potential; right eye early Fuchs' dystrophy. Plan: DSEK or DMEK.**
>
> **With an ACIOL: "pseudophakic bullous keratopathy with anterior chamber IOL; plan EK (or PK) with IOL exchange."**

Viva questions and answers

Basics

Q How does the endothelium keep the cornea clear?
Barrier function (tight junctions limit aqueous entry) and pump function (sodium–potassium ATPase and bicarbonate-linked transport move fluid out) balance the swelling pressure of stromal glycosaminoglycans. Stromal water stays near 78% and central thickness about 540 µm.

Q What are the important endothelial cell counts?
About 5,000–6,000 cells/mm² at birth, 3,000 in young adults and 2,500 in older adults, with loss of about 0.6% a year. Intraocular surgery is risky below 1,000 cells/mm²; oedema usually appears below 400–500. Uncomplicated phacoemulsification loses roughly 5–10%. Cells do not divide in vivo; neighbours enlarge and spread.

Q What causes PBK?
Pre-existing endothelial disease (Fuchs' guttae, low count, herpes); surgical trauma (hard nucleus, phaco energy near the cornea, shallow chamber, long or complicated surgery, DMD); IOL-related (ACIOL, especially closed-loop or iris-fixated, sulcus haptic touching the cornea); vitreous touch; inflammation, raised IOP and TASS after surgery.

Q Why does an ACIOL damage the endothelium?
Intermittent lens–cornea touch, especially on eye rubbing; chronic low-grade inflammation and the uveitis–glaucoma–hyphaema (UGH) syndrome; iris tuck and PAS. Old closed-loop designs were the worst.

Q Why is vision worst on waking?
With the lids closed there is no evaporation, so the tear film becomes less hypertonic and draws less water from the cornea; a weak endothelium cannot clear the overnight load. Vision improves through the day as the surface dries.

Q Why are bullae painful, and why does the pain later settle?
Fluid lifts the epithelium into bullae; when they rupture, exposed nerve endings cause pain, watering and photophobia, and the defect may become infected. With chronic oedema, subepithelial fibrosis forms: pain settles but vision falls further.

Examination and signs

Q What are the signs of corneal decompensation?
Loss of lustre, epithelial microcysts and bullae, stromal thickening with ground-glass haze, Descemet's folds (striate keratopathy), raised pachymetry, and in long-standing cases subepithelial fibrosis and vascularisation; vision is worst in the morning.

Q What do you check in the anterior chamber and IOL?
ACIOL: type, tilt, haptics in the angle, iris tuck, UGH syndrome, patent iridectomy. Posterior chamber IOL: in bag or sulcus, decentration, haptic touching the cornea. Vitreous strands to the wound (peaked pupil), cells and PAS.

Q Why examine the fellow eye?
Oedema hides the endothelium of the affected eye. Guttae in the other eye point to Fuchs' dystrophy decompensated by surgery; vesicles or bands suggest posterior polymorphous dystrophy; a hammered-silver endothelium with iris atrophy suggests iridocorneal endothelial (ICE) syndrome.

Q What is the differential diagnosis of oedema after cataract surgery?
Transient: striate keratopathy from surgical trauma (clears in a week), raised IOP, DMD, TASS, herpetic endotheliitis (keratic precipitates; responds to antiviral and steroid). Persistent: PBK, Brown–McLean syndrome, vitreous touch, epithelial downgrowth.

Q What is DMD and how is it managed?
Descemet's membrane stripped from the stroma at the incision by blunt instruments or repeated entry; localised oedema with a membrane visible at the slit lamp or on AS-OCT. Small ones reattach; larger ones need air or long-acting gas injected into the anterior chamber (descemetopexy); if that fails, EK.

Q How do you differentiate PBK from Fuchs' dystrophy and congenital hereditary endothelial dystrophy (CHED)?
Fuchs': bilateral, older women, autosomal dominant, central guttae with beaten-metal look, morning blur. CHED: bilateral oedema from birth or early childhood, thick blue-grey cornea of normal diameter, normal IOP, nystagmus in the recessive type (SLC4A11 gene). PBK: unilateral after surgery, fellow eye normal or with guttae.

Investigations

Q What does pachymetry tell you?
Normal central thickness is about 520–540 µm; rising thickness is the earliest objective sign of endothelial stress; over 640 µm suggests the cornea will not tolerate more intraocular surgery; a difference of over 10% between morning and evening shows early failure. Serial values monitor a graft. Ultrasonic, Scheimpflug or AS-OCT.

Q What does specular microscopy show?
Reflected light images the endothelial mosaic: cell density (cells/mm^2); coefficient of variation of cell area (polymegathism; healthy under about 30%, abnormal over 40%); percentage of hexagonal cells (pleomorphism; normal over 60%); guttae as dark spots. Not possible through dense oedema; then use confocal microscopy or examine the fellow eye.

Q What is the role of confocal microscopy and AS-OCT?
Confocal microscopy images the endothelium and stroma through oedema where specular microscopy fails. AS-OCT shows corneal thickness, bullae, stromal scarring (EK or PK), DMD, IOL and angle, and after EK the lenticule position and attachment.

Q What other tests are needed before surgery?
Visual potential (projection of rays, Maddox rod, two-point discrimination, entoptic phenomenon), B-scan for retinal detachment or vitreous disease, AS-OCT for stromal scarring, angle and IOL position, macular optical coherence tomography (OCT) for CMO when the view allows, and gonioscopy where possible.

Management

Q Outline the medical treatment.
Hypertonic saline (sodium chloride 5%) drops 4–8 times a day and 5% ointment at night; warm air from a hair dryer at arm's length in the morning; lower IOP with timolol or brimonidine, avoiding topical carbonic anhydrase inhibitors (they impair the pump); BCL for painful bullae; cycloplegic and lubricants; steroid if inflamed. These are temporary measures.

Q Why does a bandage contact lens help?
It splints the bullae so they do not rupture, covers exposed nerve endings and lets the epithelium heal beneath it. Give prophylactic antibiotic and watch for keratitis.

Q What do you do when visual potential is poor?
Aim for a comfortable eye: BCL; anterior stromal puncture (25–26 G needle; the scars anchor the epithelium); phototherapeutic keratectomy (PTK); bullectomy with amniotic membrane graft; Gundersen conjunctival flap; cross-linking (temporary); as last resorts retrobulbar alcohol or evisceration.

Q What is the definitive treatment?
EK when the stroma is clear: DSEK or Descemet's stripping automated endothelial keratoplasty (DSAEK), with donor posterior stroma, Descemet's membrane and endothelium held by an air bubble, or DMEK with Descemet's membrane and endothelium only. PK when there is stromal scarring or heavy vascularisation. Treat the cause at the same sitting: anterior vitrectomy for vitreous touch, IOL exchange for an ACIOL.

Q How do DSEK and DMEK compare?
DMEK gives better and faster vision, no hyperopic shift and the least rejection (1–3%), but the scroll is hard to unfold, rebubbling is commoner and it is difficult with an ACIOL, aphakia, vitrectomy or poor view. DSEK is easier in complex PBK eyes, with a hyperopic shift of about 1 D and rejection of 8–14%. Both beat PK: closed chamber, little astigmatism, stronger globe.

Q What are the steps of DSEK?
Donor lenticule cut on an artificial anterior chamber (or by microkeratome) and trephined at 7.5–8 mm from the endothelial side. Host: side ports, trypan blue, 8–8.5 mm epithelial mark, score and strip Descemet's membrane, inferior peripheral iridectomy. Insert through a 4.5–5 mm incision (folded "taco", glide or injector), fill with air, and keep the patient supine for 24 hours.

Q When is IOL exchange done and with what?
When an ACIOL or a malpositioned lens touches the endothelium or causes UGH syndrome: remove it and place a sulcus IOL on a capsular rim, a retropupillary iris-claw lens or a scleral-fixated lens, combined with EK or PK.

Complications and prognosis

Q What are the complications of endothelial keratoplasty?
Graft detachment (commonest; rebubble), pupillary block from air (inferior iridectomy, partial air release), primary graft failure, endothelial rejection, interface haze, epithelial ingrowth, raised IOP from steroid or PAS, late endothelial failure.

Q What happens if PBK is left untreated?
Recurrent painful erosions, microbial keratitis, subepithelial fibrosis and vascularisation; once the stroma scars, PK rather than EK is needed and the prognosis falls.

Q What is the prognosis?
Good with EK when the stroma is clear and the macula healthy (Brightbill grade II). Vision is limited by CMO, stromal scarring, glaucoma and macular disease; PBK grafts lose endothelium faster than grafts for Fuchs' dystrophy, and eyes with an ACIOL, a glaucoma tube or vitrectomy do worse.

Recent advances and favourite examiner questions

Q What are Rho-kinase inhibitors and what is their role?
Ripasudil 0.4% and netarsudil 0.02% drops promote endothelial proliferation, adhesion and migration and reduce apoptosis; tried in oedema after cataract surgery and in Fuchs' dystrophy (after Descemet stripping without a graft), and used with cultured endothelial cells.

Q Can Descemet stripping without a graft be used in PBK?
No. Descemet stripping only (Descemetorhexis without EK) suits Fuchs' dystrophy with central guttae and a healthy periphery (over about 1,000 cells/mm²), from which cells migrate in, helped by a Rho-kinase inhibitor. In PBK the whole endothelium is depleted, so a graft is needed.

Q What is cultured endothelial cell therapy?
Endothelial cells grown from one donor cornea are injected into the anterior chamber with a Rho-kinase inhibitor after the diseased endothelium is scraped off; the patient lies face down for about 3 hours so the cells settle. One donor can treat many eyes; it is already approved for bullous keratopathy in Japan.

Q How do you prevent PBK at cataract surgery?
Specular count when guttae or a low count are suspected; dispersive viscoelastic (soft-shell technique); phaco in the bag, away from the cornea, with low energy; avoid an ACIOL; manage vitreous well; control IOP and inflammation after surgery. With Fuchs' dystrophy: cataract surgery alone if the count is over 1,000 cells/mm² and thickness under 600 µm; combined EK and cataract surgery if the count is under 800 or thickness over 640 µm.

Q What is TASS and how does it differ from endophthalmitis?
Toxic anterior segment syndrome is sterile inflammation within 12–24 hours of surgery from contaminated instruments or solutions (detergent or glutaraldehyde residue): limbus-to-limbus oedema, fibrin, sometimes hypopyon, little pain, no vitritis; it responds to intensive steroid. Endophthalmitis begins after 2–7 days, with pain, vitritis and positive cultures.

Q What is Brown–McLean syndrome?
Peripheral corneal oedema with a clear centre, years after intracapsular cataract extraction, often with orange pigment on the endothelium; it seldom progresses and usually needs no treatment.

Examiner traps

- Not examining the fellow eye, and so missing Fuchs' dystrophy as the cause.
- Reading Goldmann IOP at face value: it is falsely low on an oedematous cornea.
- Not naming the IOL type and position: an ACIOL changes the plan (IOL exchange).
- Promising vision without checking projection, B-scan and the macula (CMO).
- Prescribing dorzolamide or brinzolamide to a decompensating cornea.
- Saying PK is the treatment of choice: EK is preferred when the stroma is clear.
- Missing an infiltrate under a bandage contact lens.

- PBK = endothelial pump failure after IOL surgery; oedema below about 400–500 cells/mm².
- Morning blur, bullae, Descemet's folds, thick cornea; always look for guttae in the fellow eye.
- Medical treatment is temporary: 5% saline, BCL, IOP lowering without topical carbonic anhydrase inhibitors.
- Clear stroma: DSEK or DMEK; scarred stroma: PK; ACIOL: exchange it.
- DMEK: best vision, rejection 1–3%.
- Rho-kinase inhibitors and cultured endothelial cells are the newer options.
- Causes, mnemonic **FAVOURITE**: Fuchs', ACIOL, Vitreous touch, Operative trauma, UGH syndrome, Raised IOP, Inflammation (uveitis, herpes), TASS, Endothelial (Descemet's) detachment.

Read more: Baidya p.55–59, 71–75, 81, 109–110 · Namrata p.128–133 · Aravind 2.15–2.16 (p.117–124)

EC3 Corneal opacity: nebula, macula, leucoma and adherent leucoma, including a leucomatous cornea with band keratopathy

Cornea · On the list: S.No 19, 40 · ★ Patient already arrived: none

At a glance

- A **corneal opacity** is loss of corneal transparency from a scar, most often after healed keratitis or trauma. By density: **nebula** (faint; iris detail seen through it), **macula** (moderate; iris pattern lost but pupil margin seen), **leucoma** (dense white; iris and pupil hidden).
- **Adherent leucoma**: a leucoma with iris incarcerated in it after a perforation. **Anterior staphyloma**: an ectatic scar lined by iris.
- Why kept: quick to examine, stable signs; tests how you describe an opacity, judge visual potential and choose between optical iridectomy, deep anterior lamellar keratoplasty (DALK), penetrating keratoplasty (PK) and cosmetic options.
- Diagnosis lines:
  - S.No 19: “Right eye central leucomatous corneal opacity (size, depth, vessels), non-adherent; left eye adherent leucoma with pupil drawn towards the scar, intraocular pressure (IOP) __, projection __; probably healed corneal ulcers, perforated on the left.”
  - S.No 40: “Right eye leucomatous cornea with secondary band-shaped keratopathy; globe size, IOP and projection __.” Band keratopathy in detail: card EC6.
- Opacities in both eyes from childhood: ask about keratomalacia (vitamin A deficiency), measles, trachoma and repeated ulcers.

What to look for and present

Look for	What to say or record
Onset and cause	Since childhood after “eye infection”, injury, surgery, chemical, trachoma, measles, malnutrition; from birth (congenital)
History of perforation	Sudden relief of pain with a gush of warm fluid during the ulcer: suggests adherent leucoma
Present complaint	Vision, glare, cosmetic concern, pain or redness (active disease, glaucoma)
Site, number, size	Central, paracentral or peripheral; relation to the pupil; size in mm in two meridians
Density	Nebular, macular or leucomatous (slit beam, retroillumination); chalky calcium, pigment
Depth and thickness	Slit section: superficial, mid-stromal, full thickness; thinning, facet, ectasia
Surface	Flat or raised; fluorescein for epithelial defect; Seidel test if thin
Vascularisation	Superficial or deep, number of quadrants
Iris adherence	Iris strand or tissue in the scar, pupil peaked or drawn towards it, brown pigment on its back, anterior chamber (AC) shallow or irregular
Ectasia	Kerectasia (bulging thin scar) or anterior staphyloma (bluish, iris-lined); partial or total
Corneal sensation	Reduced over scars; markedly reduced in herpetic and neurotrophic scars
IOP	Digital or Tono-Pen; raised in adherent leucoma and staphyloma
Lens and glow	Anterior polar or complicated cataract; red glow through a clear zone
Visual potential	Projection of rays in four quadrants, colour perception, Maddox rod, B-scan ultrasonography; fixation and amblyopia in children
Classification to quote	Nebula, macula, leucoma; adherent or not; anterior staphyloma, partial or total

Diagnosis and summary to say

> **Say it**
>
> Model for S.No 19: "Mr X, a 35-year-old man, has had poor vision in both eyes since childhood after repeated eye infections, and recalls a sudden gush of fluid from the left eye during one of them. The right eye has a central 5 × 4 mm dense white leucoma involving the anterior two-thirds of the stroma, with deep vessels in two quadrants; the iris and pupil are hidden behind it, the AC is of normal depth and the periphery is clear. The left eye has a 4 mm paracentral leucoma with iris incarcerated in it, the pupil drawn up towards the scar and the AC shallow at that site; the eye is firm on digital tonometry. Projection of rays is accurate in all quadrants in the right eye and defective temporally in the left; the B-scan is normal in both eyes."
>
> **Diagnosis: Right eye central leucomatous corneal opacity with deep vascularisation; left eye adherent leucoma with secondary glaucoma; probably healed corneal ulcers, perforated on the left. The right eye has the better prognosis for keratoplasty.**
>
> **S.No 40: "Right eye leucomatous cornea with secondary band-shaped keratopathy; globe normal or phthisical, projection accurate or absent."**

Viva questions and answers

Basics

Q Define nebula, macula and leucoma.
Grades of density of a corneal scar. Nebula: a faint haze through which iris detail is seen. Macula: moderately dense; iris pattern is lost but the pupil margin can be made out. Leucoma: dense and white; iris and pupil cannot be seen through it.

Q Why can a nebula blur vision more than a leucoma?
A nebula over the pupil lets light through but scatters it irregularly, causing glare and irregular astigmatism, so the image is degraded; a dense leucoma blocks light completely, and if clear cornea remains beside it a sharp, though dimmer, image can still form.

Q Why does a healed ulcer leave an opacity?
Bowman's layer does not regenerate, and activated keratocytes (myofibroblasts) lay down new collagen of irregular diameter and spacing, which scatters light; the deeper and wider the stromal loss, the denser the scar. Superficial epithelial lesions heal without a scar.

Q What is an adherent leucoma and how does it form?
A leucoma with iris incarcerated in it. A weak ulcer floor gives way (often on coughing or straining), aqueous escapes, the iris–lens diaphragm moves forward and the iris plugs the hole; scar tissue then incorporates the iris. Signs: iris tissue in the scar, brown pigment on its back, pupil drawn towards it, irregular AC.

Q What are pseudocornea, anterior staphyloma and kerectasia?
Pseudocornea: the fibrous scar that forms over prolapsed iris after a large perforation. Anterior staphyloma: an ectatic pseudocornea lined by iris that bulges under raised IOP, looking bluish because uvea shows through; partial or total. Kerectasia: bulging of a thinned scar without iris in it.

Q How do anterior synechia, leucoma, adherent leucoma and anterior staphyloma differ?
Anterior synechia: iris stuck to the back of the cornea (perforated ulcer, angle closure), surface flat, vision may be normal. Leucoma: dense scar, flat surface, normal AC and pupil. Adherent leucoma: flat scar with iris in it, pupil drawn up, AC irregular, IOP normal or raised. Anterior staphyloma: ectatic, iris-lined scar, AC very shallow or absent, IOP raised.

Q What causes corneal opacity?
Acquired: healed bacterial, fungal, herpetic or Acanthamoeba keratitis; trauma and surgery; chemical injury; trachoma; keratomalacia; interstitial keratitis; degenerations (band keratopathy, spheroidal degeneration); dystrophies; failed graft; long-standing oedema. Congenital, mnemonic STUMPED: Sclerocornea, Tears in Descemet's membrane (congenital glaucoma, forceps), Ulcer, Metabolic (mucopolysaccharidosis), Peters' anomaly, Endothelial dystrophy, Dermoid.

Examination and signs

Q How do you describe a corneal opacity?
Eye, number, site (central, paracentral, peripheral; relation to the pupil), size in two meridians, shape, density, depth on slit section, margins, surface (flat, facet, ectatic), vessels, calcium or pigment, iris adherence, sensation, AC depth and IOP.

Q How do you tell superficial from deep corneal vessels?
Superficial: bright red, wavy and branching, continuous with conjunctival vessels across the limbus. Deep: straight, parallel, brush-like, ending abruptly at the limbus (from anterior ciliary vessels), dull and seen in the stroma. Ghost vessels are empty deep vessels left after inflammation settles.

Q What does iris incarceration imply?
A past perforation; a high risk of secondary glaucoma; and a harder keratoplasty (synechiolysis needed, more peripheral anterior synechiae (PAS) and rejection). Confirm with the slit lamp, and with anterior segment optical coherence tomography (AS-OCT) or ultrasound biomicroscopy (UBM) when the view is poor.

Q Why is IOP important in adherent leucoma?
Incarcerated iris drags the iris root forward and PAS close the angle, so secondary angle-closure glaucoma is common; raised IOP stretches the scar into an ectasia or anterior staphyloma and damages the optic nerve. Measure digitally or with a Tono-Pen (Goldmann is unreliable on a scar) and control IOP before any keratoplasty.

Q How do you assess visual potential when the fundus cannot be seen?
Perception and projection of light in four quadrants, colour perception, Maddox rod (an unbroken line), two-point light discrimination, the Purkinje vascular entoptic test, a potential acuity meter or laser interferometer through a clear area, and B-scan for retinal detachment, vitreous disease and globe length; visual evoked potentials and electroretinography when in doubt or in children.

Q What is a facet?
A depressed scar where stroma was lost but the epithelium has healed over it; fluorescein pools in the dip without staining, and it causes irregular astigmatism.

Q How do you perform the Seidel test?
Paint a moistened fluorescein strip (or 2% drops) over the suspected leak and watch under cobalt blue light: aqueous dilutes the dye into a bright green stream (positive). Used for a thin or ectatic scar or a suspected fistula.

Investigations

Q Which investigations help plan surgery?
AS-OCT for scar depth (DALK if Descemet's membrane and endothelium are spared, PK if full thickness), thinning, iris adhesion and angle; UBM through dense scars; B-scan; specular microscopy of a clear area or the fellow eye; gonioscopy where possible; examination under anaesthesia in children.

Q What can a B-scan show in these eyes?
Retinal or choroidal detachment, vitreous exudates or haemorrhage, a disc cup, axial length (a long eye in staphyloma; a short eye with thick choroid and calcification in phthisis bulbi), and any intraocular tumour in a blind eye.

Q When are systemic tests needed?
In band keratopathy without an ocular cause: serum calcium, phosphate, uric acid, creatinine, parathyroid hormone, vitamin D and angiotensin-converting enzyme (sarcoidosis). In a child with bilateral scars: nutrition and vitamin A status.

Management

Q Outline the management of a corneal opacity.
It depends on site, depth, visual potential and the patient's need. Peripheral nebula or macula: spectacles or rigid gas-permeable (RGP) contact lens for irregular astigmatism. Superficial scar: phototherapeutic keratectomy (PTK). Central opacity with good potential: DALK if Descemet's membrane and endothelium are healthy, PK if full thickness or adherent. Small central opacity with clear periphery: optical iridectomy. Blind eye: cosmetic contact lens, prosthetic shell or tattooing; blind painful eye: evisceration.

Q What is PTK and what are its limits?
Excimer laser (193 nm) ablation of the anterior stroma to remove superficial scars, band keratopathy remnants and dystrophies, and to smooth the surface. Useful for opacities within about the anterior 100 µm; deeper scars need DALK. Risks: hyperopic shift, haze, recurrence and thinning, so check pachymetry first.

Q What is an optical iridectomy and when is it done?
A sector iridectomy behind clear cornea to make a new pupil, usually inferonasal, for a small central opacity with a clear periphery and a clear lens when keratoplasty is unavailable or risky (for example Peters' anomaly in infants). First check that vision improves when the pupil is dilated; phenylephrine can bridge the wait.

Q What is the role of keratoplasty in adherent leucoma?
PK with synechiolysis, pupil reconstruction and IOP control, with cataract extraction and intraocular lens implantation if the lens is opaque (triple procedure). The prognosis is only fair because PAS, vessels and glaucoma increase rejection and failure.

Q How is corneal tattooing done?
For a disfiguring leucoma in a blind, quiet, non-ectatic eye. Under topical anaesthesia the epithelium over the scar is scraped off; filter paper soaked in 2% gold chloride (brown) or 2% platinum chloride (black) is applied for 2–3 minutes; then 2% hydrazine hydrate turns the salt to dark metal; wash, antibiotic and pad. Alternatives: India ink, keratopigmentation, a coloured cosmetic contact lens or prosthetic shell. Avoid in a thin or ectatic scar or a painful glaucomatous eye.

Q How do you manage an anterior staphyloma?
Control IOP. If vision is possible, PK (often large) with glaucoma surgery; if the eye is blind and painful or disfiguring, staphylectomy, evisceration or enucleation with a prosthesis.

Q How would you manage band keratopathy over a leucomatous cornea (S.No 40)?
Find the cause (chronic uveitis, phthisis, silicone oil, long-standing glaucoma or oedema). A blind, comfortable eye needs nothing more than a cosmetic shell. For irritation or before a graft: remove the epithelium and chelate the calcium with disodium ethylenediaminetetraacetic acid (EDTA) about 3%, then a bandage contact lens. Details on card EC6.

Q When is a keratoprosthesis considered?
For bilateral corneal blindness where a donor graft is bound to fail (multiple failures, heavy vascularisation, chemical injury, Stevens–Johnson syndrome): Boston type I, or osteo-odonto-keratoprosthesis for very dry eyes, usually in the better-potential eye only.

Complications and prognosis

Q What are the complications of an adherent leucoma?
Secondary glaucoma, ectatic scar and anterior staphyloma, corneal fistula, recurrent ulceration in a degenerated scar (atheromatous ulcer), endophthalmitis through a thin scar, cataract and phthisis bulbi.

Q What is the prognosis of keratoplasty for corneal opacity?
By Brightbill's grading: excellent for avascular central and traumatic scars (grade I), good for small vascularised scars (grade II), fair for heavily vascularised corneas (grade III), guarded for staphyloma and paediatric grafts (grade IV). Adherent leucoma with PAS and glaucoma does worse.

Q What is special about corneal opacity in children?
An opacity in the visual axis in the first years causes deprivation amblyopia and nystagmus, so treat early: optical iridectomy or PK (by 2–12 months in Peters' anomaly), then refraction and patching of the better eye. Paediatric grafts reject more, sutures loosen early (all out by about 6 months) and follow-up needs examinations under anaesthesia.

Recent advances and favourite examiner questions

Q Which eye would you operate first at S.No 19?
The eye with better visual potential and graft prognosis, usually the non-adherent leucoma with normal IOP and accurate projection (here the right). The adherent leucoma with raised IOP needs glaucoma control first and has a guarded prognosis.

Q What is Peters' anomaly?
A congenital central leucoma from a defect of posterior stroma, Descemet's membrane and endothelium, with a clear periphery. Type I: iris strands from the collarette to the scar; type II: lens adhesion or cataract, usually bilateral. Glaucoma in 50–70%. Genes include PAX6, PITX2 and FOXC1; Peters' plus syndrome adds short stature, cleft lip and developmental delay.

Q What newer options exist for corneal scars?
DALK by big-bubble (no endothelial rejection), femtosecond laser-assisted keratoplasty, PTK for superficial scars, AS-OCT-guided planning, keratopigmentation for cosmesis, keratoprosthesis; and regression of vessels before grafting by fine-needle diathermy or bevacizumab (subconjunctival or topical).

Examiner traps

- Describing only "a white opacity" without size, depth, vessels, iris adherence and IOP.
- Missing iris incarceration: look for a peaked pupil and brown pigment on the back of the scar.
- Forgetting IOP in adherent leucoma, and using Goldmann on a scar.
- Proposing keratoplasty without checking projection of rays and B-scan.
- Calling an ectatic iris-lined scar a leucoma: it is an anterior staphyloma.
- In a child, forgetting amblyopia and the need for early surgery.
- At S.No 40, offering a graft to a phthisical, blind eye: treat for comfort and cosmesis.

- Nebula (iris detail seen), macula (pupil margin seen), leucoma (nothing seen); a nebula can blur vision more than a leucoma.
- Adherent leucoma = leucoma with iris incarcerated after perforation; watch IOP.
- Anterior staphyloma = ectatic scar lined by iris.
- Visual potential first: projection of rays, B-scan.
- Superficial scar: PTK or DALK; full thickness or adherent: PK; small central scar: optical iridectomy; blind eye: cosmetic lens or tattoo.
- Congenital opacities, mnemonic **STUMPED**: Sclerocornea, Tears in Descemet's, Ulcer, Metabolic, Peters', Endothelial dystrophy, Dermoid.

Read more: Aravind 2.10–2.11 (p.88–93) · Baidya p.22–23, 59–62, 67–70, 81–85 · Namrata p.148–150, 160–162

EC4 Bacterial corneal ulcer with hypopyon

Cornea · On the list: S.No 53 · ★ Patient already arrived: none

At a glance

- **Bacterial keratitis**: an epithelial defect with stromal suppuration caused by bacteria. It is an ocular emergency: Pseudomonas and pneumococcus can perforate the cornea within 48–72 hours.
- **Hypopyon**: leucocytes and fibrin settled in the anterior chamber (AC), produced by toxin-driven iritis. In a bacterial ulcer it is **sterile and mobile** until the cornea perforates.
- Why kept: a classic case; examiners test the ulcer description, bacterial versus fungal, scraping and media, fortified drops, steroids and the management of perforation.
- Diagnosis line, S.No 53: "Left eye central (or paracentral) bacterial corneal ulcer, __ × __ mm, dense stromal infiltrate to __ depth, surrounding oedema, __ mm mobile hypopyon, no perforation, Seidel negative, lacrimal sac __; probably pneumococcal (or Pseudomonas); severe (or moderate)."
- Always give fungal keratitis as the main differential and say that the scraping report decides.

What to look for and present

Look for	What to say or record
Onset and speed	Hours to 2 days: Pseudomonas, Staphylococcus aureus, pneumococcus; slow and indolent: fungus, Acanthamoeba, Moraxella, Nocardia, atypical mycobacteria
Injury and agent	Fingernail, dust, metal: bacterial; vegetable matter: fungal; no injury but contact lens: Pseudomonas or Acanthamoeba
Contact lens, swimming, pond water	Pseudomonas; Acanthamoeba (pain out of proportion, ring infiltrate)
Treatment so far	Drugs, frequency, compliance; topical steroid or native medicines worsen infection and favour fungus
Eye predisposition	Chronic dacryocystitis (pneumococcus), trichiasis, entropion, lagophthalmos, dry eye, bullous keratopathy, graft, herpes
Systemic	Diabetes, alcoholism (Moraxella), malnutrition, immunosuppression
Lacrimal sac and lids	Regurgitation on pressure over the lacrimal sac (ROPLAS); lid and lash position
Ulcer	Site; size of epithelial defect and of infiltrate in two meridians; shape; margins (fluffy and active, or sharp and healing); depth; slough; surrounding oedema; thinning; satellites; pigment
Fluorescein and Seidel	Extent of the defect; leak over a thin floor or descemetocele
Anterior chamber	Hypopyon height in mm, level and mobility, colour (haemorrhagic in pneumococcus and herpes), cells, fibrin
Corneal sensation	Reduced in herpetic and neurotrophic ulcers
Intraocular pressure (IOP)	Gentle digital or Tono-Pen; no Goldmann or pressure on a thin cornea
Severity grade	Mild: under 2 mm, under 20% depth; moderate: 2–5 mm, 20–50%; severe: over 5 mm, over 50% or scleral spread
Fellow eye, fundus	Bilateral disease suggests an immune cause; B-scan ultrasonography if perforated or no view (endophthalmitis)

Diagnosis and summary to say

SAY IT

"Mr X, a 50-year-old farmer, has had pain, redness, watering and blurred vision in the left eye for 4 days after a fingernail injury, treated with drops from a pharmacy and no steroid. The left eye has lid oedema, ciliary congestion and a 4 × 3.5 mm central ulcer with a dense cream-white stromal infiltrate to about half the stromal depth, fluffy margins, surrounding stromal oedema and an epithelial defect of the same size. There is a 2 mm hypopyon with a horizontal level that shifts when the head is tilted; there is no thinning, satellite lesion or perforation, and the Seidel test is negative. ROPLAS is negative and corneal sensation is present. The right eye is normal."

Diagnosis: Left eye central bacterial corneal ulcer (moderate to severe) with mobile hypopyon, no perforation. Plan: scraping for Gram stain, potassium hydroxide (KOH) mount and culture, then fortified cefazolin 5% and tobramycin 1.3% with atropine.

Viva questions and answers

Basics

Q Define corneal ulcer and hypopyon.
Corneal ulcer: a break in the epithelium with loss of stromal tissue and inflammatory infiltration. Hypopyon: polymorphonuclear leucocytes and fibrin collected in the lower AC, produced by iritis from bacterial toxins.

Q Why is the hypopyon of a bacterial ulcer sterile?
Bacteria cannot cross an intact Descemet's membrane, so the hypopyon is an inflammatory response to toxins and becomes infected only after perforation. Fungal hyphae can penetrate Descemet's membrane, so a fungal hypopyon may contain organisms.

Q Which bacteria cause keratitis?
Gram-positive: Streptococcus pneumoniae, Staphylococcus aureus and epidermidis, Streptococcus pyogenes. Gram-negative: Pseudomonas aeruginosa, Moraxella, enteric bacilli (Proteus, Klebsiella, Serratia), Haemophilus, Neisseria. Others: Nocardia and atypical mycobacteria. In India pneumococcus and Pseudomonas are the commonest.

Q Which organisms can invade an intact epithelium?
Neisseria gonorrhoeae and meningitidis, Corynebacterium diphtheriae, Haemophilus influenzae (aegyptius) and Listeria.

Q What is ulcus serpens?
An acute hypopyon ulcer caused by Streptococcus pneumoniae: a central grey-white ulcer that creeps across the cornea with one undermined advancing edge while the other edge heals; early large hypopyon (sometimes haemorrhagic), marked iritis and early perforation. It is classically linked to chronic dacryocystitis.

Q What are the features of Pseudomonas, Staphylococcus, Moraxella and Nocardia ulcers?
Pseudomonas: rapid liquefactive melting, sticky greenish-yellow discharge, ground-glass oedema of the surrounding cornea, ring ulcer, perforation within 48 hours; contact lens wearers. Staphylococcus aureus: well-defined round or oval cream-white infiltrate with little surrounding oedema. Moraxella: indolent inferior ulcer in alcoholics, diabetics and debilitated patients. Nocardia: superficial wreath-like ulcer with pin-head infiltrates.

Q What are the stages of a corneal ulcer?
Infiltration, active ulceration (progression), regression and cicatrisation; the final scar may be a nebula, macula or leucoma depending on depth.

Examination and signs

Q How do you differentiate bacterial from fungal keratitis?
Bacterial: rapid, symptoms more than signs, purulent discharge, wet ulcer with sharp epithelial edge, dense suppuration and surrounding oedema, mobile hypopyon with a horizontal level. Fungal: vegetable trauma, slow, signs more than symptoms, dry raised grey-white infiltrate with feathery margins, satellite lesions, immune ring, endothelial plaque, pigment in dematiaceous fungi, thick fixed hypopyon with a convex upper border.

Q How do Acanthamoeba and herpetic keratitis differ from bacterial keratitis?
Acanthamoeba: contact lens or muddy water, pain out of proportion, pseudodendrites, radial keratoneuritis, ring infiltrate, waxing and waning course; needs calcofluor white and non-nutrient agar. Herpes simplex: recurrences, dendritic or geographic ulcer with terminal bulbs, reduced sensation, disciform keratitis with keratic precipitates; hypopyon is uncommon.

Q How do you test the mobility of a hypopyon and what does it tell you?
Note the level, ask the patient to tilt the head or lie on one side for about 10 minutes, and look again: a fluid bacterial hypopyon shifts and levels again; a thick fungal one stays fixed. Measure the height in mm at the slit lamp; a falling height is a sign of healing.

Q What signs show the ulcer is healing or worsening?
Healing: less pain, blunted, well-defined margins, smaller and less dense infiltrate, less oedema and AC reaction, falling hypopyon, re-epithelialisation, vessels growing in. Worsening: more pain, enlarging dense infiltrate, thinning, rising hypopyon, new satellites, scleral spread, perforation.

Q Why do you check the lacrimal sac and syringe?
Chronic dacryocystitis is a reservoir of pneumococcus and causes ulcus serpens and recurrence. Press over the sac and syringe; if mucopus regurgitates, treat the sac by dacryocystorhinostomy (DCR), or dacryocystectomy in an emergency, before corneal or intraocular surgery.

Investigations

Q How do you take a corneal scraping?
Before antibiotics if possible. Topical proparacaine 0.5% (least bactericidal), speculum, slit lamp or microscope; wipe away loose mucus; scrape the base and the advancing edge with a Kimura spatula, a no. 15 blade or a 26 G needle (pneumococcus lies at the edge, Moraxella in the base), avoiding lids and lashes. Make slides first, then inoculate the media in C-shaped streaks without cutting the agar; flame the spatula or change the blade between samples.

Q Which stains and culture media do you use?
Stains: Gram (bacteria), KOH 10% mount (fungi), Giemsa (cells and fungi), calcofluor white or acridine orange under fluorescence (fungi, Acanthamoeba), acid-fast stain (Nocardia, mycobacteria). Media: blood agar (most bacteria and fungi), chocolate agar (Haemophilus, Neisseria, Moraxella), thioglycollate broth (aerobes and anaerobes), brain–heart infusion broth, Sabouraud's dextrose agar (fungi, Nocardia; room temperature), non-nutrient agar with Escherichia coli overlay (Acanthamoeba), Löwenstein–Jensen medium (mycobacteria).

Q How long are cultures kept and when is a growth significant?
Bacterial cultures 7 days, anaerobic up to 14 days, fungal 2 weeks, mycobacterial 4–6 weeks. A growth is significant if the same organism grows on two media, or on one solid medium with a matching smear, or as confluent growth of a known pathogen on one solid medium (Jones criteria).

Q What do you do if the ulcer is not responding and the culture is negative?
Check compliance and drug toxicity; stop antibiotics for 24 hours and scrape again; corneal biopsy with a 2–4 mm skin-biopsy trephine at the edge, split for culture and histopathology; confocal microscopy and polymerase chain reaction (PCR); rethink the diagnosis (fungus, Acanthamoeba, Nocardia, atypical mycobacteria, Pythium, herpes, neurotrophic ulcer).

Management

Q What is the empirical treatment?
Admit severe ulcers. Fortified cefazolin 5% (50 mg/mL) with fortified tobramycin 1.3% (about 14 mg/mL), or a fourth-generation fluoroquinolone with tobramycin. Loading dose: one drop every 5 minutes for 30 minutes, then every 30–60 minutes day and night for 48 hours, the two drugs 5 minutes apart; taper on clinical response, and change drugs by the culture report only if the ulcer is not improving.

Q When is fluoroquinolone monotherapy acceptable?
In small (under 3 mm), peripheral or mid-peripheral, superficial (under 50% depth) ulcers: moxifloxacin 0.5% or gatifloxacin 0.3% hourly. Central, deep, larger or worsening ulcers need fortified combination therapy. Fluoroquinolones are cheap, stable and less toxic, but older ones cover pneumococcus poorly and resistance is rising.

Q How do you prepare fortified drops?
Cefazolin 5%: dissolve a 500 mg vial in 10 mL of sterile water or artificial tears. Tobramycin (or gentamicin): add 2 mL of injection (80 mg) to a 5 mL bottle of 0.3% drops (15 mg), giving about 13.6 mg/mL. Vancomycin 5% for methicillin-resistant Staphylococcus aureus (MRSA); ceftazidime 5% for resistant Pseudomonas. Refrigerate; use within about a week.

Q What are the adjuncts?
Atropine 1% two or three times a day (or homatropine 2%) relieves ciliary spasm and pain, prevents posterior synechiae and reduces exudation; oral analgesics; timolol 0.5% or oral acetazolamide for raised IOP; dark glasses. No eye pad (it warms the eye and traps discharge) and no topical steroid at the start.

Q When are subconjunctival and systemic antibiotics used?
Subconjunctival (cefazolin 100 mg, gentamicin 20 mg, vancomycin 25 mg) only when frequent drops are impossible or fortified drops unavailable. Systemic: perforation or impending perforation, scleral spread, endophthalmitis, gonococcal or Haemophilus infection, and children; for example oral ciprofloxacin 750 mg twice daily, or intravenous ceftriaxone for Neisseria.

Q What did the Steroids for Corneal Ulcers Trial (SCUT) show?
In 500 culture-positive bacterial ulcers, prednisolone phosphate 1% added after 48 hours of moxifloxacin gave no overall difference in 3-month best-corrected acuity, scar size or perforation. Ulcers with very poor vision at entry (counting fingers or worse) or central deep ulcers did better with steroid; Nocardia ulcers did worse. So use steroid only after 48 hours of effective antibiotic, with a known sensitive organism that is not fungus, Nocardia or Acanthamoeba, in an improving ulcer.

Q How do you manage impending or actual perforation?
Descemetocele or perforation under 2 mm: cyanoacrylate glue after debriding necrotic tissue and drying the bed, with a bandage contact lens (BCL); multilayered amniotic membrane for small perforations or thinning; tectonic patch graft for 3–5 mm; therapeutic keratoplasty (TPK) for 5 mm or more or uncontrolled infection. Add aqueous suppressants, a shield and systemic antibiotics, and avoid pressure on the eye.

Q What are the principles of therapeutic keratoplasty?
Excise all infected tissue with a 1 mm clear margin; oversize the donor by 0.75–1 mm; interrupted sutures; peripheral iridectomy; AC wash; send the host button for culture and histopathology; continue antibiotics. The aim is to eradicate infection and save the globe; an optical graft can follow later.

Complications and prognosis

Q List the complications of a bacterial corneal ulcer.
Descemetocele, perforation with iris prolapse, adherent leucoma, anterior staphyloma, corneal fistula, secondary glaucoma (inflammation, peripheral anterior synechiae, pupillary block), complicated or anterior polar cataract, purulent iridocyclitis, endophthalmitis and panophthalmitis, expulsive haemorrhage after perforation, scleritis, phthisis bulbi; on healing, a scar, vessels and irregular astigmatism.

Q What is a descemetocele and how do you recognise it?
Descemet's membrane bulging through the floor of an ulcer after the stroma has melted: a clear, dark-looking blister surrounded by grey infiltrate, which does not take fluorescein while the rim does. It signals imminent perforation: glue with a BCL or a tectonic graft, and lower IOP.

Q What are the signs of perforation?
Sudden relief of pain with a gush of fluid, a shallow or flat AC, a soft eye, iris prolapse or iris plugging the ulcer, a peaked pupil, a positive Seidel test and a falling hypopyon.

Q What is the prognosis?
It depends on the organism (Pseudomonas and pneumococcus are the worst), size, depth, site, delay in treatment, steroid misuse and host factors. Most ulcers heal in 2–3 weeks with a scar; a central ulcer leaves a leucoma that may need an optical keratoplasty once the eye has been quiet for some months; perforation and endophthalmitis carry a poor outlook.

Recent advances and favourite examiner questions

Q When do you suspect drug toxicity?
When the infiltrate is settling but the epithelial defect will not close, with punctate staining, chemosis and a quiet AC after days of hourly fortified aminoglycoside. Reduce the frequency or change the drug. Ciprofloxacin drops can leave white crystalline deposits in the ulcer bed that are not infection.

Q What newer treatments and diagnostics are there?
Photoactivated chromophore for keratitis–corneal cross-linking (PACK-CXL; riboflavin with ultraviolet A, which kills organisms and resists melting), rose bengal photodynamic therapy, fortified linezolid 0.2% for MRSA, intrastromal antibiotic injection; diagnosis by PCR, matrix-assisted laser desorption ionisation time-of-flight (MALDI-TOF) mass spectrometry, in vivo confocal microscopy, and anterior segment optical coherence tomography for depth.

Q What did the Mycotic Ulcer Treatment Trial (MUTT) show, and why does it matter here?
For filamentous fungal keratitis, natamycin 5% did better than voriconazole 1%, especially for Fusarium. If the scraping of a "bacterial" ulcer shows hyphae, switch to natamycin.

Examiner traps

- Calling every hypopyon ulcer bacterial without asking about vegetable trauma and looking for feathery margins and satellites.
- Forgetting to press over the lacrimal sac and syringe; missed dacryocystitis means pneumococcal recurrence.
- Measuring the ulcer once: give the defect and the infiltrate separately, in two meridians, with depth.
- Starting steroid at presentation; SCUT supports it only after 48 hours of effective antibiotic in a known, non-fungal infection.
- Using Goldmann tonometry or pressing on a thin, perforating cornea.
- Gluing a large perforation: glue suits holes under 2 mm; larger ones need a patch graft or TPK.
- Leaving the eye padded: no pad in an infected ulcer.

- Bacterial ulcer: rapid, purulent, wet, sharp edge, mobile sterile hypopyon.
- Scrape base and edge with proparacaine; Gram and KOH; blood, chocolate, thioglycollate and Sabouraud's media.
- Fortified cefazolin 5% and tobramycin 1.3%, loading dose every 5 minutes for 30 minutes, then hourly; fluoroquinolone alone only for small, peripheral, shallow ulcers.
- Atropine yes; pad and early steroid no.
- Glue under 2 mm, patch graft 3–5 mm, TPK 5 mm or more.
- Pneumococcus: ulcus serpens and dacryocystitis; Pseudomonas: contact lens and rapid melt.

Read more: Baidya p.10–19, 22–23, 26–30 · Namrata p.90–104 · Aravind 2.5 (p.66–71)

EC5 Corneal stromal dystrophies — granular dystrophy types 1 and 2 (Avellino), with lattice, macular and Fuchs as contrasts

Cornea · On the list: S.No 28 · ★ Patient already arrived: none

At a glance

- **What it is:** an inherited, bilateral, symmetrical, non-inflammatory build-up of abnormal protein in the corneal stroma. Granular dystrophy is **autosomal dominant** and caused by mutations in the **transforming growth factor beta-induced (TGFBI) gene** on chromosome **5q31**.
- **Why examiners keep it:** a stable, bilateral, easy-to-see case that opens the whole dystrophy viva: the International Committee for Classification of Corneal Dystrophies (IC3D) system, the granular–lattice–macular table, stains, recurrence in grafts and the risk of refractive surgery.
- **Pattern to recognise:** discrete white **breadcrumb or snowflake opacities** in the central anterior stroma, **clear stroma between them and a clear periphery**, normal thickness, and vision better than the cornea looks.
- **Two types:** granular type 1 (classic; Arg555Trp, arginine replaced by tryptophan at codon 555) and granular type 2 (**Avellino**, combined granular–lattice; Arg124His, arginine to histidine at codon 124).
- **Diagnosis line for S.No 28:** "Both eyes (BE) granular corneal dystrophy, probably type 1, with discrete anterior stromal breadcrumb opacities, clear intervening and peripheral stroma and no recurrent erosions; best-corrected visual acuity (BCVA) right eye (RE) 6/… and left eye (LE) 6/…".
- If mid-stromal lattice lines are also present: "BE granular corneal dystrophy type 2 (Avellino)".

What to look for and present

Look for	What to say or record
Age at onset, symptoms	Type 1: deposits appear in childhood, but glare, photophobia and blur usually only after 40. Type 2: earlier, more glare. Lattice and macular blur vision much earlier.
Family history (draw a pedigree)	Affected parent and siblings of both sexes: autosomal dominant (granular, lattice). Unaffected parents, consanguinity: autosomal recessive (macular). Offer to examine relatives.
Recurrent erosions	Sharp pain, watering and photophobia on waking: uncommon and mild in type 1, commoner in type 2, frequent in lattice and Reis–Bücklers.
Previous surgery	Laser in situ keratomileusis (LASIK) or photorefractive keratectomy (PRK): Avellino flare-up. Phototherapeutic keratectomy (PTK), deep anterior lamellar keratoplasty (DALK) or penetrating keratoplasty (PK): look for recurrence.
Systemic	Facial palsy, lax skin, neuropathy: lattice type 2 (gelsolin amyloidosis), never granular.
Vision and refraction	Unaided, pinhole and best-corrected, each eye; often surprisingly good.
Slit lamp: diffuse and direct focal	Number, size, shape and density of discrete white opacities; coalescence into rings, chains or lines.
Slit lamp: retroillumination and optical section	Granular deposits turn translucent on retroillumination; lattice lines are refractile with a double contour; the optical section gives depth (anterior stroma or full thickness).
Stroma between deposits; periphery	Clear between deposits and a clear peripheral 2–3 mm in granular; hazy in lattice and macular; macular reaches the limbus.
Epithelium, sensation	Fluorescein for erosions or an irregular surface; sensation near normal in granular, reduced in lattice and macular.
Thickness, Descemet membrane, endothelium	Normal in granular and lattice; thin cornea with grey Descemet membrane and guttae-like changes in macular.
Intraocular pressure (IOP), fundus	Baseline IOP before any steroid course; fundus usually visible.
Special tests	Anterior segment optical coherence tomography (AS-OCT) for depth; pachymetry; specular microscopy; confocal microscopy; TGFBI testing; histology of any excised tissue.
Classification you will be asked	IC3D 2015 group (epithelial–stromal TGFBI dystrophies), evidence category C1; type 1 or type 2.

Compare the three classic stromal dystrophies:

Feature	Granular type 1	Lattice type 1	Macular
Inheritance, gene	Autosomal dominant; TGFBI Arg555Trp	Autosomal dominant; TGFBI Arg124Cys	Autosomal recessive; carbohydrate sulphotransferase 6 (CHST6), 16q22

Feature	Granular type 1	Lattice type 1	Macular
Onset	Deposits in first decade; symptoms after 40	First decade; erosions early	First decade; vision falls by 10–30 years
Opacities	Discrete white crumbs, snowflakes, rings in anterior stroma	Refractile branching lines and dots	Grey-white ill-defined spots, superficial centrally, deeper peripherally
Stroma between	Clear	Hazy later	Hazy (ground glass) early
Periphery	Spared	Spared	Involved to the limbus
Thickness, endothelium	Normal	Normal	Thin; Descemet membrane and endothelium involved
Vision	Good till 40s–50s	Reduced by 30s–40s	Poor early
Recurrent erosions	Uncommon	Frequent	Occasional
Deposit, stain	Hyaline; Masson trichrome red	Amyloid; Congo red	Glycosaminoglycan; Alcian blue
Recurrence after keratoplasty	Common, superficial	Common (commonest of the three)	Least

Diagnosis and summary to say

SAY IT

Mr X, a 38-year-old clerk, has mild glare and slowly increasing blur in both eyes for five years; his father and one sister have "white spots" on the cornea. He has no pain or watering on waking, no refractive surgery and no systemic illness. Best-corrected vision is 6/9 in each eye and intraocular pressure is 14 and 15 mm Hg. Both corneas show many discrete, sharply outlined, white crumb-like opacities in the central anterior stroma, opaque on direct illumination and translucent on retroillumination, with clear stroma between them and a clear peripheral zone of about 2–3 mm. There are no lattice lines; the epithelium is intact and corneal sensation, thickness and endothelium are normal. Anterior chamber, iris, lens and fundus are normal.

Diagnosis: Both eyes granular corneal dystrophy type 1, with discrete anterior stromal opacities and clear intervening and peripheral stroma, not yet affecting vision significantly.

If lattice lines lie in the mid-stroma with anterior haze: **Both eyes granular corneal dystrophy type 2 (Avellino, combined granular–lattice).**

Viva questions and answers

Basics

Q Define a corneal dystrophy.
An inherited, usually bilateral and symmetrical, slowly progressive, non-inflammatory loss of corneal clarity; typically central and avascular, not caused by environmental or systemic disease; it starts in one layer with a characteristic deposit or pattern.

Q How does a dystrophy differ from a degeneration?
Dystrophy: inherited, early onset, bilateral and symmetrical, central, avascular, primary. Degeneration: acquired, age-related or secondary to disease or environment, often unilateral or asymmetric, peripheral, may be vascularised.

Q What is the IC3D classification?
A system (2008; second edition 2015; third edition 2024) naming each dystrophy by anatomy and genetics. The 2015 groups: epithelial and subepithelial; epithelial–stromal TGFBI; stromal; endothelial. Evidence categories: C1 gene and mutation known; C2 locus mapped, gene unknown; C3 clinical entity without a locus; C4 suspected new dystrophy.

Q What does the TGFBI gene make, and which dystrophies does it cause?
Keratoepithelin, an extracellular adhesion protein made by epithelium and keratocytes (old gene name BIGH3); the mutant protein aggregates. Hot spots at codons 124 and 555: Reis–Bücklers (Arg124Leu), Thiel–Behnke (Arg555Gln), lattice type 1 (Arg124Cys), granular type 1 (Arg555Trp), granular type 2 (Arg124His).

Q Which of these dystrophies are not autosomal dominant?
Macular dystrophy (autosomal recessive, CHST6 on 16q22); also recessive are gelatinous drop-like dystrophy, lattice type 3 and congenital hereditary endothelial dystrophy. Granular, lattice type 1, Avellino, Reis–Bücklers and Thiel–Behnke are dominant.

Q Why is type 2 called Avellino dystrophy?
It was first described in families tracing their roots to Avellino in Italy. It combines granular deposits and lattice lines in one cornea; commonest in Korea and Japan; homozygotes have severe early disease.

Q What is the deposit in granular dystrophy?
Non-collagenous hyaline protein (mutant keratoepithelin) forming sharply outlined extracellular aggregates, mainly in the anterior stroma; electron microscopy shows rod-shaped bodies. Type 2 also contains amyloid.

Examination and signs

Q Describe the slit-lamp picture of granular type 1.
Bilateral discrete grey-white crumb-, snowflake-, ring- or popcorn-shaped opacities in the central anterior stroma; white on direct illumination, translucent on retroillumination; clear stroma between them and a clear 2–3 mm periphery. Later they enlarge, coalesce and go deeper, and the stroma between turns ground-glass.

Q Why does vision stay good for so long?
The deposits are discrete, the stroma between them is clear and the surface stays regular; vision falls only with confluence, superficial spread or haze, usually after 40–50 years.

Q How does granular type 2 differ from type 1?
Fewer, larger star- or ring-shaped anterior stromal deposits come first; lattice lines follow in the mid and deep stroma; anterior haze develops with age; onset is earlier, with more erosions and glare; it worsens dramatically after LASIK.

Q How do you tell lattice lines from granular deposits?
Lattice lines are fine, branching, refractile filaments with a clear core and double contour on retroillumination, with dots on a hazy background; granular deposits are opaque discrete crumbs with clear stroma between them.

Q Which stromal dystrophy thins the cornea and reaches the limbus?
Macular dystrophy: diffuse haze, ill-defined grey-white spots out to the limbus, reduced thickness and Descemet membrane or endothelial involvement; it needs keratoplasty earliest.

Q What is lattice type 2?
Meretoja syndrome (familial amyloidosis, Finnish type): gelsolin gene on chromosome 9, autosomal dominant; a few thick lattice lines running in from the periphery; facial palsy, lax facial skin, dermatochalasis, cranial and peripheral neuropathy. IC3D says it is not a true corneal dystrophy.

Q What is recurrent corneal erosion, and in which dystrophies?
Sudden pain, watering and photophobia on opening the eyes in the morning, as the lid tears off loosely attached epithelium. Commonest with epithelial basement membrane dystrophy, Reis–Bücklers and lattice; less with granular type 2; rare and mild in granular type 1.

Q What is the differential diagnosis of granular-looking stromal opacities?
Other TGFBI dystrophies; macular, Schnyder and fleck dystrophies; paraproteinaemic keratopathy (monoclonal gammopathy or myeloma: check serum protein electrophoresis); healed nummular keratitis; drug or lipid deposits.

Investigations

Q Name the histology stains for the three stromal dystrophies.
Granular: hyaline, bright red with Masson trichrome. Lattice: amyloid, orange-pink with Congo red, apple-green birefringence and dichroism under polarised light, fluorescent with thioflavin T, metachromatic with crystal violet. Macular: glycosaminoglycan (keratan sulphate), blue with Alcian blue and colloidal iron, periodic acid–Schiff positive.

Q How is macular dystrophy subtyped?
By antigenic keratan sulphate: type I absent from cornea and serum (commonest); type IA present only in keratocytes; type II present in cornea and serum. They look identical clinically.

Q What does AS-OCT add?
Deposit depth and the clear stroma beneath; this decides PTK (disease in the anterior 100–150 µm) or DALK or PK. It also measures thickness and the depth of recurrence in a graft.

Q When is genetic testing useful?
Not routinely. It confirms the TGFBI mutation in atypical cases, screens relatives, detects homozygotes (worse prognosis) and screens refractive-surgery candidates for Avellino in high-prevalence populations.

Q What does confocal microscopy show?
Granular type 1: hyper-reflective, sharply bordered anterior stromal deposits; type 2: these plus linear branching lattice deposits; lattice: branching filaments with changing reflectivity; macular: diffuse, ill-defined reflective material.

Q Why do specular microscopy in a stromal dystrophy?
To judge the endothelium before choosing DALK or PK; essential in macular dystrophy, where subclinical endothelial disease favours PK.

Management

Q How do you manage S.No 28 if vision is good?
Explain the disease and its inheritance; observe; spectacles; tinted lenses for glare; lubricants; advise against LASIK; yearly review with BCVA and slit-lamp photographs; examine relatives.

Q How do you treat recurrent erosions?
Acute: lubricants, cycloplegic, bandage contact lens and antibiotic cover. Prevention: 5% sodium chloride ointment at night for months. Persistent cases: epithelial debridement, anterior stromal puncture outside the visual axis, or PTK.

Q What is PTK and what are its limits?
Excimer laser (argon fluoride, 193 nm) ablation of the anterior stroma to remove superficial opacities and smooth the surface; suits deposits in the anterior 100–150 µm while leaving at least 250 µm of stroma. Problems: hyperopic shift, haze, delayed healing, infection and recurrence; it can be repeated.

Q When DALK and when PK?
DALK when opacities are deep but the endothelium is healthy (granular, lattice): no endothelial rejection, early steroid taper, stronger wound. PK when Descemet membrane or endothelium is involved (often macular) or when DALK perforates.

Q Do dystrophies recur in a graft? Why, where and how treated?
Yes: host epithelium and invading host keratocytes lay down fresh protein in the donor. Recurrence is subepithelial or at the interface, first at the graft–host junction, after about 1–19 years; commonest with Reis–Bücklers and lattice, frequent in granular (earlier in type 2), least in macular. Superficial recurrence is treated with PTK.

Complications and prognosis

Q Why can LASIK worsen Avellino dystrophy?
The flap cut and stromal wound healing make epithelium and keratocytes produce more mutant TGFBI protein, which deposits at the flap interface; dense, progressive opacities appear months to years later and may need flap lift with debridement, PTK or keratoplasty. PRK can also provoke deposits, so refractive surgery is contraindicated.

Q What is the prognosis of granular type 1?
Good: most keep useful vision into middle age; keratoplasty is needed late, if at all; recurrences after surgery are usually superficial and treatable by PTK. Macular dystrophy has the worst visual prognosis but the fewest recurrences.

Recent advances and favourite examiner questions

Q Contrast Reis–Bücklers and Thiel–Behnke dystrophies.
Both are autosomal dominant Bowman layer TGFBI dystrophies with painful erosions from childhood. Reis–Bücklers (granular type 3, Arg124Leu): geographic map-like opacities, worse vision, rod-shaped bodies staining red with Masson. Thiel–Behnke (Arg555Gln): honeycomb pattern, milder, curly fibres on electron microscopy. Both recur early after keratoplasty; PTK is preferred.

Q Contrast Fuchs endothelial dystrophy with granular dystrophy.
Fuchs is posterior: endothelial cell loss with central guttae (beaten-metal look), Descemet thickening, then stromal and epithelial oedema with bullae; women 4:1, after 50; blur worst on waking; thick cornea. Granular is anterior stromal deposits with normal thickness and endothelium.

Q What is the genetics of Fuchs dystrophy?
Late-onset: mostly a cytosine–thymine–guanine (CTG) trinucleotide repeat expansion (CTG18.1) in the transcription factor 4 (TCF4) gene on chromosome 18; early-onset: mutations in collagen VIII alpha-2 (COL8A2). Autosomal dominant with variable penetrance; many cases seem sporadic.

Q How is Fuchs dystrophy managed?
Early: 5% sodium chloride drops 4–8 times a day and ointment at night; a hairdryer at arm's length in the morning; bandage contact lens for ruptured bullae. Surgery: Descemet membrane endothelial keratoplasty (DMEK) or Descemet stripping automated endothelial keratoplasty (DSAEK); PK if the stroma is scarred; combine with cataract surgery when needed.

Q What newer treatments exist for Fuchs dystrophy?
Descemet stripping only (remove the central guttae and let peripheral cells migrate), helped by the Rho-kinase inhibitor ripasudil; injection of cultured endothelial cells with a Rho-kinase inhibitor.

Q What is Schnyder corneal dystrophy?
An autosomal dominant disorder of corneal lipid metabolism (UbiA prenyltransferase domain-containing 1 gene): central subepithelial cholesterol crystals in a ring or disc, stromal haze and early arcus; about half have raised serum cholesterol, so check the lipid profile. Some have no crystals.

Q What treatments for TGFBI dystrophies are under research?
Allele-specific gene editing and small interfering ribonucleic acid, both silencing the mutant TGFBI allele, are experimental; femtosecond-assisted DALK and repeat PTK with mitomycin C are current refinements.

Examiner traps

- Calling it “macular dystrophy” because it is central: macular dystrophy is autosomal recessive, hazy between spots, reaches the limbus and thins the cornea.
- Describing only “white spots”: examiners want **clear stroma between the deposits and a clear periphery**, seen with retroillumination and an optical section.
- Forgetting to ask about LASIK or PRK, and to warn affected relatives against refractive surgery without screening for Avellino.
- Saying keratoplasty cures it: TGFBI dystrophies recur in grafts; PTK for superficial disease, DALK when the endothelium is healthy.
- Mixing up the stains: Masson trichrome red for granular, Congo red for lattice, Alcian blue for macular.
- Calling Reis–Bücklers “honeycomb”: in IC3D the honeycomb pattern belongs to Thiel–Behnke.
- Saying macular dystrophy recurs most in grafts: it recurs least, although it needs keratoplasty earliest.

- Granular: TGFBI on 5q31, autosomal dominant, hyaline, Masson red, clear between and around the crumbs.
- Avellino (granular type 2): Arg124His, granular crumbs plus lattice lines; LASIK makes it worse.
- Lattice: amyloid, Congo red with apple-green birefringence, early erosions; type 2 is gelsolin systemic amyloidosis.
- Macular: autosomal recessive, CHST6, keratan sulphate, Alcian blue, hazy to the limbus, thin; needs keratoplasty earliest, recurs least.
- Superficial disease and graft recurrence: PTK; deep disease with healthy endothelium: DALK.
- Fuchs is the endothelial contrast: guttae, oedema, TCF4; treated with endothelial keratoplasty.
- Mnemonic for stains: “Marilyn Monroe Always Gets Her Man in LA City” (Macular, Mucopolysaccharide, Alcian blue; Granular, Hyaline, Masson; Lattice, Amyloid, Congo red).

Read more: Baidya p.47–52 · Namrata p.117–133 · Aravind 2.14–2.15 (p.107–120)

EC6 Corneal degenerations — climatic droplet keratopathy and band-shaped keratopathy, with arcus, Vogt girdle, Salzmann, Terrien and pellucid

Cornea · On the list: S.No 35, 40 · ★ Patient already arrived: none

At a glance

- **What they are:** acquired, age-related or secondary changes in a previously normal cornea. Two are on the list: **climatic droplet keratopathy (CDK)**, an ultraviolet (UV)-induced protein deposit, and **band-shaped keratopathy (BSK)**, calcium in Bowman layer.
- **Why examiners keep them:** quick cases that test slit-lamp description (amber oil droplets against a chalky plaque with holes), causes, systemic work-up, and one procedure you must describe step by step: ethylenediaminetetraacetic acid (EDTA) chelation.
- **Both lie in the interpalpebral strip**, the cornea exposed between the lids, and start at 3 and 9 o'clock.
- **Diagnosis line for S.No 35:** "Both eyes (BE) climatic droplet keratopathy (spheroidal degeneration), primary corneal type, grade …, with BE pseudophakia (posterior chamber intraocular lens, PCIOL)".
- **Diagnosis line for S.No 40:** "Right eye (RE) calcific band-shaped keratopathy over a leucomatous corneal opacity, secondary to … (state the cause: chronic uveitis, phthisis, old keratitis); vision perception of light with accurate projection".
- In S.No 40 the leucoma usually limits vision more than the band: decide whether the eye has visual potential before planning surgery.

What to look for and present

Look for	What to say or record
Occupation, sun exposure	Outdoor work (farmer, fisherman, labourer), sand, wind, glare from water or snow, no sunglasses: CDK.
Ocular history	Chronic uveitis, glaucoma, injury, old keratitis or trachoma, retinal surgery with silicone oil, long-term drops (phosphate-containing steroids, mercury-preserved pilocarpine): band keratopathy; scars and keratitis also cause secondary CDK.
Systemic history	Childhood joint pain (juvenile idiopathic arthritis, JIA), kidney or parathyroid disease, vitamin D tablets, sarcoidosis, myeloma, gout.
Symptoms	Blur when central; foreign-body sensation, pain and watering when epithelium breaks over nodules or plaques; cosmetic white patch.
Visual acuity	Each eye; in a leucomatous eye record perception of light and projection in four quadrants.
CDK at the slit lamp	Golden-yellow or amber, translucent, oil-like droplets beneath the epithelium, in Bowman layer and superficial stroma; begin at the nasal and temporal limbus and spread centrally in a horizontal strip; coalescent or elevated nodules; grade it.
BSK at the slit lamp	Grey-white chalky plaque at Bowman level in the interpalpebral band; clear zone at the limbus; small round holes; flaking edges or white elevated nodules; fluorescein for epithelial defects.
The leucoma (S.No 40)	Size, density, depth, thinning, vascularisation; iris adherent (adherent leucoma); whether iris and pupil are visible.
Conjunctiva, lids	Pinguecula, pterygium, conjunctival droplets, dry eye, lagophthalmos, trachoma scars or trichiasis.
Anterior chamber, iris, pupil	Cells, flare, keratic precipitates, posterior synechiae, festooned pupil, emulsified silicone oil; pseudoexfoliation at the pupil margin.
Lens or intraocular lens	PCIOL position and capsule (S.No 35); complicated cataract in uveitis.
Intraocular pressure (IOP), globe	Soft, small globe: phthisis; stony hard: absolute glaucoma.
Fundus	Direct view if possible; otherwise B-scan ultrasound for detachment, silicone oil or phthisis.
Classification to quote	CDK: type (primary corneal, secondary corneal, conjunctival) and grade (trace to 4). BSK: calcific or non-calcific, peripheral or central, extent.

Diagnosis and summary to say

> **Say it**
>
> Mr X, a 68-year-old farmer who has worked in the sun all his life, had cataract surgery in both eyes three years ago; vision has stayed slightly hazy, with glare in bright light, but there is no pain. Best-corrected vision is 6/12 in the right eye and 6/18 in the left; intraocular pressure is 16 mm Hg in each eye. Both eyes have pingueculae and many small, translucent, golden-yellow oil-like droplets beneath the epithelium and in the superficial stroma, in a horizontal band running from the nasal and temporal limbus towards the centre; the band crosses the visual axis in the left eye only. There are no elevated nodules and no epithelial defects. Both eyes have a well-centred posterior chamber intraocular lens in the bag, with fine pseudoexfoliation material at the pupil margin; the discs are healthy and the maculae show a few dry drusen.
>
> **Diagnosis: Both eyes climatic droplet keratopathy, primary corneal type, grade 1 in the right eye and grade 3 in the left, with both eyes pseudophakia (posterior chamber intraocular lens in the bag) and pseudoexfoliation.**
>
> For S.No 40: **Right eye calcific band-shaped keratopathy over a leucomatous corneal opacity, secondary to (cause), with perception of light and accurate projection.**

Viva questions and answers

Basics

Q Define a corneal degeneration. How does it differ from a dystrophy?
Acquired deterioration of previously normal corneal tissue from age, environment or disease. Unlike a dystrophy it is not inherited, starts later, is often unilateral or asymmetric, peripheral or interpalpebral rather than central, may be vascularised, and is linked to ocular or systemic disease.

Q What is CDK, and what are its other names?
Accumulation of translucent golden globules beneath the epithelium, in Bowman layer and anterior stroma of the exposed cornea, sometimes the conjunctiva. Also called spheroidal degeneration, Labrador keratopathy, Bietti nodular degeneration, chronic actinic keratopathy, keratinoid degeneration, fisherman's keratitis and corneal elastosis.

Q What causes CDK?
Chronic UV light, direct and reflected from snow, sand or water, alters plasma proteins diffusing from limbal vessels into the exposed cornea; age, wind, sand and dust microtrauma, low humidity and dry eye add to it. Commonest in elderly men working outdoors.

Q What are the droplets made of?
Extracellular protein with elastotic features, rich in photo-oxidised tryptophan, tyrosine, cystine and cysteine; eosinophilic on haematoxylin and eosin; not calcium. They darken from yellow to brown with age.

Q Classify CDK.
Type 1, primary corneal: bilateral, interpalpebral, age and UV related, cornea otherwise normal. Type 2, secondary corneal: after scars, keratitis, trachoma, lattice dystrophy or glaucoma; may be unilateral or central. Type 3, conjunctival: at 3 and 9 o'clock with pinguecula, alone or with type 1.

Q How is CDK graded?
Johnson and Ghosh: trace, a few lesions at one end of the strip; grade 1, interpalpebral periphery only; grade 2, centre involved without visual loss; grade 3, centre involved with visual loss; grade 4, grade 3 plus elevated nodules.

Q What is band-shaped keratopathy?
Deposition of calcium hydroxyapatite in the epithelial basement membrane, Bowman layer and anterior stroma, forming a horizontal interpalpebral band.

Q Why is calcium deposited in the exposed strip?
Evaporation concentrates calcium and phosphate in the tear film there; loss of carbon dioxide raises the local pH, which makes calcium phosphate less soluble; inflammation adds to the alkalinity. Raised serum calcium or phosphate makes it worse.

Q List the causes of band keratopathy.
Ocular: chronic uveitis (especially JIA), phthisis bulbi, long-standing glaucoma, chronic corneal oedema, interstitial keratitis, silicone oil in the anterior chamber, dry eye. Systemic: hypercalcaemia (hyperparathyroidism, vitamin D excess, sarcoidosis, milk–alkali syndrome, myeloma, bone metastases), renal failure, gout. Chemical: mercury, phosphate-containing drops. Familial; idiopathic.

Q Why is JIA important?
The chronic anterior uveitis of oligoarticular, antinuclear antibody-positive young girls is white and painless, so band keratopathy, complicated cataract and glaucoma (the triad taught with Still disease) may be the first signs; such children need slit-lamp screening every 3–6 months.

Examination and signs

Q How do you tell CDK from band keratopathy at the slit lamp?
CDK: amber, translucent, oil-like droplets of varying size, raised, starting at the limbus, often with pinguecula. Band keratopathy: grey-white chalky flat plaque with a clear zone at the limbus and small holes, later white nodules, and gritty to a blade.

Q Why is there a clear zone at the limbus, and holes in the band?
Limbal vessels buffer the pH and Bowman layer ends near the limbus, so calcium does not settle there. The holes (the "Swiss-cheese" look) are where corneal nerves pierce Bowman layer.

Q Where does band keratopathy start?
Usually peripherally at 3 and 9 o'clock, the two ends spreading to meet centrally; in chronic uveitis and phthisis it may begin centrally.

Q What is a leucomatous opacity?
A dense white scar hiding iris details. A nebula is faint (iris seen); a macula is denser (iris margins seen); a leucoma is total; with iris incarcerated it is an adherent leucoma.

Q What else must you examine in S.No 40?
Light perception and projection, IOP, globe size (phthisis), signs of uveitis or silicone oil, the other eye, and B-scan if the fundus is hidden: this decides between optical and comfort surgery.

Q What is the differential diagnosis of an interpalpebral corneal band?
Calcific band keratopathy; CDK; Vogt limbal girdle; urate (gout) keratopathy, brown and non-calcific; Salzmann nodules; calcareous degeneration; familial band keratopathy.

Q Why look for pseudoexfoliation in a CDK patient?
Both are age and UV related and are reported together in sunny, outdoor populations, alongside cataract and pterygium. Pseudoexfoliation means weak zonules and glaucoma risk, so check the pupil margin, intraocular lens centration and IOP.

Investigations

Q What systemic work-up do you order for band keratopathy?
When no ocular cause is obvious: serum calcium and phosphate, alkaline phosphatase, uric acid, urea and creatinine, parathyroid hormone, vitamin D; angiotensin-converting enzyme, lysozyme and chest X-ray for sarcoidosis; serum protein electrophoresis for myeloma; antinuclear antibody in a child with uveitis.

Q How is calcium confirmed in tissue?
Basophilic granules on haematoxylin and eosin in the basement membrane and Bowman layer; black with von Kossa stain, red with alizarin red.

Q Which imaging helps plan treatment?
Anterior segment optical coherence tomography (AS-OCT) shows the depth of calcium or droplets: anterior disease suits chelation, keratectomy or phototherapeutic keratectomy (PTK); deep scarring needs lamellar keratoplasty. B-scan assesses the posterior segment behind an opaque cornea.

Management

Q How do you manage CDK?
Most need nothing beyond UV protection (wraparound sunglasses, hat) and lubricants; bandage contact lens for erosions. Central visual loss: superficial keratectomy with a no. 15 blade, with or without amniotic membrane; excimer PTK for lesions within about 100 µm or residual irregularity; lamellar keratoplasty for deep scars.

Q Why is penetrating keratoplasty risky in CDK?
Grafts fail often: these eyes have dry, UV-damaged surfaces, poor goblet cell function, lid problems and continuing exposure.

Q CDK and cataract: which first?
Central droplets distort keratometry and limit post-operative vision. Either PTK first and cataract surgery once keratometry is stable, or cataract surgery first and PTK about three months later; counsel that the result depends on the cornea.

Q When and how do you treat band keratopathy?
Treat the cause first. Indications: reduced vision, irritation, epithelial breakdown, cosmesis. Mild: lubricants, bandage contact lens. Definitive: EDTA chelation, superficial keratectomy, PTK to smooth, amniotic membrane, lamellar keratoplasty for deep deposits.

Q Describe EDTA chelation step by step.
Topical anaesthesia and speculum; scrape the epithelium off the plaque with a no. 15 blade; apply disodium EDTA 1.5–3% (about 0.05 mol/L) on a soaked cellulose sponge or in a corneal trephine well for 5–30 minutes, renewing it; scrape away the softened calcium with a blunt spatula; irrigate; smooth with a diamond burr if needed; bandage contact lens.

Q How do you make 3% EDTA in theatre?
Dilute 2 mL of 15% disodium EDTA with 8 mL of normal saline. Dipotassium EDTA is a cheaper alternative.

Q What is the care after chelation?
Bandage contact lens until healed, topical antibiotic, cycloplegic, lubricants, and a non-steroidal or acetate-steroid drop; avoid phosphate-containing steroid drops, which re-precipitate calcium; review for recurrence.

Q What is the role of amniotic membrane?
As a graft or patch after removal it provides a basement membrane, speeds epithelial healing and reduces inflammation, scarring and recurrence; useful when the surface is poor.

Q Why not PTK alone for band keratopathy?
Calcium ablates at a different rate from stroma, so PTK alone leaves an irregular surface; use it after chelation, with a masking fluid, to smooth what remains.

Q How do you manage S.No 40 if the eye has no visual potential?
Treat for comfort and appearance: lubricants, bandage contact lens, chelation for painful nodules, cosmetic tinted contact lens or prosthetic shell; conjunctival flap for persistent pain; evisceration for a blind, painful eye.

Complications and prognosis

Q What are the complications of CDK?
Epithelial breakdown over nodules, recurrent erosion, sterile ulceration, microbial keratitis, thinning, rarely perforation, and visual loss when central.

Q What is the outlook after EDTA chelation?
Vision and comfort improve in most otherwise healthy eyes; recurrence over years is common if the cause persists (uveitis, silicone oil, hypercalcaemia), and chelation can be repeated.

Recent advances and favourite examiner questions

Q Arcus senilis: what is it and when does it matter?
Lipid in the peripheral stroma, first above and below, later a ring with a sharp outer edge, a clear zone at the limbus and a hazy inner edge. Under about 50 (arcus juvenilis): check fasting lipids for familial hypercholesterolaemia. Unilateral arcus: carotid disease or hypotony on the side without arcus.

Q What is Vogt limbal girdle?
Chalky white crescents at the nasal and temporal limbus in older people. Type 1 has a clear zone from the limbus and holes (an early calcific band); type 2 has no clear zone (elastotic, like pinguecula). Harmless.

Q What is Salzmann nodular degeneration?
Grey-white or bluish raised nodules in the mid-periphery after chronic keratitis (trachoma, phlyctenulosis, vernal keratoconjunctivitis), contact lens wear, or in middle-aged women; collagen and hyaline beneath the epithelium. Lubricants; superficial keratectomy peels them off a cleavage plane; PTK; mitomycin C reduces recurrence.

Q How do you tell Terrien marginal degeneration from Mooren ulcer?
Terrien: painless superior peripheral thinning, mostly men, bilateral, intact epithelium, vessels crossing the gutter, lipid line at the steep central edge, against-the-rule astigmatism, rare perforation. Mooren: painful, ulcerated, overhanging edge, no clear zone, interpalpebral, needs immunosuppression.

Q What is pellucid marginal degeneration?
Bilateral inferior band of thinning 1–2 mm wide from 4 to 8 o'clock, 1–2 mm from the limbus, with protrusion above the band (a "beer-belly" profile); no vessels, lipid or inflammation; high against-the-rule astigmatism; "crab-claw" or "kissing doves" topography. Rigid or scleral lenses, cross-linking, crescentic lamellar graft.

Q How does calcareous degeneration differ from band keratopathy?
Calcareous degeneration affects all layers of a badly damaged, often phthisical and scarred cornea; band keratopathy is superficial, at Bowman layer, and confined to the exposed strip.

Examiner traps

- Calling amber oily droplets "band keratopathy": calcium is chalky white with holes and a clear limbal zone; CDK is golden and translucent.
- Forgetting occupation and sun exposure in CDK, and the systemic causes (calcium, kidneys, sarcoidosis, JIA) in band keratopathy.
- Not stating light perception and projection, IOP and B-scan in S.No 40: an eye with a leucoma, band and phthisis may have no visual potential.
- Describing chelation vaguely: examiners expect the strength (1.5–3%), the epithelium removed first, the time (5–30 minutes) and a bandage lens.
- Prescribing phosphate-containing steroid drops after chelation.
- Offering penetrating keratoplasty first for CDK: superficial keratectomy or PTK usually suffices and grafts fare badly.
- Missing pseudoexfoliation at the pupil margin or the intraocular lens in a CDK patient.

- CDK: UV, outdoor men, amber droplets at 3 and 9 o'clock spreading centrally; protein, not calcium.
- CDK grades: trace, 1 periphery, 2 centre with good vision, 3 centre with poor vision, 4 nodules.
- BSK: calcium hydroxyapatite in Bowman layer; clear limbal zone; Swiss-cheese holes are nerve channels.
- BSK causes: chronic uveitis (JIA), phthisis, absolute glaucoma, silicone oil, hypercalcaemia, renal failure.
- Chelation: remove epithelium, EDTA 1.5–3% for 5–30 minutes, scrape, bandage lens; no phosphate steroids.
- Degeneration: acquired, late, asymmetric, peripheral, vascularised; dystrophy: inherited, early, symmetric, central.
- Causes of band keratopathy, mnemonic "UV PHOSPHATE": Uveitis, Vitamin D excess, Phthisis, Hyperparathyroidism, Oil (silicone), Sarcoidosis, Phosphate drops, Hypercalcaemia of malignancy, Absolute glaucoma, Tear evaporation (dry eye), End-stage renal disease.

Read more: Baidya p.33–34, 49, 65–70 · Namrata p.125, 143–153 · Aravind 2.4 (p.63–65), 2.10 (p.88–90), 2.14 (p.107)

EC7 Keratoconus, including the listed "Lisch nodules" (neurofibromatosis type 1) and acute hydrops

Cornea · On the list: S.No 44 · ★ Patient already arrived: none

At a glance

- **What it is:** a bilateral, asymmetric, progressive, non-inflammatory ectasia: the cornea thins (usually inferior to centre), bulges into a cone and develops irregular myopic astigmatism. Onset at puberty; progression slows after 30 and usually stops by 40.
- **Why it is kept:** eponymous signs, a management ladder from spectacles to cross-linking and keratoplasty, and the numbers examiners love (47 dioptres, 400 µm, 3 mW/cm²).
- **The listed patient is unusual:** Lisch nodules are iris hamartomas of neurofibromatosis type 1 (NF1). Confirm what the iris shows (true Lisch nodules, iris naevi or freckles, or Brushfield spots of Down syndrome) and examine the skin before accepting the label.
- **Diagnosis line for S.No 44:** "Both eyes (BE) keratoconus, Amsler–Krumeich stage … right eye (RE) and … left eye (LE), with bilateral iris nodules suggestive of Lisch nodules; to rule out NF1".
- If the spots are Brushfield spots: "BE keratoconus with Brushfield spots; suspect Down syndrome". If the iris is normal, write "BE keratoconus" and say the label needs checking.

What to look for and present

Look for	What to say or record
Age, symptoms, course	Teens or twenties; blurred, distorted vision, frequent change of glasses, glare, monocular diplopia, contact lens intolerance.
Eye rubbing, allergy	Itching and knuckle rubbing; vernal keratoconjunctivitis (VKC), eczema, asthma; snoring and floppy lids (sleep apnoea).
Family and systemic	Relatives with keratoconus; Down, Marfan or Ehlers–Danlos features; Leber congenital amaurosis (LCA). For the "Lisch nodules": skin patches, lumps, learning difficulty, an affected parent.
Past treatment	Spectacles, rigid or scleral lenses, cross-linking, ring segments, keratoplasty; episodes of hydrops; laser in situ keratomileusis (LASIK) (post-LASIK ectasia).
Visual acuity	Unaided, pinhole, best-corrected with spectacles and with a rigid lens; asymmetry between eyes.
External signs	Munson sign on downgaze; Rizzuti sign; evert lids for papillae; floppy lids.
Retinoscopy, distant direct	Scissoring reflex; oil-droplet reflex.
Slit lamp: cornea	Cone position, thinning maximal at the apex, Fleischer ring (cobalt blue), Vogt striae, prominent nerves, apical scar, hydrops or post-hydrops scar.
Slit lamp: iris nodules	Number, size, colour, shape, both eyes: smooth dome-shaped tan nodules (Lisch); flat dark spot (naevus or freckle); grey-white ring of spots in the mid-periphery (Brushfield).
Rest of the eye	Lens subluxation (Marfan); intraocular pressure (IOP), underread in thin corneas; fundus for LCA, retinitis pigmentosa, optic pallor (glioma in NF1).
General examination	Café-au-lait macules, axillary freckling, neurofibromas; facies of Down syndrome; Marfan habitus; joint laxity.
Topography and tomography	Inferior steepening, asymmetric bow-tie, maximum keratometry (Kmax), thinnest pachymetry and its location, posterior elevation, Belin–Ambrosio display.
Progression	Serial maps on the same machine: rising Kmax, thinning, rising astigmatism: the trigger for cross-linking.
Classification you will be asked	Amsler–Krumeich stage; cone type (nipple, oval, globus); severity by keratometry.

Diagnosis and summary to say

> **Say it**
>
> Mr X, a 19-year-old student, has gradually blurred, distorted vision in both eyes for four years, with frequent changes of glasses, night glare and constant rubbing of itchy eyes since childhood; there has been no sudden painful whitening of either eye. He has several light-brown skin patches, and his mother has similar ones. Best-corrected spectacle vision is 6/12 in the right eye and 6/36 in the left, improving to 6/6 and 6/9 with rigid lenses; retinoscopy shows scissoring in both eyes. Both corneas show inferocentral thinning and protrusion with a Fleischer ring and Vogt striae, the left more advanced with a positive Munson sign; both irides carry several small, smooth, dome-shaped light-brown nodules. Tomography shows mean central keratometry of 49 and 54 dioptres (D) and thinnest pachymetry of 455 and 390 µm, with inferior steepening and abnormal posterior elevation in both eyes.
>
> **Diagnosis: Both eyes keratoconus, Amsler–Krumeich stage 2 in the right eye and stage 3 in the left, with bilateral Lisch nodules and café-au-lait macules suggestive of neurofibromatosis type 1, and allergic eye disease with eye rubbing.**
>
> If the spots are Brushfield spots: **Both eyes keratoconus with Brushfield spots; Down syndrome to be confirmed.**

Viva questions and answers

Basics

Q Define keratoconus.
A bilateral, usually asymmetric, progressive, non-inflammatory ectasia with central or paracentral stromal thinning, apical protrusion and irregular astigmatism (the classic triad). The 2015 global consensus says truly unilateral keratoconus does not exist.

Q Who gets it?
Teenagers, usually noticed around 15; more males; about 400 per 100,000, more in South Asians, who present younger, often with vernal disease; 6–18% have an affected relative.

Q What is the pathogenesis?
A genetically susceptible cornea plus mechanical stress, chiefly rubbing: raised matrix metalloproteinases, reduced inhibitors, keratocyte apoptosis driven by interleukin-1, oxidative stress and loss of collagen lamellae.

Q What is the histological triad?
Stromal thinning, breaks in Bowman layer, and iron in the basal epithelium (Fleischer ring); Descemet membrane and endothelium are normal unless hydrops occurs.

Q Name the associations.
Atopy and VKC with rubbing; Down syndrome; Marfan, Ehlers–Danlos, osteogenesis imperfecta, mitral valve prolapse; LCA (oculodigital poking) and retinitis pigmentosa; floppy eyelid syndrome; Crouzon, Turner; NF1 (rare).

Q What are Lisch nodules?
Melanocytic iris hamartomas of NF1: multiple, bilateral, smooth, dome-shaped, tan to brown nodules on the iris surface; appear in childhood, present in nearly all adults with NF1; harmless; two or more is a diagnostic criterion.

Q How do you handle "keratoconus with Lisch nodules"?
Confirm at the slit lamp: Lisch nodules are raised, multiple, bilateral; naevi usually single, flat, darker; Brushfield spots grey-white in a mid-peripheral ring (Down syndrome, a strong keratoconus association); mammillations tiny uniform villi. If true Lisch nodules, assess NF1, then treat the keratoconus on its merits; the link is weak.

Q What are the diagnostic criteria for NF1?
Two or more of: six or more café-au-lait macules (over 5 mm before puberty, over 15 mm after); axillary or inguinal freckling; two neurofibromas or one plexiform; optic pathway glioma; two or more Lisch nodules or choroidal abnormalities; sphenoid wing dysplasia or tibial bowing; a pathogenic NF1 variant (17q11.2, autosomal dominant). One suffices with an affected parent.

Examination and signs

Q What is the earliest sign?
Scissoring of the retinoscopy reflex; at the slit lamp the Fleischer ring is the earliest. Subclinical disease shows only on topography or tomography.

Q Describe Munson and Rizzuti signs.
Munson: V-shaped bulge of the lower lid on downgaze, in moderate to advanced disease. Rizzuti: a penlight from the temporal side is focused by the cone into a sharp beam just inside the nasal limbus.

Q What is the Fleischer ring?
Yellow-brown iron (haemosiderin) in the basal epithelium around the base of the cone; partial or complete; best seen with the cobalt-blue filter; outlines the cone and guides graft size.

Q What are Vogt striae?
Fine vertical stress lines in the posterior stroma and Descemet membrane at the apex, parallel to the cone axis, which vanish briefly on gentle pressure.

Q Explain the oil-droplet reflex and scissoring.
Oil droplet (Charleux): through a dilated pupil the cone shows a dark ring round a bright centre, from total internal reflection. Scissoring: the cone and its periphery differ in power, so near neutralisation two reflex bands move in opposite directions.

Q What causes prominent corneal nerves?
Ocular: keratoconus, Fuchs dystrophy, congenital glaucoma, dry eye, trauma, ageing. Systemic: NF1, leprosy, multiple endocrine neoplasia type 2B, Refsum disease, ichthyosis; a useful link to the listed nodules.

Q What types of cone are there?
Nipple: 5 mm or less, central or inferonasal, easiest to fit. Oval: 5–6 mm, inferotemporal, more hydrops. Globus: over 75% of the cornea, hardest to fit.

Q How do you tell keratoconus from pellucid degeneration and keratoglobus?
Keratoconus: puberty, thinning maximal at the apex, Fleischer ring, Vogt striae. Pellucid: 20–40 years, inferior 1–2 mm band of thinning with protrusion above it, against-the-rule astigmatism, "crab-claw" map. Keratoglobus: from birth, limbus-to-limbus thinning, globular cornea, easy perforation.

Investigations

Q What are the Placido topography criteria?
Rabinowitz: central keratometry (K) over 47.2 D; inferior–superior asymmetry (I–S) over 1.4 D (1.2–1.4 suspect); skew of the steepest radial axes (SRAX) over 21° (asymmetric bow-tie); more than 1 D difference in central K between eyes.

Q What is KISA%?
K × I–S × AST × SRAX × 100 ÷ 300, where AST is the regular corneal astigmatism. Over 100% is keratoconus; 60–100% suspect.

Q Why use Pentacam, and what does it show?
The rotating Scheimpflug camera maps the posterior surface and thickness limbus to limbus, correcting for eye movement. Keratoconus: abnormal posterior elevation (often first), thinnest point under about 480 µm and displaced inferiorly, fast peripheral thickness increase, high Kmax.

Q What is the Belin–Ambrosio enhanced ectasia display?
Elevation against a best-fit sphere that excludes the 3.5–4 mm zone around the thinnest point, exaggerating the cone, plus thickness-progression indices; deviation values are white, yellow (suspect) or red (abnormal), and the final D summarises ectasia risk. Used to screen refractive-surgery candidates.

Q What is forme fruste keratoconus?
Subclinical disease: no clinical signs but abnormal topography or tomography, typically the better eye of a keratoconic patient. It contraindicates LASIK.

Q Give the Amsler–Krumeich classification.
Stage 1: eccentric steepening, myopia or astigmatism under 5 D, mean K 48 D or less. Stage 2: 5–8 D, K 53 D or less, no scar, thinnest 400 µm or more. Stage 3: 8–10 D, K over 53 D, no scar, 300–400 µm. Stage 4: refraction impossible, K over 55 D, central scar, about 200 µm.

Q How do you document progression?
Serial tomography on one machine: Kmax up 1 D or more in 12 months (or 0.5 D in 6), thinning, or rising astigmatism or myopia; also falling best-corrected vision or new contact lens intolerance.

Management

Q Outline the management ladder.
Stop rubbing and treat allergy; spectacles; contact lenses (soft toric, rigid gas-permeable, hybrid, piggyback, scleral); corneal collagen cross-linking (CXL) once progression is shown; intracorneal ring segments for lens intolerance with a clear centre; keratoplasty: deep anterior lamellar (DALK) preferred, penetrating (PK) for Descemet scarring.

Q Which contact lens and which fit?
Rigid gas-permeable first, fitted by three-point touch (light apical bearing with mid-peripheral touch) for vision, comfort and less scarring; Rose K designs for steep cones; scleral lenses for advanced or intolerant eyes.

Q How does CXL work?
Ultraviolet A (UVA) light excites riboflavin to release reactive oxygen species that bond collagen fibrils in the anterior 300 µm, stiffening the cornea; anterior segment optical coherence tomography (AS-OCT) shows a demarcation line at about that depth.

Q Describe the Dresden protocol.
Remove 7–9 mm of central epithelium; riboflavin 0.1% in 20% dextran every 2–5 minutes for 30 minutes until seen in the anterior chamber; UVA 370 nm at 3 mW/cm² for 30 minutes (5.4 J/cm²), riboflavin continued; bandage contact lens and antibiotic.

Q Why 400 µm, and what if the cornea is thinner?
Riboflavin absorbs most UVA anteriorly, so at 400 µm the endothelium is safe. For 320–400 µm, hypo-osmolar riboflavin (in saline, no dextran) swells the stroma above 400 µm before irradiation.

Q What are accelerated and epithelium-on CXL?
Accelerated: same 5.4 J/cm² in less time (9 mW/cm² for 10 minutes, 18 for 5, 30 for 3; Bunsen–Roscoe law), slightly weaker; pulsed light restores oxygen. Epithelium-on: less pain and infection, but less riboflavin penetration and effect.

Q Give contraindications and complications of CXL.
Contraindications: thinnest under 400 µm (standard protocol), past herpetic keratitis, active infection, severe dry eye, central scar, pregnancy; Kmax over 58 D fails more. Complications: pain, delayed healing, infection, herpes reactivation, sterile infiltrates, haze, endothelial damage, failure.

Q How do intracorneal ring segments work, and in whom?
Polymethyl methacrylate arcs (Intacs, hexagonal; Ferrara, triangular) at about 75% depth shorten the arc and flatten the centre 2–3 D. Need a clear centre, at least 450 µm at the track, K 46–60 D, contact lens failure. Risks: extrusion, migration, infection, melt, vascularisation.

Q DALK or PK?
DALK (big bubble: air at 60–80% depth bares Descemet membrane) keeps the host endothelium: no endothelial rejection, shorter steroids, stronger globe. PK for post-hydrops Descemet scars or perforated DALK; success 90–95%, as the cornea is avascular, uninflamed, with healthy endothelium.

Q What is Urrets-Zavalia syndrome?
A fixed dilated pupil with iris atrophy after keratoplasty for keratoconus, attributed to iris ischaemia from raised IOP, air or gas in the anterior chamber, or mydriatics.

Complications and prognosis

Q What is acute hydrops?
A break in Descemet membrane lets aqueous into the stroma: sudden painful blur, a white swollen cone, intrastromal clefts on AS-OCT. Commoner with large cones, rubbing, vernal disease and Down syndrome.

Q How do you treat acute hydrops?
5% sodium chloride drops and ointment, cycloplegic, topical steroid, antibiotic if epithelium breaks, IOP-lowering drops, shield, no rubbing; clears in 2–4 months. Faster: intracameral air or sulphur hexafluoride 20% or perfluoropropane 14%, with or without compression sutures; graft for a residual scar after 2–3 months.

Q Why is keratoconus a contraindication to LASIK, and what else excludes it?
Flap and stromal ablation weaken an already weak cornea and cause ectasia; photorefractive keratectomy (PRK) is also avoided. Also excluded: forme fruste or abnormal Belin–Ambrosio display, thin cornea, residual bed under 250–300 µm or over 40% tissue altered, unstable refraction, rubbing.

Recent advances and favourite examiner questions

Q Name newer options for keratoconus.
Topography-guided photorefractive keratectomy with CXL (Athens protocol); corneal allogenic intrastromal ring segments (donor tissue); Bowman layer transplantation for very steep cones; toric phakic lenses once stable; epithelial thickness maps and Corvis biomechanics for early diagnosis.

Q What is posterior keratoconus?
An unrelated congenital, usually unilateral, non-progressive excavation of the posterior corneal surface with stromal haze: anterior segment dysgenesis.

Examiner traps

- Accepting “Lisch nodules” without looking: say what you actually see and how you separated Lisch nodules from naevi and Brushfield spots.
- Missing Brushfield spots: Down syndrome is a strong association and its patients get hydrops.
- Saying Fleischer ring is in Descemet membrane: it is iron in the basal epithelium (the Kayser–Fleischer ring is at Descemet).
- Recommending cross-linking without documented progression or without checking the thinnest pachymetry against 400 µm.
- Calling acute hydrops an emergency for keratoplasty: treat medically or with air or gas first.
- Forgetting to examine the tarsal conjunctiva and to advise against eye rubbing.
- Offering LASIK, PRK or ring segments to a scarred stage 4 cornea.

- Triad: thinning, protrusion, irregular astigmatism; histology: thin stroma, Bowman breaks, epithelial iron.
- Earliest sign: scissoring; earliest slit-lamp sign: Fleischer ring.
- Topography: K over 47.2 D, I–S over 1.4 D, SRAX over 21°, KISA% over 100.
- CXL Dresden: riboflavin 0.1%, UVA 370 nm, 3 mW/cm² for 30 minutes, 5.4 J/cm², thinnest 400 µm.
- Ladder: stop rubbing, spectacles, rigid lens, CXL if progressing, rings, DALK, PK.
- Lisch nodules mean NF1 (17q11.2); Brushfield spots mean Down syndrome.
- Acute hydrops: Descemet break; medical treatment for 2–4 months, air or gas to speed it; DALK or PK later for the scar.

Read more: Baidya p.37–47, 487–488, 719–722 · Namrata p.106–116 · Aravind 2.13 (p.98–106), 3.1 (p.145)

EO1 Optic atrophy, including post-neuritic optic atrophy

Other — neuro-ophthalmology · On the list: S.No 2, 37 · ★ Patient already arrived: S.No 2

At a glance

- **What it is:** pallor of the optic disc *plus* evidence of optic nerve dysfunction — the end stage of retinal ganglion cell axon loss anywhere from the retina to the lateral geniculate body. It is a sign, not a diagnosis: always name the type and the cause.
- **Post-neuritic optic atrophy:** follows optic neuritis. After papillitis it has the secondary pattern (dirty grey disc, blurred margins, gliosis, sheathed vessels); after retrobulbar neuritis it looks primary, often with temporal pallor.
- **Why examiners keep it:** optic nerve function testing, the look of each type, causes and work-up, the Optic Neuritis Treatment Trial (ONTT), neuromyelitis optica and rehabilitation.
- **S.No 2 (patient already here):** "Both eyes post-neuritic (secondary) optic atrophy following optic neuritis; best-corrected acuity __ and __; relative afferent pupillary defect (RAPD) __; red–green dyschromatopsia; field defects __; cause — multiple sclerosis (MS), neuromyelitis optica spectrum disorder (NMOSD) or myelin oligodendrocyte glycoprotein (MOG) antibody-associated disease (MOGAD) — to be established."
- **S.No 37:** "Both eyes retinitis pigmentosa (RP) with consecutive optic atrophy (waxy disc pallor), arteriolar attenuation and bone-spicule pigmentation."
- **Rule:** pallor alone is not atrophy — prove dysfunction with acuity, RAPD, colour vision and fields.

What to look for and present

Look for	What to say or record
Onset and course	Sudden and painful (neuritis); sudden and painless (ischaemia); gradual (compression, toxic, hereditary, glaucoma); one eye, both, or sequential; recovery and recurrences.
Features of past neuritis	Pain on eye movement, colours "washed out", recovery over weeks, steroids given; Uhthoff phenomenon.
Neurological history	Numbness, weakness, bladder symptoms, diplopia, Lhermitte sign (MS); myelitis, intractable hiccups or vomiting (NMOSD); recent infection or vaccination in a child (MOGAD).
Drugs, toxins, family	Ethambutol, isoniazid, linezolid, amiodarone; methanol; tobacco and alcohol, B12 deficiency; family history (Leber hereditary optic neuropathy, dominant optic atrophy, RP).
Other symptoms	Headache and transient obscurations (post-papilloedema atrophy); endocrine symptoms (pituitary tumour); night blindness (RP); deafness and diabetes (Wolfram).
Visual acuity	Distance and near best-corrected visual acuity (BCVA); often worse than the disc suggests in compressive and neuritic disease; central acuity may be kept in RP despite a pale disc.
Pupils	RAPD by swinging flashlight, graded or measured with neutral density filters; may be absent when both nerves are equally damaged (likely in S.No 2).
Colour, brightness, contrast	Ishihara plates (red–green loss); red desaturation and light brightness comparison between eyes; Pelli–Robson contrast.
Fields	Confrontation, then automated perimetry: central, centrocaecal, altitudinal, arcuate, bitemporal or homonymous; ring scotoma and tubular field in RP.
Disc	Colour (chalky white, dirty grey, waxy yellow); margins (sharp or blurred); cup (saucer, filled, excavated); lamina seen or hidden; vessels (normal, attenuated, sheathed); capillary count (Kestenbaum); pattern (diffuse, temporal, band, altitudinal).
Rest of fundus	Nerve fibre layer loss on red-free light; macula; bone-spicule pigment; chorioretinal scars; optociliary shunt vessels (meningioma).
Neurological examination	Ocular motility (internuclear ophthalmoplegia, skew), nystagmus, cerebellar signs, limb power, sensory level, reflexes.
Classification to state	Ophthalmoscopic — primary, secondary, consecutive, glaucomatous, partial; pathological — ascending or descending; aetiological.

Diagnosis and summary to say

> **SAY IT**
>
> Mrs X, 32, had painful loss of vision in the right eye three years ago and in the left eye one year ago; each recovered partly over six to eight weeks, and she received intravenous steroids for the second episode. She has no other neurological symptoms and no relevant drug or family history. Best-corrected vision is 6/24 in the right eye and 6/12 in the left, with a right relative afferent pupillary defect, Ishihara plates read 4 of 21 on the right and 12 of 21 on the left, and red desaturation on the right. Both discs are dirty greyish-white with slightly blurred margins, cups filled with glial tissue, lamina not visible, few capillaries on the disc, and narrowed, sheathed arterioles near the disc; the maculae and peripheries are normal. Fields show a right central scotoma and a left paracentral scotoma, and the neurological examination is normal.
>
> **Diagnosis (S.No 2):** both eyes post-neuritic (secondary) optic atrophy after sequential optic neuritis, right worse than left; the cause — multiple sclerosis, neuromyelitis optica spectrum disorder or myelin oligodendrocyte glycoprotein antibody disease — to be established by brain and orbit imaging and antibody tests.
>
> **S.No 37:** both eyes retinitis pigmentosa with consecutive optic atrophy (waxy disc pallor), arteriolar attenuation and bone-spicule pigmentation.

Viva questions and answers

Basics

Q Define optic atrophy.
Not a disease but the morphological end point of any process that irreversibly damages retinal ganglion cells and their axons in the anterior visual pathway, seen as disc pallor with loss of nerve function.

Q Why does an atrophic disc look pale?
Healthy axon bundles conduct light into the disc, where it picks up the pink of the capillaries. After axon loss, astrocytes lie across the incoming light and reflect it white; capillaries are fewer; and loss of tissue exposes the white lamina cribrosa.

Q What is the Kestenbaum capillary index?
The number of small capillaries (not the main vessels) crossing the disc margin — normally about 10–12; 6 or fewer indicates atrophy. A fall in the number of disc vessels is called Kestenbaum's sign.

Q Classify optic atrophy.
Ophthalmoscopic — primary, secondary, consecutive, glaucomatous (cavernous) and partial (temporal, band, altitudinal, wedge). Pathological — ascending (anterograde) or descending (retrograde), plus transsynaptic. Aetiological — hereditary, consecutive, post-inflammatory, pressure or traction, ischaemic, toxic or nutritional, metabolic, traumatic.

Q What are ascending and descending atrophy?
Ascending — the lesion is in the retina or disc and degeneration travels towards the lateral geniculate body (central retinal artery occlusion, RP, glaucoma, extensive panretinal photocoagulation). Descending — the lesion is in the nerve, chiasm or tract and degeneration travels towards the eye (compression, trauma, retrobulbar neuritis); pallor appears after 4–6 weeks and is complete by 6–8 weeks.

Q Describe primary optic atrophy and give causes.
Orderly axon loss replaced by glial columns without disturbing the disc architecture: chalky white disc, sharp margins, lamina cribrosa well seen (laminar dots), shallow saucer cup, normal vessels and retina. Causes: retrobulbar neuritis, compression (pituitary tumour, meningioma, glioma, aneurysm), trauma, hereditary, toxic and nutritional optic neuropathy.

Q Describe secondary optic atrophy and give causes.
Follows long-standing disc swelling, with marked glial proliferation: dirty grey disc, blurred margins, cup filled with fibroglial tissue hiding the lamina, peripapillary sheathing and narrowing of arteries, tortuous veins, sometimes drusen-like bodies. Causes: papilloedema, papillitis, anterior ischaemic optic neuropathy.

Q Describe consecutive optic atrophy and give causes.
Ascending atrophy from retinal or choroidal disease: waxy yellow pallor, normal margins, normal cup, marked arteriolar attenuation, with the causative retinal disease visible. Causes: RP, central retinal artery occlusion, extensive photocoagulation, diffuse chorioretinitis, long-standing retinal detachment, high myopia.

Q What exactly is "post-neuritic" optic atrophy?
The atrophy that follows optic neuritis. After papillitis or neuroretinitis it is secondary in pattern (grey, blurred margins, gliosis, sheathed vessels); after retrobulbar neuritis it is primary in pattern, typically temporal pallor from loss of the papillomacular bundle. Say which pattern you see.

Q Why is the disc waxy in RP (S.No 37)?
Ganglion cell loss secondary to outer retinal degeneration plus a layer of astroglial tissue over the disc, which gives a waxy, yellowish sheen, with marked arteriolar attenuation.

Examination and signs

Q Which tests show optic nerve dysfunction?
Reduced acuity, RAPD, dyschromatopsia (mainly red–green), red desaturation, reduced brightness and contrast sensitivity, field defects (central, centrocaecal, arcuate, altitudinal), prolonged visual evoked potential (VEP), and a normal photostress recovery (prolonged in macular disease).

Q How do you grade an RAPD?
Swinging flashlight test: grade 1 — weak initial constriction then greater redilation; 2 — initial stall then dilation; 3 — immediate dilation; 4 — no reaction (amaurotic pupil). Quantify with neutral density filters (0.3–1.8 log units) before the normal eye. If one pupil cannot move, judge the other pupil (reverse RAPD).

Q How do you test red desaturation and brightness?
Show a red bottle top to each eye in turn and ask how red it looks in the bad eye if the good eye scores 100%; do the same for the brightness of a torch.

Q How does colour vision separate optic nerve from retinal disease?
Optic nerve disease gives a red–green defect early and out of proportion to acuity; macular disease gives a blue–yellow defect. Exceptions: glaucoma and dominant optic atrophy cause blue–yellow (tritan) loss.

Q What do patterns of pallor tell you?
Temporal pallor — papillomacular bundle (MS, toxic and nutritional, Leber hereditary optic neuropathy, dominant optic atrophy); band or bow-tie pallor — loss of nasal retinal fibres (chiasmal lesion, or the eye opposite an optic tract lesion); altitudinal — ischaemic optic neuropathy; wedge — branch retinal artery occlusion; diffuse — most others.

Q How do you tell glaucomatous from non-glaucomatous atrophy?
Glaucoma — cupping exceeds pallor, rim stays pink, vertical cup, bayoneting, nasal shift of vessels, laminar dots, acuity and colour preserved until late. Non-glaucomatous — pallor exceeds cupping, pale rim, shallow cup, acuity, colour and RAPD affected early.

Q What is the Foster Kennedy syndrome?
Optic atrophy in one eye from direct compression (frontal lobe or olfactory groove meningioma) with papilloedema in the other from raised intracranial pressure. Pseudo-Foster Kennedy is commoner: sequential ischaemic optic neuropathy, with atrophy in the old eye and swelling in the new.

Q What pale-looking discs are not optic atrophy?
Normal temporal pallor, pseudophakia, myopic or tilted discs, a large physiological cup, hypoplasia, coloboma, morning glory, optic pit and drusen — function is normal or otherwise explained.

Investigations

Q How would you investigate S.No 2?
Magnetic resonance imaging (MRI) of brain and orbits with gadolinium and fat suppression; optical coherence tomography (OCT) of the retinal nerve fibre layer (RNFL) and ganglion cell layer; VEP; serum aquaporin-4 (AQP4) and MOG antibodies by cell-based assay; blood count, B12, syphilis serology, angiotensin-converting enzyme, antinuclear antibody, erythrocyte sedimentation rate; cerebrospinal fluid oligoclonal bands if MS is suspected.

Q What does OCT add?
Measures RNFL and ganglion cell–inner plexiform layer thinning (temporal after neuritis, bow-tie in chiasmal disease) and detects loss long before pallor — pallor needs about 50% axon loss, whereas OCT detects 7–9 µm thinning. Severe atrophy reaches a "floor" (about 40–50 µm), beyond which RNFL cannot track change.

Q What does the VEP show?
Demyelination — prolonged P100 latency with relatively preserved amplitude, a delay that often persists after recovery and marks past neuritis; axonal, ischaemic or compressive loss — reduced amplitude with near-normal latency. It also helps in suspected non-organic loss.

Q How does MRI separate MS, NMOSD and MOGAD?
MS — short posterior optic nerve lesion, ovoid periventricular plaques perpendicular to the ventricles; NMOSD — long posterior lesion reaching the chiasm, spinal lesion over 3 or more vertebral segments; MOGAD — long anterior lesion with perineural sheath enhancement and disc swelling.

Q What confirms consecutive atrophy in S.No 37?
Full-field electroretinogram — rod responses reduced or extinguished first, then cones; fields show ring scotoma and later a tubular field; OCT shows outer retinal loss and any cystoid macular oedema.

Management

Q Can optic atrophy be treated?
Lost axons do not regenerate. Treat the cause to save what remains (decompress a tumour, stop ethambutol, give B12, lower intracranial or intraocular pressure, prevent further neuritis), protect the other eye, and rehabilitate.

Q Describe the Optic Neuritis Treatment Trial.
455 patients aged 18–46 with acute unilateral optic neuritis within 8 days, randomised to intravenous methylprednisolone 250 mg 6-hourly for 3 days then oral prednisone 1 mg/kg/day for 11 days; oral prednisone 1 mg/kg/day for 14 days; or placebo — each with a short taper.

Q What did the ONTT show?
Intravenous steroid speeded recovery but did not change vision at 1 year; oral prednisone alone increased recurrences, so it is not used; MS conversion fell at 2 years in the intravenous group with an abnormal MRI, but not after 3 years. Fifteen-year risk of MS: 50% overall — 25% with a normal baseline MRI, 72% with one or more lesions. Ten-year recurrence: 35%.

Q When would you give steroids today?
For faster recovery, severe or bilateral loss, an only eye or occupational need: intravenous methylprednisolone 1 g daily for 3 days, then oral prednisolone 1 mg/kg/day for 11 days and taper — after excluding tuberculosis, diabetes and peptic ulcer. Refer to a neurologist for MS disease-modifying therapy when MRI shows lesions.

Q How are NMOSD and MOGAD treated?
NMOSD (AQP4 antibody, women 9:1, severe, poor recovery): intravenous methylprednisolone 1 g for 5 days with early plasma exchange; prevent relapse with azathioprine, mycophenolate or rituximab, or newer eculizumab, inebilizumab or satralizumab — MS drugs such as interferon-beta and natalizumab can worsen it. MOGAD (often bilateral with marked disc swelling, good recovery): steroids with a slow taper over months, as relapses follow rapid withdrawal; intravenous immunoglobulin for relapsing disease.

Q Outline low-vision rehabilitation.
Best refraction; hand, stand and electronic magnifiers and telescopes; large print, contrast and lighting; tints for glare; screen readers; eccentric viewing training for central scotomas; mobility training and a white cane for field loss; disability certification, vocational support and counselling.

Complications and prognosis

Q What is the visual prognosis after typical optic neuritis?
Recovery begins within 3 weeks in about 80% and within 5 weeks in over 90%; two-thirds regain 6/6 and fewer than 15% stay worse than 6/12. Colour, contrast and depth often remain subtly impaired, and RNFL thinning persists.

Q Which forms carry a poor prognosis?
NMOSD (severe, bilateral, recurrent), arteritic ischaemic neuropathy, Leber hereditary optic neuropathy with the 11778 mutation, long-standing compression and severe toxic damage. Established atrophy itself does not recover.

Q What are the features of Leber hereditary optic neuropathy?
Mitochondrial (maternal) inheritance; young men; painless sequential central loss weeks to months apart; acute disc hyperaemia with peripapillary telangiectatic vessels that do not leak, later temporal atrophy; centrocaecal scotoma. Mutations 11778 (commonest, worst recovery), 3460 and 14484 (best). Idebenone 900 mg/day; avoid tobacco and alcohol.

Q How does ethambutol cause optic atrophy?
Dose-related toxicity (risk rises above 15 mg/kg/day), usually after 2–8 months: bilateral central or centrocaecal scotomas, dyschromatopsia, later temporal pallor. Stop the drug at once; check vision and colour at baseline and monthly; recovery is partial.

Recent advances and favourite examiner questions

Q What is the Uhthoff phenomenon?
Transient blurring in a previously affected eye when body temperature rises — exercise, a hot bath, fever — recovering within minutes to an hour on cooling. Heat worsens conduction block in demyelinated axons. It predicts MS and recurrence.

Q What is the Pulfrich phenomenon?
Slower conduction in one optic nerve makes a pendulum swinging in a flat plane appear to move in an ellipse — a false depth cue from the latency difference. A neutral density filter over the normal eye can equalise it.

Q How is the Lhermitte sign different?
An electric-shock feeling down the spine or limbs on bending the neck, from demyelination in the cervical cord — not a visual symptom. Do not swap it with Uhthoff.

Examiner traps

- Diagnosing optic atrophy from a pale disc alone — pseudophakia and normal temporal pallor fool many; prove dysfunction.
- Calling every post-neuritic disc “primary” — after papillitis it is secondary in pattern; describe colour, margins, cup and vessels.
- Saying “no RAPD, so the nerves are fine” in S.No 2 — symmetrical bilateral damage abolishes the RAPD.
- Swapping Uhthoff (heat-induced blur) and Lhermitte (electric shock on neck flexion).
- Quoting the ONTT as proof that steroids improve final vision — they only speed recovery; oral prednisone alone increases recurrences.
- Forgetting NMOSD and MOGAD in bilateral or severe neuritis, and treating NMOSD with MS drugs.
- In S.No 37, calling the waxy disc “primary” — it is consecutive (ascending) atrophy from RP.

- Optic atrophy = pallor plus dysfunction; name the type and the cause.
- Primary: chalky white, sharp margins, lamina seen. Secondary: dirty grey, blurred, gliosis, sheathed vessels. Consecutive: waxy, attenuated arterioles, retinal disease.
- Kestenbaum: normal 10–12 disc capillaries; 6 or fewer in atrophy.
- Descending atrophy shows pallor at 4–6 weeks, complete by 6–8 weeks.
- ONTT: intravenous steroid speeds recovery only; oral prednisone alone raises recurrence; MS risk 25% (normal MRI) versus 72% (lesions) at 15 years.
- Bilateral or severe neuritis: test AQP4 and MOG antibodies.
- Mnemonic for the disc: **"Colour, Contour, Cup, Capillaries, Calibre"** — colour, margin, cup, Kestenbaum count, vessel calibre.

Read more: Baidya p.630–634 (optic atrophy), p.607–615 (optic neuritis), p.320 and p.324 (waxy disc in retinitis pigmentosa) · Namrata p.377–383 · Aravind 7.5 (p.464–473), 7.3 (p.442–448)

EO2 Congenital anterior segment anomalies — microcornea, corectopia, and limbal dermoid and lipodermoid

Other · On the list: S.No 6, 49 · ★ Patient already arrived: S.No 6

At a glance

- **Microcornea**: horizontal corneal diameter **less than 10 mm in an adult** (less than 9 mm at birth). **Corectopia**: a pupil displaced from its normal, almost central position. Both are developmental anomalies and travel with **coloboma, glaucoma and cataract**.
- **Epibulbar choristomas**: the **limbal (solid) dermoid** sits astride the **inferotemporal limbus**; the **lipodermoid (dermolipoma)** lies under the conjunctiva in the **superotemporal fornix**. Either may be part of **Goldenhar syndrome**.
- Kept because the signs are stable and the viva can range over embryology, anterior segment dysgenesis, glaucoma and amblyopia.
- **S.No 6** ★: "Right eye microcornea with corectopia; left eye typical iris coloboma — bilateral congenital anterior segment anomalies, probably of the coloboma spectrum; to rule out microphthalmos, fundal coloboma and glaucoma." The iris coloboma is covered in full on card ER5.
- **S.No 49**: "Right (or left) eye epibulbar lipodermoid in the superotemporal fornix with optic disc coloboma; Goldenhar (oculo-auriculo-vertebral) spectrum to be excluded."
- Examiners want **measured** corneal diameters, the pupil position, intraocular pressure (IOP), gonioscopy, the fundus, refraction and amblyopia, and a systemic check (ears, face, spine, teeth, heart).

What to look for and present

Look for	What to say or record
Onset and course	Static since birth: dysgenesis; a mass **enlarging at puberty**: dermoid; an acquired displaced pupil in a young woman: iridocorneal endothelial (ICE) syndrome
Watering, photophobia, large or cloudy eye in infancy	Developmental glaucoma; microcornea, aniridia and Axenfeld–Rieger eyes carry a lifelong glaucoma risk
Poor vision, squint, nystagmus	Amblyopia (anisometropic, deprivation, strabismic); nystagmus means poor vision from infancy (aniridia, foveal hypoplasia, macular coloboma)
Family history, consanguinity; maternal rubella, alcohol	Coloboma and Axenfeld–Rieger are often autosomal dominant; aniridia two-thirds familial; microphthalmos with rubella and fetal alcohol syndrome
Systemic: ears, face, spine, teeth, umbilicus, heart, kidneys	Preauricular tags, microtia (Goldenhar); missing teeth, periumbilical skin folds (Rieger syndrome); heart and ear defects with coloboma; Wilms tumour with sporadic aniridia
Corneal diameter, horizontal and vertical, both eyes	Normal adult 11.5–12 mm horizontally; microcornea less than 10 mm; megalocornea 13 mm or more. Record "right 9 mm, left 11.5 mm"
Eyeball size; axial length by A-scan	Microphthalmos: axial length 2 standard deviations below the mean for age (adult below about 21 mm); nanophthalmos below about 20.5 mm with thick sclera
Pupil: position, distance from centre, shape, reaction	"Pupil displaced superotemporally, 2 mm from the centre, round, reacting" = corectopia. Keyhole pupil pointing inferonasally = typical iris coloboma
Iris: collarette strands, hypoplasia, transillumination, posterior embryotoxon	Persistent pupillary membrane (strands from the collarette); Axenfeld–Rieger (strands to a prominent Schwalbe's line); aniridia stump
Lens: position, notch, cataract	Ectopia lentis et pupillae (lens and pupil displaced in opposite directions); lens notch in coloboma; cataract
IOP and gonioscopy	Narrow angle in a small eye; high iris insertion, iris processes, peripheral anterior synechiae (PAS)
Epibulbar mass: site, size, colour, consistency, mobility, hairs, lipid arc, posterior limit	Limbal dermoid: firm, white-yellow, dome-shaped, fine hairs, fixed. Lipodermoid: soft, yellow, mobile, posterior edge not seen
Cycloplegic refraction and keratometry	Astigmatism from a dermoid; hypermetropia with microcornea and nanophthalmos; anisometropia drives amblyopia
Dilated fundus	Disc or retinochoroidal coloboma; optic nerve hypoplasia; foveal hypoplasia in aniridia
Lids and ocular motility	Upper-lid coloboma and **Duane retraction syndrome** (Goldenhar)
Special tests	Ultrasound biomicroscopy (UBM) and anterior segment optical coherence tomography (AS-OCT) for depth of a dermoid; orbital magnetic resonance imaging (MRI) for a deep lipodermoid

Diagnosis and summary to say

> **Say it**
>
> Mr X, a 20-year-old man, has had poor vision in the right eye since early childhood, with no watering or photophobia in infancy, no injury or surgery, and no similar illness in the family. The right cornea measures 9 mm horizontally against 11.5 mm on the left; the right anterior chamber is of normal depth and the pupil is displaced superotemporally about 2 mm from the centre, round and reacting. The left eye has a keyhole-shaped full-thickness iris defect reaching the iris root inferonasally, with a flat notch in the lens edge in the same meridian. Intraocular pressure is 14 and 16 millimetres of mercury (mmHg) by applanation, gonioscopy shows open angles, and the discs are healthy; dilated fundus examination shows a retinochoroidal coloboma below the left disc. Right best-corrected vision is 6/24 with a hypermetropic correction, suggesting anisometropic amblyopia; systemic examination shows no ear, facial, dental or spinal anomaly.
>
> **Diagnosis (S.No 6):** Right eye microcornea with corectopia and amblyopia; left eye typical iris coloboma with retinochoroidal coloboma — bilateral congenital anomalies of the coloboma spectrum, with no glaucoma at present.
>
> **Diagnosis (S.No 49):** Right (or left) eye epibulbar lipodermoid in the superotemporal fornix with optic disc coloboma; Goldenhar syndrome to be excluded.

Viva questions and answers

Basics

Q Define microcornea. What are the normal corneal diameters?
Horizontal diameter below 10 mm in an adult (below 9 mm in a term newborn). Normal adult cornea is 11.5–12 mm horizontally and 10.5–11 mm vertically; the newborn cornea is about 9.5–10 mm and reaches adult size by about 2 years. Megalocornea is 13 mm or more in an adult.

Q How do microphthalmos, nanophthalmos and isolated microcornea differ?
Isolated microcornea: small cornea, normal axial length. Microphthalmos: the whole eye is small; simple, or complex (with coloboma, cataract, persistent fetal vasculature). Nanophthalmos: simple microphthalmos with axial length below about 20.5 mm, thick sclera, crowded anterior segment, hypermetropia of +8 to +20 dioptres (D), and angle closure and uveal effusion after surgery.

Q What are the associations of microcornea?
Ocular: hypermetropia; angle closure, and later open-angle glaucoma (about 20%); congenital cataract; uveal coloboma; persistent fetal vasculature; Axenfeld–Rieger, Peters anomaly; amblyopia. Systemic: rubella, fetal alcohol syndrome, CHARGE (coloboma, heart defects, choanal atresia, retarded growth, genital and ear anomalies; CHD7 gene), Nance–Horan, Weill–Marchesani, Ehlers–Danlos.

Q How do you tell megalocornea from buphthalmos?
Megalocornea: bilateral, clear, non-progressive cornea of 13 mm or more, X-linked recessive (CHRDL1) so mostly males, normal IOP and disc, deep chamber, iridodonesis, later lens subluxation and early cataract. Buphthalmos: raised IOP, photophobia, watering, corneal oedema, horizontal or concentric Haab's striae, an enlarging eye, cupping and axial myopia.

Q What is corectopia? Give its causes.
Displacement of the pupil from its normal position (normally very slightly inferonasal). Congenital: isolated, ectopia lentis et pupillae, Axenfeld–Rieger, iris coloboma, partial aniridia, Peters anomaly, persistent pupillary membrane. Acquired: trauma, surgery (iris or vitreous in the wound, peaked pupil), synechiae, adherent leucoma, ICE syndrome, iris tumour, iris ischaemia after angle closure.

Q What is ectopia lentis et pupillae?
Autosomal recessive (ADAMTSL4), bilateral; the pupil and lens are displaced in **opposite** directions. The pupil is small, oval and dilates poorly; a persistent pupillary membrane is seen in about 87%, with iridodonesis, transillumination defects, cataract and risk of retinal detachment; there are no systemic features.

Q Describe the Axenfeld–Rieger spectrum.
Autosomal dominant, bilateral anterior segment dysgenesis (PITX2 on 4q25, FOXC1 on 6p25). **Axenfeld anomaly**: posterior embryotoxon (prominent, anteriorly placed Schwalbe's line) with iris strands to it. **Rieger anomaly**: adds iris hypoplasia, corectopia, polycoria and ectropion uveae. **Rieger syndrome**: adds hypodontia, maxillary hypoplasia, redundant periumbilical skin, hypospadias. Glaucoma in about 50%.

Q How is ICE syndrome different from Axenfeld–Rieger?
ICE is acquired and unilateral, in women aged 20–50: abnormal endothelium (ICE cells on specular microscopy) grows over angle and iris, giving PAS beyond Schwalbe's line that pull the pupil towards them; variants are progressive iris atrophy, Chandler and Cogan–Reese. Axenfeld–Rieger is congenital, bilateral and familial with normal endothelium.

Examination and signs

Q How do you measure the corneal diameter at the bedside?
Callipers or a clear ruler against the cornea, or the slit-beam length on the slit lamp, limbus to limbus horizontally and vertically in both eyes; in infants, callipers during examination under anaesthesia (EUA).

Q What is a persistent pupillary membrane, and how do you tell it from posterior synechiae?
A remnant of the anterior tunica vasculosa lentis, which normally regresses in the last trimester. Its strands arise from the **collarette** and cross the pupil to the opposite collarette, the lens (pigment stars on the capsule) or the cornea; synechiae arise from the **pupil margin** after inflammation. Most need nothing; a dense membrane may need mydriatics, laser or excision to prevent amblyopia.

Q Describe aniridia.
Bilateral near-total absence of iris (a stump on gonioscopy); PAX6 on 11p13; two-thirds autosomal dominant. Nystagmus, foveal and optic nerve hypoplasia, aniridic keratopathy (limbal stem cell deficiency), glaucoma (50–75%), cataract, ectopia lentis. Sporadic cases may have WAGR (Wilms tumour, aniridia, genitourinary anomalies, retardation): genetic testing and 3-monthly renal ultrasound in early childhood.

Q What is Peters anomaly?
A central corneal leucoma over a defect in posterior stroma, Descemet's membrane and endothelium. Type 1: iris strands from the collarette to the edge of the opacity, clear lens; type 2: keratolenticular adhesion or cataract, more severe. About 80% bilateral; glaucoma in 50–70%. Peters plus: autosomal recessive, short stature, cleft lip or palate, developmental delay.

Q How do you tell a limbal dermoid from a lipodermoid?
Limbal dermoid: inferotemporal limbus, half on cornea and half on sclera, firm, white-yellow, dome-shaped, keratinised, fine hairs, fixed, sometimes with a lipid arc in the cornea ahead of it. Lipodermoid: superotemporal fornix between the superior and lateral recti, soft, yellow, mobile with the conjunctiva, its posterior edge merging with orbital fat and never seen.

Q How do you tell a lipodermoid from prolapsed orbital fat?
Prolapsed orbital fat appears in the elderly, is often bilateral, yellow, smooth, without hairs, and can be pushed back into the orbit. A lipodermoid is congenital, may bear fine hairs, has a whitish keratinised surface and cannot be reposited.

Q What is Goldenhar syndrome?
Oculo-auriculo-vertebral spectrum (first and second branchial arch), mostly sporadic. Triad: epibulbar dermoid or lipodermoid (bilateral in about 25%), preauricular appendages, pretragal fistulae; also upper-lid coloboma, Duane syndrome, microtia, deafness, hemifacial microsomia, vertebral, cardiac and renal defects. Differential: Treacher Collins (bilateral, lower-lid coloboma, no dermoid).

Q Why does a limbal dermoid reduce vision?
It flattens the cornea in the meridian of the lesion, giving large, often oblique astigmatism, with irregular astigmatism from its edge and lipid; it may encroach on the visual axis; the result is anisometropic amblyopia, sometimes with strabismus.

Investigations

Q What investigations would you do for S.No 6?
Cycloplegic refraction, keratometry and A-scan axial length; applanation IOP, central corneal thickness and gonioscopy; dilated fundus with optical coherence tomography (OCT) of disc and macula; fields if glaucoma is suspected; UBM of angle and ciliary body; genetic testing if syndromic.

Q Which investigations help before excising a dermoid?
UBM and AS-OCT to show the depth of corneal and scleral involvement and whether Descemet's membrane is involved; MRI of the orbit for posterior extension of a lipodermoid into fat, muscle or lacrimal gland. For Goldenhar: spine X-ray, audiometry, echocardiography, renal ultrasound. Biopsy is rarely needed.

Q What is the histology of a dermoid and a lipodermoid?
Both are choristomas (normal tissue at an abnormal site). Dermoid: keratinised stratified squamous epithelium over dense collagen with hair follicles, sebaceous and sweat glands, sometimes cartilage, bone or lacrimal tissue (complex choristoma). Lipodermoid: mostly adipose tissue with few adnexal structures.

Q How are corneal dermoids graded (Mann)?
Grade I: superficial limbal (epibulbar) dermoid, the commonest. Grade II: larger, covering part or all of the central cornea, extending into stroma but sparing Descemet's membrane and endothelium. Grade III: whole anterior segment involved, often with microphthalmos.

Management

Q How would you manage S.No 6?
Full cycloplegic refraction and glasses; occlusion or atropine penalisation if still in the amblyopic age; tinted or painted-iris contact lens for glare from the coloboma; protective glasses; lifelong IOP and disc checks; watch for cataract; regular fundus review for retinal detachment (card ER5); genetic counselling.

Q When and how would you excise a limbal dermoid?
Indications: visual axis involvement, amblyopia not responding to glasses and occlusion, dellen, irritation, rapid growth, cosmesis. Lamellar sclerokeratectomy (shave excision) from the corneal side in a clean plane, not chasing full depth; cover with a lamellar patch graft or amniotic membrane; mitomycin C reduces recurrence and pseudopterygium; send for histopathology.

Q Will excision cure the astigmatism?
Usually not: the astigmatism often persists or even increases, so surgery is mainly for cosmesis, comfort or a lesion entering the visual axis, and amblyopia treatment must continue afterwards. Tell the parents this before surgery.

Q How do you manage a lipodermoid?
Usually leave it alone. If it is cosmetically troublesome, excise only the visible anterior part under the conjunctiva and preserve the conjunctiva. Deep dissection risks damage to the lacrimal ductules (dry eye), lateral rectus and levator (restriction, ptosis), symblepharon and orbital fat prolapse.

Q What are the complications of dermoid surgery?
Corneal scarring and vascularisation, persistent epithelial defect, pseudopterygium, perforation with deep lesions, recurrence, limbal stem cell deficiency, increased astigmatism.

Q How is glaucoma managed in anterior segment dysgenesis?
Medical first: beta-blockers, carbonic anhydrase inhibitors, prostaglandin analogues (no brimonidine under 2 years). Angle surgery works less well than in primary congenital glaucoma, so trabeculectomy with mitomycin C or a drainage device is often needed; cyclophotocoagulation for refractory eyes.

Complications and prognosis

Q What must be watched for in an eye with microcornea?
Angle-closure and later open-angle glaucoma, cataract, amblyopia, and retinal detachment if coloboma coexists. Cataract surgery is harder: little working space, shallow chamber, zonular weakness, high and less predictable lens power, and uveal effusion in nanophthalmos.

Q What is the visual prognosis?
Isolated microcornea or corectopia with corrected refraction and treated amblyopia gives useful vision; macular or disc coloboma, foveal hypoplasia, nystagmus or glaucoma limit it. Limbal dermoids do well if amblyopia is treated early; grade III lesions do poorly.

Recent advances and favourite examiner questions

Q Which genes are linked with these anomalies?
PAX6 (aniridia, some Peters anomaly and coloboma); PITX2 and FOXC1 (Axenfeld–Rieger, some Peters); CYP1B1 (primary congenital glaucoma, some Peters); ADAMTSL4 (ectopia lentis et pupillae); SOX2 and OTX2 (severe microphthalmos); MFRP (nanophthalmos); CHRDL1 (X-linked megalocornea); CHD7 (CHARGE).

Q Give the key points of the left iris coloboma in S.No 6 (details on card ER5).
Inferonasal keyhole pupil, from failure of closure of the embryonic fissure; complete (full thickness, to the root) or partial (margin only); incomplete forms show only on transillumination. Look for a lens notch, ciliary body, retinochoroidal and disc coloboma, and CHARGE. Tinted or iris-printed contact lens for glare; screen for retinal detachment (23–40% with retinochoroidal coloboma).

Q What are the causes of congenital corneal opacity?
STUMPED: Sclerocornea; Tears in Descemet's membrane (congenital glaucoma, forceps); Ulcer (herpes, bacterial, neurotrophic); Metabolic (mucopolysaccharidoses); Peters anomaly (posterior corneal defect); Endothelial dystrophy (congenital hereditary endothelial dystrophy, posterior polymorphous dystrophy); Dermoid.

Q Where does microcornea sit in Hoskins' classification of developmental glaucoma?
Under corneotrabeculodysgenesis, with the other corneal defects: peripheral (Axenfeld), mid-peripheral (Rieger), central (Peters) and abnormal corneal size (microcornea, megalocornea). The other groups are isolated trabeculodysgenesis and iridotrabeculodysgenesis (for example aniridia).

Examiner traps

- Saying "microcornea" without a number: give the horizontal diameter of **both** corneas in millimetres.
- Skipping IOP and gonioscopy: small eyes and dysgenesis carry a glaucoma risk.
- Missing the fundus: microcornea, corectopia or iris coloboma oblige you to look for retinochoroidal and disc coloboma.
- Promising that dermoid excision will correct the astigmatism.
- Advising complete removal of a lipodermoid: it risks ptosis, dry eye and restriction.
- Forgetting the systemic examination: ears and spine (Goldenhar), teeth and umbilicus (Rieger), kidneys (sporadic aniridia).
- Calling an acquired unilateral corectopia in an adult "congenital": think of ICE syndrome, trauma or surgery.

- Microcornea: under 10 mm in an adult; megalocornea: 13 mm or more.
- Dermoid: inferotemporal limbus, firm, hairs. Lipodermoid: superotemporal fornix, soft, yellow, posterior edge unseen.
- Goldenhar triad: epibulbar dermoid, preauricular tags, pretragal fistula.
- Axenfeld (embryotoxon with strands), then Rieger (plus iris), then Rieger syndrome (plus teeth, face, umbilicus); glaucoma in half.
- Ectopia lentis et pupillae: pupil and lens go opposite ways.
- Sporadic aniridia: look for Wilms tumour (WAGR, 11p13).
- Mnemonic for congenital corneal opacity: **STUMPED**.

Read more: Baidya p.1–4, 55, 59–63, 125, 172, 340–342 · Namrata p.157–169, 191, 319–322 · Aravind p.115, 122

EO3 Aphakia and zonular (lamellar) cataract

Other · On the list: S.No 29, 48 · ★ Patient already arrived: S.No 29

At a glance

- **Aphakia**: absence of the crystalline lens from the pupillary area — surgical (commonest), traumatic, posterior dislocation (functional aphakia) or, rarely, congenital. Optically the eye is **highly hypermetropic (about +10 dioptres (D) at the spectacle plane)** and cannot accommodate.
- **Zonular (lamellar) cataract**: the commonest developmental cataract; a shell of opacity around a clear centre, with clear cortex outside and spoke-like **riders**; usually bilateral and autosomal dominant. Being partial, it often allows fair vision.
- **S.No 29** ★: "Right eye aphakia, presumably after lens aspiration in childhood for developmental cataract (confirm), with amblyopia; left eye zonular (lamellar) cataract — bilateral developmental cataract." Look for squint, nystagmus and aphakic glaucoma.
- **S.No 48**: "Right eye pseudophakia (posterior chamber intraocular lens, PCIOL) with a superior trabeculectomy bleb and surgical peripheral iridectomy; left eye aphakia; both eyes advanced glaucomatous cupping (cup–disc ratio 0.9)." The bleb is covered on card G2.
- Examiners ask: signs of aphakia, problems with aphakic glasses, correction with and without capsular support, the morphology and causes of lamellar cataract, timing of surgery, and the intraocular lens (IOL) power and posterior capsule in children.

What to look for and present

Look for	What to say or record
How, when and why the lens was removed; age at surgery	Childhood lens aspiration (S.No 29); adult cataract surgery with capsule rupture or nucleus drop; old intracapsular extraction; trauma
Glasses or contact lens worn; diplopia; "things look big and close"	Aphakic glasses in one eye only cause aniseikonia and diplopia; poor compliance means amblyopia
Cataract eye: since when, school work, glare	Lamellar cataract is often noticed at school age and progresses slowly
Family history; neonatal fits, tetany, rickets, prematurity; maternal rash	Autosomal dominant inheritance; hypocalcaemia, vitamin D deficiency; rubella
Glaucoma: drops, laser, surgery (S.No 48)	Bleb and drugs; aphakic glaucoma after childhood surgery
Vision unaided, with pinhole and with +10 D; near	An aphakic eye sees about 6/60 unaided and improves with +10 to +12 D
Signs of aphakia	**Deep anterior chamber, iridodonesis, jet-black pupil, Purkinje images 3 and 4 absent**, surgical scar, peripheral iridectomy, vitreous face
Capsule and vitreous	Posterior capsule or anterior rim, Soemmering's ring, Elschnig pearls, vitreous in the chamber or wound (peaked pupil) — decides the secondary IOL
Retinoscopy and refraction	About +10 D in a previously emmetropic adult, with +3 D more for near
Lamellar cataract: undilated and dilated, distant direct ophthalmoscopy, slit lamp, retroillumination	Disc-shaped opacity in one lamella, clear centre, clear cortex, riders at its edge; record size in mm, density and the clear zone
Intraocular pressure (IOP) and gonioscopy	Aphakic glaucoma; bleb function and angle in S.No 48
Cornea	Microcornea; vitreous touch, aphakic bullous keratopathy
Alignment, nystagmus, amblyopia	Sensory squint in unilateral aphakia; nystagmus means early dense bilateral cataract and a poor outlook
Fundus with +90 D and indirect ophthalmoscope	Disc cupping (0.9 in S.No 48), retinal detachment, cystoid macular oedema
Systemic	Rickets, dental enamel defects, Chvostek and Trousseau signs, dysmorphism
Before a secondary IOL	Keratometry, axial length, chamber depth, endothelial count, capsular support

Diagnosis and summary to say

Say it

Mr X, a 16-year-old boy, had surgery on his right eye at the age of 3 for a "white spot" and wears a +11 D glass for that eye only; his left eye vision has been slowly falling for school work. The right anterior chamber is deep, the iris is tremulous, the pupil is black with a thin posterior capsular remnant, and only two Purkinje images are seen; best-corrected vision is 6/36 with +11.00 D sphere, suggesting amblyopia. The left lens has a 5 mm disc-shaped opacity in the fetal nucleus with a clear centre, clear cortex and spoke-like riders at its edge; vision is 6/18 undilated, improving to 6/12 after dilatation. Intraocular pressure is 16 and 14 mmHg (millimetres of mercury) with open angles, the discs are healthy, and there is a small right exotropia without nystagmus. There are no signs of rickets or hypocalcaemia.

Diagnosis (S.No 29): Right eye surgical aphakia after childhood lens aspiration, with amblyopia and sensory exotropia; left eye zonular (lamellar) cataract — bilateral developmental cataract.

Diagnosis (S.No 48): Right eye pseudophakia with a superior trabeculectomy bleb and surgical peripheral iridectomy; left eye aphakia; both eyes advanced glaucoma (cup–disc ratio 0.9).

Viva questions and answers

Basics

Q Define aphakia and give its causes.
Absence of the crystalline lens from the pupillary area. Surgical: cataract extraction without IOL, lensectomy in children, nucleus drop. Traumatic: extrusion through a wound, absorption after capsule rupture in a child. Posterior dislocation into the vitreous (functional aphakia). Congenital: primary (lens never formed) or secondary (absorbed in utero, for example rubella).

Q What is the optical state of an aphakic eye?
High hypermetropia, because the lens supplies about +16 to +19 D of the eye's power; at the spectacle plane a previously emmetropic eye needs about +10 to +11 D, plus about +3 D for near. Accommodation is lost.

Q What is a zonular (lamellar) cataract?
An opacity confined to one lamella of the lens, usually in the fetal nucleus around a clearer embryonic nucleus, with clear cortex outside, and short spoke-like opacities (riders) straddling its equator. It is the commonest visually significant developmental cataract, usually bilateral and symmetrical.

Q Why does it form a shell, and what are riders?
An insult acting for a short period of lens growth makes only the fibres laid down then opaque; later fibres are clear and bury the opacity. Its diameter dates the insult (larger means later). Riders are small arcuate cortical opacities at the edge of the zone, like riders astride a horse.

Q What causes lamellar cataract?
Autosomal dominant heredity (commonest; crystallin, connexin and HSF4 genes); hypocalcaemia (neonatal tetany, hypoparathyroidism); vitamin D deficiency (rickets); neonatal hypoglycaemia; malnutrition in late fetal or early infant life; maternal rubella. Look for dental enamel defects formed at the same time.

Examination and signs

Q What are the signs of aphakia?
Deep anterior chamber; iridodonesis (tremulous iris on eye movement); jet-black pupil; only Purkinje images 1 and 2; surgical scar and peripheral iridectomy; vitreous face in the pupil or vitreous in the chamber; capsular remnants; retinoscopy showing high hypermetropia, with vision improving on +10 D.

Q Describe the Purkinje–Sanson images.
Image 1: anterior cornea, bright, erect, virtual. Image 2: posterior cornea, faint, erect. Image 3: anterior lens surface, largest, erect, moves with the light. Image 4: posterior lens surface, small, inverted, real, moves against the light. Aphakia shows only 1 and 2; in pseudophakia the IOL gives 3 and 4.

Q How do you tell surgical aphakia from a lens dislocated into the vitreous?
Both give a deep chamber, iridodonesis and a black pupil, but a dislocated lens is seen in the vitreous on indirect ophthalmoscopy or B-scan, there is usually no surgical scar, and it may cause uveitis or glaucoma (card EO4).

Q How do you examine a lamellar cataract?
Distant direct ophthalmoscopy: a dark disc with a red glow around it. Slit lamp with dilated pupil: the optical section shows an opaque lamella with clear zones inside and outside; retroillumination shows the riders. Record size, density and vision undilated and dilated.

Q When is a lamellar cataract visually significant?
A central opacity of 3 mm or more, dense, vision worse than about 6/18 or falling school performance, no clear fundus view through it, a poor retinoscopic reflex, squint or nystagmus. Good vision through a dilated pupil means it can wait.

Q What are the problems of aphakic spectacles?
Magnification of about 25–30% (objects look nearer, misjudged distances); ring scotoma from the prism at the lens edge, causing the "jack-in-the-box" phenomenon; pincushion distortion; spherical and chromatic aberration; restricted field; heavy, thick lenses that magnify the eyes; sensitivity to vertex distance; no accommodation.

Q What is the jack-in-the-box phenomenon?
The peripheral prism of a strong plus lens creates a ring-shaped blind zone between the lens field and the peripheral field. It moves with eye movement (roving ring scotoma), so objects suddenly pop out of it or vanish into it.

Q Why are spectacles unsuitable for unilateral aphakia?
With a normal fellow eye, the 25–30% image size difference (aniseikonia) cannot be fused (tolerance is about 5%), causing diplopia. A contact lens (about 7–10%) or an IOL (about 2–3%) is used instead.

Investigations

Q What would you do before a secondary IOL?
Keratometry and axial length; chamber depth; specular microscopy; gonioscopy (before an anterior chamber IOL, ACIOL); dilated check of capsular support and vitreous; B-scan if media are hazy; macular optical coherence tomography (OCT); peripheral retina for breaks.

Q How do you investigate bilateral developmental cataract?
A familial autosomal dominant lamellar cataract needs nothing more. Otherwise: TORCH serology (toxoplasma, rubella, cytomegalovirus, herpes) and syphilis serology; urine for reducing substances (galactosaemia) and amino acids (Lowe syndrome); serum calcium, phosphorus and glucose; paediatric and genetic review. Ocular: age-appropriate acuity, B-scan, biometry under anaesthesia, visual evoked potentials.

Management

Q What are the options for correcting aphakia?
Spectacles (bilateral aphakia, elderly); contact lenses (rigid gas-permeable, silicone hydrogel, extended-wear silicone in infants); secondary IOL — sulcus three-piece IOL if there is capsular support, otherwise an ACIOL, iris-claw IOL or scleral-fixated IOL (sutured, or sutureless intrascleral: glued IOL, Yamane flanged technique).

Q How do you choose a secondary IOL?
Capsule or anterior rim present: three-piece IOL in the sulcus, with optic capture if possible. No support: healthy angle, deep chamber, good endothelium and no glaucoma, an open-loop ACIOL; healthy iris, a retropupillary iris-claw IOL; glaucoma, synechiae, a child or damaged iris, a scleral-fixated or glued IOL. Clear the anterior vitreous first.

Q Which option for the left eye of S.No 48?
Advanced glaucoma with a bleb in the other eye: avoid an ACIOL (angle damage, IOP rise, endothelial loss). Use a sulcus IOL if a capsular rim exists, otherwise a scleral-fixated or retropupillary iris-claw IOL through a temporal approach, sparing the superior conjunctiva for filtering surgery; control IOP first. With 0.9 cups the visual gain may be small; a contact lens is an alternative. Glasses fail because of aniseikonia with the pseudophakic right eye.

Q Describe the glued IOL and the Yamane technique.
Glued IOL (Agarwal): two partial-thickness scleral flaps 180° apart about 1.5 mm behind the limbus, sclerotomies beneath, haptics of a three-piece IOL externalised and tucked into scleral tunnels, flaps sealed with fibrin glue, with vitrectomy. Yamane: haptics drawn out through 30-gauge needles and their ends melted into flanges, without flaps or sutures.

Q What are the complications of ACIOL and iris-claw lenses?
ACIOL: corneal decompensation, uveitis–glaucoma–hyphaema syndrome, peripheral anterior synechiae, cystoid macular oedema, pupil block. Iris-claw: oval pupil, iris chafing, uveitis, hyphaema, disenclavation and dislocation. Scleral-fixated: suture erosion or breakage, tilt, vitreous haemorrhage, retinal detachment, endophthalmitis.

Q How do you manage a lamellar cataract?
Partial, with good dilated vision: observe; a mydriatic (phenylephrine 2.5%) may help; correct refraction and treat amblyopia. Visually significant: lens aspiration with IOL. In bilateral cases operate the denser eye first and the second within about a week.

Q What is the timing of surgery for a dense congenital cataract?
Unilateral: before 6 weeks of age; bilateral: before 8 weeks, before deprivation amblyopia becomes fixed. Operating in the first 4 weeks carries a higher glaucoma risk. Lamellar cataracts usually appear later and are partial, so their timing is decided by visual function.

Q What are the steps of paediatric cataract surgery?
Small incisions, sutured because of low scleral rigidity; high-viscosity viscoelastic; anterior continuous curvilinear capsulorhexis (CCC) or vitrectorhexis (elastic capsule); aspiration of the soft lens; primary posterior capsulorhexis with anterior vitrectomy in young children; in-the-bag single-piece hydrophobic acrylic IOL; intensive topical steroid and cycloplegic afterwards.

Q When do you do a primary posterior capsulotomy and anterior vitrectomy?
Visual axis opacification is almost certain in young children if the capsule is left. Under 5 years: posterior capsulorhexis with anterior vitrectomy; 5–8 years: posterior capsulorhexis with or without vitrectomy; over 8 years: keep the capsule and use neodymium-doped yttrium aluminium garnet (Nd:YAG) laser capsulotomy later. The anterior hyaloid otherwise acts as a scaffold for lens epithelial cells.

Q How do you choose IOL power in a child?
Keratometry and immersion A-scan under anaesthesia; SRK/T, Holladay 1 or Hoffer Q for short eyes; implant if axial length is over 17 mm and corneal diameter over 10 mm. Under-correct for the myopic shift: 20% under 2 years, 15% at 2–4, 10% at 4–8, emmetropia after 8. Rough guide (rule of 7): age in years plus target hypermetropia is about 7.

Q What did the Infant Aphakia Treatment Study (IATS) show?
In infants aged 1–6 months with unilateral cataract, a primary IOL gave no better vision at 4.5 years than contact lens aphakia, with more adverse events and reoperations (mostly visual axis opacification). So contact lens correction is reasonable in early infancy, with an IOL later.

Q How do you correct aphakia in an infant without an IOL?
An extended-wear silicone or rigid gas-permeable contact lens, over-corrected by about +2 to +3 D because the infant's world is near; spectacles in bilateral aphakia; secondary IOL later (often after 2 years); patching of the better eye for amblyopia.

Complications and prognosis

Q What is aphakic glaucoma, and how is it managed?
Glaucoma after lens removal, commonest after childhood surgery (3–32%). Early: pupil block by vitreous, air or membrane (iridectomy, membranectomy). Late: open-angle, years later, so check IOP, corneal diameter, axial length and disc for life. Treat with beta-blockers, carbonic anhydrase inhibitors, prostaglandin analogues (no brimonidine under 2 years); then a drainage device; cyclophotocoagulation if refractory.

Q What are the risk factors for glaucoma after paediatric cataract surgery?
Surgery in the first month of life, microcornea or microphthalmos, persistent fetal vasculature, retained lens matter, repeated surgery and postoperative inflammation.

Q What other complications follow aphakia or paediatric cataract surgery?
Amblyopia, visual axis opacification, inflammatory membranes, IOL decentration, retinal detachment (about 1–1.5%, more after intracapsular surgery), cystoid macular oedema, endophthalmitis, aphakic bullous keratopathy from vitreous touch, strabismus.

Q What is the visual prognosis?
Lamellar cataract does well, being partial and later in onset; dense early bilateral cataract with nystagmus, unilateral cataract with poor occlusion, microphthalmos and other anomalies do worse. Most failures are from amblyopia, not surgery.

Recent advances and favourite examiner questions

Q What is the bag-in-the-lens IOL?
An IOL with a peripheral groove into which matched anterior and posterior capsulorhexis edges are captured, sealing the bag and keeping the visual axis clear without vitrectomy; used in children.

Q Why is optic capture used in children?
Pushing the optic through the posterior capsulorhexis behind the capsule keeps the IOL centred and the visual axis clear; in older children it can replace an anterior vitrectomy.

Examiner traps

- Saying "aphakia" without the cause and without looking for capsular remnants: the capsule decides the secondary IOL.
- Forgetting IOP and the disc: aphakic glaucoma after childhood surgery is silent, and S.No 48 has 0.9 cups.
- Prescribing glasses for unilateral aphakia: aniseikonia makes them unwearable.
- Calling a lamellar cataract "nuclear": the lamellar one has a clearer centre, clear cortex and riders.
- Operating on every lamellar cataract: a partial one with good dilated vision can be observed.
- Offering an anterior chamber IOL to a glaucomatous eye or a child.
- Forgetting amblyopia treatment after childhood surgery.

- Aphakia: deep chamber, tremulous iris, black pupil, two Purkinje images, about +10 D.
- Aphakic glasses: 25–30% magnification, ring scotoma, jack-in-the-box, pincushion; never for unilateral aphakia.
- Lamellar cataract: opaque shell, clearer centre, riders; heredity, low calcium, low vitamin D.
- Dense congenital cataract: operate before 6 weeks (unilateral) or 8 weeks (bilateral).
- Under-correct the IOL: 20% under 2 years, 15% to 4, 10% to 8, emmetropia after 8.
- Posterior capsulorhexis with anterior vitrectomy under 5 years.
- Mnemonic for aphakia signs, **DIPPS**: Deep chamber, Iridodonesis, Pupil jet-black, Purkinje 3 and 4 absent, Scar and iridectomy.

Read more: Baidya p.119–121, 137–142 · Namrata p.345, 409–414 · Aravind p.309, 399

EO4 Ectopia lentis — subluxated and dislocated lens

Other · On the list: S.No 39 · ★ Patient already arrived: S.No 39

At a glance

- **Ectopia lentis**: displacement of the crystalline lens from its normal place behind the pupil (the patellar fossa). **Subluxation**: partly displaced, some zonules intact, lens still in the pupil area. **Dislocation (luxation)**: all zonules gone, lens wholly out of the fossa — in the anterior chamber, in the vitreous (floating or on the retina), under the conjunctiva (phacocele) or out of the eye.
- **Trauma** causes about half; **Marfan syndrome** is the commonest heritable cause. Bilateral, symmetrical, non-traumatic ectopia lentis means a systemic cause until proved otherwise.
- Kept because the signs are vivid and the viva covers systemic disease, glaucoma mechanisms and the choice of surgery.
- **S.No 39** ★: "Left eye posterior dislocation of the crystalline lens into the vitreous cavity (clear or cataractous, mobile or fixed), with functional aphakia; cause — traumatic, or the systemic cause found; right eye normal (or subluxated, with direction)."
- If the lens lies in front of the iris: "Left eye anterior dislocation of the lens into the anterior chamber with pupil-block glaucoma" — an emergency.

What to look for and present

Look for	What to say or record
Onset: sudden after injury or gradual; one or both eyes	Trauma is one-sided; heritable ectopia lentis is bilateral and symmetrical
Fluctuating vision, uniocular diplopia, frequent change of glasses, glare	Looking through the phakic and aphakic parts; lenticular myopia and astigmatism
Pain, redness, haloes, vomiting	Pupil-block or lens-induced glaucoma; lens-induced uveitis
Previous eye surgery; pseudoexfoliation; high myopia	Dropped nucleus; zonular weakness
Heart, joints, intellect, fits, thrombosis, stroke; family deaths; consanguinity	Marfan (aortic root dilatation, mitral valve prolapse); homocystinuria (learning difficulty, fits, thrombosis); sulphite oxidase deficiency (fits); recessive disease
General examination	Marfan: tall, arm span to height over 1.05, long fingers, thumb and wrist signs, pectus, high arched palate, scoliosis. Homocystinuria: tall, fair hair, malar flush, stiff joints. Weill–Marchesani: short, stubby fingers
Vision and refraction through the phakic and aphakic zones	Myopic astigmatism through the lens; about +10 dioptres (D) through the aphakic part
Anterior chamber depth	Deep in posterior dislocation; shallow in anterior dislocation or forward subluxation; uneven in subluxation
Iris and pupil	**Iridodonesis**; displaced pupil (ectopia lentis et pupillae); poor dilation in Marfan (hypoplastic dilator)
Lens: direction, clock hours of zonular loss, visible edge, zonules, **phacodonesis**, cataract, size	"Lens equator seen superotemporally in the undilated pupil, 4 clock hours of loss, stretched zonules, clear lens." Small round lens: microspherophakia
Vitreous in the anterior chamber	Changes the surgery (anterior vitrectomy)
Intraocular pressure (IOP) and gonioscopy	Pupil block; angle recession after trauma; open angle with high IOP in phacolytic glaucoma
Fundus	Lens in the vitreous, mobile or resting inferiorly on the retina; lattice, breaks, retinal detachment (RD)
Distant direct ophthalmoscopy	Dark crescent of the lens edge across the red glow: subluxation
Fellow eye	Subluxated in heritable disease; normal after trauma
Special tests	B-scan if the media are hazy; ultrasound biomicroscopy (UBM) of zonules; urine nitroprusside; echocardiography; plasma homocysteine

Diagnosis and summary to say

> **Say it**
>
> Mr X, a 24-year-old man, has had painless blurring of the left eye for 6 months; he denies injury and there is no family history of heart disease or sudden death. He is tall with long fingers and positive thumb and wrist signs. The left anterior chamber is deep, the iris tremulous and the pupil black; indirect ophthalmoscopy shows a clear, mobile lens in the inferior vitreous, with an attached retina and lattice degeneration inferotemporally. Corrected vision is 6/9 with +10.50 D sphere, IOP is 16 mmHg (millimetres of mercury) with an open angle, and the eye is quiet. The right lens is subluxated superotemporally, with stretched zonules visible at its edge.
>
> **Diagnosis (S.No 39):** Left eye posterior dislocation of a clear crystalline lens into the vitreous with functional aphakia, and right eye superotemporal subluxation — bilateral ectopia lentis, probably Marfan syndrome; no glaucoma or uveitis at present.
>
> **If traumatic:** Left eye traumatic posterior dislocation of the lens into the vitreous, with (angle recession or other injuries), right eye normal.

Viva questions and answers

Basics

Q Define ectopia lentis, subluxation and dislocation.
Ectopia lentis: displacement of the lens from its normal position. Subluxation: partial displacement with some zonules intact, the lens still within the lens space. Dislocation (luxation): complete separation, the lens lying outside the patellar fossa — anterior chamber, vitreous, retinal surface, under the conjunctiva or extruded.

Q Classify the causes of ectopia lentis.
Trauma (commonest). Heritable without systemic disease: familial ectopia lentis, ectopia lentis et pupillae, aniridia. With systemic disease: Marfan, homocystinuria, Weill–Marchesani, sulphite oxidase deficiency, hyperlysinaemia, Ehlers–Danlos. Eye disease: pseudoexfoliation, high myopia, buphthalmos, megalocornea, hypermature cataract, uveitis, tumour, syphilis.

Q Compare Marfan syndrome and homocystinuria.
Marfan: autosomal dominant, FBN1 (fibrillin-1) on 15q21; superotemporal subluxation; zonules stretched but present; normal intelligence; aortic root dilatation, mitral valve prolapse; nitroprusside negative. Homocystinuria: autosomal recessive, cystathionine beta-synthase deficiency; inferonasal subluxation; zonules broken; learning difficulty, fits, fair hair, osteoporosis, thromboembolism; nitroprusside positive.

Q How often does ectopia lentis occur in these conditions?
About 50–80% of people with Marfan syndrome, bilateral, usually stable after childhood. In untreated homocystinuria about 90% by the mid-teens, and often progressive. Microspherophakia is almost constant in Weill–Marchesani syndrome.

Q What is Weill–Marchesani syndrome, and what is inverse glaucoma?
Short stature, brachydactyly, stiff joints; autosomal recessive (ADAMTS10) or dominant (FBN1); a small spherical lens with high lenticular myopia that moves forward and causes pupil block. Miotics worsen this (inverse glaucoma) by relaxing the zonules; cycloplegics tighten them and pull the lens back. Treat with laser iridotomy, then lens removal.

Q Why does a subluxated lens cause myopia and astigmatism?
Loss of zonular tension lets the lens round up and gain power (lenticular myopia); uneven loss tilts it and curves it unequally, giving lenticular astigmatism, often irregular. When the edge crosses the pupil there is a phakic and an aphakic zone, so the patient has uniocular diplopia.

Examination and signs

Q What are the signs of a subluxated lens?
Uneven anterior chamber depth; iridodonesis and phacodonesis; the lens edge visible in the pupil with stretched or broken zonules; a dark crescent on the red reflex; vitreous in the anterior chamber; lens tilt; high myopic astigmatism.

Q What are iridodonesis and phacodonesis?
Iridodonesis is quivering of the iris on eye movement, from loss of lens support (subluxation, dislocation, aphakia). Phacodonesis is quivering of the lens itself, seen at the slit lamp with the pupil dilated, from zonular weakness.

Q What does a lens dislocated into the anterior chamber look like?
A clear lens looks like a large oil droplet with a golden rim in front of the iris; a cataractous lens looks like a white disc. It causes pupil block with high IOP, pain, and corneal oedema from endothelial touch.

Q What are the signs of a lens dislocated into the vitreous?
Deep anterior chamber, iridodonesis, black pupil, aphakic refraction (+10 to +12 D), Purkinje images 3 and 4 absent, no surgical scar, and the lens seen in the vitreous — mobile in a liquid vitreous or lying on the inferior retina.

Q What is the usual direction of displacement?
Marfan: superotemporal; homocystinuria: inferonasal; Weill–Marchesani: forward into the chamber or downward; trauma: any direction, away from the torn zonules; ectopia lentis et pupillae: opposite to the pupil. Direction suggests the cause but does not prove it.

Investigations

Q Which systemic investigations would you order?
Urine cyanide–nitroprusside test (purple-red colour with homocystine), confirmed by plasma total homocysteine and amino acids (high methionine); echocardiography (aortic root size, mitral valve prolapse) and electrocardiogram (ECG); FBN1 genetic testing; urine sulphite and S-sulphocysteine for sulphite oxidase deficiency; syphilis serology in adults; examination of relatives.

Q What are the revised Ghent criteria?
Without family history, Marfan is diagnosed by: aortic root dilatation (Z-score 2 or more) with ectopia lentis; or aortic dilatation with an FBN1 mutation; or aortic dilatation with a systemic score of 7 or more; or ectopia lentis with an FBN1 mutation linked to aortic disease. With a family history: ectopia lentis, or a systemic score of 7 or more, or aortic dilatation.

Q What is in the systemic score?
Out of 20: wrist and thumb signs 3 (one alone 1); pectus carinatum 2; hind-foot deformity 2; pneumothorax 2; dural ectasia 2; protrusio acetabuli 2; reduced upper-to-lower segment ratio with increased arm span 1; scoliosis 1; myopia over 3 D 1; mitral valve prolapse 1; facial features, skin striae and reduced elbow extension 1 each.

Q Which eye investigations help?
B-scan (lens position, retinal detachment) when the media are hazy — the lens appears as an oval echogenic body; UBM for zonules and ciliary body; biometry, remembering readings may be unreliable; specular microscopy if the lens touched the cornea; macular optical coherence tomography.

Management

Q How is a subluxated lens managed without surgery?
Careful refraction through both zones; when the edge crosses the pupil, correcting the aphakic part (about +10 to +12 D) often gives the best vision; contact lens; occlusion for amblyopia in children; IOP control; regular retinal review.

Q What are the indications for surgery?
Lens in the anterior chamber; pupil-block or lens-induced glaucoma; lens-induced uveitis; cataract; uncorrectable vision or amblyopia not responding to treatment; lens touching the cornea; RD needing surgery.

Q How do you manage a lens in the anterior chamber?
As an emergency: lower IOP (intravenous mannitol, oral acetazolamide); constrict the pupil with pilocarpine to trap the lens in front of the iris; remove it whole through a limbal incision (intracapsular extraction with cryoprobe or vectis), with anterior vitrectomy and peripheral iridectomy; place a secondary intraocular lens (IOL) later.

Q How do you manage a lens dislocated into the vitreous?
A clear lens in a quiet eye with normal IOP may be watched, with aphakic correction. Operate for inflammation, glaucoma, cataract, RD, or to fit an IOL: three-port pars plana vitrectomy, perfluorocarbon liquid (PFCL) to float the lens off the retina, fragmentation with a fragmatome (or limbal removal of a hard lens), laser to breaks, then a scleral-fixated, glued or iris-claw IOL.

Q How is a nucleus dropped during cataract surgery managed?
The cataract surgeon should not chase it: anterior vitrectomy, an IOL if support allows, watertight wound closure, steroids and IOP control, and referral. Vitrectomy with fragmentation (PFCL to protect the macula) is best within about 2 weeks, sooner for uncontrolled inflammation, IOP or RD; small cortical pieces may be watched.

Q How do you plan surgery for a subluxated cataractous lens?
By clock hours of zonular loss: up to 3, phacoemulsification with an in-the-bag foldable IOL; 3–5, add a capsular tension ring (CTR); 5–7, a Cionni ring sutured to the sclera or capsular tension segments; 7 or more, intracapsular extraction or lensectomy with anterior vitrectomy and a scleral-fixated or anterior chamber IOL.

Q What precautions apply during surgery on a subluxated lens?
Dispersive viscoelastic to plug the dialysis; capsulorhexis started away from it; capsular hooks to support the bag; low flow, vacuum and bottle height ("slow-motion" phacoemulsification); cortex stripped tangentially; little IOL rotation; ready to do an anterior vitrectomy.

Q What does a capsular tension ring do?
An open polymethyl methacrylate ring placed in the bag after capsulorhexis; it spreads force to the intact zonules, keeps the bag round, and reduces decentration and capsular contraction. It is not enough for progressive zonulopathy without scleral fixation, and is contraindicated with a torn rhexis or posterior capsule.

Q How do you treat ectopia lentis in a child with Marfan syndrome?
Mostly optically, with careful refraction (often aphakic correction through the clear zone) and occlusion. Surgery if the edge bisects the pupil and vision is uncorrectable: lensectomy with anterior vitrectomy, aphakic contact lens or later a scleral-fixated IOL; intralenticular aspiration keeps a capsular rim for a later sulcus IOL.

Q What are the anaesthetic risks in homocystinuria and Marfan syndrome?
Homocystinuria: thromboembolism (stroke, pulmonary embolism, myocardial infarction) around general anaesthesia — prefer local anaesthesia, lower homocysteine first (pyridoxine, folate, vitamin B12, betaine), avoid nitrous oxide, hydrate well, compression stockings, early mobilisation. Marfan: aortic dissection and valve disease — cardiology review, continue beta-blockers, avoid hypertension and tachycardia.

Complications and prognosis

Q What are the complications of ectopia lentis?
Amblyopia; glaucoma (pupil block, phacolytic, lens-particle, phacoanaphylactic, angle recession); lens-induced uveitis; corneal decompensation from lens touch; cataract; RD (5–11% in Marfan, 8–38% when ectopia lentis is present); vitreous haemorrhage.

Q What is phacolytic glaucoma?
A hypermature lens, often dislocated, leaks high-molecular-weight protein through an intact capsule; protein-laden macrophages block the trabecular meshwork. Open angle, very high IOP, white or iridescent particles in the aqueous, no keratic precipitates. Lower IOP, then remove the lens.

Q What is phacoanaphylactic uveitis?
A granulomatous immune reaction to lens protein released through a breached capsule (trauma, retained fragments): mutton-fat keratic precipitates, flare, sometimes hypopyon. Treat with steroids and complete removal of lens matter.

Q Why is retinal detachment common, and how do you watch for it?
Axial myopia, vitreous liquefaction, lattice degeneration, and traction from a mobile lens on the vitreous base. Examine the periphery with indentation at each visit, laser any breaks, and repair RD by vitrectomy, often with lensectomy.

Q What is the prognosis?
Vision is good if amblyopia is prevented and RD is caught early. Life expectancy in Marfan depends on the aorta (beta-blockers or losartan, elective root replacement); in homocystinuria on thrombosis control (about half respond to pyridoxine).

Recent advances and favourite examiner questions

Q How is homocystinuria treated?
High-dose pyridoxine (vitamin B6), with folate and vitamin B12; betaine for non-responders; a low-methionine diet with cystine supplements, aiming to keep plasma total homocysteine low. Early treatment reduces lens dislocation, thrombosis and learning difficulty.

Q What newer devices help with weak zonules?
Cionni ring, Ahmed capsular tension segments, capsular anchors, sutureless intrascleral IOL fixation (glued IOL, Yamane flanged technique), and retropupillary iris-claw lenses.

Q What is sulphite oxidase deficiency?
A rare autosomal recessive disorder (alone or as molybdenum cofactor deficiency) with intractable infantile fits, developmental delay and early ectopia lentis; diagnosed by sulphite in fresh urine and raised S-sulphocysteine; the outlook is poor.

Q Why do miotics harm and cycloplegics help in microspherophakia?
Miotics contract the ciliary muscle, relaxing the zonules so the round lens moves forward and tightens the pupil block; cycloplegics relax the muscle, tighten the zonules and flatten and pull the lens back.

Examiner traps

- Saying “dislocated lens” without saying where it lies (anterior chamber or vitreous) and whether it is clear, mobile or inflamed.
- Skipping the general examination and the fellow eye: bilateral non-traumatic ectopia lentis means a systemic cause.
- Giving pilocarpine for pupil block in microspherophakia.
- Planning general anaesthesia in homocystinuria without thrombosis precautions, or Marfan surgery without an echocardiogram.
- Treating the direction of displacement as diagnostic.
- Rushing to remove a clear, quiet lens in the vitreous.
- Not examining the peripheral retina.
- Forgetting to refract through the aphakic zone.

- Subluxation is partial; dislocation is complete.
- Marfan: up and out, zonules stretched, aorta. Homocystinuria: down and in, zonules broken, thrombosis.
- Weill–Marchesani: small round lens, short fingers, pupil block; miotics harm.
- Lens in the anterior chamber is an emergency; a quiet clear lens in the vitreous may be watched.
- Clock hours: under 3, phaco and IOL; 3–5, CTR; 5–7, Cionni ring; over 7, lensectomy with a scleral-fixated or anterior chamber IOL.
- Every non-traumatic case: urine nitroprusside and echocardiography.
- Mnemonic: **“Marfan looks Up, Homocystinuria looks Down.”**

Read more: Baidya p.119–128 · Namrata p.341–344, 414–420, 428–431 · Aravind p.207, 247–248, 309–310

EO5 Ciliary staphyloma

Other · On the list: S.No 30 · ★ Patient already arrived: none

At a glance

- **Staphyloma**: a localised bulge (ectasia) of a thinned, weakened outer coat of the eye — cornea or sclera — **lined by uvea**, which shows through as a bluish-black colour. An ectasia without uveal lining (keratoconus) is not a staphyloma.
- **Ciliary staphyloma**: the bulge lies over the ciliary body, about **2–8 mm behind the limbus**. It is usually the end result of **long-standing raised intraocular pressure (IOP)** — absolute or secondary glaucoma, or buphthalmos — or of **scleritis** or **trauma**.
- Kept because the sign is striking, and the viva covers the types of staphyloma, glaucoma, the blind painful eye and the differential from **ciliary body melanoma**.
- **S.No 30**: "Both eyes ciliary staphyloma (give the clock hours), secondary to long-standing uncontrolled glaucoma (or buphthalmos, or old scleritis — state the cause), with absolute glaucoma (state vision)."
- A **bilateral** staphyloma points to a bilateral cause: late buphthalmos (primary congenital glaucoma), bilateral chronic glaucoma, or necrotising scleritis in rheumatoid arthritis.

What to look for and present

Look for	What to say or record
Large, watering, light-shy eyes in infancy	Buphthalmos: an infant's sclera stretches under pressure
Glaucoma history: drops, laser, surgery, pain, haloes	Long-standing uncontrolled IOP (angle closure, neovascular, uveitic, traumatic glaucoma)
Painful red eyes; joint disease, vasculitis	Necrotising scleritis (rheumatoid arthritis, granulomatosis with polyangiitis); scleromalacia perforans is painless
Injury, eye surgery, infection (herpes zoster, tuberculosis)	Thin scleral scar; surgically induced necrotising scleritis
Pain and vision	A blind painful eye decides management
Vision with projection of light	Often no perception of light (absolute glaucoma)
The bulge: eye, clock hours, distance from limbus, size, shape, colour, vessels	"Bluish-black, lobulated scleral bulge from 10 to 2 o'clock, 2 to 8 mm behind the limbus, thin sclera with uvea showing through, no nodule or sentinel vessel"
Gentle palpation and transillumination	Soft; transmits light (thin wall) — a solid pigmented tumour casts a shadow
Cornea: diameter, Haab's striae, oedema, sensation	Diameter over 12 mm with horizontal striae: buphthalmos; oedema and reduced sensation in absolute glaucoma
Anterior chamber, iris, pupil	Deep chamber in buphthalmos; iris atrophy, ectropion uveae, fixed mid-dilated pupil, new vessels
Lens	Cataract; subluxation from stretched zonules
IOP: rebound tonometer or Tono-Pen; no pressure on the bulge	Usually high; indentation tonometry reads falsely low in a stretched, thin-walled eye
Gonioscopy, if the cornea allows	Angle closure with peripheral anterior synechiae (PAS), recession, dysgenesis
Fundus or B-scan	Total cupping, optic atrophy; retinal or choroidal detachment; any mass
Ultrasound biomicroscopy (UBM)	Thin sclera bowing out with a thin uveal lining and no solid mass
Fellow eye and systemic	Bilateral cause; joints, skin, vasculitis; port-wine stain (Sturge–Weber)

Diagnosis and summary to say

SAY IT

Mr X, a 40-year-old man, has had no useful vision in either eye for several years, with bouts of pain and redness; he was told in childhood that his eyes were large and watery and has used drops only irregularly. Both eyes show bluish-black, lobulated scleral bulges about 2–8 mm behind the limbus, from 9 to 3 o'clock superiorly in the right eye and 10 to 2 o'clock in the left; they transmit light, and no nodule or sentinel vessel is seen. The corneas measure 14 mm with horizontal Haab's striae and mild oedema, the anterior chambers are deep, and the pupils are mid-dilated and fixed with iris atrophy. Vision is no perception of light in the right eye and perception of light with inaccurate projection in the left; IOP is 42 and 38 millimetres of mercury (mmHg) by rebound tonometry. B-scan shows total disc cupping and no intraocular mass in either eye.

Diagnosis (S.No 30): Both eyes ciliary staphyloma secondary to long-standing uncontrolled glaucoma from late buphthalmos (primary congenital glaucoma), with absolute glaucoma in the right eye and near-absolute glaucoma in the left.

If due to scleritis: Both eyes ciliary staphyloma after necrotising anterior scleritis (scleromalacia perforans) in rheumatoid arthritis, with normal IOP.

Viva questions and answers

Basics

Q Define staphyloma. How does it differ from ectasia?
A staphyloma is a localised bulge of a thinned, weakened outer coat (cornea or sclera) lined by uveal tissue, which shows through as bluish-black. An ectasia is a bulge of the outer coat without uveal lining, such as keratoconus or keratoglobus. Both form when a weak wall yields to normal or raised IOP.

Q Classify staphylomas by site.
Anterior (corneal): an ectatic corneal scar lined by iris after a perforated ulcer. Intercalary: at the limbus, lined by the iris root. Ciliary: 2–8 mm behind the limbus, lined by ciliary body. Equatorial: at the equator, lined by choroid, where the vortex veins leave the eye. Posterior: behind the equator, usually at the posterior pole in pathological myopia.

Q Why does a staphyloma look blue?
The thin, stretched sclera lets the dark uveal pigment show through, and light scattered by the thin collagen adds a bluish tint.

Q What causes ciliary staphyloma?
Long-standing raised IOP: absolute glaucoma after chronic angle closure, neovascular, uveitic or traumatic glaucoma; buphthalmos. Scleritis: necrotising scleritis, scleromalacia perforans. Trauma: perforating injury, scleral scar. Surgery: surgically induced necrotising scleritis, a thin scleral flap. Rarely herpes zoster or tuberculous scleritis.

Q Why does the ciliary region give way?
The sclera is thinnest (about 0.3 mm) just behind the rectus insertions, close to this zone; the anterior ciliary vessels pierce it here, leaving weak canals; and this zone is the usual site of anterior scleritis. In an infant the whole sclera is distensible.

Q Where does the ciliary body lie on the surface?
From the scleral spur, just behind the limbus, to the ora serrata about 6 mm (nasal) to 7 mm (temporal) behind the limbus: the pars plicata in front (about 2 mm) and the pars plana behind (about 4 mm). A bulge 2–8 mm behind the limbus therefore overlies it.

Q How do ciliary and intercalary staphylomas differ?
An intercalary staphyloma sits at the limbus, lined by the iris root, usually after a peripheral corneal perforation or limbal wound. A ciliary staphyloma lies further back over the ciliary body, usually from raised IOP or scleritis.

Examination and signs

Q How would you describe a staphyloma?
Eye; site in clock hours; distance from the limbus; size; shape (single, multiple, lobulated, ring-shaped); colour; surface vessels; consistency on gentle touch; transillumination; relation to the rectus insertions; and the associated corneal, IOP and fundus findings.

Q How do you measure IOP in such an eye?
Applanation if the cornea is regular; otherwise a rebound tonometer, Tono-Pen, or digital estimate. Avoid Schiotz indentation and pressure on the thin area: it risks rupture, and low scleral rigidity makes indentation readings falsely low.

Q What does transillumination show?
In a dark room a bright light on the globe or through the pupil passes through a thin staphyloma, which glows. A ciliary body melanoma or other solid pigmented mass blocks it and casts a shadow, often larger than the visible lesion.

Q What are the signs of absolute glaucoma?
No perception of light; stony-hard eye; dilated circumcorneal vessels; corneal oedema with reduced sensation; iris atrophy and ectropion uveae; fixed, mid-dilated pupil; glaucomatous cataract; total cupping. Late results: ciliary and equatorial staphyloma, bullous keratopathy, corneal ulcer, endophthalmitis, phthisis.

Q Which signs point to buphthalmos as the cause?
Corneal diameter over 12 mm in infancy (often 14 mm or more), horizontal or concentric Haab's striae, deep anterior chamber, thin bluish sclera, axial myopia, subluxated lens, cupped disc, and a history of watering and photophobia in infancy.

Investigations

Q Which investigations would you do?
B-scan (cupping, detachment, intraocular mass, axial length); UBM (scleral thickness, uveal lining, ciliary body, angle); gonioscopy if possible; fields if vision allows. For scleritis: blood count, erythrocyte sedimentation rate, C-reactive protein, rheumatoid factor, anti-cyclic citrullinated peptide, antineutrophil cytoplasmic antibody (ANCA), syphilis serology, chest X-ray. Magnetic resonance imaging if a tumour is suspected.

Q How do ultrasound and UBM separate a staphyloma from a ciliary body melanoma?
Staphyloma: thin sclera bowing outward with a thin uveal lining and an echo-free cavity continuous with the eye; no mass. Melanoma: a solid dome or mushroom-shaped mass with low-to-medium internal reflectivity, acoustic hollowness and choroidal excavation, sometimes with extrascleral extension.

Q What is the differential diagnosis of a bluish-black scleral bulge or patch?
Ciliary body melanoma with extrascleral spread (sentinel vessels, sector cataract); scleral thinning after scleritis or scleromalacia perforans (flat unless IOP is high); uveal prolapse after injury or surgery; intercalary or equatorial staphyloma; ocular melanocytosis (flat, slate-grey); Axenfeld nerve loop; senile scleral plaque (flat, grey, near the horizontal recti); blue sclera (diffuse); conjunctival naevus or melanoma (moves with the conjunctiva).

Q How do you tell scleromalacia perforans from a staphyloma?
Scleromalacia perforans is a painless necrotising scleritis in older women with long-standing rheumatoid arthritis: a yellow necrotic plaque separates and leaves bare, dark uvea, usually flat. It bulges only if IOP rises — then it has become a staphyloma.

Management

Q What are the principles of management?
Find and treat the cause; control IOP to stop further stretching; protect the eye from rupture with polycarbonate glasses; relieve pain; reinforce the sclera if perforation threatens; give cosmetic rehabilitation; remove the eye only if it is blind and painful, or if a tumour cannot be excluded.

Q How would you lower the IOP?
Aqueous suppressants: timolol 0.5% twice daily, dorzolamide 2% three times daily, brimonidine 0.2% three times daily (not under 2 years); oral acetazolamide 250 mg four times daily for short periods; intravenous mannitol 20% 1–2 g/kg for an acute rise; a prostaglandin analogue (latanoprost 0.005% at night). Topical atropine 1% and prednisolone acetate 1% for comfort in an inflamed blind eye.

Q Which glaucoma surgery suits an eye with a ciliary staphyloma?
In a blind or near-blind eye, cyclodestruction. Trabeculectomy over thin sclera risks perforation and bleb failure, and pressure swings may enlarge the staphyloma. If the eye still has useful vision, a glaucoma drainage device placed away from the thin zone, with a patch graft over the tube, is safer.

Q How is transscleral cyclophotocoagulation done here?
Diode laser (810 nm), contact probe about 1.2 mm behind the limbus; 1.5–2 s at 1,500–2,000 mW; 12–24 burns, sparing 3 and 9 o'clock (long posterior ciliary vessels and nerves). Find the ciliary body by transillumination because the anatomy is distorted, and do not fire over the thinned bulge. Steroid and cycloplegic drops afterwards.

Q What about cyclocryotherapy?
A 3.5 mm probe at about −60 to −80 °C placed over the ciliary body, usually 180° at a time sparing 3 and 9 o'clock; effective but painful, with more inflammation, hypotony and phthisis than laser. It is now used mainly when laser is unavailable.

Q When is a scleral patch graft indicated?
Impending or actual perforation, or a thin, enlarging staphyloma in a seeing eye: donor sclera, processed pericardium, fascia lata or cornea is sutured over the thin area after IOP control. In scleritis, control the inflammation with systemic immunosuppression first.

Q How do you manage a blind painful eye with a staphyloma?
Atropine, steroid and aqueous suppressant drops; cyclodestruction; retrobulbar injection of absolute alcohol (about 1 ml, after local anaesthetic) or chlorpromazine for pain; finally evisceration or enucleation with an orbital implant and a prosthesis.

Q Evisceration or enucleation?
Evisceration (contents removed, sclera and muscles kept) gives better motility and is simpler, but must never be done if an intraocular tumour has not been excluded, and a very thin sclera may need partial excision. Enucleation is chosen when a tumour is possible or tissue is needed; place an implant (silicone, polymethyl methacrylate, hydroxyapatite or porous polyethylene) and fit the prosthesis at about 6 weeks.

Q What is a cosmetic shell?
A thin, custom-painted scleral shell worn over a blind, disfigured but comfortable eye to match the fellow eye. A large staphylomatous globe may not fit one, and the shell may irritate; such eyes often need evisceration first.

Complications and prognosis

Q What complications can follow?
Rupture after trivial trauma, with uveal prolapse or expulsive haemorrhage; sympathetic ophthalmia after rupture or surgery; enlargement; pain; exposure keratopathy and corneal ulcer; endophthalmitis; phthisis bulbi; disfigurement.

Q What is the prognosis?
Visual prognosis is poor: a staphyloma signals years of uncontrolled IOP with optic atrophy, so the aim is a comfortable, safe and acceptable-looking eye. After scleritis, the prognosis depends on control of the systemic vasculitis, which can be life-threatening if untreated.

Q What is sympathetic ophthalmia, and why does it matter here?
A bilateral granulomatous panuveitis following penetrating injury or surgery to one eye, from autoimmunity to uveal antigens; a ruptured staphyloma with uveal prolapse can trigger it. Treat with systemic steroids and immunosuppressants; enucleating a hopeless injured eye within about 2 weeks of injury is the classic preventive step.

Recent advances and favourite examiner questions

Q What is micropulse transscleral cyclophotocoagulation?
Diode energy delivered in short repeated "on" pulses (duty cycle about 31%) with a sweeping probe, so tissue cools between pulses; it causes less inflammation, hypotony and phthisis than continuous-wave treatment and suits eyes with some vision.

Q What is endoscopic cyclophotocoagulation?
Laser ablation of the ciliary processes under direct endoscopic view, through a limbal or pars plana route, often combined with cataract surgery; it spares the overlying sclera.

Q What is the treatment of an anterior (corneal) staphyloma?
In a blind eye, evisceration or enucleation with a prosthesis; staphylectomy (excision with closure) is historical. In an eye with potential, keratoplasty after IOP control.

Q Where else in the course do staphylomas appear?
Posterior staphyloma in pathological myopia (with lacquer cracks and myopic maculopathy); peripapillary staphyloma among excavated disc anomalies; staphyloma with limbal dermoids and in congenital corneal opacity; and as a complication of absolute glaucoma and necrotising scleritis.

Examiner traps

- Calling it "scleral thinning" without describing the bulge: a staphyloma is an ectasia **lined by uvea**.
- Not excluding ciliary body melanoma: transilluminate, and do B-scan or UBM before any surgery; never eviscerate an eye whose contents are unknown.
- Pressing on the globe or using a Schiotz tonometer.
- Recommending trabeculectomy over thin sclera, or firing laser over the bulge.
- Missing the cause: measure the cornea and look for Haab's striae; ask about rheumatoid arthritis and scleritis.
- Forgetting protective glasses and the fellow eye.

- Staphyloma: bulge of a thinned coat lined by uvea; ectasia has no uveal lining.
- Sites from front to back: anterior, intercalary, ciliary (2–8 mm, ciliary body), equatorial (vortex veins), posterior (myopia).
- Causes: glaucoma (absolute, buphthalmos), scleritis, trauma, surgery.
- Transillumination: a staphyloma glows; a melanoma casts a shadow.
- Treat: lower IOP, cyclodestruction, patch graft, protect; blind and painful: retrobulbar alcohol, evisceration or enucleation, shell.
- Mnemonic for the sites: **"All Ink Comes Evenly Packed"** — Anterior, Intercalary, Ciliary, Equatorial, Posterior.

Read more: Baidya p.101–105, 169–172, 441 · Namrata p.188–190 · Aravind p.92–93, 207, 225–227